

## Lenovo ThinkSystem SR650 V4 Server

### Product Guide

The Lenovo ThinkSystem SR650 V4 is an ideal 2-socket 2U rack server for customers that need industry-leading reliability, management, and security, as well as maximizing performance and flexibility for future growth. The SR650 V4 is based on two Intel Xeon 6700-series or Xeon 6500-series processors, with Performance-cores (P-cores), formerly codenamed "Granite Rapids-SP".

The SR650 V4 is designed to handle a wide range of workloads, such as databases, virtualization and cloud computing, virtual desktop infrastructure (VDI), infrastructure security, systems management, enterprise applications, collaboration/email, streaming media, web, and HPC.



Figure 1. Lenovo ThinkSystem SR650 V4 with optional security bezel

[360° View](#)
[Full 3D Tour](#)

### Did you know?

The SR650 V4 server supports a variety of cooling options including the new Lenovo Compute Complex Neptune Core module which uses open-loop liquid cooling to remove the heat from processors, memory, and voltage regulators. This Neptune Core module can remove more than 80% of heat from the server using liquid cooling, resulting in up to 35.9% power savings at the data center level. See the [Lenovo Compute Complex Neptune Core Module](#) section for details.

For additional GPU support in a 2U rack form factor, with support for 4 double-wide GPUs with NVLink, consider the [ThinkSystem SR650a V4 server](#).

## Key features

Combining performance and flexibility, the SR650 V4 server is a great choice for enterprises of all sizes. The server offers a broad selection of drive and slot configurations and offers numerous high performance features. Outstanding reliability, availability, and serviceability (RAS) and high-efficiency design can improve your business environment and can help save operational costs.

## Scalability and performance

The ThinkSystem SR650 V4 offers numerous features to boost performance, improve scalability and reduce costs:

- Supports one or two Intel Xeon 6700-series or 6500-series processors with Performance-cores (P-cores)
  - Up to 86 cores and 172 threads
  - Core speeds of up to 4 GHz
  - TDP ratings of up to 350 W
- Support for DDR5 memory DIMMs to maximize the performance of the memory subsystem:
  - Up to 32 DDR5 memory DIMMs, 16 DIMMs per processor
  - 8 memory channels per processor (2 DIMMs per channel)
  - Supports 1 DIMM per channel operating at 6400 MHz
  - Supports 2 DIMMs per channel operating at 5200 MHz
  - Using 256GB 3D RDIMMs, the server supports up to 8TB of system memory
- Support for MRDIMMs for increased memory bandwidth with memory bus speeds of up to 8000 MHz.
- Support for Compute Express Link (CXL) memory modules in an E3.S 2T form factor, up to 12x CXL modules with two processors. With CXL 2.0 for next-generation workloads, you can reduce compute latency in the data center and lower TCO. CXL is a protocol that runs across the standard PCIe physical layer and can support both standard PCIe devices as well as CXL devices on the same link.
- Supports up to 10x single-width GPUs or 2x double-wide GPUs, for substantial processing power in a 2U system. For support for 4x double-wide GPUs, consider the [ThinkSystem SR650a V4 server](#).
- Supports up to 40x 2.5-inch drive bays, by using combinations of front-accessible (up to 24 hot-swap bays), mid bays (8 simple-swap bays) and rear-accessible (8 hot-swap bays).
- Supports 36x NVMe drives without oversubscription of PCIe lanes (1:1 connectivity). The use of NVMe drives maximizes drive I/O performance, in terms of throughput and latency.
- Supports up to 16x 3.5-inch drive bays for lower-cost high-capacity HDD storage. 2.5-inch and 3.5-inch drive bays can be mixed if desired.
- Support for up to 32x E3.S 1T NVMe drives, a new form factor drive for high-density and high-performance storage.
- Supports high-speed RAID controllers providing 12 Gb SAS connectivity to the drive backplanes. A variety of PCIe 3.0 and PCIe 4.0 RAID adapters are available.
- Supports M.2 drives for convenient operating system boot functions. M.2 drives can be internally mounted or can be mounted at the front or rear of the server as hot-swap drives. Optional RAID-0 or RAID-1.
- Up to 10x PCIe slots (rear accessible), plus two slots dedicated to OCP 3.0 adapters. 2.5-inch drive configurations also support an additional internal bay for a cabled RAID adapter or HBA.
- The server has up to two dedicated industry-standard OCP 3.0 slots supporting a variety of Ethernet network adapters. A simple-swap mechanism with a thumbscrew and pull-tab enables tool-less installation and removal of the adapter. The adapter supports shared BMC network sideband connectivity to enable out-of-band systems management.

- The server offers PCI Express 5.0 I/O expansion capabilities that doubles the theoretical maximum bandwidth of PCIe 4.0 (32GT/s in each direction for PCIe Gen 5, compared to 16 GT/s with PCIe Gen 4 and 8 GT/s with PCIe Gen 3). A PCIe 5.0 x16 slot provides 128 GB/s bandwidth, enough to support a dual-port 200GbE network connection or a single-port 400GbE connection.

### **Availability and serviceability**

The SR650 V4 provides many features to simplify serviceability and increase system uptime:

- Designed to run 24 hours a day, 7 days a week
- The server offers Single Device Data Correction (SDDC, also known as Chipkill), Adaptive Double-Device Data Correction (ADDDC, also known as Redundant Bit Steering or RBS), and memory mirroring for redundancy in the event of a non-correctable memory failure.
- The server offers hot-swap drives, supporting RAID redundancy for data protection and greater system uptime.
- Available M.2 boot adapters support RAID-1 (using onboard hardware RAID or using Intel VROC) which can enable two M.2 drives to be configured as a redundant pair.
- The server has up to two hot-swap redundant power supplies and up to six hot-swap redundant fans to provide availability for business-critical applications.
- The light path diagnostics feature uses LEDs to lead the technician to failed (or failing) components, which simplifies servicing, speeds up problem resolution, and helps improve system availability.
- Solid-state drives (SSDs) offer more reliability and performance than traditional mechanical HDDs for greater uptime.
- Proactive Platform Alerts (including PFA and SMART alerts): Processors, voltage regulators, memory, internal storage (SAS/SATA HDDs and SSDs, NVMe SSDs, M.2 storage), fans, power supplies, RAID controllers, server ambient and subcomponent temperatures. Alerts can be surfaced through the XClarity Controller to managers such as Lenovo XClarity One, VMware vCenter, and Microsoft System Center. These proactive alerts let you take appropriate actions in advance of possible failure, thereby increasing server uptime and application availability.
- The built-in XClarity Controller continuously monitors system parameters, triggers alerts, and performs recovery actions in case of failures to minimize downtime.
- Built-in diagnostics in UEFI, using Lenovo XClarity Provisioning Manager, speed up troubleshooting tasks to reduce service time.
- Lenovo XClarity Provisioning Manager supports diagnostics and can save service data to a USB key drive or remote CIFS share folder for troubleshooting and reduce service time.
- Auto restart in the event of a momentary loss of AC power (based on power policy setting in the XClarity Controller service processor)
- Offers a diagnostics port on the front of the server to allow you to attach an external diagnostics handset for enhanced systems management capabilities.
- Support for the XClarity Mobile app running on a supported smartphone or tablet and connected to the server through the service-enabled USB port, enables additional local systems management functions.
- Three-year or one-year customer-replaceable unit and onsite limited warranty (varies by geography), 9 x 5 next business day. Optional service upgrades are available.

### **Manageability and security**

Systems management features simplify local and remote management of the SR650 V4:

- The server includes XClarity Controller 3 (XCC3) to monitor server availability. Optional upgrade to XCC3 Premier to provide remote control (keyboard video mouse) functions, support for the mounting of remote media files (ISO and IMG image files), boot capture and power capping. XCC3 Premier also

offers additional features such as Neighbor Groups, System Guard, a CNSA-compliant security mode, a FIPS 140-3-compliant mode, and enhanced NIST 800-193 support.

- Dedicated Ethernet port at the rear of the server for remote management (BMC management).
- Lenovo XClarity One offers comprehensive hardware management tools that help to increase uptime, reduce costs and improve productivity through advanced server management capabilities.
- UEFI-based Lenovo XClarity Provisioning Manager, accessible from F1 during boot, provides system inventory information, graphical UEFI Setup, platform update function, RAID Setup wizard, operating system installation function, and diagnostic functions.
- Support for Lenovo XClarity Energy Manager which captures real-time power and temperature data from the server and provides automated controls to lower energy costs.
- An integrated industry-standard Unified Extensible Firmware Interface (UEFI) enables improved setup, configuration, and updates, and simplifies error handling.
- Support for industry standard management protocols, IPMI 2.0, SNMP 3.0, Redfish REST API, serial console via IPMI
- Enhanced security for storage connectivity with Fibre Channel adapters such as the Emulex SecureHBA, which provides autonomous in-flight encryption with integrated post-quantum cryptography (PQC) algorithms and CNSA 1.0 and 2.0 compliance.
- An integrated hardware Trusted Platform Module (TPM) supporting TPM 2.0 enables advanced cryptographic functionality, such as digital signatures and remote attestation.
- Administrator and power-on passwords help protect from unauthorized access to the server.
- Supports Secure Boot to ensure only a digitally signed operating system can be used. Supported with HDDs and SSDs, as well as M.2 drives.
- Industry-standard Advanced Encryption Standard (AES) NI support for faster, stronger encryption.
- Intel Execute Disable Bit functionality can prevent certain classes of malicious buffer overflow attacks when combined with a supported operating system.
- Intel Trusted Execution Technology provides enhanced security through hardware-based resistance to malicious software attacks, allowing an application to run in its own isolated space, protected from all other software running on a system.
- Additional physical security features are an available chassis intrusion switch and available lockable front bezel.

## **Energy efficiency**

The SR650 V4 offers the following energy-efficiency features to save energy, reduce operational costs, and increase energy availability:

- The server supports advanced Lenovo Neptune Core direct-water cooling (DWC) capabilities, where heat from key components is removed from the rack and the data center using an open loop and coolant distribution units, resulting in lower energy costs:
  - Processor Neptune Core Module uses liquid cooling to remove heat from the processors
  - Compute Complex Neptune Core Module uses liquid cooling to remove heat from the processors, memory and voltage regulators
- Energy-efficient system board components help lower operational costs.
- High-efficiency power supplies with 80 PLUS Titanium certifications
- Solid-state drives (SSDs) consume as much as 80% less power than traditional spinning 2.5-inch HDDs.
- Support for Lenovo XClarity Energy Manager provides advanced data center power notification, analysis, and policy-based management to help achieve lower heat output and reduced cooling needs.

- The server uses hexagonal ventilation holes, which can be grouped more densely than round holes, providing more efficient airflow through the system and thus keeping your system cooler.

## Performance benchmarks

At Lenovo, we know customers care most about how their workloads perform in real-world environments, which is why we carefully tune our server hardware, firmware, and drivers to help deliver strong benchmark results that reflect practical business needs. The goal is simple: to help customers run their critical applications more efficiently and achieve better outcomes.

Lenovo performance engineers have achieved world record performance benchmarks on many ThinkSystem servers. See the complete list of current world records in the Lenovo Press article [Performance Benchmark World Record Summary for Lenovo ThinkSystem Servers](#).

The Lenovo Server Performance Team also publishes reports on key performance world records at the time the world record is announced. See all published [Performance Benchmark Reports](#).

## Comparing the SR650 V4 to the SR650 V3

The ThinkSystem SR650 V4 improves on the previous generation SR650 V3, as summarized in the following table.

Table 1. Comparing the SR650 V4 to the previous generation SR650 V3

| Feature   | SR650 V3                                                                                                                                                                                                                                                                       | SR650 V4                                                                                                                                                                                                                                                                                                                                                                    | Benefits                                                                                                                                                                                                                                                                                                        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Processor | <ul style="list-style-type: none"> <li>• 2x 4th Gen Intel Xeon Scalable Processors</li> <li>• Up to 60 cores</li> <li>• TDP up to 350W</li> <li>• 80x PCIe 5.0 lanes per processor</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>• 2x Intel Xeon 6700 or 6500-series processors with Performance cores (P-cores)</li> <li>• up to 86 cores and 172 threads</li> <li>• TDP up to 350W</li> <li>• 88x PCIe 5.0 lanes per processor</li> </ul>                                                                                                                           | <ul style="list-style-type: none"> <li>• Significant increase in cores per processor</li> <li>• Increased performance</li> <li>• Consolidation of more apps on same number of servers, reducing costs</li> <li>• Increased number of lanes means more higher performance networking and NVMe storage</li> </ul> |
| Memory    | <ul style="list-style-type: none"> <li>• DDR5 memory operating up to 4800 MHz</li> <li>• 8 channels per CPU</li> <li>• 32 DIMMs (16 per processor), 2 DIMMs per channel</li> <li>• Supports RDIMMs, 3DS RDIMMs and 9x4 RDIMMs</li> <li>• Up to 8TB of system memory</li> </ul> | <ul style="list-style-type: none"> <li>• DDR5 memory operating up to 6400 MHz</li> <li>• 8 channels per CPU</li> <li>• 32 DIMMs (16 per processor), 2 DIMMs per channel</li> <li>• Supports RDIMMs, 3DS RDIMMs, and MRDIMMs</li> <li>• Supports CXL 2.0 memory in E3.S 2T form factor, up to 12x DIMMs with two processors</li> <li>• Up to 8TB of system memory</li> </ul> | <ul style="list-style-type: none"> <li>• Support for new memory technologies</li> <li>• 1.7x increased memory bandwidth</li> <li>• 2.3x increased memory bandwidth with MRDIMM</li> <li>• Support for CXL memory enables additional memory support using E3.S drive bays</li> </ul>                             |

| Feature          | SR650 V3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SR650 V4                                                                                                                                                                                                                                                                                                                                                                                                                                    | Benefits                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internal storage | <ul style="list-style-type: none"> <li>Up to 20x 3.5", 40x 2.5" drive bays</li> <li>Front: up to 12x3.5" / 24x2.5", hot-swap</li> <li>Mid (internal): up to 4x3.5" / 8x2.5", hot-swap</li> <li>Rear: up to 4x3.5" / 8x2.5", 2x7mm boot drive, hot-swap</li> <li>2x Internal M.2 with optional RAID 1</li> <li>2x rear 7mm hot swap SAS/SATA/NVMe (RAID support via VROC)</li> <li>PCIe 4.0 and PCIe 5.0 NVMe drive support</li> <li>20x Onboard NVMe ports</li> <li>NVMe Retimer adapters for 32x NVMe</li> </ul> | <ul style="list-style-type: none"> <li>Up to 16x 3.5", 40x 2.5" drive bays, 32x E3.S</li> <li>Front: up to 12x3.5" / 24x2.5"/ 32x E3.S, hot-swap</li> <li>Mid (internal): 8x2.5", simple swap</li> <li>Rear: up to 4x3.5" / 8x2.5", hot-swap</li> <li>2x Internal M.2 with optional RAID 1</li> <li>2x front or rear hot-swap M.2 with RAID 1</li> <li>PCIe 5.0 NVMe drive support</li> <li>36x Onboard NVMe ports</li> </ul>               | <ul style="list-style-type: none"> <li>Flexible storage offerings</li> <li>Up to 32x NVMe drives for high-performance storage</li> <li>Support E3.S NVMe drives up to 32</li> <li>M.2 for effective boot drive support</li> <li>Support rear or front hot swap M.2 drives for increased uptime</li> <li>Additional NVMe ports means no need for Retimer adapters, freeing up slots for other adapters</li> </ul> |
| RAID             | <ul style="list-style-type: none"> <li>8-, 16- and 32-port RAID adapters with up to 8GB flash</li> <li>Support for Lenovo and Broadcom adapters</li> <li>Support for PCIe or Internal cabled (CFF) form factor adapters</li> <li>Support for NVMe drives connected to 940 RAID adapters (Tri-Mode)</li> <li>Storage HBAs available</li> <li>PCIe 3.0 and PCIe 4.0 adapter choices with support for Gen 5 adapters when available</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>8-, 16- and 32-port RAID adapters with up to 8GB flash</li> <li>Support for Lenovo and Broadcom adapters</li> <li>Support for PCIe or Internal cabled (CFF) form factor adapters</li> <li>Support for NVMe drives connected to 940 RAID adapters (Tri-Mode)</li> <li>Storage HBAs available</li> <li>PCIe 3.0 and PCIe 4.0 adapter choices with support for Gen 5 adapters when available</li> </ul> | <ul style="list-style-type: none"> <li>Consistent RAID/HBA support</li> <li>Flexible config solution</li> <li>PCIe Gen 5 allows for greater storage performance</li> </ul>                                                                                                                                                                                                                                       |
| Networking       | <ul style="list-style-type: none"> <li>1x OCP 3.0 slot with PCIe Gen 5 x16 interface</li> <li>OCP slot can be rear or front-accessible</li> <li>Additional PCIe adapters supported</li> <li>1GbE dedicated Management port</li> </ul>                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>2x OCP 3.0 rear-accessible slots with PCIe Gen 5 interface</li> <li>Additional PCIe adapters supported</li> <li>1GbE dedicated Management port</li> </ul>                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>2x OCP slots for increased networking ports</li> </ul>                                                                                                                                                                                                                                                                                                                    |
| PCIe             | <ul style="list-style-type: none"> <li>Supports PCIe 5.0</li> <li>Up to 10x rear PCIe slots (6x full height, 4x low-profile) or 8x rear PCIe slots (all full-height slots)</li> <li>Up to 9x slots can be PCIe 5.0</li> <li>Optional 2x additional front PCIe slots</li> <li>Supports a RAID/HBA in CFF form factor (does not occupy a PCIe slot)</li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li>Supports PCIe 5.0</li> <li>Up to 10x rear PCIe slots (6x full height, 4x low-profile)</li> <li>All slots are PCIe 5.0</li> <li>Riser cards are modular for easy upgrades</li> <li>Supports a RAID/HBA in CFF form factor (does not occupy a PCIe slot)</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>PCIe Gen 5 allows for greater I/O performance</li> <li>Flexible PCIe offerings</li> <li>More PCIe connections for greater PCIe/NVMe support</li> <li>Modular riser card design minimizes initial investment if fewer slots are needed</li> </ul>                                                                                                                          |

| Feature                 | SR650 V3                                                                                                                                                                                                                                                                                                           | SR650 V4                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Benefits                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPU support             | <ul style="list-style-type: none"> <li>Supports up to 8x single-wide GPUs</li> <li>Supports up to 3x double-wide GPUs</li> </ul>                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Supports up to 10x single-wide GPUs</li> <li>Supports up to 2x double-wide GPUs</li> </ul>                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>High performance GPU support</li> </ul>                                                                                                                                                                                                                                                                                                                     |
| Management and security | <ul style="list-style-type: none"> <li>Integrated XClarity Controller 2</li> <li>Support for full XClarity toolset</li> <li>Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT)</li> <li>Tamper Switch security solution (intrusion switch)</li> <li>Support for 2nd remote management port</li> </ul> | <ul style="list-style-type: none"> <li>Integrated XClarity Controller 3</li> <li>Support for full XClarity toolset</li> <li>Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT)</li> <li>Tamper Switch security solution (intrusion switch)</li> <li>Support for 4-in-1 daisy-chain remote management connectivity</li> <li>For Fibre Channel storage connectivity, available in-flight encryption adapters such as Emulex SecureHBAs</li> </ul> | <ul style="list-style-type: none"> <li>New XCC3 offers improved management capabilities including OpenBMC support</li> <li>Same system management tool with previous generation</li> <li>Silicon-level security solution</li> <li>Daisy-chain option to reduce the number of management switch ports required</li> <li>Storage connectivity encryption ensures that data remains secure</li> </ul> |
| Power                   | <ul style="list-style-type: none"> <li>Choice of 750W, 1100W, 1800W, 2400W, 2600W AC Hot Plug PSUs</li> <li>Available in Titanium and Platinum efficiency levels</li> <li>1100W -48VDC Platinum general support</li> <li>240V HVDC support for PRC customers</li> <li>Active-Standby mode</li> </ul>               | <ul style="list-style-type: none"> <li>Choice of 800W, 1300W, 2000W, 2700W, 3200W AC Hot Plug PSUs</li> <li>Available in Titanium and Platinum efficiency levels</li> <li>1300W -48VDC general support</li> <li>1300W HVAC/HVDC general support</li> <li>240V HVDC support for PRC customers</li> <li>Active-Standby mode</li> </ul>                                                                                                                         | <ul style="list-style-type: none"> <li>Higher capacity power supplies to support high-powered components</li> <li>Support Industrial CRPS form factor PSUs</li> <li>Support for Telco customers with -48V requirements</li> <li>Support for customers with HVDC/HVAC power requirements</li> </ul>                                                                                                 |

## Components and connectors

The following figure shows the front of the SR650 V4.

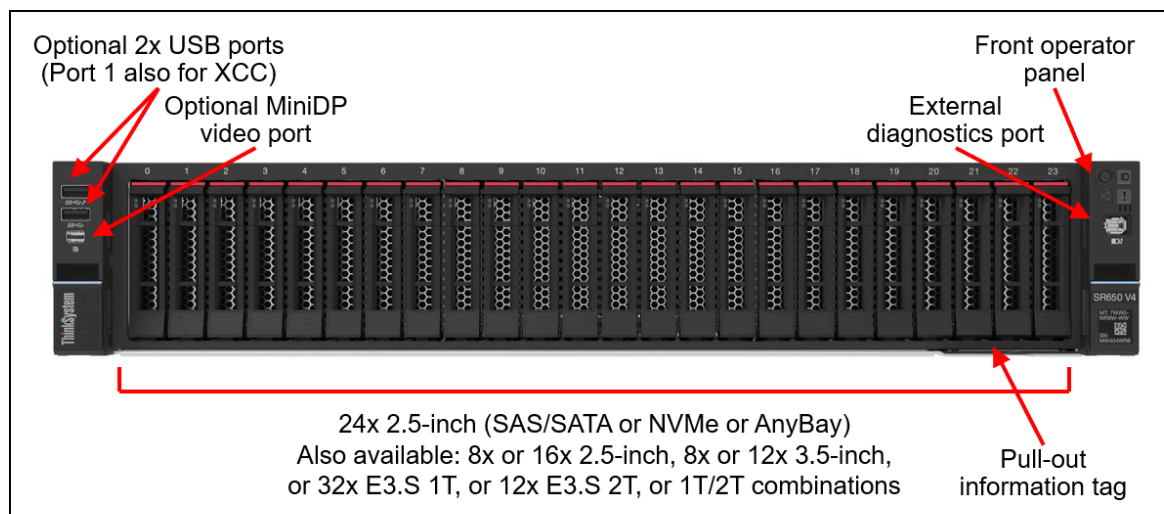


Figure 2. Front view of the ThinkSystem SR650 V4 with 2.5-inch drive bays

For details on the front ports, including the optional front USB and Mini DP video port, see the [Local management](#) section.

The following figure shows the front configurations of the SR650 V4. The server supports either 2.5-inch hot-swap drives (8, 16 or 24 drive bays) or 3.5-inch hot-swap drives (8 or 12 bays) at the front. The server also supports up to 32 E3.S 1T hot-swap drive bays or up to 12 E3.S 2T bays for CXL memory.

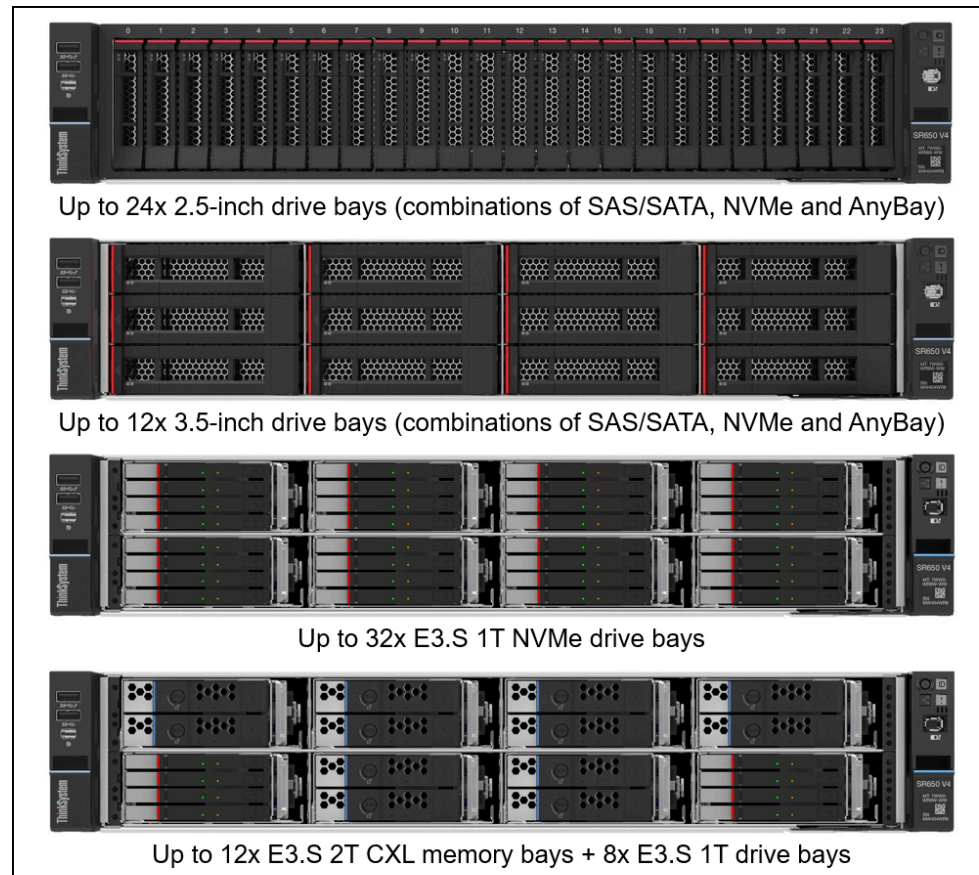


Figure 3. Front view configurations of the ThinkSystem SR650 V4

The following figure shows the components visible from the rear of the server. The figure shows one configuration, with ten PCIe slots, however there are additional rear configurations which include 3.5-inch drive bays or 2.5-inch drive bays, and rear-accessible hot-swap M.2 drive bays.

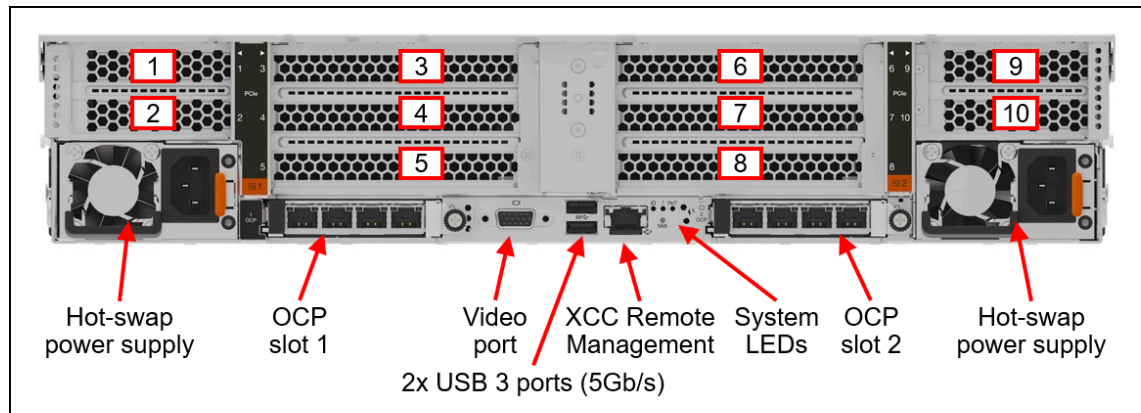


Figure 4. Rear view of the ThinkSystem SR650 V4 (configuration with ten PCIe slots)

The following figure shows the rear of the server when configured with open-loop cooling, either using the Lenovo Processor Neptune Core Module or the Lenovo Compute Complex Neptune Core Module. The inlet and outlet water hoses are routed through slot 8.

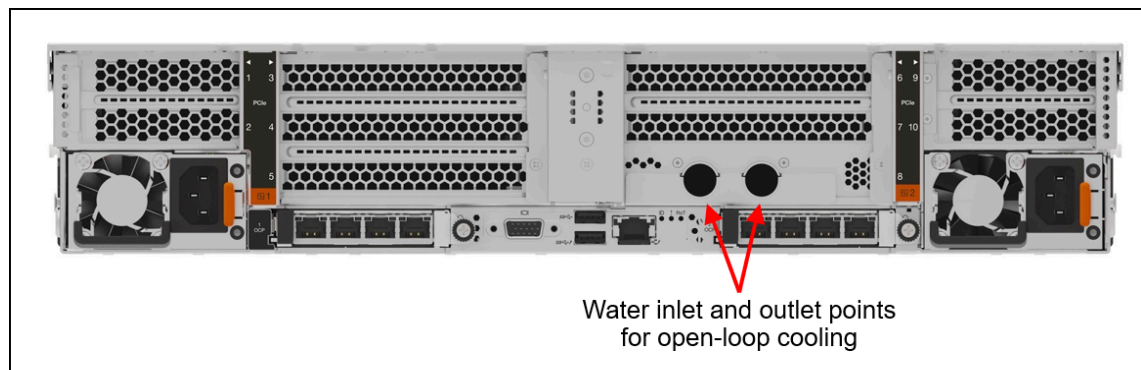


Figure 5. Rear of the SR650 V4 with open-loop water cooling

The following figure shows the locations of key components inside the server when configured with 4x 2.5-inch rear drive bays.

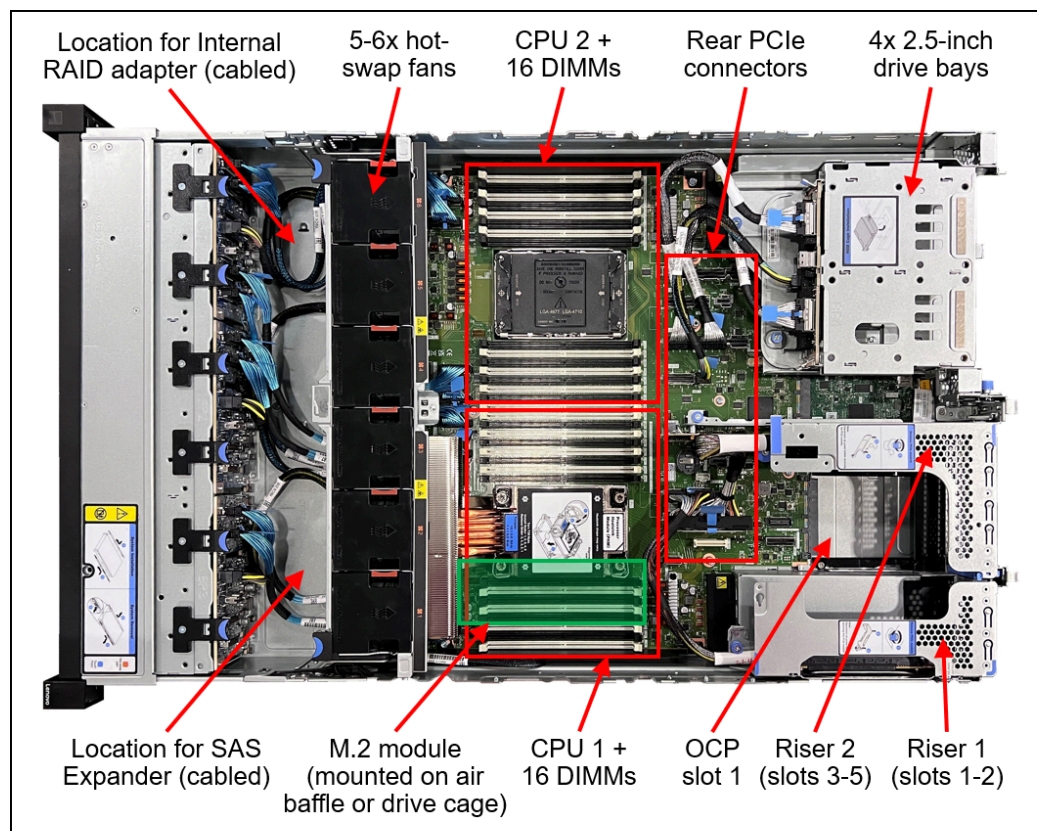


Figure 6. Internal view of the ThinkSystem SR650 V4

## System architecture

The following figure shows the architectural block diagram of the SR650 V4, showing the major components and their connections.

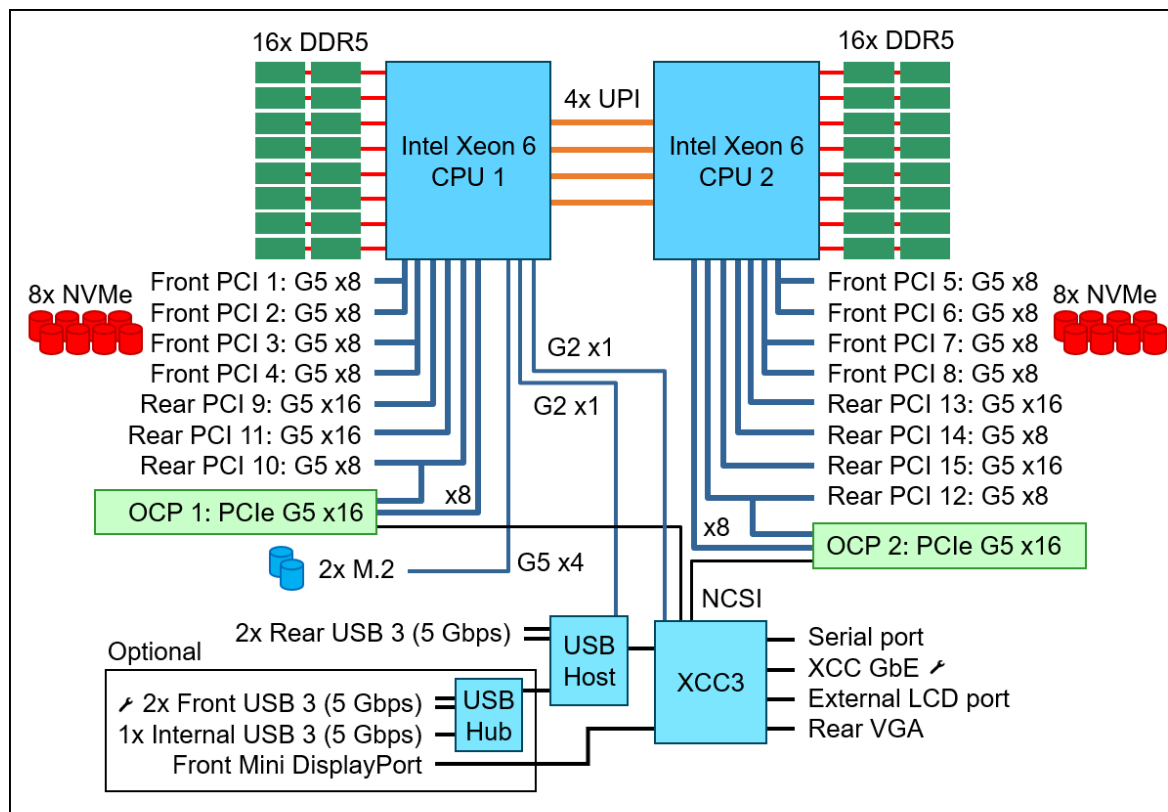


Figure 7. SR650 V4 system architectural block diagram

Most of the PCIe connections are implemented using cables, which maximizes the flexibility in how the server can be configured. For example, the block diagram below shows how a configuration with 10 rear slots is connected.

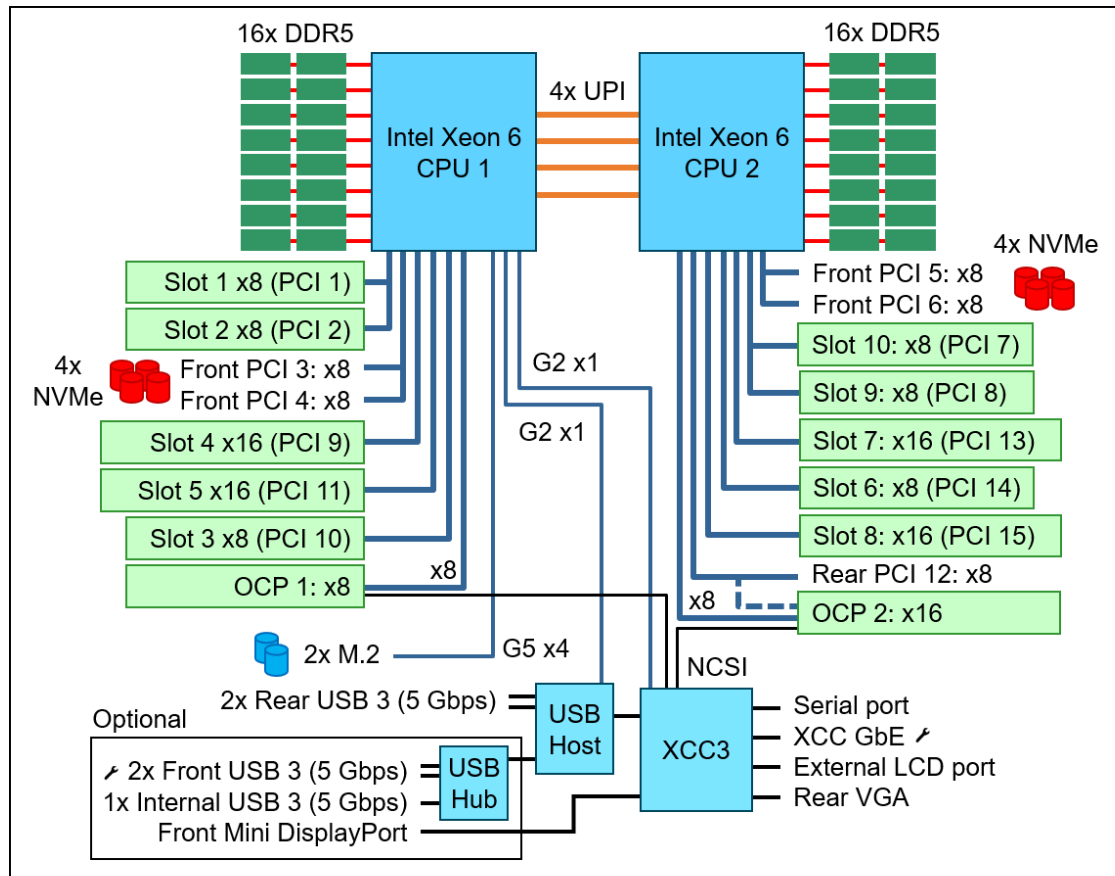


Figure 8. SR650 V4 system architectural block diagram with 10 slots

## Standard specifications

The following table lists the standard specifications.

Table 2. Standard specifications

| Components        | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine types     | <ul style="list-style-type: none"> <li>• 7DGC - 1 year warranty</li> <li>• 7DGD - 3 year warranty</li> <li>• 7DK2 - 3 year warranty - SR650 V4 with Compute Complex Neptune Core liquid cooling</li> <li>• 7DLN - 3 year warranty - SR650 V4 with SAP HANA</li> <li>• - 1 year warranty - SR650 V4 for vSAN</li> <li>• - 3 year warranty - SR650 V4 for vSAN</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Form factor       | 2U rack                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Processor         | One or two Intel Xeon 6700P-series or 6500P-series processors (formerly codenamed "Granite Rapids"). Supports processors up to 86 cores and 172 threads, core speeds of up to 4.0 GHz, and TDP ratings of up to 350 W.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Chipset           | None. Integrated into the processor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Memory            | 32 DIMM slots with two processors (16 DIMM slots per processor). Each processor has 8 memory channels, with 2 DIMMs per channel (DPC). 16 DIMM slots with 2 processors when the Compute Complex Neptune Core liquid cooling is used (machine type 7DK2). Lenovo TruDDR5 RDIMMs and MRDIMMs are supported. RDIMMs operate at up to 6400 MHz at 1 DPC and up to 5200 MHz at 2 DPC. MRDIMMs are supported up to 8000 MHz at 1 DPC (no support for 2 DPC). CXL 2.0 memory is also supported, up to 12 DIMMs (6 per processor) installed in E3.S 2T drive bays.                                                                                                                                                                                                                                                                                           |
| Memory maximum    | Up to 8TB by using 32x 256GB 3DS RDIMMs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Memory protection | ECC, SDDC (for x4-based memory DIMMs), ADDDC (for x4-based memory DIMMs), and memory mirroring.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Disk drive bays   | <p>Up to 16x 3.5-inch or 40x 2.5-inch or 32x E3.S hot-swap drive bays:</p> <ul style="list-style-type: none"> <li>• Front bays can be 3.5-inch (8 or 12 bays), or 2.5-inch (8, 16 or 24 bays), or E3.S (32 bays)</li> <li>• Middle bays are 2.5-inch (8 bays) simple-swap</li> <li>• Rear bays can be 3.5-inch (4 bays) or 2.5-inch (4 or 8 bays)</li> <li>• Combinations of SAS/SATA, NVMe, or AnyBay (supporting SAS, SATA or NVMe) are available</li> </ul> <p>M.2 support, for OS boot and drive storage support:</p> <ul style="list-style-type: none"> <li>• 2x front or rear hot-swap M.2 drive bays, or</li> <li>• Internal M.2 module supporting up to two M.2 drives</li> </ul> <p>See <a href="#">Supported drive bay combinations</a> for details. AnyBay bays support SAS, SATA or NVMe drives. NVMe bays only support NVMe drives.</p> |

| Components               | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum internal storage | <ul style="list-style-type: none"> <li>2.5-inch drives: <ul style="list-style-type: none"> <li>1228.8TB using 40x 30.72TB 2.5-inch SAS SSDs</li> <li>307.2TB using 40x 7.68TB 2.5-inch SATA SSDs</li> <li>2211.84TB using 36x 61.44TB 2.5-inch NVMe SSDs</li> <li>96TB using 40x 2.4TB 2.5-inch SAS HDDs</li> </ul> </li> <li>EDSFF drives <ul style="list-style-type: none"> <li>1966.08TB using 32x 61.44TB E3.S EDSFF NVMe SSDs</li> </ul> </li> <li>3.5-inch drives: <ul style="list-style-type: none"> <li>384TB using 16x 24TB 3.5-inch SAS HDDs</li> <li>384TB using 16x 24TB 3.5-inch SATA HDDs</li> <li>245.76TB using 16x 15.36TB 3.5-inch SAS SSDs</li> <li>122.88TB using 16x 7.68TB 3.5-inch SATA SSDs</li> <li>61.44TB using 4x 15.36TB 3.5-inch NVMe SSDs</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                        |
| Storage controllers      | <ul style="list-style-type: none"> <li>Onboard NVMe ports with optional RAID support using Intel VROC</li> <li>NVMe RAID support using a Tri-mode RAID adapter</li> <li>12 Gb SAS/SATA RAID adapters, PCIe 4.0 or PCIe 3.0 host interface</li> <li>12 Gb SAS/SATA HBA (non-RAID), PCIe 4.0 or PCIe 3.0 host interface</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Optical drive bays       | No internal optical drive.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Tape drive bays          | No internal backup drive.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Network interfaces       | Two dedicated OCP 3.0 SFF slots with a PCIe 5.0 host interface, either x8 or x16. Support a variety of 2-port and 4-port adapters with up to 400 GbE network connectivity. One port of each installed OCP adapter can optionally be shared with the XClarity Controller (XCC) management processor for Wake-on-LAN and NC-SI support.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| PCI Expansion slots      | <p>Up to 10x slots, all at the rear, plus 2x OCP slots. All slots are PCIe 5.0. The use of Riser 3 and 4 requires 2 processors.</p> <ul style="list-style-type: none"> <li>Riser 1: 2x low-profile slots, x8 or x16 (CPU 1)</li> <li>Riser 2: 3x full-height slots, two x16 and one x8 (CPU 1)</li> <li>Riser 3: 3x full-height slots, two x16 and one x8 (CPU 2)</li> <li>Riser 4: 2x low-profile slots, x8 or x16 (CPU 2)</li> </ul> <p>Other slot configurations are available. See the <a href="#">I/O expansion</a> for details.</p> <p>The server also supports rear 2.5" or 3.5" drive bays, and rear hot-swap M.2 drives, instead of some slots. If the server has 8x 2.5-inch rear drive bays, then Riser 1 slots are full-height.</p> <p>All configurations include at the rear of the server:</p> <ul style="list-style-type: none"> <li>2x OCP slots with PCIe 5.0 x16 or x8 connection</li> </ul> <p>For 2.5-inch front drive configurations, the server supports the installation of a CFF RAID adapter or HBA in a dedicated area that does not consume any of the rear PCIe slots.</p> |
| GPU support              | Support for up to 10x single-wide GPUs or 2x double-wide GPUs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Ports                    | <p>Front: External diagnostics port, optional 2x USB 3 (5 Gb/s) port, one supports XCC local management, optional Mini DisplayPort (miniDP) v1.1a video port.</p> <p>Rear: 2x USB 3 (5 Gb/s) ports, 1x VGA video port, 1x RJ-45 1GbE systems management port for XCC remote management. Optional DB-9 COM serial port (installs in a slot). Support for an optional adapter to share an incoming remote management network connection across 4 servers (installs in an OCP slot).</p> <p>Internal: Optional 1x USB 3 (5 Gb/s) connector for operating system or license key purposes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Components                  | Specification                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cooling                     | 6x (with two processors installed) or 5x (with one processor installed) single-rotor or dual-rotor hot swap 60 mm fans, configuration dependent. Fans are N+1 redundant, tolerating a single-rotor failure. One fan integrated in each power supply. For customers with water infrastructure in their data center, the server also supports open-loop water cooling for efficient heat removal.                                                      |
| Power supply                | Up to two hot-swap redundant AC power supplies, 80 PLUS Platinum or 80 PLUS Titanium certification. 800W, 1300W, 2000W, 2700W and 3200W AC options. All AC power supplies support 230V power; some also support 115V input supply. In China only, all power supply options support 240 V DC. Support for HVDC and -48V DC power supply options.                                                                                                      |
| Video                       | Embedded graphics with 16 MB memory with 2D hardware accelerator, integrated into the XClarity Controller 3 management controller. Two video ports (rear VGA and optional front Mini DisplayPort); both can be used simultaneously if desired. Maximum resolution is 1920x1200 32bpp at 60Hz.                                                                                                                                                        |
| Hot-swap parts              | Drives, power supplies, and fans.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Systems management          | Operator panel with status LEDs. Optional External Diagnostics Handset with LCD display. Clarity Controller 3 (XCC3) embedded management based on the ASPEED AST2600 baseboard management controller (BMC) and OpenBMC, XClarity One centralized infrastructure delivery, XClarity Integrator plugins, and XClarity Energy Manager centralized server power management. Optional XCC3 Premier to enable remote control functions and other features. |
| Security features           | Chassis intrusion switch, Power-on password, administrator's password, Root of Trust module supporting TPM 2.0 and Platform Firmware Resiliency (PFR). Optional lockable front security bezel.                                                                                                                                                                                                                                                       |
| Operating systems supported | Microsoft Windows Server, Red Hat Enterprise Linux, SUSE Linux Enterprise Server, VMware ESXi, Ubuntu Server. See the <a href="#">Operating system support</a> section for specifics.                                                                                                                                                                                                                                                                |
| Limited warranty            | Three-year or one-year (model dependent) customer-replaceable unit and onsite limited warranty with 9x5 next business day (NBD).                                                                                                                                                                                                                                                                                                                     |
| Service and support         | Optional service upgrades are available through Lenovo Services: 4-hour or 2-hour response time, 6-hour fix time, 1-year or 2-year warranty extension, software support for Lenovo hardware and some third-party applications.                                                                                                                                                                                                                       |
| Dimensions                  | Width: 445 mm (17.5 in.), height: 87 mm (3.4 in.), depth: 796 mm (31.3 in.). See <a href="#">Physical and electrical specifications</a> for details.                                                                                                                                                                                                                                                                                                 |
| Weight                      | Maximum weight: 38.8 kg (85.5 lb)                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Lenovo Top Choice

Lenovo offers two “quick-ship” programs to make it easy for our customers to get quick delivery of our offerings:

- **Lenovo Top Choice Express:** A Configure-to-Order (CTO) model that uses a "best of" subset of components. It allows for customization while maintaining significantly faster lead times than standard CTO.

Top Choice Express (TCE) is the way to get custom configurations as quickly as possible. It allows DCSC configurator users to configure systems in **CTO Mode**, selecting the required hardware components while still benefiting from faster turnaround than traditional CTO. These configurations are built and fully integrated by Lenovo Manufacturing, with simple configuration steps, instant pricing, and quicker delivery.

To see which components are TCE, check the TCE column in the various component tables in this product guide. **Green cells** mean that the component is enabled for ordering with Top Choice Express in all regions world-wide. **Orange cells** with the ⓘ symbol mean that the component is enabled for Top Choice Express in only some regions. Mouse over or tap on the cell to see which regions have it enabled for TCE.

Note: Top Choice Express is for configure-to-order builds only, not for standalone option part numbers.

- **Lenovo Top Choice Stock:** Pre-built, fixed-configuration units held in distribution. They are the "Ready-to-Ship" inventory options for customers who need a server quickly.

Top Choice Stock (TCS) is used in **Preconfigured Mode** in DCSC, starting from a standard pre-built configuration and adding option components as needed. Systems are assembled and fulfilled by Lenovo partners, enabling quick order processing and fast shipment through distribution. This path supports rapid delivery while maintaining flexibility through add-on options. For details of TCS models, see the [Preconfigured Models](#) section.

Note: Ship dates are estimates only; actual ship times may vary based on platform and component availability, order processing, logistics, and external factors beyond Lenovo's control. Lenovo is not liable for delays.

## Models

ThinkSystem SR650 V4 models can be configured by using the [Lenovo Data Center Solution Configurator \(DCSC\)](#).

Topics in this section:

- [CTO models](#)
- [CTO models for Windows 10 and Windows 11](#)
- [Base feature codes](#)
- [Preconfigured models](#)

### CTO models

ThinkSystem SR650 V4 models can be configured by using the [Lenovo Data Center Solution Configurator \(DCSC\)](#).

Configure-to-order (CTO) models are used to create models with factory-integrated server customizations. For CTO models, two types of base CTO models are available for the SR650 V4 as listed in the columns in the following table:

- General purpose base CTO models are for general business (non-HPC) and is selectable by choosing **General Purpose** mode in DCSC.
- AI and HPC base models are intended for Artificial Intelligence (AI) and High Performance Computing (HPC) configurations and solutions are enabled using the **AI & HPC Hardware - ThinkSystem Hardware** mode in DCSC. These configurations, along with Lenovo EveryScale Solutions, can also be built using [System x and Cluster Solutions Configurator \(x-config\)](#). **Tip:** Some HPC and AI models are not listed in DCSC and can only be configured in x-config.

Preconfigured server models may also be available for the SR650 V4, however these are region-specific; that is, each region may define their own server models, and not all server models are available in every region.

The following table lists the base CTO models of the ThinkSystem SR650 V4 server.

Table 3. Base CTO models

| Machine Type/Model<br>General purpose | Machine Type/Model<br>for AI and HPC | Description                                                              |
|---------------------------------------|--------------------------------------|--------------------------------------------------------------------------|
| 7DGDCTO1WW                            | 7DGDCTOLWW                           | ThinkSystem SR650 V4 – 3-year base warranty                              |
| 7DGCCTO1WW                            | 7DGCCTOLWW                           | ThinkSystem SR650 V4 – 1-year base warranty                              |
| 7DLNCTO1WW                            | 7DLNCTOLWW                           | ThinkSystem SR650 V4 – SAP HANA configurations with 3-year base warranty |

The SR650 V4 uses an additional machine type of 7DK2 for water-cooled servers using the Compute Complex Neptune Core Module, where memory is water-cooled in addition to the two CPUs, as listed in the following table.

Servers of machine type 7DK2 share components with machine types 7DGD, however the system board is different. Specifically, there are fewer DIMM slots and the DIMM slots are spaced wider apart to accommodate the heat-transfer components.

Table 4. Base CTO models for CPU+Memory water cooling

| Machine Type/Model<br>General purpose | Machine Type/Model<br>for AI and HPC | Description                                      |
|---------------------------------------|--------------------------------------|--------------------------------------------------|
| 7DK2CTO1WW                            | -                                    | ThinkSystem SR650 V4 CPU+DIMM DWC - 3yr Warranty |

## CTO models for Windows 10 and Windows 11

The SR650 V4 can run Windows 10 and Windows 11, however only a subset of adapters and drives can be installed. For ease of configuration, the following Base CTO models have been announced to assist building a configuration that can be used with the client operating systems.

Table 5. Base CTO models for SR650 V4 with Windows 10 and Windows 11

| Machine Type/Model | Description                                        |
|--------------------|----------------------------------------------------|
| 7DGDCTO3WW         | ThinkSystem SR650 V4 Workstation - 3 year Warranty |
| 7DGCCTO3WW         | ThinkSystem SR650 V4 Workstation - 1 year Warranty |

## Base feature codes

Models of the SR650 V4 are defined based on the configuration of front drives. The feature codes for these chassis bases are as listed in the following table.

Table 6. Chassis base feature codes

| Feature code | Description                            | Top Choice Express | Purpose                                            |
|--------------|----------------------------------------|--------------------|----------------------------------------------------|
| C3QL         | ThinkSystem SR650 V4 12x3.5" Chassis   | TCE                | Configurations with front 3.5-inch hot-swap drives |
| C3QK         | ThinkSystem SR650 V4 24x2.5" Chassis   | TCE                | Configurations with front 2.5-inch hot-swap drives |
| C3QM         | ThinkSystem SR650 V4 EDSFF 3.S Chassis | Not TCE            | Configurations with front E3.S drive bays          |

## Preconfigured models

The following tables list the available preconfigured models, grouped by region.

- [Models for Asia Pacific region](#)
- [Models for South East Asian countries \(ASEAN\)](#)
- [Models for Brazil](#)
- [Models for EMEA region](#)
- [Models for Hong Kong, Taiwan, Korea \(HTK\)](#)
- [Models for Latin American countries \(except Brazil\)](#)

Refer to the Specifications section for information about standard features of the server.

Common to all models:

- Power supplies are Platinum unless otherwise stated
- All models include a Toolless Slide Rail Kit

## Models for Asia Pacific region

The following table lists the models for the Asia Pacific region: Australia, Bangladesh, Brunei, Hong Kong, India, Japan, Korea, Sri Lanka, Malaysia, New Zealand, Philippines, Singapore, Thailand, Taiwan, Vietnam

Table 7. Models for Asia Pacific markets

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory      | RAID        | Drive bays                    | OCP          | Slots                             | Power supply | Fans   | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|-------------|-------------|-------------------------------|--------------|-----------------------------------|--------------|--------|--------------|--------------|------|--------------|
| 7DGCA00RAP | N                | 1x 6517P 16C 190W 3.2GHz       | 1x64GB      | Option      | 2.5" Open backplanes Open bay | Open         | Open                              | 2x 1300W TT  | 5x Std | Opt          | Opt          | Std  | Opt          |
| 7DGDA01QAP | Y                | 1x 6505P 12C 150W 2.2GHz       | 1x16GB      | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16          | 2x 800W PT   | 5x Std | Y            | Y            | Std  | Opt          |
| 7DGDA01TAP | Y                | 1x 6505P 12C 150W 2.2GHz       | 1x16GB      | 940-16i 4GB | 12x 3.5" SAS Open bay         | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16          | 2x 800W PT   | 5x Std | Y            | Y            | Std  | Opt          |
| 7DGDA02GAP | Y                | 1x 6507P 8C 150W 3.5GHz        | 1x16GB      | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slot 5 x16                        | 2x 800W PT   | 5x Std | Opt          | Opt          | Std  | Opt          |
| 7DGDA02LAP | Y                | 1x 6507P 8C 150W 3.5GHz        | 1x32GB 2Rx8 | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slot 3 x8, Slot 4 x16, Slot 5 x16 | 2x 800W PT   | 5x Std | Opt          | Opt          | Pre  | Opt          |
| 7DGDA02QAP | Y                | 1x 6507P 8C 150W 3.5GHz        | 1x16GB      | 940-8i 4GB  | 8x 2.5" SAS; 3x 1.2TB 10K     | 1x 4-1G 5719 | Slot 3 x8, Slot 4 x16, Slot 5 x16 | 2x 800W PT   | 5x Std | Opt          | Opt          | Pre  | Opt          |
| 7DGDA01RAP | Y                | 1x 6515P 16C 150W 2.3GHz       | 1x16GB      | 940-16i 4GB | 12x 3.5" SAS Open bay         | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16          | 2x 800W PT   | 5x Std | Y            | Y            | Std  | Opt          |
| 7DGDA01SAP | Y                | 1x 6515P 16C 150W 2.3GHz       | 1x16GB      | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16          | 2x 800W PT   | 5x Std | Y            | Y            | Std  | Opt          |
| 7DGDA02HAP | Y                | 1x 6515P 16C 150W 2.3GHz       | 1x16GB      | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slot 5 x16                        | 2x 800W PT   | 5x Std | Opt          | Opt          | Std  | Opt          |
| 7DGDA02WAP | N                | 1x 6515P 16C 150W 2.3GHz       | 1x16GB      | 940-16i     | 12x 3.5" SAS Open bay         | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16          | 2x 800W PT   | 5x Std | Y            | Y            | Std  | Opt          |
| 7DGDA04UAP | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slot 3 x8, Slot 4 x16, Slot 5 x16 | 2x 800W PT   | 5x Std | Opt          | Opt          | Pre  | Opt          |

† Processor description: Processor model, number of cores, thermal design power (TDP), core frequency

## Models for South East Asian countries (ASEAN)

Table 8. Models for South East Asian countries (ASEAN)

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory      | RAID       | Drive bays                       | OCP             | Slots      | Power supply | Fans   | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|-------------|------------|----------------------------------|-----------------|------------|--------------|--------|--------------|--------------|------|--------------|
| 7DGDA023SG | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 2x 1.2TB 10K        | 1x 4-1G 5719    | Slot 5 x16 | 2x 1300W PT  | 5x Std | Opt          | Opt          | Pre  | Opt          |
| 7DGDA02USG | Y                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 2x 1.2TB 10K        | 1x 4-1G 5719    | Slot 5 x16 | 2x 1300W PT  | 5x Std | Opt          | Opt          | Pre  | Opt          |
| 7DGDA01XSG | N                | 2x 6517P 16C 190W 3.2GHz       | 2x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 4x 2.4TB 10K HDD    | 1x 2-10GT 57416 | Slot 5 x16 | 2x 1300W PT  | 6x Std | Opt          | Opt          | Pre  | Opt          |
| 7DGDA029SG | N                | 1x 6517P 16C 190W 3.2GHz       | 4x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 2x 480GB VA RI SATA | 1x 2-10GT 57416 | Slot 5 x16 | 2x 1300W PT  | 5x Std | Opt          | Opt          | Pre  | Opt          |
| 7DGDA02XSG | N                | 2x 6517P 16C 190W 3.2GHz       | 2x16GB      | 940-8i 4GB | 8x 2.5" SAS; 4x 960GB VA RI SATA | 1x E610-T2      | Slot 5 x16 | 2x 1300W PT  | 6x Std | Opt          | Opt          | Pre  | Opt          |

† Processor description: Processor model, number of cores, thermal design power (TDP), core frequency

## Models for Brazil

Table 9. Models for Brazil

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory      | RAID       | Drive bays                                                  | OCP          | Slots                    | Power supply | Fans    | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|-------------|------------|-------------------------------------------------------------|--------------|--------------------------|--------------|---------|--------------|--------------|------|--------------|
| 7DGD100BBR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID                   | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGD100EBR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 2.5" Open backplanes Open bay                               | Open         | Open                     | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGD100XBR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID                   | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA01KBR | Y                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay                                        | 1x 4-1G 5719 | Slot 5 x16               | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA03NBR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA03PBR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 2x 2.4TB 10K HDD                               | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA049BR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 1x 1.2TB 10K                                   | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA05JBR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Yes          |
| 7DGDA057BR | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGD100GBR | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID, 2x 960GB ER3 M.2 | Open         | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGDA03QBR | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA05ABR | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA05HBR | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Yes          |

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory | RAID       | Drive bays                                                  | OCP  | Slots                    | Power supply | Fans    | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|--------|------------|-------------------------------------------------------------|------|--------------------------|--------------|---------|--------------|--------------|------|--------------|
| 7DGD100HBR | N                | 1x 6520P 24C 210W 2.4GHz       | 1x64GB | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID, 2x 960GB ER3 M.2 | Open | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGD100JBR | N                | 1x 6520P 24C 210W 2.4GHz       | 1x64GB | 940-8i 4GB | 8x 2.5" SAS Open bay                                        | Open | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |

† Processor description: Processor model, number of cores, thermal design power (TDP), core frequency

#### Models for EMEA region

Table 10. Models for EMEA region

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory      | RAID       | Drive bays                    | OCP  | Slots      | Power supply     | Fans    | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|-------------|------------|-------------------------------|------|------------|------------------|---------|--------------|--------------|------|--------------|
| 7DGDA00NEA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay          | Open | Slot 5 x16 | 1x1300W TT Prem  | 5x Std  | Opt          | Opt          | Pre  | Yes          |
| 7DGDA031EA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 2.5" Open backplanes Open bay | Open | Open       | 2x 1300W TT Prem | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGDA04AEA | N                | 1x 6507P 8C 150W 3.5GHz        | 1x32GB 2Rx8 | Option     | 2.5" Open backplanes Open bay | Open | Open       | 2x 1300W TT Prem | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGDA030EA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay          | Open | Slot 5 x16 | 2x 1300W TT Prem | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA010EA | N                | 1x 6527P 24C 255W 3.0GHz       | 4x64GB      | 940-8i 4GB | 8x 2.5" SAS Open bay          | Open | Slot 5 x16 | 1x1300W TT Prem  | 5x Std  | Opt          | Opt          | Pre  | Yes          |

† Processor description: Processor model, number of cores, thermal design power (TDP), core frequency

## Models for Hong Kong, Taiwan, Korea (HTK)

**AP models:** Customers in Hong Kong, Taiwan, and Korea also have access to the [Asia Pacific region](#) models.

Table 11. Models for Hong Kong, Taiwan, Korea (HTK)

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory      | RAID        | Drive bays                    | OCP          | Slots                             | Power supply | Fans    | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|-------------|-------------|-------------------------------|--------------|-----------------------------------|--------------|---------|--------------|--------------|------|--------------|
| 7DGDA01NCN | N                | 1x 6507P 8C 150W 3.5GHz        | 1x16GB      | 940-8i 4GB  | 8x 2.5" SAS Open bay          | 1x 4-1G 5719 | Slot 3 x8, Slot 4 x16, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA01PCN | N                | 1x 6507P 8C 150W 3.5GHz        | 1x16GB      | 940-16i 4GB | 12x 3.5" SAS Open bay         | 1x 4-1G 5719 | Slot 3 x8, Slot 4 x16, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA048CN | N                | 1x 6517P 16C 190W 3.2GHz       | 4x32GB 2Rx8 | 940-8i 4GB  | 8x 2.5" SAS; 2x 2.4TB 10K HDD | 1x 4-1G 5719 | Slot 5 x16                        | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Std  | Opt          |

† Processor description: Processor model, number of cores, thermal design power (TDP), core frequency

## Models for Latin American countries (except Brazil)

Table 12. Models with a 3-year warranty for Latin American countries (except Brazil)

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory      | RAID       | Drive bays                                                  | OCP          | Slots                    | Power supply | Fans    | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|-------------|------------|-------------------------------------------------------------|--------------|--------------------------|--------------|---------|--------------|--------------|------|--------------|
| 7DGD100ELA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 2.5" Open backplanes Open bay                               | Open         | Open                     | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGD100WLA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID                   | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA01ULA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID                   | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA039LA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay                                        | Open         | Slot 5 x16               | 2x 1300W PT  | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA03ULA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 1x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA058LA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA05KLA | N                | 1x 6505P 12C 150W 2.2GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGD100GLA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID, 2x 960GB ER3 M.2 | Open         | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGD100RLA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 12x 3.5" (8xSAS, 4xNVMe) Open bay; M.2 B540p Rear RAID      | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGD100ULA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID                   | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGDA03VLA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | 940-8i 4GB | 8x 2.5" SAS; 1x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA059LA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |
| 7DGDA05LLA | N                | 1x 6515P 16C 150W 2.3GHz       | 1x32GB 2Rx8 | Option     | 8x 2.5" SAS; 2x 480GB VA RI SATA                            | 1x 4-1G 5719 | Slots 1&2 x8, Slot 5 x16 | 2x 1300W PT  | 5x Std  | Opt          | Opt          | Pre  | Opt          |

| Model      | Top Choice Stock | Intel Xeon Scalable processor† | Memory | RAID       | Drive bays                                                  | OCP             | Slots                    | Power supply | Fans    | Internal USB | Front USB/DP | XCC3 | Intru switch |
|------------|------------------|--------------------------------|--------|------------|-------------------------------------------------------------|-----------------|--------------------------|--------------|---------|--------------|--------------|------|--------------|
| 7DGD1005LA | N                | 1x 6520P 24C 210W 2.4GHz       | 1x64GB | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID                   | 1x 2-10GT 57416 | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |
| 7DGD100HLA | N                | 1x 6520P 24C 210W 2.4GHz       | 1x64GB | 940-8i 4GB | 8x 2.5" SAS Open bay; M.2 B540p Rear RAID, 2x 960GB ER3 M.2 | Open            | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGD100JLA | N                | 1x 6520P 24C 210W 2.4GHz       | 1x64GB | 940-8i 4GB | 8x 2.5" SAS Open bay                                        | Open            | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Yes          |
| 7DGD100SLA | N                | 1x 6520P 24C 210W 2.4GHz       | 1x64GB | 940-8i 4GB | 12x 3.5" (8xSAS, 4xNVMe) Open bay; M.2 B540p Rear RAID      | 1x 2-10GT 57416 | Slots 1&2 x8, Slot 5 x16 | 2x 800W PT   | 5x Perf | Opt          | Opt          | Pre  | Opt          |

† Processor description: Processor model, number of cores, thermal design power (TDP), core frequency

## Processors

The SR650 V4 supports one or two of the following Intel processors:

- Intel Xeon 6500-series with P-cores (formerly "Granite Rapids" or GNR)
- Intel Xeon 6700-series with P-cores (formerly "Granite Rapids" or GNR)

Topics in this section:

- [Processor options](#)
- [Processor features](#)
- [One-processor configurations](#)
- [Processor cooling](#)
- [Lenovo Processor Neptune Core Module - Open-loop liquid cooling](#)
- [Lenovo Compute Complex Neptune Core Module - Open-loop liquid cooling](#)
- [UEFI operating modes](#)

## Processor options

The following table lists the Intel Xeon 6 processors with P-cores that are supported by the SR650 V4.

**Compute Complex Neptune Core Module support:** As described in the [Lenovo Compute Complex Neptune Core Module - Open-loop liquid cooling](#) section, two processors are required. As a result, if you are configuring the SR650 V4 with machine type 7DK2, then the CPUs with a maximum quantity of only 1 processor are not supported.

Processor option part numbers include a heatsink, however they do not include a system fan.

The table includes a Top Choice Express column; for CTO orders, select a Top Choice Express component for faster delivery. See the [Lenovo Top Choice](#) section for more information.

Table 13. Intel Xeon 6 P-core processor support

| Part number                                | Feature code | SKU   | Description                                                               | Top Choice Express | Maximum quantity |
|--------------------------------------------|--------------|-------|---------------------------------------------------------------------------|--------------------|------------------|
| <b>Intel Xeon 6500-series with P-cores</b> |              |       |                                                                           |                    |                  |
| 4XG7B17492                                 | CGVJ         | 6503P | ThinkSystem SR650/a V4 Intel Xeon 6503P 8C 135W 2.8GHz Processor w/o Fan  | Not TCE            | 2                |
| 4XG7B03703                                 | C5QQ         | 6505P | ThinkSystem SR650/a V4 Intel Xeon 6505P 12C 150W 2.2GHz Processor w/o fan | Not TCE            | 2                |
| 4XG7B03700                                 | C5R6         | 6507P | ThinkSystem SR650/a V4 Intel Xeon 6507P 8C 150W 3.5GHz Processor w/o fan  | Not TCE            | 2                |
| CTO only                                   | C5RB         | 6511P | Intel Xeon 6511P 16C 150W 2.3GHz Processor                                | Not TCE            | 1*               |
| 4XG7B03705                                 | C5RD         | 6515P | ThinkSystem SR650/a V4 Intel Xeon 6515P 16C 150W 2.3GHz Processor w/o fan | TCE                | 2                |
| 4XG7B03701                                 | C5QV         | 6517P | ThinkSystem SR650/a V4 Intel Xeon 6517P 16C 190W 3.2GHz Processor w/o fan | Not TCE            | 2                |
| 4XG7B03707                                 | C5QR         | 6520P | ThinkSystem SR650/a V4 Intel Xeon 6520P 24C 210W 2.4GHz Processor w/o fan | TCE                | 2                |
| CTO only                                   | C5R9         | 6521P | Intel Xeon 6521P 24C 225W 2.6GHz Processor                                | Not TCE            | 1*               |
| 4XG7B03696                                 | C659         | 6527P | ThinkSystem SR650/a V4 Intel Xeon 6527P 24C 255W 3.0GHz Processor w/o fan | TCE                | 2                |

| Part number                                | Feature code | SKU   | Description                                                                                | Top Choice Express | Maximum quantity |
|--------------------------------------------|--------------|-------|--------------------------------------------------------------------------------------------|--------------------|------------------|
| 4XG7B03691                                 | C5QT         | 6530P | ThinkSystem SR650/a V4 Intel Xeon 6530P 32C 225W 2.3GHz Processor w/o fan                  | TCE                | 2                |
| <b>Intel Xeon 6700-series with P-cores</b> |              |       |                                                                                            |                    |                  |
| 4XG7B03702                                 | C5R7         | 6714P | ThinkSystem SR650/a V4 Intel Xeon 6714P 8C 165W 4.0GHz Processor w/o fan                   | TCE                | 2                |
| 4XG7B03704                                 | C5R5         | 6724P | ThinkSystem SR650 V4/SR650a V4/SR850 V4 Intel Xeon 6724P 16C 210W 3.6GHz Processor w/o Fan | TCE                | 2                |
| CTO only                                   | CE89         | 6725P | Intel Xeon 6725P 16C 235W 3.7GHz Processor                                                 | Not TCE            | 2                |
| 4XG7B03693                                 | C5R4         | 6730P | ThinkSystem SR650/a V4 Intel Xeon 6730P 32C 250W 2.5GHz Processor w/o fan                  | TCE                | 2                |
| CTO only                                   | C5QN         | 6731P | Intel Xeon 6731P 32C 245W 2.5GHz Processor                                                 | Not TCE            | 1*               |
| 4XG7B09322                                 | CARB         | 6732P | ThinkSystem SR650/a V4 Intel Xeon 6732P 32C 350W 3.8GHz Processor w/o fan                  | Not TCE            | 2                |
| 4XG7B03698                                 | C5R0         | 6736P | ThinkSystem SR650/a V4 Intel Xeon 6736P 36C 205W 2.0GHz Processor w/o fan                  | Not TCE            | 2                |
| 4XG7B03706                                 | C5QX         | 6737P | ThinkSystem SR650/a V4 Intel Xeon 6737P 32C 270W 2.9GHz Processor w/o fan                  | TCE                | 2                |
| 4XG7B03695                                 | C5R3         | 6740P | ThinkSystem SR650/a V4 Intel Xeon 6740P 48C 270W 1.9GHz Processor w/o fan                  | Not TCE            | 2                |
| CTO only                                   | C5QU         | 6741P | Intel Xeon 6741P 48C 300W 2.5GHz Processor                                                 | Not TCE            | 1*               |
| 4XG7B09321                                 | CARA         | 6745P | ThinkSystem SR650/a V4 Intel Xeon 6745P 32C 300W 3.1GHz Processor w/o fan                  | Not TCE            | 2                |
| 4XG7B03690                                 | C5R8         | 6747P | ThinkSystem SR650/a V4 Intel Xeon 6747P 48C 330W 2.7GHz Processor w/o fan                  | Not TCE            | 2                |
| 4XG7B03697                                 | C5R1         | 6760P | ThinkSystem SR650/a V4 Intel Xeon 6760P 64C 330W 2.2GHz Processor w/o fan                  | Not TCE            | 2                |
| CTO only                                   | C5QW         | 6761P | Intel Xeon 6761P 64C 350W 2.5GHz Processor                                                 | Not TCE            | 1*               |
| 4XG7B03692                                 | C5QY         | 6767P | ThinkSystem SR650/a V4 Intel Xeon 6767P 64C 350W 2.4GHz Processor w/o fan                  | Not TCE            | 2                |
| CTO only                                   | C5QP         | 6781P | Intel Xeon 6781P 80C 350W 2.0GHz Processor                                                 | Not TCE            | 1*               |
| 4XG7B03694                                 | C5QM         | 6787P | ThinkSystem SR650/a V4 Intel Xeon 6787P 86C 350W 2.0GHz Processor w/o fan                  | Not TCE            | 2                |

\* These processors are only supported in 1-socket configurations; configure-to-order (CTO) only, no field upgrades are supported, not supported with Compute Complex Neptune Core (machine type 7DK2)

## Processor features

Processors supported by the SR650 V4 include embedded accelerators to add even more processing capability:

- QuickAssist Technology (Intel QAT)

Help reduce system resource consumption by providing accelerated cryptography, key protection, and data compression with Intel QuickAssist Technology (Intel QAT). By offloading encryption and decryption, this built-in accelerator helps free up processor cores and helps systems serve a larger number of clients.

- Intel Dynamic Load Balancer (Intel DLB)

Improve the system performance related to handling network data on multi-core Intel Xeon Scalable processors. Intel Dynamic Load Balancer (Intel DLB) enables the efficient distribution of network processing across multiple CPU cores/threads and dynamically distributes network data across multiple CPU cores for processing as the system load varies. Intel DLB also restores the order of networking data packets processed simultaneously on CPU cores.

- Intel Data Streaming Accelerator (Intel DSA)

Drive high performance for storage, networking, and data-intensive workloads by improving streaming data movement and transformation operations. Intel Data Streaming Accelerator (Intel DSA) is designed to offload the most common data movement tasks that cause overhead in data center-scale deployments. Intel DSA helps speed up data movement across the CPU, memory, and caches, as well as all attached memory, storage, and network devices.

- Intel In-Memory Analytics Accelerator (Intel IAA)

Run database and analytics workloads faster, with potentially greater power efficiency. Intel In-Memory Analytics Accelerator (Intel IAA) increases query throughput and decreases the memory footprint for in-memory database and big data analytics workloads. Intel IAA is ideal for in-memory databases, open source databases and data stores like RocksDB, Redis, Cassandra, and MySQL.

- Intel Advanced Matrix Extensions (Intel AMX)

Intel Advanced Matrix Extensions (Intel AMX) is a built-in accelerator in Xeon 6 P-core processors that significantly improves deep learning training and inference. With Intel AMX, you can fine-tune deep learning models or train small to medium models in just minutes. Intel AMX offers discrete accelerator performance without added hardware and complexity.

The processors also support a separate and encrypted memory space, known as the SGX Enclave, for use by Intel Software Guard Extensions (SGX). The size of the SGX Enclave supported varies by processor model. Intel SGX offers hardware-based memory encryption that isolates specific application code and data in memory. It allows user-level code to allocate private regions of memory (enclaves) which are designed to be protected from processes running at higher privilege levels.

The following table summarizes the key features of the Intel Xeon 6 P-core processors that are supported in the SR650 V4.

Table 14. Intel 6 P-core processor features

| CPU model                           | Die | Cores/ threads | Core speed (Base / TB max) | L3 cache | Mem. chan | Max RDIMM speed | Max MRDIMM speed | UPI 2.0 links & speed | PCIe lanes | TDP  | Accelerators |     |     |     | SGX Enclave Size |
|-------------------------------------|-----|----------------|----------------------------|----------|-----------|-----------------|------------------|-----------------------|------------|------|--------------|-----|-----|-----|------------------|
|                                     |     |                |                            |          |           |                 |                  |                       |            |      | QAT          | DLB | DSA | IAA |                  |
| Intel Xeon 6500-series with P-cores |     |                |                            |          |           |                 |                  |                       |            |      |              |     |     |     |                  |
| 6503P                               | LCC | 8 / 16         | 2.8GHz / 4.3 GHz           | 48 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 135W | 2            | 2   | 2   | 2   | 512GB            |
| 6505P                               | LCC | 12 / 24        | 2.2GHz / 4.1 GHz           | 48 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 150W | 2            | 2   | 2   | 2   | 128GB            |
| 6507P                               | LCC | 8 / 16         | 3.5GHz / 4.3 GHz           | 48 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 150W | 2            | 2   | 2   | 2   | 512GB            |
| 6511P                               | LCC | 16 / 32        | 2.3GHz / 4.2 GHz           | 72 MB    | 8         | 6400 MHz        | None             | None†                 | 88         | 150W | 2            | 2   | 2   | 2   | 128GB            |
| 6515P                               | LCC | 16 / 32        | 2.3GHz / 3.8 GHz           | 72 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 150W | 2            | 2   | 2   | 2   | 128GB            |
| 6517P                               | LCC | 16 / 32        | 3.2GHz / 4.2 GHz           | 72 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 190W | 2            | 2   | 2   | 2   | 512GB            |
| 6520P                               | HCC | 24 / 48        | 2.4GHz / 4 GHz             | 144 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 210W | 2            | 2   | 2   | 2   | 128GB            |

| CPU model                           | Die | Cores/ threads | Core speed (Base / TB max) | L3 cache | Mem. chan | Max RDIMM speed | Max MRDIMM speed | UPI 2.0 links & speed | PCIe lanes | TDP  | Accelerators |     |     |     | SGX Enclave Size |
|-------------------------------------|-----|----------------|----------------------------|----------|-----------|-----------------|------------------|-----------------------|------------|------|--------------|-----|-----|-----|------------------|
|                                     |     |                |                            |          |           |                 |                  |                       |            |      | QAT          | DLB | DSA | IAA |                  |
| 6521P                               | HCC | 24 / 48        | 2.6GHz / 4.1 GHz           | 144 MB   | 8         | 6400 MHz        | None             | None‡                 | 88         | 225W | 3            | 3   | 3   | 3   | 128GB            |
| 6527P                               | HCC | 24 / 48        | 3.0GHz / 4.2 GHz           | 144 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 255W | 4            | 4   | 4   | 4   | 512GB            |
| 6530P                               | HCC | 32 / 64        | 2.3GHz / 4.1 GHz           | 144 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 225W | 2            | 2   | 2   | 2   | 128GB            |
| Intel Xeon 6700-series with P-cores |     |                |                            |          |           |                 |                  |                       |            |      |              |     |     |     |                  |
| 6714P                               | LCC | 8 / 16         | 4.0GHz / 4.3 GHz           | 48 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 165W | 2            | 2   | 2   | 2   | 512GB            |
| 6724P                               | LCC | 16 / 32        | 3.6GHz / 4.3 GHz           | 72 MB    | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 210W | 2            | 2   | 2   | 2   | 512GB            |
| 6725P                               | LCC | 16 / 32        | 3.7GHz / 4.8 GHz           | 192 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 235W | 2            | 2   | 2   | 2   | 512GB            |
| 6730P                               | HCC | 32 / 64        | 2.5GHz / 3.8 GHz           | 288 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 250W | 4            | 4   | 4   | 4   | 512GB            |
| 6731P                               | HCC | 32 / 64        | 2.5GHz / 4.1 GHz           | 144 MB   | 8         | 6400 MHz        | None             | None‡                 | 88         | 245W | 3            | 3   | 3   | 3   | 128GB            |
| 6732P                               | HCC | 32 / 64        | 3.8GHz / 4.3 GHz           | 144 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 350W | 4            | 4   | 4   | 4   | 512GB            |
| 6736P                               | HCC | 36 / 72        | 2.0GHz / 4.1 GHz           | 144 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 205W | 4            | 4   | 4   | 4   | 512GB            |
| 6737P                               | HCC | 32 / 64        | 2.9GHz / 4 GHz             | 144 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 270W | 4            | 4   | 4   | 4   | 512GB            |
| 6740P                               | XCC | 48 / 96        | 2.1GHz / 3.8 GHz           | 288 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 270W | 2            | 2   | 2   | 2   | 128GB            |
| 6741P                               | XCC | 48 / 96        | 2.5GHz / 3.8 GHz           | 288 MB   | 8         | 6400 MHz        | None             | None‡                 | 88         | 300W | 3            | 3   | 3   | 3   | 128GB            |
| 6745P                               | HCC | 32 / 64        | 3.1GHz / 4.3 GHz           | 336 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 300W | 4            | 4   | 4   | 4   | 512GB            |
| 6747P                               | XCC | 48 / 96        | 2.7GHz / 3.9 GHz           | 288 MB   | 8         | 6400 MHz        | 8000 MHz         | 4 / 24 GT/s           | 88         | 330W | 4            | 4   | 4   | 4   | 512GB            |
| 6760P                               | XCC | 64 / 128       | 2.2GHz / 3.8 GHz           | 320 MB   | 8         | 6400 MHz        | None             | 4 / 24 GT/s           | 88         | 330W | 2            | 2   | 2   | 2   | 128GB            |
| 6761P                               | XCC | 64 / 128       | 2.5GHz / 3.9 GHz           | 336 MB   | 8         | 6400 MHz        | 8000 MHz         | None‡                 | 88         | 350W | 3            | 3   | 3   | 3   | 128GB            |
| 6767P                               | XCC | 64 / 128       | 2.4GHz / 3.9 GHz           | 336 MB   | 8         | 6400 MHz        | 8000 MHz         | 4 / 24 GT/s           | 88         | 350W | 4            | 4   | 4   | 4   | 512GB            |
| 6781P                               | XCC | 80 / 160       | 2.0GHz / 3.8 GHz           | 336 MB   | 8         | 6400 MHz        | 8000 MHz         | None‡                 | 88         | 350W | 3            | 3   | 3   | 3   | 128GB            |
| 6787P                               | XCC | 86 / 172       | 2.0GHz / 3.8 GHz           | 336 MB   | 8         | 6400 MHz        | 8000 MHz         | 4 / 24 GT/s           | 88         | 350W | 4            | 4   | 4   | 4   | 512GB            |

‡ These processors not have UPI links and are rich one-socket (R1S) processors

## One-processor configurations

The SR650 V4 can be used with one processor installed. Most core functions of the server (including the XClarity Controller) are connected to processor 1 as shown in the [System architecture](#) section.

With only one processor, the server has the following capabilities:

- 16 memory DIMMs
- Riser 1 (slots 1 & 2) and Riser 2 (slots 3, 4, 5)
- OCP slot 1
- Up to 16x 2.5-inch NVMe drives using onboard NVMe connections (24x NVMe supported using Tri-Mode)
- Up to 16x E3.S 1T NVMe drives
- Up to 6x CXL memory modules in E3.S 2T drive bays
- Internal RAID adapter or HBA (CFF form factor)
- M.2 drives

## Processor cooling

The SR650 V4 offers four implementations to remove heat from the processors:

- Entry heatsinks, suitable for configurations that generate lower heat levels, and processors  $\leq 205\text{W}$
- Standard heatsinks with integrated heat pipes, suitable for configurations that generate lower heat levels, and processors  $\leq 330\text{W}$
- Performance heatsinks with separate radiator connected heat pipes, suitable for most configurations
- Open-loop liquid cooling of the processors, as described in the [Lenovo Processor Neptune Core Module](#) section
- Open-loop liquid cooling of the processors and memory, as described in the [Lenovo Compute Complex Neptune Core Module](#) section

For details about what configurations are supported with each, see the Thermal Rules section in the Lenovo Docs site for the SR650 V4:

[https://pubs.lenovo.com/sr650-v4/thermal\\_rules](https://pubs.lenovo.com/sr650-v4/thermal_rules)

Ordering information is listed in the following table.

Table 15. Processor cooling options

| Feature code                                                     | Description                                              | Top Choice Express | Purpose                                                                                                                                                        |
|------------------------------------------------------------------|----------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cooling options for machine types 7DGD and 7DGC                  |                                                          |                    |                                                                                                                                                                |
| BPDQ                                                             | ThinkSystem SR650 V3 AL Extrusion Entry Heatsink         | TCE                | Entry 2U heatsink. Automatically selected based on the server configuration.                                                                                   |
| BPDR                                                             | ThinkSystem V4 2U Standard Heatsink                      | TCE                | Standard 2U heatsink. Automatically selected based on the server configuration.                                                                                |
| C3QR                                                             | ThinkSystem 2U V4 Performance Heatsink                   | TCE                | Performance 2U heatsink. Automatically selected based on the server configuration.                                                                             |
| C1XH                                                             | ThinkSystem V4 1U/2U Processor Neptune Core Module       | Not TCE            | Enables open-loop liquid cooling of the processors. See the <a href="#">Lenovo Processor Neptune Core Module</a> section.                                      |
| Cooling options for machine type 7DK2 (CPU+Memory water cooling) |                                                          |                    |                                                                                                                                                                |
| C6AY                                                             | ThinkSystem V4 1U/2U Compute Complex Neptune Core Module | Not TCE            | Enables open-loop liquid cooling of the processors, memory and voltage regulators. See the <a href="#">Lenovo Compute Complex Neptune Core Module</a> section. |

## Lenovo Processor Neptune Core Module - Open-loop liquid cooling

The SR650 V4 also supports advanced direct-water cooling (DWC) capability with the Lenovo Processor Neptune Core Module. This module implements a liquid cooling solution where heat from the processors is removed from the rack and the data center using an open loop and coolant distribution units.

With the Processor Neptune Core Module, all heat generated by the processors is removed from the server using water. This means that the server fans and data center air conditioning units only need to remove the heat generated by the other components. This results in lower air conditioning costs and it enables the use of slower fans which results in lower overall power consumption.

Internal testing has shows a 63% fan power saving per node and 24.7% rack-level power saving with the use of open-loop liquid cooling. Power savings are configuration dependent.

The following figure shows the Lenovo Processor Neptune Core Module.

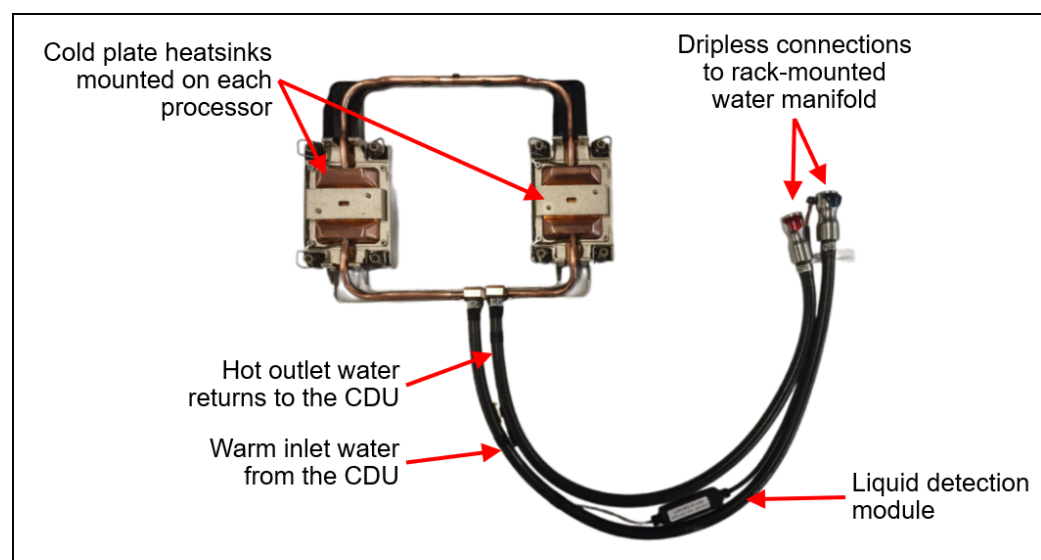


Figure 9. Lenovo Processor Neptune Core Module

The Processor Neptune Core Module also includes a leak detection module which can detect a leakage of more than 0.5ml (about 10 drops) along the length of the tube and then issue an event to the XClarity Controller. XCC will then post an error to the System Event Log and enable further actions. Once the liquid evaporates, a further event is issue to XCC.

The Processor Neptune Core Module is only available in CTO orders, not as a field upgrade. Ordering information is listed in the following table.

Table 16. Lenovo Processor Neptune Core Module

| Part number | Feature code | Description                                        | Top Choice Express |
|-------------|--------------|----------------------------------------------------|--------------------|
| CTO only    | C1XH*        | ThinkSystem V4 1U/2U Processor Neptune Core Module | Not TCE            |

\* In DCSC, this feature code is listed in the Processor tab

Configuration notes:

- The Processor Neptune Core Module requires water infrastructure be available in the rack cabinet and data center, as described in the [Water infrastructure](#) section.
- All processor SKUs are supported

- Either one or two CPUs are supported
- All front drive bay configurations are supported
- Slot 8 is not available for adapters - the water loop is routed through the space otherwise occupied by slot 8
- OCP slot 2 can only have a PCIe x8 connection.
- Rear drive bays are supported
- M.2 adapters are supported based on the configurations in the [Storage configurations](#) section
- Standard fans can be configured in most configurations
- The use of a cable management arm (CMA) is not supported

For more information, see the Thermal Rules page:

[https://pubs.lenovo.com/sr650-v4/thermal\\_rules](https://pubs.lenovo.com/sr650-v4/thermal_rules)

## Lenovo Compute Complex Neptune Core Module - Open-loop liquid cooling

The SR650 V4 also supports advanced direct-water cooling (DWC) of the processors, memory DIMMs, and voltage regulators (collectively called the compute complex) with the Lenovo Compute Complex Neptune Core Module. This heat is removed from the rack and the data center using an open loop and coolant distribution units.

With this solution, all heat generated by the compute complex is removed from the server using water, which means that the server fans and data center air conditioning units only need to remove the heat generated by the other components. This results in lower air conditioning costs and it enables the use of slower fans which results in lower overall power consumption.

Internal testing has shows an 92% fan power saving per node and 35.9% rack-level power saving with the use of open-loop liquid cooling. Power savings are configuration dependent. A rack cabinet with 20x SR650 V4 servers using 100% air cooling (PUE 1.6) requires 44.8 kW of data center power, including data center cooling costs, whereas 20x servers with Compute Complex Neptune Core Module with 80% liquid cooling (PUE 1.05) and 20% air cooling (PUE 1.6) requires 28.7 kW of data center power. This represents a 35.9% power saving at the rack level.

The following figure shows the Lenovo Compute Complex Neptune Core Module. The water flows past all DIMMs in parallel and then passes across both processors.

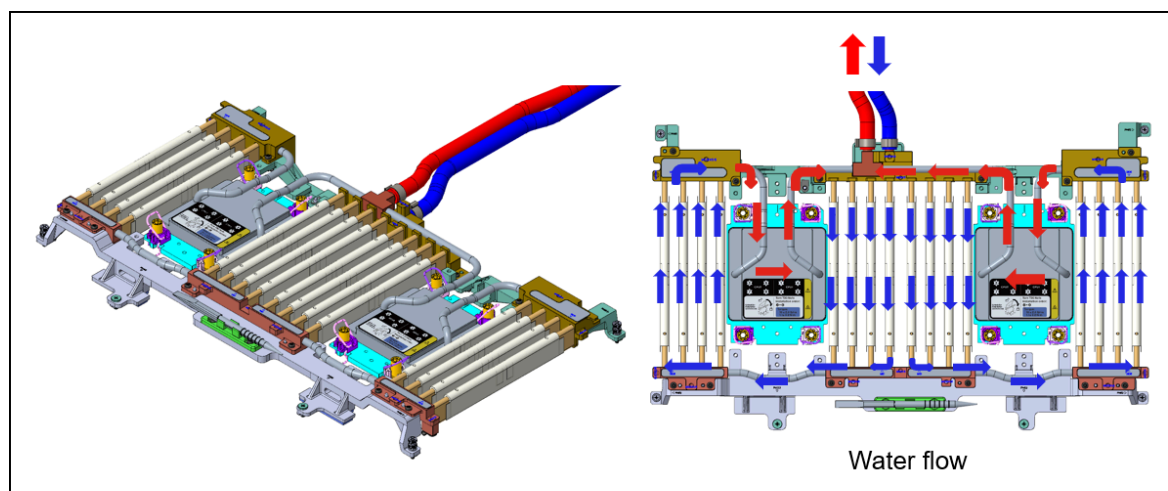


Figure 10. Lenovo Compute Complex Neptune Core Module

The Compute Complex Neptune Core Module also includes a leak detection module which can detect a leakage of more than 0.5ml (about 10 drops) along the length of the tube and then issue an event to the XClarity Controller. XCC will then post an error to the System Event Log and enable further actions. Once the liquid evaporates, a further event is issue to XCC.

The Compute Complex Neptune Core Module is only available in CTO orders, not as a field upgrade. The component listed in the following table will be automatically derived by the configurator when you build a server using machine type 7DK2 (see [Models](#) section).

Table 17. Compute Complex Neptune Core open-loop cooling

| Part number | Feature code | Description                                              |
|-------------|--------------|----------------------------------------------------------|
| CTO only    | C6AY         | ThinkSystem V4 1U/2U Compute Complex Neptune Core Module |

Configuration notes:

- The SR650 V4 must be configured using machine type 7DK2.

- The Compute Complex Neptune Core Module requires water infrastructure be available in the rack cabinet and data center, as described in the [Water infrastructure](#) section.
- Two processors are required; single-processor configurations are not supported
- All processor SKUs are supported except those that are only supported in 1-socket configurations
- Up to 16 DIMMs are supported, 8 per processor (1 DIMM per channel)
- Drive bay configurations are supported as listed in the [Storage configurations](#) section.
- Slot 8 is not available for adapters - the water loop is routed through the space otherwise occupied by slot 8
- OCP slot 2 can only have a PCIe x8 connection.
- M.2 adapters are supported based on the configurations in the [Storage configurations](#) section
- Standard fans can be configured in most configurations
- The use of a cable management arm (CMA) is not supported

For more information, see the Thermal Rules page for the direct water cooling module:

[https://pubs.lenovo.com/sr650-v4/thermal\\_rules](https://pubs.lenovo.com/sr650-v4/thermal_rules)

## UEFI operating modes

The SR650 V4 offers preset operating modes that affect energy consumption and performance. These modes are a collection of predefined low-level UEFI settings that simplify the task of tuning the server to suit your business and workload requirements.

The following table lists the feature codes that allow you to specify the mode you wish to preset in the factory for CTO orders.

**Limited choice for LCC processors:** If you select a processor with an LCC topology (see the Die column in the [Processor features](#) table), you will only be able to select General Computing - Power Efficiency (C3JB) in DCSC. The other modes are still supported, however, they can only be set in the field, not in the factory. Note that this is the only mode that is ERP Lot9-compliant for EU and UK customers.

Table 18. UEFI operating mode presets in DCSC

| Feature code | Description                                                | Top Choice Express |
|--------------|------------------------------------------------------------|--------------------|
| C3JB         | ThinkSystem General Computing - Power Efficiency (default) | TCE                |
| C3JA         | ThinkSystem General Computing - Peak Frequency             | TCE                |
| C3J9         | ThinkSystem General Computing - Max Performance            | TCE                |
| C3J8         | ThinkSystem High Performance Computing (HPC)               | TCE                |
| C9U8         | ThinkSystem Low Latency                                    | TCE                |
| C9UA         | ThinkSystem Virtualization - Power Efficiency              | TCE                |
| C9U9         | ThinkSystem Virtualization - Max Performance               | TCE                |
| C9U7         | ThinkSystem DataBase - Transaction Processing              | TCE                |
| CCVW         | ThinkSystem Database Memory                                | Not TCE            |

The preset modes for the SR650 V4 are as follows:

- **ThinkSystem General Computing - Power Efficiency** (feature C3JB): This workload profile optimizes the performance per watt efficiency with a bias towards performance. This workload profile is analogous to “Efficiency – Favor Performance” operating mode on ThinkSystem V3 servers. This profile contains settings for ENERGY STAR® and ERP Lot9 compliance.

- **ThinkSystem General Computing - Peak Frequency** (feature C3JA): This workload profile is defined by the requirement to drive the highest core frequencies out of a processor across a subset of cores available – not for all cores active. This workload profile benefits workloads requiring either high per core and / or overall CPU package frequency. These workloads may have variable resource demands, are relatively insensitive to overall platform latency, and are generally CPU clock constrained. Tuning a system for highest possible core frequency may mean allowing inactive cores to transfer in and out of sleep states (C-states), which allows active cores to run at higher frequency for different durations of time. Allowing cores to go into low power states allows for higher per core frequency but can introduce “jitter” in the systems clock frequency.
- **ThinkSystem General Computing - Max Performance** (feature C3J9): This workload profile maximizes the absolute performance of the system without regard for power savings. Power savings features are disabled. This operating mode should be used when an application can sustain work across all cores simultaneously and is Non-uniform Memory Access (NUMA) aware.
- **ThinkSystem High Performance Computing (HPC)** (feature C3J8): This profile is for customers running large-scale scientific and engineering workloads. These environments tend to be clustered environments where each node performs at maximum utilization for extended periods of time, and the application is Non-Uniform Memory Access (NUMA) aware.
- **ThinkSystem Low Latency** (feature C9U8): This workload profile seeks to minimize overall transaction latency. Low wait times for core-to-core, core-to-cache, CPU-to-memory, and CPU-to-adaptor communication are critical. Clock frequency is only important in so far as it minimizes intra- and inter-CPU latency. This profile is characterized by maintaining low processor clock variability and maintaining all external CPU links, memory and PCIe subsystems at maximum frequency.
- **ThinkSystem Virtualization - Power Efficiency** (feature C9UA): This workload profile is for virtualization environments. The profile ensures that all available virtualization options are enabled. Power saving features are enabled.
- **ThinkSystem Virtualization - Max Performance** (feature C9U9): This workload profile is for virtualization environments. The profile ensures that all available virtualization options are enabled. Power saving features are disabled.
- **ThinkSystem DataBase - Transaction Processing** (feature C9U7): This workload profile is for online transaction processing (OLTP) applications that require a database back-end.

## Memory

The SR650 V4 uses Lenovo TruDDR5 memory operating at up to 8000 MHz. The server supports up to 32 DIMMs with 2 processors. The processors have 8 memory channels and support 2 DIMMs per channel (DPC). The server supports up to 8TB of memory using 32x 256GB RDIMMs and two processors. The server also supports up to 12x CXL memory modules (6 per CPU) which are installed in E3.S 2T drive bays.

DIMMs operate at the following speeds, up to the memory bus speed of the processor selected. See the [Processor features](#) section for specifics.

- RDIMMs and 3DS RDIMMs:
  - 1 DIMM per channel: Up to 6400 MHz
  - 2 DIMMs per channel using RDIMMs: Up to 5200 MHz
- MRDIMMs (support for MRDIMMs is processor-model dependent)
  - 1 DIMM per channel: 8000 MHz
- CXL 2.0 memory modules - installs in E3.S 2T front drive bays
  - Up to 12 modules per server (6 per CPU)

**Water-cooled memory:** The SR650 V4 can also be configured to implement water-cooling for installed memory DIMMs using the Lenovo Compute Complex Neptune Core Module. The server is limited to 16 DIMM slots (8 DIMMs per processor, 1 DIMM per channel). For more information see the [Lenovo Compute Complex Neptune Core Module](#) section.

Lenovo TruDDR5 memory uses the highest quality components that are sourced from Tier 1 DRAM suppliers and only memory that meets the strict requirements of Lenovo is selected. It is compatibility tested and tuned to maximize performance and reliability. From a service and support standpoint, Lenovo TruDDR5 memory automatically assumes the system warranty, and Lenovo provides service and support worldwide.

- [RDIMM and MRDIMM memory](#)
- [CXL memory](#)
- [Memory rules](#)
- [Memory DIMM filler](#)

### RDIMM and MRDIMM memory

The following table lists the RDIMMs, 3DS RDIMMs, and MRDIMMs memory that are currently supported by the SR650 V4. These DIMMs are installed in the DIMM slots adjacent to the processors. The table also lists the supported quantities. The "CPU+Mem water cooled" DIMM quantity column is when the server is configured to use Lenovo Compute Complex Neptune Core Module for water-cooled memory.

The table includes a Top Choice Express column; for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

Table 19. Memory options for DIMM slots

| Part number | Feature code | Description                                   | Top Choice Express | DRAM technology | Quantities supported per CPU (double for 2 CPUs) |                              |
|-------------|--------------|-----------------------------------------------|--------------------|-----------------|--------------------------------------------------|------------------------------|
|             |              |                                               |                    |                 | P-cores CPUs                                     | P-cores CPU+Mem water cooled |
| x4 RDIMMs   |              |                                               |                    |                 |                                                  |                              |
| 4X77A90964  | C0U9         | ThinkSystem 32GB TruDDR5 6400MHz (1Rx4) RDIMM | TCE                | 16Gb            | 4, 8 per CPU                                     | 4, 8 per CPU                 |

| Part number                                                                                                                        | Feature code | Description                                        | Top Choice Express | DRAM technology | Quantities supported per CPU (double for 2 CPUs) |                              |
|------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------|--------------------|-----------------|--------------------------------------------------|------------------------------|
|                                                                                                                                    |              |                                                    |                    |                 | P-cores CPUs                                     | P-cores CPU+Mem water cooled |
| 4X77A90966                                                                                                                         | C0TQ         | ThinkSystem 64GB TruDDR5 6400MHz (2Rx4) RDIMM      | TCE                | 16Gb            | 1, 4, 8, 12, 16 per CPU                          | 1, 4, 8 per CPU              |
| 4X77A90997                                                                                                                         | BZ7D         | ThinkSystem 96GB TruDDR5 6400MHz (2Rx4) RDIMM      | Not TCE            | 24Gb            | 8, 16 per CPU                                    | 8 per CPU                    |
| 4X77A90993                                                                                                                         | C0U1         | ThinkSystem 128GB TruDDR5 6400MHz (2Rx4) RDIMM     | Not TCE            | 32Gb            | 8, 16 per CPU                                    | 8 per CPU                    |
| x8 RDIMMs                                                                                                                          |              |                                                    |                    |                 |                                                  |                              |
| 4X77A90963                                                                                                                         | C0U2         | ThinkSystem 16GB TruDDR5 6400MHz (1Rx8) RDIMM      | Not TCE            | 16Gb            | 1, 4, 8 per CPU                                  | 1, 4, 8 per CPU              |
| 4X77A90965                                                                                                                         | BYTJ         | ThinkSystem 32GB TruDDR5 6400MHz (2Rx8) RDIMM      | TCE                | 16Gb            | 1, 4, 8, 12, 16 per CPU                          | 1, 4, 8 per CPU              |
| 3DS RDIMMs                                                                                                                         |              |                                                    |                    |                 |                                                  |                              |
| 4X77A90994                                                                                                                         | C0U0         | ThinkSystem 256GB TruDDR5 6400MHz (4Rx4) 3DS RDIMM | Not TCE            | 32Gb            | 8, 12*, 16 per CPU                               | 8 per CPU                    |
| MRDIMMs (operate at 8000 MHz in the SR650 V4) (Note: Not all processors support MRDIMMs - see <a href="#">Processor features</a> ) |              |                                                    |                    |                 |                                                  |                              |
| 4X77A90999                                                                                                                         | C0TX         | ThinkSystem 64GB TruDDR5 8800MHz (2Rx4) MRDIMM     | Not TCE            | -               | 8 per CPU                                        | 8 per CPU                    |

\* 12x 256GB per processor support planned for November 2026

For more information on this memory, see the following Lenovo Press papers,

- [Introduction to DDR5 Memory](#)
- [Introduction to MRDIMM Memory Technology](#)

## CXL memory

The following table lists the CXL memory that are currently supported by the SR650 V4. These memory options are installed in the front E3.S 2T drive bays at the front of the server.

**OS support for CXL:** CXL memory is not supported with Windows Server and VMware ESXi. See OSIG for specifics:

[https://lenovopress.lenovo.com/osig#servers=sr650-v4-xeon-6-p-cores-7dgc-7dgd-7dk2&os\\_families=microsoft-windows-server&os\\_families=vmware-esxi&support=all](https://lenovopress.lenovo.com/osig#servers=sr650-v4-xeon-6-p-cores-7dgc-7dgd-7dk2&os_families=microsoft-windows-server&os_families=vmware-esxi&support=all)

Table 20. CXL memory options

| Part number | Feature code | Description | Top Choice Express | Quantities supported per CPU (double for 2 CPUs) |                              |
|-------------|--------------|-------------|--------------------|--------------------------------------------------|------------------------------|
|             |              |             |                    | P-cores CPUs                                     | P-cores CPU+Mem water cooled |
| CXL DIMMs   |              |             |                    |                                                  |                              |

| Part number | Feature code | Description                        | Top Choice Express | Quantities supported per CPU (double for 2 CPUs) |                              |
|-------------|--------------|------------------------------------|--------------------|--------------------------------------------------|------------------------------|
|             |              |                                    |                    | P-cores CPUs                                     | P-cores CPU+Mem water cooled |
| 4X77A91000  | C0TW         | ThinkSystem 96GB E3.S 2T CXL DIMM  | Not TCE            | 2, 4, or 6 per CPU*                              | No support                   |
| 4X77A91001  | C0TV         | ThinkSystem 128GB E3.S 2T CXL DIMM | Not TCE            | 2, 4, or 6 per CPU*                              | No support                   |

\* Quantities supported depends on DIMMs installed and the CXL interleaving setting. See the [Memory module installation rules](#) page in the User Guide

For servers that have a combination of DDR5 memory and CXL memory, you can specify how you want to the memory spaces to be presented to the operating system. For CTO orders, you can specify the factory to set the memory mode, as described in the table below. The memory mode can also be changed in UEFI at a later stage.

Table 21. Interleaving choices for CTO orders (Memory tab in DCSC)

| Part number | Feature code | Description                                                                                       | Top Choice Express | Purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|--------------|---------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CTO only    | C8VB*        | DDR5 and (volatile) CXL Memory interleaved together in one 12-way set                             | TCE                | Hetero or Heterogeneous mode. This mode supports memory interleave between CXL memory and DDR memory. Interleaving of memory requests across a combination of native attach DDR5 channels and CXL-connected memory to increase aggregate bandwidth. The entire combined capacity of DDR memory and CXL memory is visible to the software as a single NUMA domain. As a result, no software changes are needed in the system to use heterogeneous interleave mode.             |
| CTO only    | C8VC         | Native DDR5(1LM) and CXL Memory(volatile) visible to SW as separate tiers, separately interleaved | TCE                | 1LM+Vol mode. This is a 2-tier memory mode, where the DDR5 memory and CXL memory are different address spaces and separate NUMA nodes. This mode is also referred to as Software Managed tiering because the application must manage the placement of data in separate tiers and must manage any desired movement of data between tiers. This management of placement and movement may be performed by the OS, or by a higher-level middleware or directly by an application. |

\* Not supported with LCC processors. See the [Processor features](#) section to see which processors have an LCC die.

For more information on this memory, see the following Lenovo Press paper,

- [Introduction to CXL 2.0 Memory](#)

## Memory rules

The following rules apply when specifying the memory configuration:

- The tables above list the supported quantities per processor. For two processors, install the same number of DIMMs to each processor. Other quantities are not supported.
- Only a subset of processors support MRDIMMs - see the table in the [Processor features](#) section for specifics.

- All installed RDIMMs must be identical part numbers; mixing not supported

CXL-related rules:

- See the SR650 V4 User Guide for the supported combinations of CXL memory and RDIMM memory: [https://pubs.lenovo.com/sr650-v4/memory\\_module\\_installation\\_order\\_cxl](https://pubs.lenovo.com/sr650-v4/memory_module_installation_order_cxl)
- CXL heterogeneous mode (feature C8VB) is not supported with processors with an LCC die. See the [Processor features](#) section to see which processors have an LCC die.
- In the DCSC configurator, you must first select a CXL memory mode (Memory tab) before CXL memory modules can be selected (Storage tab)
- CXL memory is installed in E3.S 2T drive bays, however hot-swap functionality is not supported

For best performance, consider the following:

- Ensure the memory installed is at least the same speed as the memory bus of the selected processor.
- Populate all memory channels.

The following memory protection technologies are supported:

- ECC detection/correction
- Bounded Fault detection/correction
- SDDC (for 10x4-based memory DIMMs; look for "x4" in the DIMM description)
- ADDDC (for 10x4-based memory DIMMs)
- Memory mirroring

See the Lenovo Press article "RAS Features of the Lenovo ThinkSystem Intel Servers" for more information about memory RAS features: <https://lenovopress.lenovo.com/lp1711-ras-features-of-the-lenovo-thinksystem-intel-servers>

If memory channel mirroring is used, then DIMMs must be installed in pairs (minimum of one pair per processor), and both DIMMs in the pair must be identical in type and size. 50% of the installed capacity is available to the operating system.

Memory rank sparing is implemented using ADDDC/ADC-SR/ADDDC-MR to provide DRAM-level sparing feature support.

## Memory DIMM filler

When a GPU (or another high-thermal component) is added as a field upgrade, all empty DIMM slots must have a DIMM filler (also called a dummy DIMM or DIMM blank) installed in each DIMM slot, to ensure the optimal air flow. Ordering information is in the following table. For CTO orders, the DIMM fillers are derived by the configurator.

Table 22. ThinkSystem Dummy DIMM Package Kit

| Part number | Feature code | Description                                                       |
|-------------|--------------|-------------------------------------------------------------------|
| 4M27A11810  | AURS*        | ThinkSystem Dummy DIMM Package Kit<br>(contains 24x DIMM fillers) |

\* In CTO orders, feature code AURS is derived and is 1x DIMM filler

## Internal storage

The SR650 V4 has three drive bay zones and supports up to 16x 3.5-inch or 40x 2.5-inch hot-swap drive bays or a combination of drive bays, depending on the selected chassis and backplane configuration. The server also supports configurations with E3.S 1T and E3.S 2T drive bays including support for CXL memory in the E3.S 2T form factor. The server supports configurations without any drive bays if desired.

The three drive bay zones are as follows:

- Front:
  - Up to 12x 3.5-inch hot-swap bays, or
  - Up to 24x 2.5-inch hot-swap bays, or
  - Up to 32x E3.S 1T or 12x E3.S 2T hot-swap drive bays (or a mix of 1T/2T drive bays)
- Middle:
  - 8x 2.5-inch simple-swap bays
- Rear:
  - 4x 3.5-inch hot-swap bays, or
  - Up to 8x 2.5-inch hot-swap bays

The server also supports one or two M.2 drives, in three possible locations, configuration dependent:

- Installed in an M.2 adapter internal to the server (non-hot-swap)
- Hot-swap in the rear of the server
- Hot-swap in the front of the server

In this section:

- [NVMe drive support](#)
- [Tri-Mode support - RAID 940 adapters](#)
- [Front drive bays](#)
- [Mid drive bays](#)
- [Rear drive bays](#)
- [Storage configurations](#)
- [Field upgrades](#)
- [M.2 drives](#)
- [SED encryption key management with SKLM](#)
- [Encryption Enablement](#)

## NVMe drive support

The SR650 V4 supports NVMe drives to maximize storage performance.

- In a 2.5-inch drive configuration, the server supports up to 36 NVMe drives without oversubscription (that is, each x4 drive has a dedicated x4 (4 lanes) connection to the processor)
  - Up to 24 installed in front bays
  - Up to 28 installed in the front and rear bays
  - Up to 32 installed in front and mid bays
  - Up to 36 installed in front, mid and rear bays
- In a 3.5-inch drive configuration, the server supports up to 4x 3.5-inch NVMe or 8x 2.5-inch NVMe drives without oversubscription
- In a E3.S front drive configuration, the server supports up to 32x NVMe drives without oversubscription:
  - 32x E3.S 1T NVMe drives at the front
  - 12x E3.S 2T NVMe drives at the front
  - Combinations of E3.S 1T and 2T drive bays

In addition, the SR650 V4 supports two M.2 NVMe drives for use as boot drives, as described in the [M.2 drives](#) section

The specifics of these configurations are covered in the [Storage configurations](#) section. The tables in those sections indicate the number of NVMe drives in each configuration. The storage configuration tables also include a column that indicates the number of PCIe slots that are available.

The number of NVMe drives affects the number of PCIe slots that are available, as listed in the following tables. This is because the onboard PCIe MCIO connectors are used for both slots and NVMe drives.

Table 23. PCIe slot availability based on 3.5-inch NVMe or AnyBay drive bays configured

| 3.5-inch NVMe/AnyBay drive bays selected | Maximum rear PCIe slots  |
|------------------------------------------|--------------------------|
| 8 drive bays                             | 10 slots (+ 2 OCP slots) |
| 4 drive bays                             | 10 slots (+ 2 OCP slots) |

Table 24. PCIe slot availability based on 2.5-inch NVMe or AnyBay drive bays configured

| 2.5-inch NVMe/AnyBay drive bays selected | Maximum rear PCIe slots  |
|------------------------------------------|--------------------------|
| 36 drive bays (24 front, 8 mid, 4 rear)  | 1 slot (+ 2 OCP slots)   |
| 32 drive bays (24 front, 8 mid)          | 2 slots (+ 2 OCP slots)  |
| 28 drive bays (24 front, 4 rear)         | 2 slots (+ 2 OCP slots)  |
| 24 drive bays (all front)                | 4 slots (+ 2 OCP slots)  |
| 16 drive bays (all front)                | 6 slots (+ 2 OCP slots)  |
| 8 drive bays (all front)                 | 10 slots (+ 2 OCP slots) |

Table 25. PCIe slot availability based on E3.S 1T drive bays configured

| E3.S 1T drive bays selected | Maximum rear PCIe slots |
|-----------------------------|-------------------------|
| 32 drive bays               | 2 slots (+ 2 OCP slots) |
| 28 drive bays               | 2 slots (+ 2 OCP slots) |
| 24 drive bays               | 4 slots (+ 2 OCP slots) |
| 20 drive bays               | 4 slots (+ 2 OCP slots) |
| 16 drive bays               | 6 slots (+ 2 OCP slots) |

| E3.S 1T drive bays selected | Maximum rear PCIe slots  |
|-----------------------------|--------------------------|
| 12 drive bays               | 6 slots (+ 2 OCP slots)  |
| 8 drive bays                | 10 slots (+ 2 OCP slots) |

### Tri-Mode support - RAID 940 adapters

The RAID 940 adapters support NVMe through a feature named Tri-Mode support (or Trimode support). This feature enables the use of NVMe U.3 drives at the same time as SAS and SATA drives. Tri-Mode requires an AnyBay backplane. Cabling of the controller to the backplanes is the same as with SAS/SATA drives, and the NVMe drives are connected via a PCIe x1 link to the controller.

NVMe drives connected using Tri-Mode support provide better performance than SAS or SATA drives: A SATA SSD has a data rate of 6Gbps, a SAS SSD has a data rate of 12Gbps, whereas an NVMe U.3 Gen 4 SSD with a PCIe x1 link will have a data rate of 16Gbps. NVMe drives typically also have lower latency and higher IOPS compared to SAS and SATA drives. Tri-Mode is supported with U.3 NVMe drives and requires an AnyBay backplane.

**Tri-Mode requires U.3 drives:** Only NVMe drives with a U.3 interface are supported. U.2 drives are not supported. See the [Internal drive options](#) section for the U.3 drives supported by the server.

**New as of July 2026:** To enable Tri-Mode in CTO orders of the SR650 V4, use the following feature code:

Table 26. Tri-Mode Enablement

| Feature code | Description         |
|--------------|---------------------|
| CH5Z         | Tri-Mode Enablement |

The process to configure Tri-Mode for a CTO order in DCSC is as follows:

1. In the **Storage** tab, select one of the 940-8i or 940-16i controllers from the **SAS/SATA RAID Controller** group
2. In the **Internal Storage** tab, select select **Tri-Mode Enablement** (feature CH5Z)
3. You can then select either NVMe or SATA or SATA drives or any combination of these drives, up to the number supported by the server and the backplanes selected.

#### Tips:

- Feature CH5Z is not selectable until one of the 940-8i or 940-16i adapters in **SAS/SATA RAID Controller** group is selected.
- Feature CH5Z is not selectable if **Non RAID NVMe** (BC4V) is selected.
- The existing Tri-Mode "for U.3" adapter feature codes (features BGM1, BGM0, BM36, BDY4) in the **Tri-Mode Controller** group are for existing orders only and are being phased out (deprecated)

## Front drive bays

The front drive bay zone supports the following configurations:

- 8x or 12x 3.5-inch drive bays (hot-swap)
- 8x, 16x or 24x 2.5-inch drive bays (hot-swap)
- Up to 32x E3.S 1T drive bays (hot-swap)
- Up to 12x E3.S 2T drive bays for CXL memory (non-hot-swap)
- No backplanes and no drives (supports [field upgrades](#))

The specific combinations that are supported in the SR650 V4 are shown in the following figures. The feature codes listed are the backplane feature codes when ordering CTO and correspond to the feature codes listed in the table below the figure.

The following figure shows the supported 3.5-inch drive bay configurations.

**Tip:** The 8x 3.5-inch backplane (C4DB) is for GPU configurations only.

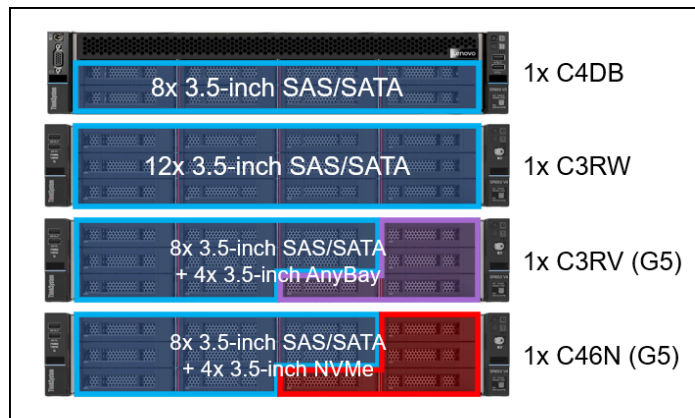


Figure 11. SR650 V4 front drive bay configurations - 3.5-inch drive bays\

The following two figures show the supported 2.5-inch drive bay configurations.

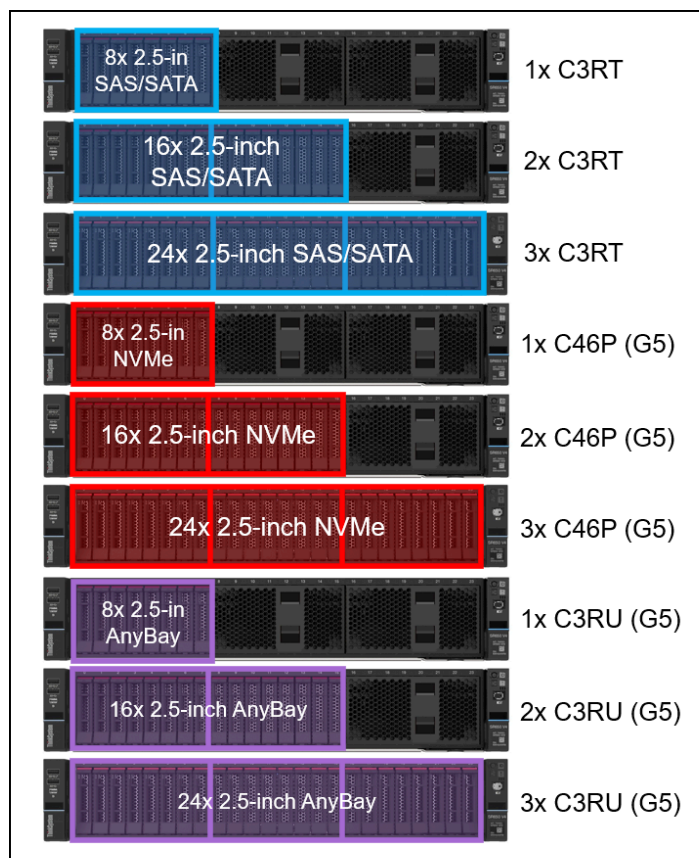


Figure 12. SR650 V4 front drive bay configurations - 2.5-inch drive bays, all the same drive type

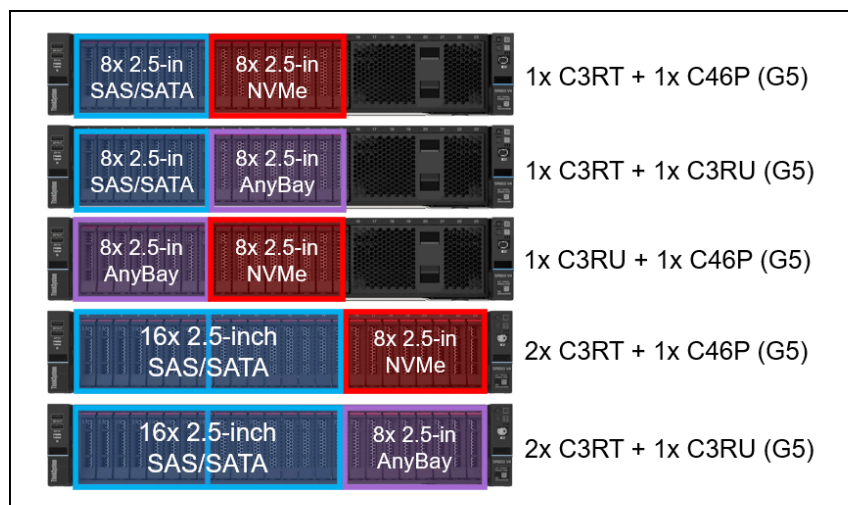


Figure 13. SR650 V4 front drive bay configurations - 2.5-inch drive bays, combinations

The following two figures show the supported E3.S drive bay configurations.

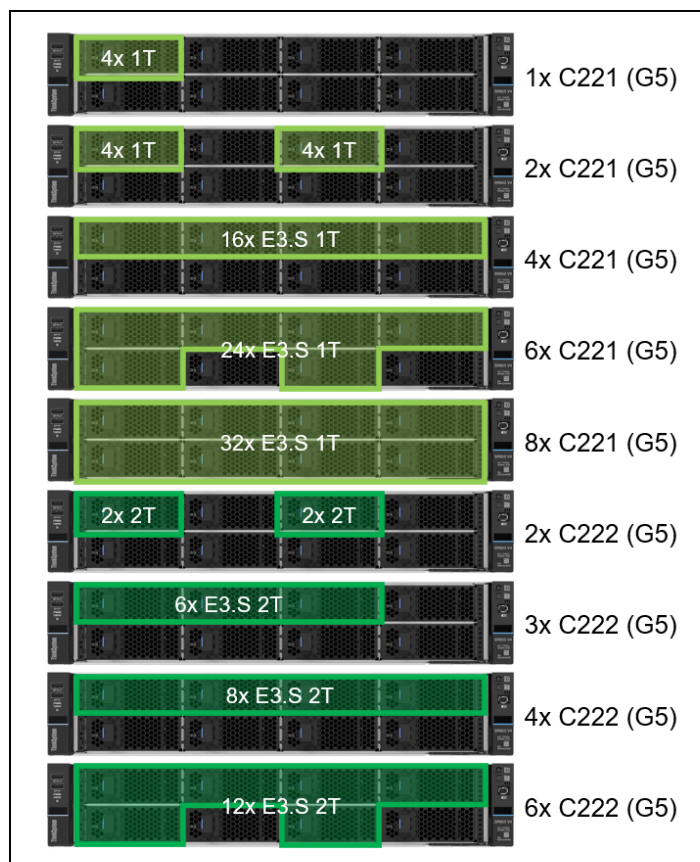


Figure 14. SR650 V4 front drive bay configurations - E3.S drive bays, all the same drive type

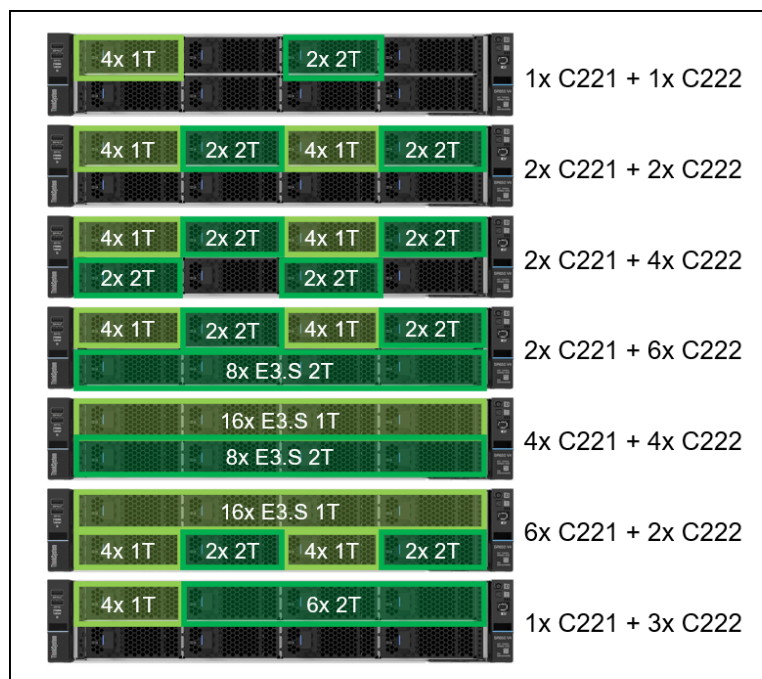


Figure 15. SR650 V4 front drive bay configurations - E3.S drive bays, combinations

The backplanes used to provide these drive bays are listed in the following table. The table includes a Top

Choice Express column: for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

**GPU configurations with 3.5-inch drives:** The 8x 3.5-inch backplane (C4DB) is for use with GPUs only (see configurations 117-1 and 117-2 in the [Storage configurations](#) section). The use of a GPU with 3.5-inch front drive bays requires this backplane. In DCSC, the selection of this backplane requires a GPU.

**Field upgrades:** All front backplanes are available as part numbers for field upgrades along with require cable option kits, as described in the [Field upgrades](#) section below.

Table 27. Backplanes for front drive bays

| Feature                                                                                                       | Description                                                                              | Top Choice Express | Bays | PCIe Gen | SAS Gen | Max qty |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------|------|----------|---------|---------|
| Front 3.5-inch drive backplanes (requires base feature C3QL - see <a href="#">Base feature codes</a> section) |                                                                                          |                    |      |          |         |         |
| C4DB                                                                                                          | ThinkSystem 2U V4 12x3.5" SAS/SATA Backplane (For GPU Configurations Only, 8 Drives Max) | TCE                | 8    | -        | 12Gb    | 1       |
| C3RW                                                                                                          | ThinkSystem 2U V4 12x3.5" SAS/SATA Backplane                                             | TCE                | 12   | -        | 12Gb    | 1       |
| C3RV                                                                                                          | ThinkSystem 2U V4 8x3.5" SAS/SATA+4x 3.5" AnyBay Backplane                               | Not TCE            | 12   | Gen5     | 24Gb    | 1       |
| C46N                                                                                                          | ThinkSystem 2U V4 8x3.5" SAS/SATA+4x 3.5" NVMe Backplane                                 | TCE                | 12   | Gen5     | 24Gb    | 1       |
| Front 2.5-inch drive backplanes (requires base feature C3QK - see <a href="#">Base feature codes</a> section) |                                                                                          |                    |      |          |         |         |
| C3RT                                                                                                          | ThinkSystem 2U V4 8x2.5" SAS/SATA Backplane                                              | TCE                | 8    | -        | 12Gb    | 3       |
| C3RU                                                                                                          | ThinkSystem 2U V4 8x2.5" AnyBay Backplane                                                | TCE                | 8    | Gen5     | 24Gb    | 3       |
| C46P                                                                                                          | ThinkSystem 2U V4 8x2.5" NVMe Backplane                                                  | TCE                | 8    | Gen5     | -       | 3       |
| Front E3.S drive backplanes (requires base feature C3QM - see <a href="#">Base feature codes</a> section)     |                                                                                          |                    |      |          |         |         |
| C221                                                                                                          | ThinkSystem V4 EDSFF E3.S 4x1T NVMe Gen5 Backplane                                       | Not TCE            | 4    | Gen5     | -       | 8       |
| C222                                                                                                          | ThinkSystem V4 EDSFF E3.S 2x2T NVMe Gen5 Backplane                                       | Not TCE            | 2    | Gen5     | -       | 8       |

The use of front drive bays has the following configuration rules:

- The SR650 V4 also supports configurations without any drive bays, allowing for drive bay upgrades as described in the [field upgrades](#) section.
- If 3.5-inch front drive bays are used, an internal (CFF) RAID adapter or HBA is not supported as the adapter and bays occupy the same physical space
- Backplane C4DB (ThinkSystem 2U V4 12x3.5" SAS/SATA Backplane (For GPU Configurations Only, 8 Drives Max)) only supports 8x drives even though it is a 12-bay backplane. This is so that the backplane can be configured with GPUs which require additional airflow. Since this is a 12-bay backplane, a 16i adapter is required for the proper cabling of the 8x active drive bays. An 8i adapter may be supported through special bid - see your sales representative for details.
- Any of the following front drive configurations can be configured with front-mounted hot-swap M.2 drives, as described in the [M.2 drives](#) section:
  - Any 8x 2.5-inch and 16x 2.5-inch drive configuration (SAS/SATA, AnyBay, or NVMe)
  - Any E3.S configuration with 6 or fewer backplanes

## Mid drive bays

The SR650 V4 supports simple-swap drives installed in the middle of the server chassis. The drive bays are accessible by removing the top lid of the server and levering the mid drive chassis up at the front.

The following configurations are supported:

- 8x 2.5-inch simple-swap SAS/SATA drive bays
- 8x 2.5-inch simple-swap NVMe drive bays

The drive bays in the open position are shown in the following figure.

**Simple-swap drive bays:** The drives that are installed in the mid-chassis drive bays are simple-swap, not hot-swap, even though the drives use hot-swap drive trays.

**M.2 support:** When mid drive bays are configured, the M.2 adapter is installed on the mid drive bay mechanical as shown in the images.

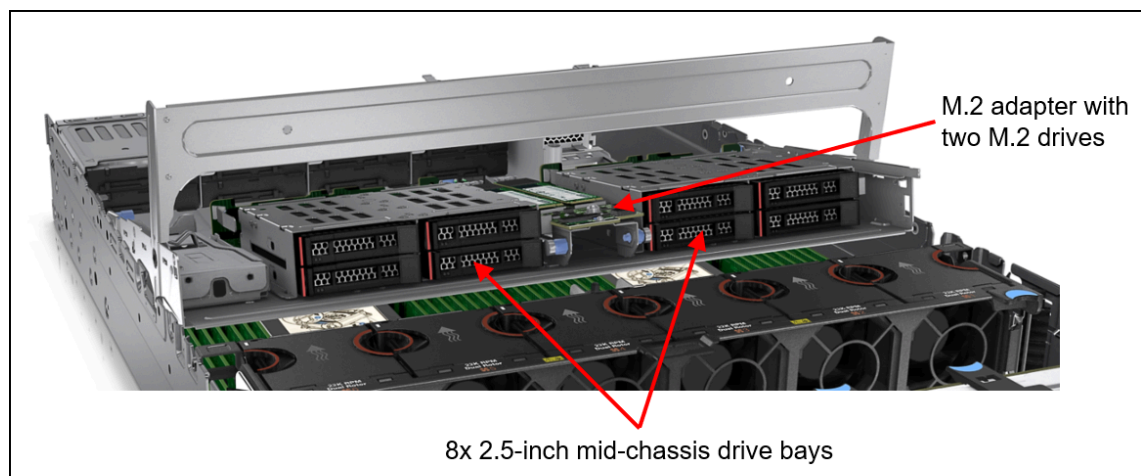


Figure 16. Mid-chassis drive bays

The backplanes used to provide these drive bays are listed in the following table.

Table 28. Backplanes for mid-chassis drive bays

| Feature                        | Description                                         | Top Choice Express | Bays | PCIe Gen | SAS Gen | Max qty |
|--------------------------------|-----------------------------------------------------|--------------------|------|----------|---------|---------|
| Mid - 2.5-inch drive backplane |                                                     |                    |      |          |         |         |
| C46L                           | ThinkSystem 2U V4 Mid 4x2.5" SAS/SATA Backplane     | Not TCE            | 4    | -        | 24Gb    | 2       |
| C46J                           | ThinkSystem 2U V4 Middle 4x2.5" NVMe Gen5 Backplane | TCE                | 4    | Gen5     | -       | 2       |

‡ 2.5-inch drive backplanes for the mid-chassis area must be installed in pairs. NVMe and SAS/SATA cannot be mixed.

**Field upgrades:** Backplanes are available as part numbers for field upgrades along with require cable option kits, as described in the [Field upgrades](#) section below.

The use of drive bays in the mid-chassis area has the following configuration rules:

- All processors are supported. Higher TDP processors will require the performance heatsinks.
- Full-length adapter cards are not supported

- GPUs (including low profile GPUs such as the NVIDIA L4) are not supported
- The mid drive bays and slots 2 & 10 share the same power connectors, so mid drive bays are mutually exclusive with slots 2 and 10. The only exception to this is config 27 with 8x mid drives and 8x rear drives - this configuration also supports slots 1 & 2.

## Rear drive bays

The SR650 V4 supports hot-swap drives installed at the rear of the server chassis. Supported configurations are as follows:

- 3.5-inch hot-swap drives
  - 4x SAS/SATA drive bays
- 2.5-inch hot-swap drives
  - 4x SAS/SATA drive bays
  - 4x NVMe Gen5 drive bays
  - 4x AnyBay Gen5 drive bays
  - 8x SAS/SATA drive bays

The configurations are shown in the following figure.

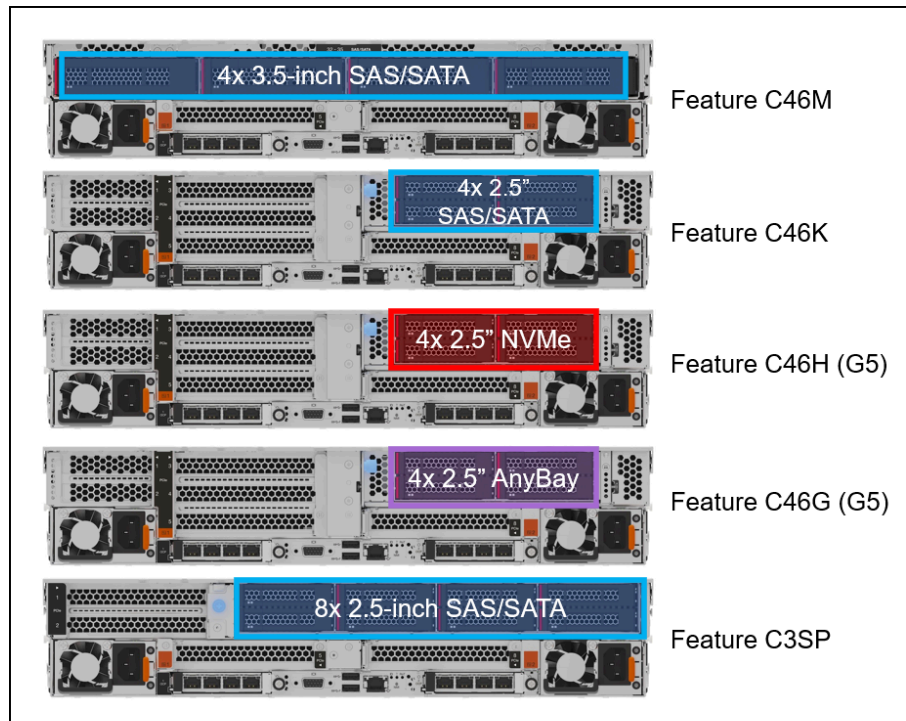


Figure 17. Rear 2.5-inch and 3.5-inch drive bay configurations

The backplanes used to provide these drive bays are listed in the following table.

Table 29. Backplanes for rear drive bays

| Feature                          | Description                                      | Top Choice Express | Bays | PCIe Gen | SAS Gen | Max qty |
|----------------------------------|--------------------------------------------------|--------------------|------|----------|---------|---------|
| Rear - 3.5-inch drive backplanes |                                                  |                    |      |          |         |         |
| C46M                             | ThinkSystem 2U V4 Rear 4x3.5" SAS/SATA Backplane | Not TCE            | 4    | -        | 24Gb    | 1       |
| Rear - 2.5-inch drive backplanes |                                                  |                    |      |          |         |         |

| Feature | Description                                         | Top Choice Express | Bays | PCIe Gen | SAS Gen | Max qty |
|---------|-----------------------------------------------------|--------------------|------|----------|---------|---------|
| C46K    | ThinkSystem 2U V4 Rear 4x2.5" SAS/SATA Backplane    | Not TCE            | 4    | -        | 24Gb    | 1       |
| C46H    | ThinkSystem 2U V4 Rear 4x2.5" NVMe Gen5 Backplane   | TCE                | 4    | Gen5     | 24Gb    | 1       |
| C46G    | ThinkSystem 2U V4 Rear 4x2.5" AnyBay Gen5 Backplane | TCE                | 4    | Gen5     | 24Gb    | 1       |
| C3SP    | ThinkSystem 2U V4 Rear 8x2.5" SAS/SATA Backplane    | Not TCE            | 8    | -        | 12Gb    | 1       |

**Field upgrades:** Backplanes are available as part numbers for field upgrades along with require cable option kits, as described in the [Field upgrades](#) section below.

The use of rear drive bays has the following configuration rules:

- The use of rear bays restricts the number of slots and the choice of risers that are supported. See the [I/O expansion](#) section for details.

## Storage configurations

This section describes the various combinations of front and rear drives that the server supports, as well as M.2 support.

**Tip:** These tables are based on Storage Configs v2.9

In this section the divided in two groups, configurations for machine types 7DGD and 7DGC (air cooled, closed loop cooling, open loop CPU-only cooling) and configurations for machine type 7DK2 (Compute Complex Neptune open-loop CPU+Memory cooling):

- [Overview of storage configs - 3.5-inch front drive bays](#)
- [Overview of storage configs - 2.5-inch front drive bays](#)
- [Overview of storage configs - E3.S front drive bays](#)
- [Overview of storage configs - Compute Complex Neptune Core](#)
- [Details of storage configs - 3.5-inch front drives](#)
- [Details of storage configs - 2.5-inch front drives](#)
- [Details of storage configs - E3.S front drives](#)
- [Details of storage configs - Compute Complex Neptune Core](#)

The following tables summarize the storage configurations for the SR650 V4. For details, including processor requirements, cooling options, M.2 support, and controller selections, see the Details tables.

## Overview of storage configs - 3.5-inch front drive bays

The following table summarizes the storage configurations with 3.5-inch front drives bays.

Jump down to the [details of the configurations](#).

Return to [Storage configurations](#).

Table 30. Overview of storage configs - 3.5-inch front drive bays

| Config | Total drives (NVMe) | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                                                    |
|--------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|-------------------------------------------------------------------------------|
|        |                     | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                               |
| 35     | 12 (0)              | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" SAS/SATA (C3RW)                                                       |
| 36     | 12 (4)              | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" (8x SAS/SATA, 4x AnyBay G5) (C3RV)                                    |
| 37     | 12 (4)              | 8                | 0      | 4    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" (8x SAS/SATA, 4x NVMe G5) (C46N)                                      |
| 38     | 16 (0)              | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 4             | 0             | 0         | 0           | Front: 12x3.5" SAS/SATA (C3RW) Rear: 4x3.5" SAS/SATA (C46M)                   |
| 41     | 16 (4)              | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 4           | Front: 12x3.5" SAS/SATA (C3RW) Rear: 4x2.5" AnyBay G5 (C46G)                  |
| 42     | 20 (8)              | 12               | 0      | 0    | 0       | 0       | 0             | 8         | 0             | 0             | 0         | 0           | Front: 12x3.5" SAS/SATA (C3RW) Mid: 2x 4x2.5" NVMe G5 (2x C46J)               |
| 117    | 8 (0)               | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x3.5" SAS/SATA (C4DB)                                                        |
| 118    | 16 (4)              | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | Front: 12x3.5" SAS/SATA (C3RW) Rear: 4x2.5" NVMe G5 (C46H)                    |
| 122    | 16 (8)              | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | Front: 12x3.5" (8x SAS/SATA, 4x AnyBay G5) (C3RV) Rear: 4x2.5" NVMe G5 (C46H) |

## Overview of storage configs - 2.5-inch front drive bays

The following table summarizes the storage configurations with 2.5-inch front drives bays.

Jump down to the [details of the configurations](#).

Return to [Storage configurations](#).

Table 31. Overview of storage configs - 2.5-inch front drive bays

| Config | Total drives (NVMe) | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                                                                                           |
|--------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|----------------------------------------------------------------------------------------------------------------------|
|        |                     | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                                                                      |
| 1      | 8 (0)               | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT)                                                                                               |
| 2      | 8 (8)               | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)                                                                                                |
| 3      | 8 (8)               | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU)                                                                                              |
| 5      | 8 (8)               | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) (Tri-mode)                                                                                   |
| 6      | 16 (0)              | 16               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" SAS/SATA (2x C3RT)                                                                                         |
| 7      | 16 (16)             | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                                                                                          |
| 8      | 16 (16)             | 0                | 8      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) + 8x2.5" NVMe G5 (C46P)                                                                      |
| 9      | 16 (8)              | 8                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 8x2.5" AnyBay G5 (C3RU)                                                                     |
| 10     | 16 (8)              | 8                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 8x2.5" NVMe G5 (C46P)                                                                       |
| 11     | 16 (16)             | 0                | 16     | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" AnyBay G5 (2x C3RU) (Tri-mode)                                                                             |
| 12     | 24 (0)              | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 3x 8x2.5" SAS/SATA (3x C3RT)                                                                                         |
| 13     | 24 (24)             | 0                | 0      | 24   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 3x 8x2.5" NVMe G5 (3x C46P)                                                                                          |
| 14     | 24 (8)              | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU)                                                               |
| 15     | 24 (8)              | 16               | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" NVMe G5 (C46P)                                                                 |
| 16     | 24 (24)             | 0                | 24     | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 3x 8x2.5" AnyBay G5 (3x C3RU) (Tri-mode)                                                                             |
| 17     | 28 (0)              | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 4             | 0         | 0           | <b>Front:</b> 3x 8x2.5" SAS/SATA (3x C3RT) <b>Rear:</b> 4x2.5" SAS/SATA (C46K)                                       |
| 18     | 28 (4)              | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 4           | <b>Front:</b> 3x 8x2.5" SAS/SATA (3x C3RT) <b>Rear:</b> 4x2.5" AnyBay G5 (C46G)                                      |
| 19     | 28 (8)              | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 4             | 0         | 0           | <b>Front:</b> 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) <b>Rear:</b> 4x2.5" SAS/SATA (C46K)             |
| 20     | 32 (0)              | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0             | 0             | 0         | 0           | <b>Front:</b> 3x 8x2.5" SAS/SATA (3x C3RT) <b>Mid:</b> 2x 4x2.5" SAS/SATA (2x C46L)                                  |
| 21     | 32 (8)              | 16               | 8      | 0    | 0       | 0       | 8             | 0         | 0             | 0             | 0         | 0           | <b>Front:</b> 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) <b>Mid:</b> 2x 4x2.5" SAS/SATA (2x C46L)        |
| 22     | 32 (8)              | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 8             | 0         | 0           | <b>Front:</b> 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) <b>Rear:</b> 8x2.5" SAS/SATA (C3SP)             |
| 23     | 32 (0)              | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 8             | 0         | 0           | <b>Front:</b> 3x 8x2.5" SAS/SATA (3x C3RT) <b>Rear:</b> 8x2.5" SAS/SATA (C3SP)                                       |
| 24     | 32 (32)             | 0                | 0      | 24   | 0       | 0       | 0             | 8         | 0             | 0             | 0         | 0           | <b>Front:</b> 3x 8x2.5" NVMe G5 (3x C46P) <b>Mid:</b> 2x 4x2.5" NVMe G5 (2x C46J)                                    |
| 25     | 36 (36)             | 0                | 0      | 24   | 0       | 0       | 0             | 8         | 0             | 0             | 4         | 0           | <b>Front:</b> 3x 8x2.5" NVMe G5 (3x C46P) <b>Mid:</b> 2x 4x2.5" NVMe G5 (2x C46J) <b>Rear:</b> 4x2.5" NVMe G5 (C46H) |

| Config | Total drives (NVMe) | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                                                                    | Backplanes |
|--------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|----------------------------------------------------------------------------------------------------|------------|
|        |                     | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                                                    |            |
| 26     | 36 (0)              | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0             | 4             | 0         | 0           | Front: 3x 8x2.5" SAS/SATA (3x C3RT) Mid: 2x 4x2.5" SAS/SATA (2x C46L) Rear: 4x2.5" SAS/SATA (C46K) |            |
| 27     | 40 (0)              | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0             | 8             | 0         | 0           | Front: 3x 8x2.5" SAS/SATA (3x C3RT) Mid: 2x 4x2.5" SAS/SATA (2x C46L) Rear: 8x2.5" SAS/SATA (C3SP) |            |
| 101    | 28 (28)             | 0                | 0      | 24   | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | Front: 3x 8x2.5" NVMe G5 (3x C46P) Rear: 4x2.5" NVMe G5 (C46H)                                     |            |
| 121    | 24 (16)             | 8                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 2x 8x2.5" NVMe G5 (2x C46P)                                               |            |
| 151    | 8 (8)               | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)                                                                              |            |
| 152    | 16 (16)             | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                                                                        |            |

## Overview of storage configs - E3.S front drive bays

The following table summarizes the storage configurations with E3.S front drives bays.

Jump down to the [details of the configurations](#).

Return to [Storage configurations](#).

Table 32. Overview of storage configs - E3.S front drive bays

| Config | Total drives (NVMe) | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                                 | Backplanes |
|--------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|-----------------------------------------------------------------|------------|
|        |                     | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                 |            |
| 59     | 8 (8)               | 0                | 0      | 0    | 8       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221)                                  |            |
| 60     | 16 (16)             | 0                | 0      | 0    | 16      | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 4xE3.S 1T NVMe G5 (4x C221)                                  |            |
| 61     | 24 (24)             | 0                | 0      | 0    | 24      | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 6x 4xE3.S 1T NVMe G5 (6x C221)                                  |            |
| 62     | 32 (32)             | 0                | 0      | 0    | 32      | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x 4xE3.S 1T NVMe G5 (8x C221)                                  |            |
| 63     | 4 (4)               | 0                | 0      | 0    | 0       | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 2xE3.S 2T NVMe G5 (2x C222)                                  |            |
| 64     | 8 (8)               | 0                | 0      | 0    | 0       | 8       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 2xE3.S 2T NVMe G5 (4x C222)                                  |            |
| 65     | 12 (12)             | 0                | 0      | 0    | 0       | 12      | 0             | 0         | 0             | 0             | 0         | 0           | 6x 2xE3.S 2T NVMe G5 (6x C222)                                  |            |
| 66     | 6 (6)               | 0                | 0      | 0    | 4       | 2       | 0             | 0         | 0             | 0             | 0         | 0           | 4xE3.S 1T NVMe G5 (C221) + 2xE3.S 2T NVMe G5 (C222)             |            |
| 67     | 12 (12)             | 0                | 0      | 0    | 8       | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221) + 2x 2xE3.S 2T NVMe G5 (2x C222) |            |
| 68     | 16 (16)             | 0                | 0      | 0    | 8       | 8       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221) + 4x 2xE3.S 2T NVMe G5 (4x C222) |            |
| 69     | 20 (20)             | 0                | 0      | 0    | 8       | 12      | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221) + 6x 2xE3.S 2T NVMe G5 (6x C222) |            |
| 70     | 20 (20)             | 0                | 0      | 0    | 16      | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 4xE3.S 1T NVMe G5 (4x C221) + 2x 2xE3.S 2T NVMe G5 (2x C222) |            |
| 71     | 24 (24)             | 0                | 0      | 0    | 16      | 8       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 4xE3.S 1T NVMe G5 (4x C221) + 4x 2xE3.S 2T NVMe G5 (4x C222) |            |
| 72     | 28 (28)             | 0                | 0      | 0    | 24      | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 6x 4xE3.S 1T NVMe G5 (6x C221) + 2x 2xE3.S 2T NVMe G5 (2x C222) |            |
| 102    | 2 (2)               | 0                | 0      | 0    | 0       | 2       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 2xE3.S 2T NVMe G5 (2x C222)                                  |            |
| 119    | 6 (6)               | 0                | 0      | 0    | 0       | 6       | 0             | 0         | 0             | 0             | 0         | 0           | 3x 2xE3.S 2T NVMe G5 (3x C222)                                  |            |
| 120    | 10 (10)             | 0                | 0      | 0    | 4       | 6       | 0             | 0         | 0             | 0             | 0         | 0           | 4xE3.S 1T NVMe G5 (C221) + 3x 2xE3.S 2T NVMe G5 (3x C222)       |            |

## Overview of storage configs - Compute Complex Neptune Core

The following table summarizes the storage configurations for servers with the Lenovo Compute Complex Neptune Core open-loop water cooling. Drive configurations include 2.5-inch and E3.S front drives bays.

Jump down to the [details of the configurations](#).

Return to [Storage configurations](#).

Table 33. Overview of storage configs - Compute Complex Neptune Core

| Config | Total drives (NVMe) | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                       |
|--------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|--------------------------------------------------|
|        |                     | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                  |
| 87     | 8 (8)               | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)                            |
| 88     | 8 (8)               | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU)                          |
| 89     | 16 (16)             | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                      |
| 90     | 16 (16)             | 0                | 8      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) + 8x2.5" NVMe G5 (C46P)  |
| 91     | 16 (8)              | 8                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 8x2.5" AnyBay G5 (C3RU) |
| 92     | 16 (8)              | 8                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 8x2.5" NVMe G5 (C46P)   |
| 103    | 8 (0)               | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT)                           |
| 104    | 8 (8)               | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) (Tri-mode)               |
| 105    | 16 (0)              | 16               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" SAS/SATA (2x C3RT)                     |
| 106    | 16 (16)             | 0                | 16     | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" AnyBay G5 (2x C3RU) (Tri-mode)         |
| 155    | 8 (8)               | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)                            |
| 156    | 16 (16)             | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                      |

## Details of storage configs - 3.5-inch front drives

The following table lists the detailed storage configurations with 3.5-inch front drives bays.

Go back to the [overview of the configurations](#).

Return to [Storage configurations](#).

The table also includes a **PCIe slots** column that indicates the number of slots available with each configuration. Mouse over the slot count to see the slot numbers. For details of the slots supported with each configuration, refer to the [Slot configurations](#) section.

Table 34. Details of storage configs - 3.5-inch front drives

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                                    | M.2          |             |              | Controllers                                 | PCIe slots ? |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|--------------------------------------------------------------------|--------------|-------------|--------------|---------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | Backplanes                                                         | M.2 Internal | M.2 Rear HS | M.2 Front HS |                                             |              |
| 35-1   | 1 or 2 | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" SAS/SATA (C3RW)                                            | Y            | Y           | N            | (4350-16i or 9350-16i)                      | 10           |
| 35-2   | 1 or 2 | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" SAS/SATA (C3RW)                                            | Y            | Y           | N            | (440-16i or 940-16i)                        | 10           |
| 36-1   | 1 or 2 | Y           | Y              | N                  | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" (8x SAS/SATA, 4x AnyBay G5) (C3RV)                         | Y            | Y           | N            | OB NVMe + (4350-16i or 9350-16i)            | 10           |
| 36-2   | 1 or 2 | Y           | Y              | N                  | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" (8x SAS/SATA, 4x AnyBay G5) (C3RV)                         | Y            | Y           | N            | OB NVMe + (440-16i or 940-16i)              | 10           |
| 37-1   | 1 or 2 | Y           | Y              | N                  | 8                | 0      | 4    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" (8x SAS/SATA, 4x NVMe G5) (C46N)                           | Y            | Y           | N            | OB NVMe + (5350-8i or 9350-8i)              | 10           |
| 37-2   | 1 or 2 | Y           | Y              | N                  | 8                | 0      | 4    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 12x3.5" (8x SAS/SATA, 4x NVMe G5) (C46N)                           | Y            | Y           | N            | OB NVMe + (545-8i or 940-8i)                | 10           |
| 38-1   | 1 or 2 | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 4             | 0             | 0         | 0           | Front: 12x3.5" SAS/SATA (C3RW)<br>Rear: 4x3.5" SAS/SATA (C46M)     | Y            | Y           | N            | (4350-16i or 9350-16i)                      | 2            |
| 38-2   | 1 or 2 | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 4             | 0             | 0         | 0           | Front: 12x3.5" SAS/SATA (C3RW)<br>Rear: 4x3.5" SAS/SATA (C46M)     | Y            | Y           | N            | (440-16i or 940-16i)                        | 2            |
| 41-1   | 2 only | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 4           | Front: 12x3.5" SAS/SATA (C3RW)<br>Rear: 4x2.5" AnyBay G5 (C46G)    | Y            | Y           | N            | Front: (4350-16i or 9350-16i) Rear: OB NVMe | 6            |
| 41-2   | 2 only | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 4           | Front: 12x3.5" SAS/SATA (C3RW)<br>Rear: 4x2.5" AnyBay G5 (C46G)    | Y            | Y           | N            | Front: (440-16i or 940-16i) Rear: OB NVMe   | 6            |
| 42-1   | 2 only | Y           | N              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 8         | 0             | 0             | 0         | 0           | Front: 12x3.5" SAS/SATA (C3RW)<br>Mid: 2x 4x2.5" NVMe G5 (2x C46J) | Y            | Y           | N            | Front: (4350-16i or 9350-16i) Mid: OB NVMe  | 7            |
| 42-2   | 2 only | Y           | N              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 8         | 0             | 0             | 0         | 0           | Front: 12x3.5" SAS/SATA (C3RW)<br>Mid: 2x 4x2.5" NVMe G5 (2x C46J) | Y            | Y           | N            | Front: (440-16i or 940-16i) Mid: OB NVMe    | 7            |
| 117-1  | 1 or 2 | Y           | Y              | N                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x3.5" SAS/SATA (C4DB)                                             | Y            | Y           | N            | (4350-16i or 9350-16i)                      | 10           |
| 117-2  | 1 or 2 | Y           | Y              | N                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x3.5" SAS/SATA (C4DB)                                             | Y            | Y           | N            | (440-16i or 940-16i)                        | 10           |
| 118-1  | 2 only | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | Front: 12x3.5" SAS/SATA (C3RW)<br>Rear: 4x2.5" NVMe G5 (C46H)      | Y            | Y           | N            | Front: (4350-16i or 9350-16i) Rear: OB NVMe | 6            |
| 118-2  | 2 only | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | Front: 12x3.5" SAS/SATA (C3RW)<br>Rear: 4x2.5" NVMe G5 (C46H)      | Y            | Y           | N            | Front: (440-16i or 940-16i) Rear: OB NVMe   | 6            |

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                                                                     | M.2          |             |              | Controllers                                                            | PCIe slots ? |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|------------------------------------------------------------------------------------------------|--------------|-------------|--------------|------------------------------------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                                                | M.2 Internal | M.2 Rear HS | M.2 Front HS |                                                                        |              |
| 122-1  | 2 only | Y           | Y              | N                  | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | <b>Front:</b> 12x3.5" (8x SAS/SATA, 4x AnyBay G5) (C3RV)<br><b>Rear:</b> 4x2.5" NVMe G5 (C46H) | Y            | Y           | N            | <b>Front:</b> OB NVMe + (4350-16i or 9350-16i)<br><b>Rear:</b> OB NVMe | 6            |
| 122-2  | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                | Y            | Y           | N            |                                                                        | 6            |

## Details of storage configs - 2.5-inch front drives

The following table lists the detailed storage configurations with 2.5-inch front drives bays.

Go back to the [overview of the configurations](#).

Return to [Storage configurations](#).

The table also includes a **PCIe slots** column that indicates the number of slots available with each configuration. Mouse over the slot count to see the slot numbers. For details of the slots supported with each configuration, refer to the [Slot configurations](#) section.

Table 35. Details of storage configs - 2.5-inch front drives

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                         | M.2          |             |              | Controllers                                            | PCIe slots ? |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|------------------------------------|--------------|-------------|--------------|--------------------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                    | M.2 Internal | M.2 Rear HS | M.2 Front HS |                                                        |              |
| 1-1    | 1 or 2 | Y           | Y              | N                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT)             | Y            | Y           | Y            | (5350-8i or 9350-8i or 4350-16i or 9350-16i)           | 10           |
| 1-2    | 1 or 2 | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | (545-8i or 940-8i or 440-16i or 940-16i)               | 10           |
| 1-3    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | (440-16i CFF or 940-16i CFF)                           | 10           |
| 1-4    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | (440-16i CFF or 940-16i CFF)                           | 5            |
| 2-1    | 2 only | Y           | Y              | N                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)              | Y            | Y           | Y            | OB NVMe                                                | 10           |
| 2-2    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | OB NVMe                                                | 3            |
| 3-1    | 2 only | Y           | Y              | N                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU)            | Y            | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 10           |
| 3-2    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 10           |
| 3-3    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 9            |
| 3-4    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 3            |
| 3-5    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 3            |
| 3-6    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 2            |
| 5-1    | 1 or 2 | Y           | Y              | N                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) (Tri-mode) | Y            | Y           | Y            | (940-8i or 940-16i) Tri-mode                           | 10           |
| 5-2    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | 940-16i CFF Tri-mode                                   | 10           |
| 5-3    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                    | Y            | Y           | Y            | 940-16i CFF Tri-mode                                   | 5            |

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                 | Backplanes | M.2         |              |                                                        | Controllers                  | PCIe slots ? |  |  |  |  |  |   |   |   |                                                        |   |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|-------------------------------------------------|------------|-------------|--------------|--------------------------------------------------------|------------------------------|--------------|--|--|--|--|--|---|---|---|--------------------------------------------------------|---|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | M.2 Internal                                    |            | M.2 Rear HS | M.2 Front HS |                                                        |                              |              |  |  |  |  |  |   |   |   |                                                        |   |
| 6-1    | 1 or 2 | Y           | Y              | N                  | 16               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" SAS/SATA (2x C3RT)                    | Y          | Y           | Y            | 2x (5350-8i or 9350-8i)                                | 10                           |              |  |  |  |  |  |   |   |   |                                                        |   |
| 6-2    | 1 or 2 | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            | Y           | Y            | Y                                                      | 2x (545-8i or 940-8i)        | 10           |  |  |  |  |  |   |   |   |                                                        |   |
| 6-3    | 1 or 2 | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            | Y           | Y            | Y                                                      | (4350-16i or 9350-16i)       | 10           |  |  |  |  |  |   |   |   |                                                        |   |
| 6-4    | 1 or 2 | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            | Y           | Y            | Y                                                      | (440-16i or 940-16i)         | 10           |  |  |  |  |  |   |   |   |                                                        |   |
| 6-5    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            | Y           | Y            | Y                                                      | (440-16i CFF or 940-16i CFF) | 10           |  |  |  |  |  |   |   |   |                                                        |   |
| 6-6    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            | Y           | Y            | Y                                                      | (440-16i CFF or 940-16i CFF) | 5            |  |  |  |  |  |   |   |   |                                                        |   |
| 7-1    | 2 only | Y           | Y              | N                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                     | Y          | Y           | Y            | OB NVMe                                                | 6                            |              |  |  |  |  |  |   |   |   |                                                        |   |
| 7-2    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            |             |              |                                                        |                              |              |  |  |  |  |  | Y | Y | Y | OB NVMe                                                | 1 |
| 8-1    | 2 only | Y           | Y              | N                  | 0                | 8      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) + 8x2.5" NVMe G5 (C46P) | Y          | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 6                            |              |  |  |  |  |  |   |   |   |                                                        |   |
| 8-2    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            |             |              |                                                        |                              |              |  |  |  |  |  | Y | Y | Y | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 6 |
| 8-3    | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            |             |              |                                                        |                              |              |  |  |  |  |  | Y | Y | Y | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 5 |
| 8-4    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            |             |              |                                                        |                              |              |  |  |  |  |  | Y | Y | Y | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 1 |
| 8-5    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            |             |              |                                                        |                              |              |  |  |  |  |  | Y | Y | Y | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 1 |
| 8-6    | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                 |            |             |              |                                                        |                              |              |  |  |  |  |  | Y | Y | Y | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 0 |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           | Backplanes                                     | M.2                                              |              |             | Controllers                                            | PCIe slots ?                           |              |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|------------------------------------------------|--------------------------------------------------|--------------|-------------|--------------------------------------------------------|----------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe |                                                | 2.5" AnyBay                                      | M.2 Internal | M.2 Rear HS |                                                        |                                        | M.2 Front HS |
| 9-1    | 2 only | Y           | Y              | N                   | 8                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0                                              | 8x2.5" SAS/SATA (C3RT) + 8x2.5" AnyBay G5 (C3RU) | Y            | Y           | Y                                                      | OB NVMe + 2x (5350-8i or 9350-8i)      | 10           |
| 9-2    | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + 2x (545-8i or 940-8i)        | 10           |
| 9-3    | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + (4350-16i or 9350-16i)       | 10           |
| 9-4    | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + (440-16i or 940-16i)         | 10           |
| 9-5    | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + (440-16i CFF or 940-16i CFF) | 9            |
| 9-6    | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + 2x (5350-8i or 9350-8i)      | 3            |
| 9-7    | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + 2x (545-8i or 940-8i)        | 3            |
| 9-8    | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + (4350-16i or 9350-16i)       | 3            |
| 9-9    | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + (440-16i or 940-16i)         | 3            |
| 9-10   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                |                                                  | Y            | Y           | Y                                                      | OB NVMe + (440-16i CFF or 940-16i CFF) | 2            |
| 10-1   | 2 only | Y           | Y              | N                   | 8                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 8x2.5" SAS/SATA (C3RT) + 8x2.5" NVMe G5 (C46P) | Y                                                | Y            | Y           | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 10                                     |              |
| 10-2   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 10                                     |              |
| 10-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 9                                      |              |
| 10-4   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 3                                      |              |
| 10-5   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 3                                      |              |
| 10-6   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 2                                      |              |
| 11-1   | 1 or 2 | Y           | Y              | N                   | 0                | 16     | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 2x 8x2.5" AnyBay G5 (2x C3RU) (Tri-mode)       | Y                                                | Y            | Y           | 2x 940-8i Tri-mode                                     | 10                                     |              |
| 11-2   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | 940-16i Tri-mode                                       | 10                                     |              |
| 11-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | 940-16i CFF Tri-mode                                   | 10                                     |              |
| 11-4   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                | Y                                                | Y            | Y           | 940-16i CFF Tri-mode                                   | 5                                      |              |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |                                                        |                              | Backplanes | M.2         |                                                             |                                                   | Controllers | PCIe slots ? |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|--------------------------------------------------------|------------------------------|------------|-------------|-------------------------------------------------------------|---------------------------------------------------|-------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay                                            | M.2 Internal                 |            | M.2 Rear HS | M.2 Front HS                                                |                                                   |             |              |
| 12-1   | 1 or 2 | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0                                                      | 3x 8x2.5" SAS/SATA (3x C3RT) | Y          | Y           | N                                                           | 3x (5350-8i or 9350-8i)                           | 10          |              |
| 12-2   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        |                              | Y          | Y           | N                                                           | 3x (545-8i or 940-8i)                             | 10          |              |
| 12-3   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        |                              | Y          | Y           | N                                                           | (4350-16i or 9350-16i) + (5350-8i or 9350-8i)     | 10          |              |
| 12-4   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        |                              | Y          | Y           | N                                                           | (440-16i or 940-16i) + (545-8i or 940-8i)         | 10          |              |
| 12-5   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        |                              | Y          | Y           | N                                                           | (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) | 10          |              |
| 12-6   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        |                              | Y          | Y           | N                                                           | (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) | 5           |              |
| 13-1   | 2 only | Y           | Y              | N                   | 0                | 0      | 24   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0                                                      | 3x 8x2.5" NVMe G5 (3x C46P)  | Y          | Y           | N                                                           | OB NVMe                                           | 4           |              |
| 13-2   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        |                              | Y          | Y           | N                                                           | OB NVMe                                           | 4           |              |
| 14-1   | 2 only | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) | Y                            | Y          | N           | OB NVMe + 3x (5350-8i or 9350-8i)                           | 10                                                |             |              |
| 14-2   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + 3x (545-8i or 940-8i)                             | 10                                                |             |              |
| 14-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + (4350-16i or 9350-16i) + (5350-8i or 9350-8i)     | 10                                                |             |              |
| 14-4   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + (440-16i or 940-16i) + (545-8i or 940-8i)         | 10                                                |             |              |
| 14-5   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) | 9                                                 |             |              |
| 14-6   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + 3x (5350-8i or 9350-8i)                           | 3                                                 |             |              |
| 14-7   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + 3x (545-8i or 940-8i)                             | 3                                                 |             |              |
| 14-8   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + (4350-16i or 9350-16i) + (5350-8i or 9350-8i)     | 3                                                 |             |              |
| 14-9   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + (440-16i or 940-16i) + (545-8i or 940-8i)         | 3                                                 |             |              |
| 14-10  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                                        | Y                            | Y          | N           | OB NVMe + (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) | 2                                                 |             |              |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           | Backplanes                               | M.2                                                  |              |             | Controllers                            | PCIe slots ?                           |              |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|------------------------------------------|------------------------------------------------------|--------------|-------------|----------------------------------------|----------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe |                                          | 2.5" AnyBay                                          | M.2 Internal | M.2 Rear HS |                                        |                                        | M.2 Front HS |
| 15-1   | 2 only | Y           | Y              | N                   | 16               | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0                                        | 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" NVMe G5 (C46P) | Y            | Y           | N                                      | OB NVMe + 2x (5350-8i or 9350-8i)      | 10           |
| 15-2   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + 2x (545-8i or 940-8i)        | 10           |
| 15-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + (4350-16i or 9350-16i)       | 10           |
| 15-4   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + (440-16i or 940-16i)         | 10           |
| 15-5   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + (440-16i CFF or 940-16i CFF) | 9            |
| 15-6   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + 2x (5350-8i or 9350-8i)      | 3            |
| 15-7   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + 2x (545-8i or 940-8i)        | 3            |
| 15-8   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + (4350-16i or 9350-16i)       | 3            |
| 15-9   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + (440-16i or 940-16i)         | 3            |
| 15-10  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          |                                                      | Y            | Y           | N                                      | OB NVMe + (440-16i CFF or 940-16i CFF) | 2            |
| 16-1   | 1 or 2 | Y           | Y              | N                   | 0                | 24     | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 3x 8x2.5" AnyBay G5 (3x C3RU) (Tri-mode) | Y                                                    | Y            | N           | 3x 940-8i Tri-mode                     | 10                                     |              |
| 16-2   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          | Y                                                    | Y            | N           | 940-16i Tri-mode + 940-8i Tri-mode     | 10                                     |              |
| 16-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          | Y                                                    | Y            | N           | 940-16i CFF Tri-mode + 940-8i Tri-mode | 10                                     |              |
| 16-4   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |                                          | Y                                                    | Y            | N           | 940-16i CFF Tri-mode + 940-8i Tri-mode | 5                                      |              |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                                     | Backplanes | M.2         |              |                                                                                   | Controllers | PCIe slots ? |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|---------------------------------------------------------------------|------------|-------------|--------------|-----------------------------------------------------------------------------------|-------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | M.2 Internal                                                        |            | M.2 Rear HS | M.2 Front HS |                                                                                   |             |              |
| 17-3   | 1 or 2 | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 4             | 0         | 0           | Front: 3x 8x2.5" SAS/SATA (3x C3RT)<br>Rear: 4x2.5" SAS/SATA (C46K) | Y          | Y           | N            | Front: (4350-16i or 9350-16i) + (5350-8i or 9350-8i) Rear: (5350-8i or 9350-8i)   | 6           |              |
| 17-4   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (440-16i or 940-16i) + (545-8i or 940-8i) Rear: (545-8i or 940-8i)         | 6           |              |
| 17-5   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | 2x (4350-16i or 9350-16i)                                                         | 6           |              |
| 17-6   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | 2x (440-16i or 940-16i)                                                           | 6           |              |
| 17-7   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) Rear: (545-8i or 940-8i) | 6           |              |
| 17-8   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | (440-16i CFF or 940-16i CFF) + (440-16i or 940-16i)                               | 6           |              |
| 17-9   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (5350-8i or 9350-8i or 4350-16i) + Exp CFF Rear: Exp CFF                   | 6           |              |
| 17-10  | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (545-8i or 940-8i) + Exp CFF Rear: Exp CFF                                 | 6           |              |
| 17-11  | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF                       | 6           |              |
| 17-12  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) Rear: (545-8i or 940-8i) | 5           |              |
| 17-13  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | (440-16i CFF or 940-16i CFF) + (440-16i or 940-16i)                               | 5           |              |
| 17-14  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                     | Y          | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF                       | 5           |              |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                                           | M.2          |             |              | Controllers                                                                               | PCIe slots ? |                                                                                     |   |                                                |   |                                                                          |   |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|----------------------------------------------------------------------|--------------|-------------|--------------|-------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|---|------------------------------------------------|---|--------------------------------------------------------------------------|---|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                      | M.2 Internal | M.2 Rear HS | M.2 Front HS |                                                                                           |              |                                                                                     |   |                                                |   |                                                                          |   |
| 18-3   | 2 only | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 4           | Front: 3x 8x2.5" SAS/SATA (3x C3RT)<br>Rear: 4x2.5" AnyBay G5 (C46G) | Y            | Y           | N            | Front: (4350-16i or 9350-16i) + (5350-8i or 9350-8i) Rear: (5350-8i or 9350-8i) + OB NVMe | 6            |                                                                                     |   |                                                |   |                                                                          |   |
| 18-4   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                      | Y            | Y           | N            |                                                                                           |              | Front: (440-16i or 940-16i) + (545-8i or 940-8i) Rear: (545-8i or 940-8i) + OB NVMe | 6 |                                                |   |                                                                          |   |
| 18-5   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                      | Y            | Y           | N            |                                                                                           |              |                                                                                     |   | Front: 2x (4350-16i or 9350-16i) Rear: OB NVMe | 6 |                                                                          |   |
| 18-6   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                      | Y            | Y           | N            |                                                                                           |              |                                                                                     |   |                                                |   | Front: 2x (440-16i or 940-16i) Rear: OB NVMe                             | 6 |
| 18-7   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                      | Y            | Y           | N            |                                                                                           |              |                                                                                     |   |                                                |   |                                                                          |   |
| 18-8   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                      | Y            | Y           | N            |                                                                                           |              |                                                                                     |   |                                                |   | Front: (440-16i CFF or 940-16i CFF) + (440-16i or 940-16i) Rear: OB NVMe | 6 |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           | Backplanes | M.2                                                                                        |              |             | Controllers | PCIe slots ?                                                                                |              |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|------------|--------------------------------------------------------------------------------------------|--------------|-------------|-------------|---------------------------------------------------------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe |            | 2.5" AnyBay                                                                                | M.2 Internal | M.2 Rear HS |             |                                                                                             | M.2 Front HS |
| 19-3   | 2 only | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 4             | 0         | 0          | Front: 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) Rear: 4x2.5" SAS/SATA (C46K) | Y            | Y           | N           | Front: OB NVMe + (4350-16i or 9350-16i) + (5350-8i or 9350-8i) Rear: (5350-8i or 9350-8i)   | 6            |
| 19-4   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | Front: OB NVMe + (440-16i or 940-16i) + (545-8i or 940-8i) Rear: (545-8i or 940-8i)         | 6            |
| 19-5   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | OB NVMe + 2x (4350-16i or 9350-16i)                                                         | 6            |
| 19-6   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | OB NVMe + 2x (440-16i or 940-16i)                                                           | 6            |
| 19-7   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | Front: OB NVMe + (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) Rear: (545-8i or 940-8i) | 6            |
| 19-8   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | OB NVMe + (440-16i CFF or 940-16i CFF) + (440-16i or 940-16i)                               | 6            |
| 19-9   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | Front: OB NVMe + (4350-16i or 9350-16i) + (5350-8i or 9350-8i) Rear: (5350-8i or 9350-8i)   | 3            |
| 19-10  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | Front: OB NVMe + (440-16i or 940-16i) + (545-8i or 940-8i) Rear: (545-8i or 940-8i)         | 3            |
| 19-11  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | OB NVMe + 2x (4350-16i or 9350-16i)                                                         | 3            |
| 19-12  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | OB NVMe + 2x (440-16i or 940-16i)                                                           | 3            |
| 19-13  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | Front: OB NVMe + (440-16i CFF or 940-16i CFF) + (545-8i or 940-8i) Rear: (545-8i or 940-8i) | 2            |
| 19-14  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                            | Y            | Y           | N           | OB NVMe + (440-16i CFF or 940-16i CFF) + (440-16i or 940-16i)                               | 2            |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           | Backplanes | M.2                                                                                             |              |             | Controllers | PCIe slots ?                                                    |              |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|------------|-------------------------------------------------------------------------------------------------|--------------|-------------|-------------|-----------------------------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe |            | 2.5" AnyBay                                                                                     | M.2 Internal | M.2 Rear HS |             |                                                                 | M.2 Front HS |
| 20-5   | 1 or 2 | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0             | 0             | 0         | 0          | Front: 3x 8x2.5" SAS/SATA (3x C3RT)<br>Mid: 2x 4x2.5" SAS/SATA (2x C46L)                        | Y            | Y           | N           | 2x (4350-16i or 9350-16i)                                       | 8            |
| 20-6   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | 2x (440-16i or 940-16i)                                         | 8            |
| 20-9   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | (5350-8i or 9350-8i or 4350-16i) + Exp CFF                      | 8            |
| 20-10  | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | (545-8i or 940-8i) + Exp CFF                                    | 8            |
| 20-11  | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | (440-16i CFF or 940-16i CFF) + Exp CFF                          | 8            |
| 20-12  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | (440-16i CFF or 940-16i CFF) + Exp CFF                          | 4            |
| 21-5   | 2 only | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 8             | 0         | 0             | 0             | 0         | 0          | Front: 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) Mid: 2x 4x2.5" SAS/SATA (2x C46L) | Y            | Y           | N           | OB NVMe + 2x (4350-16i or 9350-16i)                             | 8            |
| 21-6   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | OB NVMe + 2x (440-16i or 940-16i)                               | 8            |
| 21-9   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | OB NVMe + 2x (4350-16i or 9350-16i)                             | 3            |
| 21-10  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | OB NVMe + 2x (440-16i or 940-16i)                               | 3            |
| 22-5   | 2 only | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 8             | 0             | 0         | 0          | Front: 2x 8x2.5" SAS/SATA (2x C3RT) + 8x2.5" AnyBay G5 (C3RU) Rear: 8x2.5" SAS/SATA (C3SP)      | Y            | Y           | N           | OB NVMe + 2x (4350-16i or 9350-16i)                             | 4            |
| 22-6   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | OB NVMe + 2x (440-16i or 940-16i)                               | 4            |
| 22-11  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | OB NVMe + 2x (4350-16i or 9350-16i)                             | 3            |
| 22-12  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | OB NVMe + 2x (440-16i or 940-16i)                               | 3            |
| 23-5   | 1 or 2 | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 8             | 0             | 0         | 0          | Front: 3x 8x2.5" SAS/SATA (3x C3RT)<br>Rear: 8x2.5" SAS/SATA (C3SP)                             | Y            | Y           | N           | 2x (4350-16i or 9350-16i)                                       | 4            |
| 23-6   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | 2x (440-16i or 940-16i)                                         | 4            |
| 23-15  | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | Front: (5350-8i or 9350-8i or 4350-16i) + Exp CFF Rear: Exp CFF | 4            |
| 23-16  | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | Front: (545-8i or 940-8i) + Exp CFF Rear: Exp CFF               | 4            |
| 23-17  | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF     | 4            |
| 23-18  | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |            |                                                                                                 | Y            | Y           | N           | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF     | 3            |

| Config | CPUs   | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                                                                               | M.2          |             |              | Controllers                                                 | PCIe slots ? |
|--------|--------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|----------------------------------------------------------------------------------------------------------|--------------|-------------|--------------|-------------------------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                                                          | M.2 Internal | M.2 Rear HS | M.2 Front HS |                                                             |              |
| 24-1   | 2 only | Y           | N              | N                   | 0                | 0      | 24   | 0       | 0       | 0             | 8         | 0             | 0             | 0         | 0           | Front: 3x 8x2.5" NVMe G5 (3x C46P)<br>Mid: 2x 4x2.5" NVMe G5 (2x C46J)                                   | Y            | Y           | N            | Front: OB NVMe<br>Mid: OB NVMe                              | 2            |
| 25-1   | 2 only | Y           | N              | N                   | 0                | 0      | 24   | 0       | 0       | 0             | 8         | 0             | 0             | 4         | 0           | Front: 3x 8x2.5" NVMe G5 (3x C46P)<br>Mid: 2x 4x2.5" NVMe G5 (2x C46J)<br>Rear: 4x2.5" NVMe G5 (C46H)    | Y            | Y           | N            | Front: OB NVMe<br>Mid: OB NVMe Rear: OB NVMe                | 1            |
| 26-1   | 1 or 2 | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0             | 4             | 0         | 0           | Front: 3x 8x2.5" SAS/SATA (3x C3RT)<br>Mid: 2x 4x2.5" SAS/SATA (2x C46L)<br>Rear: 4x2.5" SAS/SATA (C46K) | Y            | Y           | N            | (5350-8i or 9350-8i or 4350-16i) + Exp CFF                  | 5            |
| 26-2   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                          | Y            | Y           | N            | (545-8i or 940-8i) + Exp CFF                                | 5            |
| 26-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                          | Y            | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF | 5            |
| 26-4   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                          | Y            | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF | 4            |
| 27-1   | 1 or 2 | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0             | 8             | 0         | 0           | Front: 3x 8x2.5" SAS/SATA (3x C3RT)<br>Mid: 2x 4x2.5" SAS/SATA (2x C46L)<br>Rear: 8x2.5" SAS/SATA (C3SP) | Y            | Y           | N            | (5350-8i or 9350-8i or 4350-16i) + Exp CFF                  | 4            |
| 27-2   | 1 or 2 | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                          | Y            | Y           | N            | (545-8i or 940-8i) + Exp CFF                                | 4            |
| 27-3   | 2 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                          | Y            | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF | 4            |
| 27-4   | 1 only | Y           | Y              | N                   |                  |        |      |         |         |               |           |               |               |           |             |                                                                                                          | Y            | Y           | N            | Front: (440-16i CFF or 940-16i CFF) + Exp CFF Rear: Exp CFF | 4            |
| 101-1  | 2 only | Y           | Y              | N                   | 0                | 0      | 24   | 0       | 0       | 0             | 0         | 0             | 0             | 4         | 0           | Front: 3x 8x2.5" NVMe G5 (3x C46P)<br>Rear: 4x2.5" NVMe G5 (C46H)                                        | Y            | Y           | N            | Front: OB NVMe<br>Rear: OB NVMe                             | 2            |

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                      | Backplanes | M.2         |              |                                                        | Controllers | PCIe slots ? |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|------------------------------------------------------|------------|-------------|--------------|--------------------------------------------------------|-------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | M.2 Internal                                         |            | M.2 Rear HS | M.2 Front HS |                                                        |             |              |
| 121-1  | 2 only | Y           | Y              | N                  | 8                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 2x 8x2.5" NVMe G5 (2x C46P) | Y          | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 6           |              |
| 121-2  | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                      | Y          | Y           | Y            | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 6           |              |
| 121-3  | 2 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                      | Y          | Y           | Y            | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 5           |              |
| 121-4  | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                      | Y          | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 1           |              |
| 121-5  | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                      | Y          | Y           | Y            | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 1           |              |
| 121-6  | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                      | Y          | Y           | Y            | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 0           |              |
| 151-1  | 1 or 2 | Y           | Y              | N                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)                                | Y          | Y           | Y            | 960W-32i                                               | 10          |              |
| 152-1  | 1 or 2 | Y           | Y              | N                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                          | Y          | Y           | Y            | 960W-32i                                               | 10          |              |

## Details of storage configs - E3.S front drives

The following table lists the detailed storage configurations with E3.S front drives bays.

Go back to the [overview of the configurations](#).

Return to [Storage configurations](#).

The table also includes a **PCIe slots** column that indicates the number of slots available with each configuration. Mouse over the slot count to see the slot numbers. For details of the slots supported with each configuration, refer to the [Slot configurations](#) section.

Table 36. Details of storage configs - E3.S front drives

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             | Backplanes                                                      | M.2          |             |              | Controllers | PCIe slots ? |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|-----------------------------------------------------------------|--------------|-------------|--------------|-------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                                                                 | M.2 Internal | M.2 Rear HS | M.2 Front HS |             |              |
| 59-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 8       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221)                                  | Y            | Y           | Y            | OB NVMe     | 10           |
| 59-2   | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                                 | Y            | Y           | Y            | OB NVMe     | 3            |
| 60-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 16      | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 4xE3.S 1T NVMe G5 (4x C221)                                  | Y            | Y           | Y            | OB NVMe     | 6            |
| 60-2   | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                                 | Y            | Y           | N            | OB NVMe     | 1            |
| 61-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 24      | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 6x 4xE3.S 1T NVMe G5 (6x C221)                                  | Y            | Y           | Y            | OB NVMe     | 4            |
| 62-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 32      | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x 4xE3.S 1T NVMe G5 (8x C221)                                  | Y            | Y           | N            | OB NVMe     | 2            |
| 63-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 2xE3.S 2T NVMe G5 (2x C222)                                  | Y            | Y           | Y            | OB NVMe     | 10           |
| 63-2   | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                                 | Y            | Y           | Y            | OB NVMe     | 3            |
| 64-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 8       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 2xE3.S 2T NVMe G5 (4x C222)                                  | Y            | Y           | Y            | OB NVMe     | 6            |
| 65-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 12      | 0             | 0         | 0             | 0             | 0         | 0           | 6x 2xE3.S 2T NVMe G5 (6x C222)                                  | Y            | Y           | Y            | OB NVMe     | 4            |
| 66-1   | 1 only | Y           | Y              | N                  | 0                | 0      | 0    | 4       | 2       | 0             | 0         | 0             | 0             | 0         | 0           | 4xE3.S 1T NVMe G5 (C221) + 2xE3.S 2T NVMe G5 (C222)             | Y            | Y           | Y            | OB NVMe     | 3            |
| 67-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 8       | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221) + 2x 2xE3.S 2T NVMe G5 (2x C222) | Y            | Y           | Y            | OB NVMe     | 6            |
| 67-2   | 1 only | Y           | Y              | N                  |                  |        |      |         |         |               |           |               |               |           |             |                                                                 | Y            | Y           | N            | OB NVMe     | 1            |
| 68-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 8       | 8       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221) + 4x 2xE3.S 2T NVMe G5 (4x C222) | Y            | Y           | Y            | OB NVMe     | 4            |
| 69-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 8       | 12      | 0             | 0         | 0             | 0             | 0         | 0           | 2x 4xE3.S 1T NVMe G5 (2x C221) + 6x 2xE3.S 2T NVMe G5 (6x C222) | Y            | Y           | N            | OB NVMe     | 2            |
| 70-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 16      | 4       | 0             | 0         | 0             | 0             | 0         | 0           | 4x 4xE3.S 1T NVMe G5 (4x C221) + 2x 2xE3.S 2T NVMe G5 (2x C222) | Y            | Y           | Y            | OB NVMe     | 4            |

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         | Mid drives |               | Rear drives |               |               |           | Backplanes | M.2                                                             |              |             | Controllers | PCIe slots ? |              |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|------------|---------------|-------------|---------------|---------------|-----------|------------|-----------------------------------------------------------------|--------------|-------------|-------------|--------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T    | 2.5" SAS/SATA | 2.5" NVMe   | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe |            | 2.5" AnyBay                                                     | M.2 Internal | M.2 Rear HS |             |              | M.2 Front HS |
| 71-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 16      | 8          | 0             | 0           | 0             | 0             | 0         | 0          | 4x 4xE3.S 1T NVMe G5 (4x C221) + 4x 2xE3.S 2T NVMe G5 (4x C222) | Y            | Y           | N           | OB NVMe      | 2            |
| 72-1   | 2 only | Y           | Y              | N                  | 0                | 0      | 0    | 24      | 4          | 0             | 0           | 0             | 0             | 0         | 0          | 6x 4xE3.S 1T NVMe G5 (6x C221) + 2x 2xE3.S 2T NVMe G5 (2x C222) | Y            | Y           | N           | OB NVMe      | 2            |
| 102-1  | 1 only | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 2          | 0             | 0           | 0             | 0             | 0         | 0          | 2x 2xE3.S 2T NVMe G5 (2x C222)                                  | Y            | Y           | N           | OB NVMe      | 5            |
| 119-1  | 1 only | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 6          | 0             | 0           | 0             | 0             | 0         | 0          | 3x 2xE3.S 2T NVMe G5 (3x C222)                                  | Y            | Y           | Y           | OB NVMe      | 2            |
| 120-1  | 1 only | Y           | Y              | N                  | 0                | 0      | 0    | 4       | 6          | 0             | 0           | 0             | 0             | 0         | 0          | 4xE3.S 1T NVMe G5 (C221) + 3x 2xE3.S 2T NVMe G5 (3x C222)       | Y            | Y           | N           | OB NVMe      | 1            |

## Details of storage configs - Compute Complex Neptune Core

The following table lists the detailed storage configurations for servers with the Lenovo Compute Complex Neptune Core open-loop water cooling. Drive configurations include 2.5-inch and E3.S front drives bays.

Go back to the [overview of the configurations](#).

Return to [Storage configurations](#).

The table also includes a **PCIe slots** column that indicates the number of slots available with each configuration. Mouse over the slot count to see the slot numbers. For details of the slots supported with each configuration, refer to the [Slot configurations](#) section.

Table 37. Details of storage configs - Compute Complex Neptune Core

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           |             |                                                  | Backplanes | M.2         |              |                                                        | Controllers                                        | PCIe slots ?                                       |   |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|-------------|--------------------------------------------------|------------|-------------|--------------|--------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|---|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | M.2 Internal                                     |            | M.2 Rear HS | M.2 Front HS |                                                        |                                                    |                                                    |   |
| 87-1   | 2 only | N           | N              | Y                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" NVMe G5 (C46P)                            | Y          | Y           | Y            | OB NVMe                                                | 4                                                  |                                                    |   |
| 88-1   | 2 only | N           | N              | Y                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU)                          | Y          | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 4                                                  |                                                    |   |
| 88-2   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  |            | Y           | Y            | Y                                                      | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i) | 4                                                  |   |
| 88-3   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  |            | Y           | Y            | Y                                                      | OB NVMe + (440-16i CFF or 940-16i CFF)             | 3                                                  |   |
| 89-1   | 2 only | N           | N              | Y                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 2x 8x2.5" NVMe G5 (2x C46P)                      | Y          | Y           | Y            | OB NVMe                                                | 3                                                  |                                                    |   |
| 90-1   | 2 only | N           | N              | Y                  | 0                | 8      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" AnyBay G5 (C3RU) + 8x2.5" NVMe G5 (C46P)  | Y          | Y           | Y            | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 3                                                  |                                                    |   |
| 90-2   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  |            |             | Y            | Y                                                      | Y                                                  | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i) | 3 |
| 90-3   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  |            |             | Y            | Y                                                      | Y                                                  | OB NVMe + (440-16i CFF or 940-16i CFF)             | 2 |
| 91-1   | 2 only | N           | N              | Y                  | 8                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0           | 8x2.5" SAS/SATA (C3RT) + 8x2.5" AnyBay G5 (C3RU) | Y          | Y           | Y            | OB NVMe + 2x (5350-8i or 9350-8i)                      | 4                                                  |                                                    |   |
| 91-2   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  | Y          | Y           | Y            | OB NVMe + 2x (545-8i or 940-8i)                        | 4                                                  |                                                    |   |
| 91-3   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  | Y          | Y           | Y            | OB NVMe + (4350-16i or 9350-16i)                       | 4                                                  |                                                    |   |
| 91-4   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  | Y          | Y           | Y            | OB NVMe + (440-16i or 940-16i)                         | 4                                                  |                                                    |   |
| 91-5   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |             |                                                  | Y          | Y           | Y            | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 3                                                  |                                                    |   |

| Config | CPUs   | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid drives    |           | Rear drives   |               |           | Backplanes | M.2                                            |              |             | Controllers | PCIe slots ?                                           |              |
|--------|--------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|---------------|---------------|-----------|------------|------------------------------------------------|--------------|-------------|-------------|--------------------------------------------------------|--------------|
|        |        | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA | 2.5" SAS/SATA | 2.5" NVMe |            | 2.5" AnyBay                                    | M.2 Internal | M.2 Rear HS |             |                                                        | M.2 Front HS |
| 92-1   | 2 only | N           | N              | Y                  | 8                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 8x2.5" SAS/SATA (C3RT) + 8x2.5" NVMe G5 (C46P) | Y            | Y           | Y           | OB NVMe + (5350-8i or 9350-8i or 4350-16i or 9350-16i) | 4            |
| 92-2   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | OB NVMe + (545-8i or 940-8i or 440-16i or 940-16i)     | 4            |
| 92-3   | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | OB NVMe + (440-16i CFF or 940-16i CFF)                 | 3            |
| 103-1  | 2 only | N           | N              | Y                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 8x2.5" SAS/SATA (C3RT)                         | Y            | Y           | Y           | (5350-8i or 9350-8i or 4350-16i or 9350-16i)           | 7            |
| 103-2  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | (545-8i or 940-8i or 440-16i or 940-16i)               | 7            |
| 103-3  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | (440-16i CFF or 940-16i CFF)                           | 6            |
| 104-1  | 2 only | N           | N              | Y                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 8x2.5" AnyBay G5 (C3RU) (Tri-mode)             | Y            | Y           | Y           | (940-8i or 940-16i) Tri-mode                           | 7            |
| 104-2  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | 940-16i CFF Tri-mode                                   | 6            |
| 105-1  | 2 only | N           | N              | Y                  | 16               | 0      | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 2x 8x2.5" SAS/SATA (2x C3RT)                   | Y            | Y           | Y           | 2x (5350-8i or 9350-8i)                                | 7            |
| 105-2  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | 2x (545-8i or 940-8i)                                  | 7            |
| 105-3  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | (4350-16i or 9350-16i)                                 | 7            |
| 105-4  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | (440-16i or 940-16i)                                   | 7            |
| 105-5  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | (440-16i CFF or 940-16i CFF)                           | 6            |
| 106-1  | 2 only | N           | N              | Y                  | 0                | 16     | 0    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 2x 8x2.5" AnyBay G5 (2x C3RU) (Tri-mode)       | Y            | Y           | Y           | 2x 940-8i Tri-mode                                     | 7            |
| 106-2  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | 940-16i Tri-mode                                       | 7            |
| 106-3  | 2 only | N           | N              | Y                  |                  |        |      |         |         |               |           |               |               |           |            |                                                | Y            | Y           | Y           | 940-16i CFF Tri-mode                                   | 6            |
| 155-1  | 2 only | N           | N              | Y                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 8x2.5" NVMe G5 (C46P)                          | Y            | Y           | Y           | 960W-32i                                               | 5            |
| 156-1  | 2 only | N           | N              | Y                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0             | 0             | 0         | 0          | 2x 8x2.5" NVMe G5 (2x C46P)                    | Y            | Y           | Y           | 960W-32i                                               | 5            |

## Field upgrades

The SR650 V4 is orderable without drive bays, allowing you to add a backplane, cabling and controllers as field upgrades. The server also supports upgrading some configurations by adding additional front drive bays (for example, upgrading from 8 to 16x 2.5-inch drive bays).

**Upgrade path:** The key criteria for upgrade support is to ensure that the target configuration is one of the supported drive bay configurations as listed in the [Supported drive bay combinations](#) section.

To add drive bays you will need to order both drive backplanes and cable kits. Backplane kits do not include

cables.

In this section:

- [Drive bay field upgrades - 3.5-inch chassis](#)
- [Drive bay field upgrades - 2.5-inch chassis](#)
- [Drive bay field upgrades - E3.S chassis](#)
- [Drive bay field upgrades - Compute Complex Neptune water cooling](#)
- [Adding an Internal \(CFF\) RAID adapter or HBA](#)
- [Adding an Internal SAS Expander](#)
- [Adding a RAID 960W-32i adapter](#)
- [Replacement cable routing walls](#)
- [RAID flash power module \(supercap\) support](#)
- [2.5-inch drive bay fillers](#)

When adding drive bays, you will also need to add the appropriate storage controller(s). Consult the tables in the [Storage configurations](#) section to determine what controller sections are supported and what additional controllers you will need. Controllers are described in the [Controllers for internal storage](#) section.

## Drive bay field upgrades - 3.5-inch chassis

The table below lists the backplane kits and cable kits needed to build one of the supported 3.5-inch chassis configurations.

The Config numbers listed here match the configuration listed in the Storage configuration [Overview](#) and [Details](#) sections.

Return to [Field upgrades](#).

For more information about the backplane kits and cable kits, see the Lenovo server options site: [https://serveroption.lenovo.com/cable\\_kit\\_options/](https://serveroption.lenovo.com/cable_kit_options/)

**Tip:** The configurations listed in the table below are not in numerical order.

Table 38. Drive bay field upgrade for the 3.5-inch chassis (Blue = SAS/SATA, Purple = AnyBay, Red = NVMe)

| Config | 3.5" front bays |         |      | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|-----------------|---------|------|----------|-----------|-----------|----------|-----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 117    | 8               | 0       | 0    | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03852</b>, ThinkSystem 2U V4 12x3.5" SAS/SATA Backplane (For GPU Configurations Only, 8 Drives Max)</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                                                                                          |
| 35     | 12              | 0       | 0    | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03851</b>, ThinkSystem SR650 V4 12x3.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                                                                                                                        |
| 36     | 8               | 4       | 0    | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03849</b>, ThinkSystem SR650 V4 8x3.5" SAS/SATA + 4 AnyBay Backplane Option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03864</b>, ThinkSystem SR650 V4 4x3.5" Chassis Front Backplane NVMe Cable Kit</li> </ul>                                                                                                                                                             |
| 37     | 8               | 0       | 4    | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03850</b>, ThinkSystem SR650 V4 8x3.5" SAS/SATA + 4 NVMe Backplane Option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03864</b>, ThinkSystem SR650 V4 4x3.5" Chassis Front Backplane NVMe Cable Kit</li> </ul>                                                                                                                                                               |
| 38     | 12              | 0       | 0    | 0        | 0         | 4         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03851</b>, ThinkSystem SR650 V4 12x3.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03853</b>, ThinkSystem SR650 V4 4x3.5" SAS/SATA Rear Backplane Option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03865</b>, ThinkSystem SR650 V4 3.5" Chassis Rear Backplane SAS/SATA Cable Kit</li> </ul>                                                                         |
| 41     | 12              | 0       | 0    | 0        | 0         | 0         | 0        | 0         | 4        | <ul style="list-style-type: none"> <li>• <b>4XH7B03851</b>, ThinkSystem SR650 V4 12x3.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03845</b>, ThinkSystem SR650 V4 4x2.5" Anybay Rear Backplane option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> <li>• <b>4X97B03874</b>, ThinkSystem SR650 V4 2.5" Chassis Rear NVMe Cable Kit</li> </ul> |

| Config | 3.5" front bays |         |      | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|-----------------|---------|------|----------|-----------|-----------|----------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 118    | 12              | 0       | 0    | 0        | 0         | 0         | 0        | 4         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03851</b>, ThinkSystem SR650 V4 12x3.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03846</b>, ThinkSystem SR650 V4 4x2.5" NVMe Rear Backplane option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03874</b>, ThinkSystem SR650 V4 2.5" Chassis Rear NVMe Cable Kit</li> </ul>                                                                                                            |
| 42     | 12              | 0       | 0    | 0        | 8         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03851</b>, ThinkSystem SR650 V4 12x3.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03843</b>, ThinkSystem SR650 V4 8x2.5" NVMe Middle Backplane option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03876</b>, ThinkSystem SR650 V4 2.5" Chassis Middle NVMe Cable Kit</li> </ul>                                                                                                        |
| 122    | 8               | 4       | 0    | 0        | 0         | 0         | 0        | 4         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03849</b>, ThinkSystem SR650 V4 8x3.5" SAS/SATA + 4 AnyBay Backplane Option Kit</li> <li>• <b>4XH7B03846</b>, ThinkSystem SR650 V4 4x2.5" NVMe Rear Backplane option Kit</li> <li>• <b>4X97B03862</b>, ThinkSystem SR650 V4 3.5" Chassis Front Backplane SAS/SATA Cable Kit</li> <li>• <b>4X97B03864</b>, ThinkSystem SR650 V4 4x3.5" Chassis Front Backplane NVMe Cable Kit</li> <li>• <b>4X97B03874</b>, ThinkSystem SR650 V4 2.5" Chassis Rear NVMe Cable Kit</li> </ul> |

## Drive bay field upgrades - 2.5-inch chassis

The tables below lists the backplane kits and cable kits needed to build one of the supported 2.5-inch chassis configurations.

The Config numbers listed here match the configuration listed in the Storage configuration [Overview](#) and [Details](#) sections.

Return to [Field upgrades](#).

**Note:** Front drive cable kits are based on the location of the backplane in the server. The three backplane locations are BP1, BP2 and BP3 as shown in the following figure.



Figure 18. Backplane numbering

For more information about the backplane kits and cable kits, see the Lenovo server options site: [https://serveroption.lenovo.com/cable\\_kit\\_options/](https://serveroption.lenovo.com/cable_kit_options/)

**Tip:** The configurations listed in the table below are not in numerical order.

Table 39. Drive bay field upgrade for the 2.5-inch chassis (Blue = SAS/SATA, Purple = AnyBay, Red = NVMe)

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                     |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                      |
| 1      | 8               | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>4XH7B03842, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>4X97B04117, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                      |
| 2      | 0               | 0       | 8    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>4XH7B03841, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>4X97B03867, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> </ul>                                                                                                                                                                           |
| 3      | 0               | 8       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>4XH7B03840, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>4X97B04117, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>4X97B03867, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> </ul>                                                                                   |
| 5      | 0               | 0       | 0    | 8        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>4XH7B03840, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>4X97B04117, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                   |
| 6      | 16              | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>4XH7B03842, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>4XH7B03842, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>4X97B04117, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>4X97B03870, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> </ul> |

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7      | 0               | 0       | 16   | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B03867</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> <li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li> </ul>                                                                                                                                                                                                |
| 11     | 0               | 0       | 0    | 16       | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                                  |
| 8      | 0               | 8       | 8    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03867</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> <li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li> </ul>                                                                                               |
| 9      | 8               | 8       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li> </ul>                                                                                            |
| 10     | 8               | 0       | 8    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li> </ul>                                                                                                                                                                                           |
| 12     | 24              | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> </ul> |

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 13     | 0               | 0       | 24   | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B03880</b>, ThinkSystem SR650 V4 2.5" Chassis 24 NVMe Gen5 Cable Kit</li> </ul>                                                                                                                                                                                                                                                                                            |
| 14     | 16              | 8       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03871</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 NVMe Cable Kit</li> </ul> |
| 15     | 16              | 0       | 8    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03871</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 NVMe Cable Kit</li> </ul>                                                                                                |
| 16     | 0               | 0       | 0    | 24       | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> </ul>                                                                                    |

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 17     | 24              | 0       | 0    | 0        | 0        | 0         | 0         | 4        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03847</b>, ThinkSystem SR650 V4 4x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> </ul>                                                                                             |
| 18     | 24              | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 4        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03847</b>, ThinkSystem SR650 V4 4x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> <li>• <b>4X97B03874</b>, ThinkSystem SR650 V4 2.5" Chassis Rear NVMe Cable Kit</li> </ul>         |
| 19     | 16              | 8       | 0    | 0        | 0        | 0         | 0         | 4        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03847</b>, ThinkSystem SR650 V4 4x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03871</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 NVMe Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> </ul> |

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 20     | 24              | 0       | 0    | 0        | 8        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03844</b>, ThinkSystem SR650 V4 8x2.5" SAS/SATA Middle Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03875</b>, ThinkSystem SR650 V4 2.5" Chassis Middle SAS/SATA Cable Kit</li> </ul>                                                                                             |
| 21     | 16              | 8       | 0    | 0        | 8        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03844</b>, ThinkSystem SR650 V4 8x2.5" SAS/SATA Middle Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03871</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 NVMe Cable Kit</li> <li>• <b>4X97B03875</b>, ThinkSystem SR650 V4 2.5" Chassis Middle SAS/SATA Cable Kit</li> </ul> |
| 22     | 16              | 8       | 0    | 0        | 0        | 0         | 0         | 8        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03847</b>, ThinkSystem SR650 V4 4x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03871</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 NVMe Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> </ul>     |

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 23     | 24              | 0       | 0    | 0        | 0        | 0         | 0         | 8        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03848</b>, ThinkSystem SR650 V4 8x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> </ul> |
| 101    | 0               | 0       | 24   | 0        | 0        | 0         | 0         | 0        | 4         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03846</b>, ThinkSystem SR650 V4 4x2.5" NVMe Rear Backplane option Kit</li> <li>• <b>4X97B03881</b>, ThinkSystem SR650 V4 2.5" Chassis 28 NVMe Gen5 Cable Kit</li> </ul>                                                                                                                                                                                                                                                                                            |
| 24     | 0               | 0       | 24   | 0        | 0        | 8         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03843</b>, ThinkSystem SR650 V4 8x2.5" NVMe Middle Backplane option Kit</li> <li>• <b>4X97B03882</b>, ThinkSystem SR650 V4 2.5" Chassis 32 NVMe Gen5 Cable Kit</li> </ul>                                                                                                                                                                                                                                                                                          |
| 25     | 0               | 0       | 24   | 0        | 0        | 8         | 0         | 0        | 4         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03843</b>, ThinkSystem SR650 V4 8x2.5" NVMe Middle Backplane option Kit</li> <li>• <b>4XH7B03846</b>, ThinkSystem SR650 V4 4x2.5" NVMe Rear Backplane option Kit</li> <li>• <b>4X97B03883</b>, ThinkSystem SR650 V4 2.5" Chassis 36 NVMe Gen5 Cable Kit</li> </ul>                                                                                                                                                                                                 |

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 26     | 24              | 0       | 0    | 0        | 8        | 0         | 0         | 4        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03844</b>, ThinkSystem SR650 V4 8x2.5" SAS/SATA Middle Backplane option Kit</li> <li>• <b>4XH7B03848</b>, ThinkSystem SR650 V4 8x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03875</b>, ThinkSystem SR650 V4 2.5" Chassis Middle SAS/SATA Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> </ul> |
| 27     | 24              | 0       | 0    | 0        | 8        | 0         | 0         | 8        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03844</b>, ThinkSystem SR650 V4 8x2.5" SAS/SATA Middle Backplane option Kit</li> <li>• <b>4XH7B03847</b>, ThinkSystem SR650 V4 4x2.5" SAS/SATA Rear Backplane option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> <li>• <b>4X97B03872</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 SAS/SATA Cable Kit</li> <li>• <b>4X97B03875</b>, ThinkSystem SR650 V4 2.5" Chassis Middle SAS/SATA Cable Kit</li> <li>• <b>4X97B03873</b>, ThinkSystem SR650 V4 2.5" Chassis Rear SAS/SATA Cable Kit</li> </ul> |
| 121    | 8               | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li> <li>• <b>4X97B03871</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP3 NVMe Cable Kit</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     |

## Drive bay field upgrades - E3.S chassis

The tables below lists the backplane kits and cable kits needed to build one of the supported configurations with E3.S drive bays.

The Config numbers listed here match the configuration listed in the Storage configuration [Overview](#) and [Details](#) sections.

Return to [Field upgrades](#).

**Note:** The table indicates the locations of each E3.S backplane. The locations of the 8 backplanes (BP1 to BP8) are shown in the following figure.

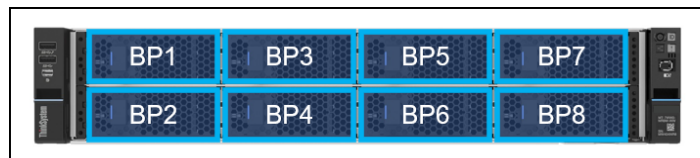


Figure 19. E3.S backplane numbering

For more information about the backplane kits and cable kits, see the Lenovo server options site: [https://serveroption.lenovo.com/cable\\_kit\\_options/](https://serveroption.lenovo.com/cable_kit_options/)

**Tip:** The configurations listed in the table below are not in numerical order.

Table 40. Field upgrades - E3.S drive bays (BP = Backplane)

| Config<br>(# of CPUs) | BP1   | BP3   | BP5   | BP7   | Backplane and cable kits required (all required)<br>Some configs also require <a href="#">replacement cable walls</a>                                                                                                                                                                                                                                                                                             |
|-----------------------|-------|-------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | BP2   | BP4   | BP6   | BP8   |                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 59 (1P)               |       |       | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |
| 59 (2P)               | 4x 1T |       | 4x 1T |       | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |
| 60 (1P)               |       |       | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03885, ThinkSystem SR650 V4 E3.S 1T Long Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul> |
|                       |       |       | 4x 1T | 4x 1T |                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 60 (2P)               | 4x 1T | 4x 1T | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>4x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul>                                                                                                                                                 |
| 61 (2P)               | 4x 1T | 4x 1T | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>4x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03885, ThinkSystem SR650 V4 E3.S 1T Long Cable Kit</li> </ul>                                                                         |
|                       | 4x 1T |       | 4x 1T |       |                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 62 (2P)               | 4x 1T | 4x 1T | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>4x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>4x 4X97B03885, ThinkSystem SR650 V4 E3.S 1T Long Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul> |
|                       | 4x 1T | 4x 1T | 4x 1T | 4x 1T |                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 102 (1P)              |       |       | 2x 2T |       | <ul style="list-style-type: none"> <li>1x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>1x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |
| 63 (1P)               |       |       | 2x 2T | 2x 2T | <ul style="list-style-type: none"> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |
| 63 (2P)               | 2x 2T |       | 2x 2T |       | <ul style="list-style-type: none"> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> </ul>                                                                                                                                                                                                                          |
| 64 (2P)               | 2x 2T | 2x 2T | 2x 2T | 2x 2T | <ul style="list-style-type: none"> <li>4x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>4x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |

| Config<br>(# of CPUs) | BP1   | BP3   | BP5   | BP7   | Backplane and cable kits required (all required)<br>Some configs also require <a href="#">replacement cable walls</a>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|-------|-------|-------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | BP2   | BP4   | BP6   | BP8   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 65 (2P)               | 2x 2T | 2x 2T | 2x 2T | 2x 2T | <ul style="list-style-type: none"> <li>4x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>4x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> <li>2x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> </ul>                                                                                                                                                                                                                          |
|                       | 2x 2T |       | 2x 2T |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 66 (1P)               |       |       | 4x 1T | 2x 2T | <ul style="list-style-type: none"> <li>1x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>1x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>1x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>1x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |
|                       |       |       |       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 67 (1P)               |       |       | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul>                                                                                                                                                  |
|                       |       |       | 2x 2T | 2x 2T |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 67 (2P)               | 4x 1T | 4x 1T | 2x 2T | 2x 2T | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> </ul>                                                                                                                                                                                                                         |
|                       |       |       |       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 68 (2P)               | 4x 1T | 2x 2T | 4x 1T | 2x 2T | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> <li>2x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> </ul>                                                                         |
|                       | 2x 2T |       | 2x 2T |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 69 (2P)               | 4x 1T | 2x 2T | 4x 1T | 2x 2T | <ul style="list-style-type: none"> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>4x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> <li>4x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul> |
|                       | 2x 2T | 2x 2T | 2x 2T | 2x 2T |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 70 (2P)               | 4x 1T | 4x 1T | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>4x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> </ul>                                                                                                                                                                                                                          |
|                       | 2x 2T |       | 2x 2T |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 71 (2P)               | 4x 1T | 4x 1T | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>4x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>4x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>4x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul>                                                                                                                                                  |
|                       | 2x 2T | 2x 2T | 2x 2T | 2x 2T |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 72 (2P)               | 4x 1T | 4x 1T | 4x 1T | 4x 1T | <ul style="list-style-type: none"> <li>4x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03854, ThinkSystem V4 1U/2U E3.S 4x1T Backplane Option Kit</li> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>4x 4X97B03884, ThinkSystem SR650 V4 E3.S 1T Short Cable Kit</li> <li>2x 4X97B03885, ThinkSystem SR650 V4 E3.S 1T Long Cable Kit</li> <li>2x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> <li>1x 4X97B04305, ThinkSystem SR650/a V4 2U Cable Wall Option Kit</li> </ul>  |
|                       | 4x 1T | 2x 2T | 4x 1T | 2x 2T |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 119 (1P)              |       |       | 2x 2T | 2x 2T | <ul style="list-style-type: none"> <li>2x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>1x 4XH7B03856, ThinkSystem V4 1U/2U E3.S 2x2T Backplane Option Kit</li> <li>2x 4X97B03888, ThinkSystem SR650 V4 E3.S 2T Short Cable Kit</li> <li>1x 4X97B03889, ThinkSystem SR650 V4 E3.S 2T Long Cable Kit</li> </ul>                                                                                                                                                                                                                          |
|                       |       |       | 2x 2T |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Drive bay field upgrades - Compute Complex Neptune water cooling

The tables below lists the backplane kits and cable kits needed to build one of the supported configurations using Lenovo Compute Complex Neptune water cooling.

The Config numbers listed here match the configuration listed in the Storage configuration [Overview](#) and [Details](#) sections.

Return to [Field upgrades](#).

For more information about the backplane kits and cable kits, see the Lenovo server options site: [https://serveroption.lenovo.com/cable\\_kit\\_options/](https://serveroption.lenovo.com/cable_kit_options/)

**Tip:** The configurations listed in the table below are not in numerical order.

Table 41. Drive bay field upgrade for the 2.5-inch chassis with Compute Complex Neptune water cooling (Blue = SAS/SATA, Purple = AnyBay, Red = NVMe)

| Config | 2.5" front bays |         |      |          | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                         |
|--------|-----------------|---------|------|----------|----------|-----------|-----------|----------|-----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/SATA        | Any Bay | NVMe | Tri-Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 103    | 8               | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                                        |
| 87     | 0               | 0       | 8    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B03867</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> </ul>                                                                                                                                                                                             |
| 88     | 0               | 8       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03867</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> </ul>                                                                                            |
| 104    | 0               | 0       | 0    | 8        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> </ul>                                                                                                                                                                                     |
| 105    | 16              | 0       | 0    | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li> <li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li> <li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li> </ul> |
| 89     | 0               | 0       | 16   | 0        | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li> <li>• <b>4X97B03867</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li> <li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li> </ul>         |

| Config | 2.5" front bays |            |      |              | Mid bays |           | Rear bays |          |           |          | Backplane and cable kits required (all required)                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------|-----------------|------------|------|--------------|----------|-----------|-----------|----------|-----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | SAS/<br>SATA    | Any<br>Bay | NVMe | Tri-<br>Mode | 2.5" SAS | 2.5" NVMe | 3.5" SAS  | 2.5" SAS | 2.5" NVMe | 2.5" Any |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 106    | 0               | 0          | 0    | 16           | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"><li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li><li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li><li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li><li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li></ul>                                                                                      |
| 90     | 0               | 8          | 8    | 0            | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"><li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li><li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li><li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li><li>• <b>4X97B03867</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP1 NVMe Cable Kit</li><li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li></ul>    |
| 91     | 8               | 8          | 0    | 0            | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"><li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li><li>• <b>4XH7B03840</b>, ThinkSystem SR650/a V4 8x2.5" AnyBay Gen5 Backplane Option Kit</li><li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li><li>• <b>4X97B03870</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 SAS/SATA Cable Kit</li><li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li></ul> |
| 92     | 8               | 0          | 8    | 0            | 0        | 0         | 0         | 0        | 0         | 0        | <ul style="list-style-type: none"><li>• <b>4XH7B03842</b>, ThinkSystem SR650/a V4 8x2.5" SAS/SATA Backplane Option Kit</li><li>• <b>4XH7B03841</b>, ThinkSystem SR650/a V4 8x2.5" NVMe Gen5 Backplane Option Kit</li><li>• <b>4X97B04117</b>, ThinkSystem SR650/a V4 2.5" Chassis Front BP1 SAS/SATA Cable Kit</li><li>• <b>4X97B03869</b>, ThinkSystem SR650 V4 2.5" Chassis Front BP2 NVMe Cable Kit</li></ul>                                                                                               |

#### Adding an Internal (CFF) RAID adapter or HBA

If you want to add an internal (CFF) storage adapter (HBA or RAID adapter) to a configuration, you will need to order the cable kit as listed in the following table. Suitable upgrades are either replacing an existing adapter in a rear PCIe slot, or adding the CFF adapter to a server without any storage adapter installed.

Table 42. Cable needed for field upgrades to add CFF adapter

| Part number | Description                                                |
|-------------|------------------------------------------------------------|
| 4X97B03877  | ThinkSystem SR650/a V4 Internal Raid/HBA Adapter Cable Kit |

#### Adding an Internal SAS Expander

If you want to add an internal (CFF) SAS Expander adapter to a configuration, you will need to order the cable kit as listed in the following table.

Table 43. Cable needed for field upgrades to add SAS Expander

| Part number | Description                                              |
|-------------|----------------------------------------------------------|
| 4X97B03878  | ThinkSystem SR650 V4 Internal Expander Adapter Cable Kit |

## Adding a RAID 960W-32i adapter

If you want to add a RAID 960W-32i adapter to an NVMe configuration, you will also need to order the cable kit as listed in the following table. Order the RAID 960W-32i adapter separately as described in the [Controllers for internal storage](#) section.

**NVMe only:** The RAID 960W-32i is only for NVMe drive configurations. See the [Storage configurations](#) section for specific configurations.

Table 44. Cable kit when adding RAID 960W-32i as a field upgrade

| Part number | Description                                            |
|-------------|--------------------------------------------------------|
| 4X97B17364  | ThinkSystem SR650 V4/SR650a V4 960W-32i RAID Cable Kit |

## Replacement cable routing walls

Cable walls are used to guide signal and power cables through the server and to maximize airflow across the components. The server comes with 1U cable walls at both side of the processor board. It is recommended to replace the 1U cable walls with 2U cable walls when there are more than five cables routed at one side.

The 2U cable walls are required for some E3.S configurations and will be automatically derived for CTO orders. For field upgrades, see the [Drive bay field upgrades - E3.S chassis](#) section, where the table indicates when replacement cable walls are required.

Table 45. Replacement cable routing walls

| Part number | Feature code | Description                                                                                                                           |
|-------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 4X97B04305  | C3RS         | ThinkSystem SR650/a V4 2U Cable Wall Option Kit <ul style="list-style-type: none"><li>• 2x 2U cable wall (SBB7A86017, C3RS)</li></ul> |

For details on installing the cable wall replacements, see the SR650 V4 User Guide:  
[https://pubs.lenovo.com/sr650-v4/cable\\_wall\\_replacement\\_2u](https://pubs.lenovo.com/sr650-v4/cable_wall_replacement_2u)

## RAID flash power module (supercap) support

High-end RAID adapters such as the RAID 940 Series include a RAID flash power module (supercap) that needs to be mounted properly. Supercap holders are integrated into the air baffles and mid-chassis drive cage. No additional components are needed.

There are up to four possible locations for supercaps, depending on the air baffle or mid-chassis drive bays installed, as follows:

- Standard air baffle: 4 supercaps
- GPU air baffle: 4 supercaps
- 2.5-inch mid-chassis drive bays: 2 supercaps

**Note:** Configurations with mid-chassis drive bays only support two supercaps. As a result, such server configurations only support two RAID adapters if the adapters include supercaps.

See the following User Guide page for the specifics on the locations:  
[https://pubs.lenovo.com/sr650-v4/supercap\\_replacement](https://pubs.lenovo.com/sr650-v4/supercap_replacement)

## 2.5-inch drive bay fillers

Backplane option kits include the necessary drive bay fillers, however if needed, additional blanks can be ordered as listed in the following table.

Table 46. Drive bay fillers for 2.5-inch bays

| Part number | Description                                                                       |
|-------------|-----------------------------------------------------------------------------------|
| 4XH7A99569  | ThinkSystem 2.5" 1x1 HDD Filler by 8 units (contains 8x single drive-bay fillers) |

## M.2 drives

The SR650 V4 supports one or two M.2 form-factor SATA or NVMe drives for use as an operating system boot solution or as additional storage.

M.2 drives can be installed in one of the following locations:

- Internal to the server (non-hot-swap) on an M.2 module which is mounted horizontally in the server. RAID is either integrated or with the use of VROC, depending on the M.2 adapter selected.

Locations of the internal M.2 are as follows:

- In servers without mid-chassis drives, the M.2 module is mounted on the air baffle
- With a mid-chassis drive cage, the M.2 module is mounted on the drive cage, as shown in the [Mid drive bays](#) section.
- Rear-mounted hot-swap M.2 drives with integrated RAID. Supported in slot 8 or slot 5.
- Front-mounted hot-swap M.2 drives with integrated RAID. Only supported with E3.S and 2.5-inch front drive configurations.

In this section:

- [M.2 ordering information](#)
- [VROC RAID support for the B340i-2i and B350-2i adapter](#)
- [M.2 adapter features](#)
- [M.2 field upgrades for internal M.2](#)
- [M.2 field upgrades for hot-swap M.2](#)

**Hot-swap tip:** The bare M.2 drives (as listed in the [Internal drive options](#) section) are non-hot-swap drives. However, when the installed in the assembly for front-mounted or rear-mounted M.2, the drives are hot-swap enabled.

The following figure shows the SR650 V4 with front and rear hot-swap M.2 drive bays.

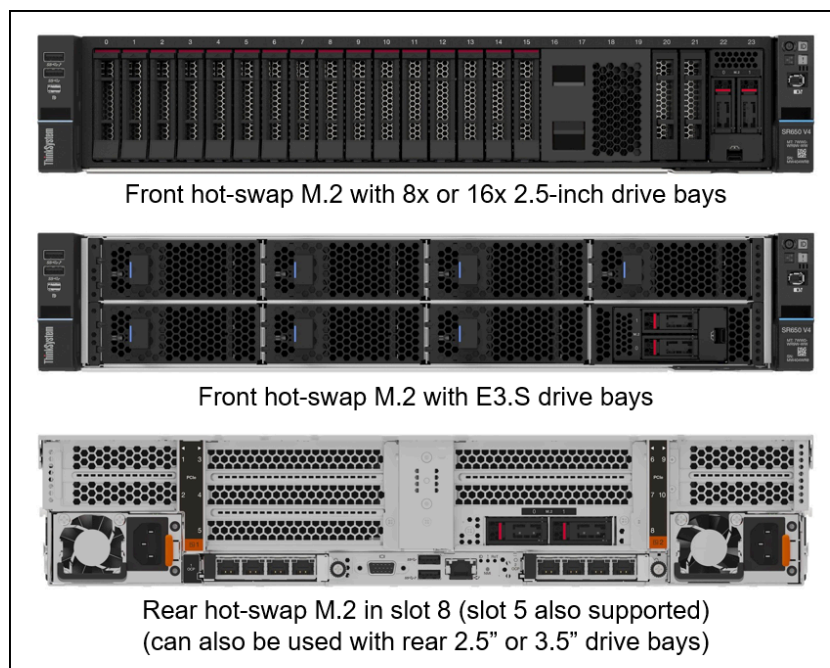


Figure 20. Hot-swap M.2 drive bays

The following figure shows the components of the rear hot-swap M.2 drive bays.

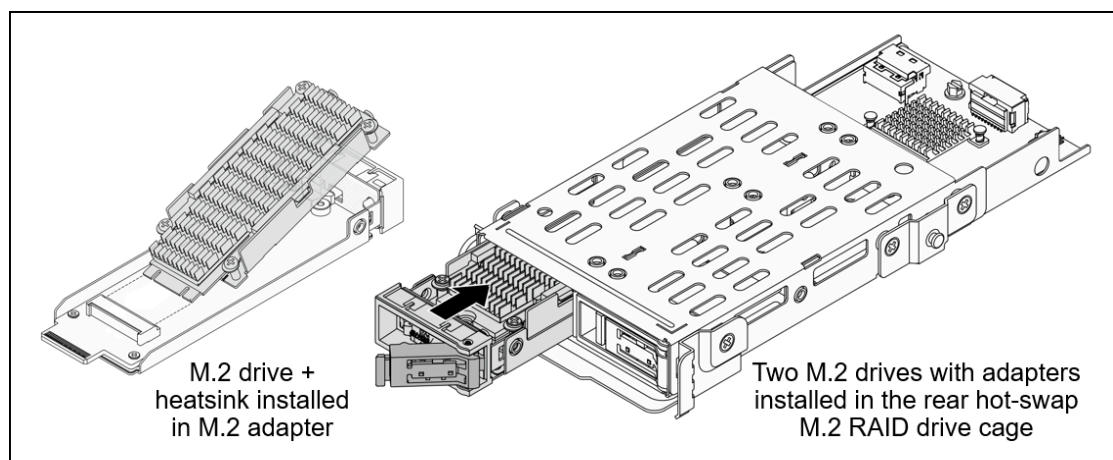


Figure 21. Rear hot-swap M.2 drive bays

The following figure shows the components of the front hot-swap M.2 drive bays.

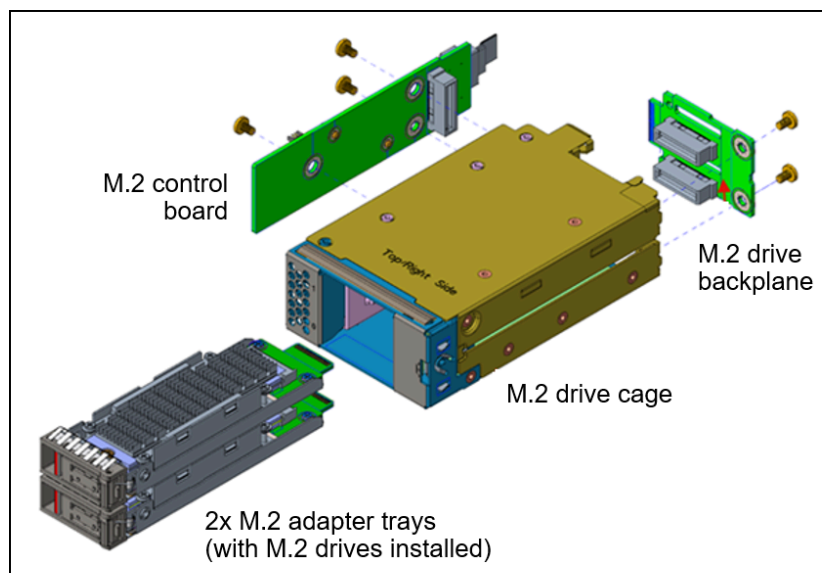


Figure 22. Front hot-swap M.2 drive bays

### M.2 ordering information

The supported M.2 module is listed in the following table. For field upgrades see the [M.2 field upgrades](#) section below.

Table 47. M.2 adapters

| Part number                 | Feature code | Description                                                                                                                                                                                                                                                                    | Top Choice Express | SATA drives | NVMe drives              | RAID                  | Max Qty |
|-----------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|--------------------------|-----------------------|---------|
| Internal M.2 (non-hot-swap) |              |                                                                                                                                                                                                                                                                                |                    |             |                          |                       |         |
| 4Y37B09898                  | CC7G         | ThinkSystem M.2 RAID B550i-2i SATA/NVMe Enablement Kit                                                                                                                                                                                                                         | TCE                | No          | Yes (x1 lane per drive)  | Integrated (Broadcom) | 1       |
| 4Y37A93746                  | C26V         | ThinkSystem M.2 RAID B545i-2i SATA/NVMe Adapter                                                                                                                                                                                                                                | Not TCE            | Yes         | Yes (x1 lane per drive)  | Integrated (Broadcom) | 1       |
| 4Y37B09897                  | CC7H         | ThinkSystem M.2 B350i-2i NVMe Enablement Kit                                                                                                                                                                                                                                   | TCE                | No          | Yes (x2 lanes per drive) | VROC                  | 1       |
| Rear hot-swap M.2           |              |                                                                                                                                                                                                                                                                                |                    |             |                          |                       |         |
| 4XH7B11661                  | CCCZ         | ThinkSystem SR650/a V4 Rear M.2 RAID B550p-2HS SATA/NVMe Enablement Kit <ul style="list-style-type: none"> <li>M.2 rear drive cage for FH slot</li> <li>M.2 rear drive cage for 2xFH slots</li> <li>M.2 2-drive hot-swap backplane</li> <li>Signal and power cables</li> </ul> | TCE                | No          | Yes (x1 lane per drive)  | Integrated (Broadcom) | 1       |

| Part number        | Feature code | Description                                                                                                                                                                                                                                                                                                                                                                                      | Top Choice Express | SATA drives | NVMe drives             | RAID                  | Max Qty |
|--------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|-------------------------|-----------------------|---------|
| 4XH7B03860         | C0JJ         | ThinkSystem SR650/a V4 M.2 RAID B540p-2HS SATA/NVMe Enablement Kit <ul style="list-style-type: none"> <li>• M.2 rear drive cage for FH slot</li> <li>• M.2 rear drive cage for 2x FH slots</li> <li>• M.2 2-drive hot-swap backplane</li> <li>• 2x empty M.2 hot-swap adapter tray</li> <li>• Signal and power cables</li> </ul>                                                                 | Not TCE            | Yes         | Yes (x1 lane per drive) | Integrated (Broadcom) | 1       |
| Front hot-swap M.2 |              |                                                                                                                                                                                                                                                                                                                                                                                                  |                    |             |                         |                       |         |
| 4XH7B11660         | CD0S         | ThinkSystem SR650/a V4 Front M.2 RAID B550d-2HS SATA/NVMe Enablement Kit <ul style="list-style-type: none"> <li>• M.2 front drive cage</li> <li>• M.2 2-drive hot-swap backplane</li> <li>• M.2 control board</li> <li>• 1x 4-bay 2.5-inch drive bay filler</li> <li>• 2x 1-bay 2.5-inch drive bay filler</li> <li>• Signal and power cables</li> </ul>                                          | TCE                | No          | Yes (x1 lane per drive) | Integrated (Broadcom) | 1       |
| 4XH7B03857         | C217         | ThinkSystem SR650/a V4 M.2 RAID B540d-2HS SATA/NVMe Enablement Kit <ul style="list-style-type: none"> <li>• M.2 front drive cage</li> <li>• M.2 2-drive hot-swap backplane</li> <li>• M.2 control board</li> <li>• 2x empty M.2 hot-swap adapter trays</li> <li>• 1x 4-bay 2.5-inch drive bay filler</li> <li>• 2x 1-bay 2.5-inch drive bay filler</li> <li>• Signal and power cables</li> </ul> | Not TCE            | Yes         | Yes (x1 lane per drive) | Integrated (Broadcom) | 1       |

Configuration notes:

- M.2 is not supported with all storage configurations - see [Storage configurations](#) for details.
- For CTO orders, all other necessary components, except for the M.2 drives themselves, will be automatically included in the order. For drives, see the [Internal drive options](#) section.
- For field upgrades of the internal M.2, an additional cable is needed as described in the [M.2 field upgrades for internal M.2](#) section below.
- For field upgrades of the B540 rear or front hot-swap M.2, one additional kit is needed for each M.2 drive you plan to install, as described in the [M.2 field upgrades for hot-swap M.2](#) section below. B550 M.2 adapters do not require these drive kits.

- The Rear M.2 cage can be installed in either slot 8 or slot 5.
- For configurations with open-loop water cooling, the Rear M.2 is installed in slot 5

#### VROC RAID support for the B340i-2i and B350-2i adapter

ThinkSystem M.2 B340i-2i NVMe Enablement Adapter (4Y37A91802) and ThinkSystem M.2 B350i-2i NVMe Enablement Kit (4Y37B09897) optionally supports RAID with the use of Intel VROC. For CTO orders, ordering information is listed in the following table.

Table 48. CTO feature codes to select VROC RAID for B340i-2i and B350-2i adapters

| Part number                                      | Feature code | Description                                 | Top Choice Express | Max Qty | RAID support |
|--------------------------------------------------|--------------|---------------------------------------------|--------------------|---------|--------------|
| VROC NVMe RAID for B340i-2i and B350-2i adapters |              |                                             |                    |         |              |
| 4L47A92670*                                      | BZ4X         | Intel VROC RAID1 Only for M.2               | TCE                | 1       | RAID-1       |
| 4L47A83669*                                      | BS7M         | Intel VROC (VMD NVMe RAID) Standard for M.2 | TCE                | 1       | RAID-0,1     |

\* The part numbers enable VROC for all installed drives, not just M.2

#### M.2 adapter features

The ThinkSystem M.2 RAID B550i-2i SATA/NVMe Enablement Kit (4Y37B09898) has the following features:

- Supports one or two NVMe M.2 drives (SATA not supported)
- Drives are not hot-swap
- Support 42mm, 60mm and 80mm drive form factors (2242, 2260 and 2280)
- RAID support via an onboard Broadcom SAS3808N RAID Controller
- With 1 drive, supports JBOD
- With 2 drives, supports 2-drive RAID-0, 2-drive RAID-1, or JBOD (default is RAID-1)
- PCIe 4.0 x2 host interface; PCIe 4.0 x1 connection to each drive
- Management and configuration support via UEFI and OS-based tools
- Supports monitoring and reporting of events and temperature
- Firmware update via Lenovo firmware update tools
- Supports SED drive encryption

The ThinkSystem M.2 RAID B545i-2i SATA/NVMe Adapter (4Y37A93746) has the following features:

- Supports one or two M.2 drives, either SATA or NVMe
- Drives are not hot-swap
- Supports M.2 2242, 2260, 2280 drive form factors (42mm, 60mm, 80mm)
- RAID support via an onboard Broadcom SAS3808N RAID Controller
- With 1 drive, supports JBOD
- With 2 drives, supports 2-drive RAID-0, 2-drive RAID-1, or JBOD (default is RAID-1)
- PCIe 4.0 x2 host interface; PCIe 4.0 x1 connection to each drive
- Management and configuration support via UEFI and OS-based tools
- Supports monitoring and reporting of events and temperature
- Firmware update via Lenovo firmware update tools
- Supports SED drive encryption

The ThinkSystem M.2 B350i-2i NVMe Enablement Kit (4Y37B09897) has the following features:

- Supports one or two NVMe M.2 drives (SATA not supported)
- Drives are not hot-swap
- Support M.2 2280 (80mm) drive form factor only
- No built-in RAID support (optionally supports Intel VROC NVMe RAID)
- PCIe 4.0 x4 host interface; PCIe 4.0 x2 connection to each drive
- Management and configuration support via UEFI and OS-based tools
- Supports monitoring and reporting of events and temperature
- Firmware update via Lenovo firmware update tools

- Supports SED drive encryption

The B550p-2HS (4XH7B11661, feature CCCZ) and B550d-2HS (4XH7B11660, feature CD0S) both have the following features:

- Supports one or two NVMe M.2 drives (SATA not supported)
- Each drive is installed in a hot-swap carrier
- Support M.2 2280 (80mm) drive form factor only
- RAID support via an onboard Broadcom SAS3808N RAID Controller
- With 1 drive, supports JBOD
- With 2 drives, supports 2-drive RAID-0, 2-drive RAID-1, or JBOD
- PCIe 4.0 x2 host interface; PCIe 4.0 x1 connection to each drive
- Management and configuration support via UEFI and OS-based tools
- Supports monitoring and reporting of events and temperature
- Firmware update via Lenovo firmware update tools
- Supports SED drive encryption

The ThinkSystem SR650/a V4 M.2 RAID B540p-2HS SATA/NVMe Enablement Kit (4XH7B03860) has the following features:

- Supports one or two M.2 drives, either SATA or NVMe
- Each drive is installed in a hot-swap carrier
- Support M.2 2280 (80mm) drive form factor only
- RAID support via an onboard Broadcom SAS3808N RAID Controller
- With 1 drive, supports JBOD
- With 2 drives, supports 2-drive RAID-0, 2-drive RAID-1, or JBOD (default is RAID-1)
- PCIe 4.0 x2 host interface; PCIe 4.0 x1 connection to each drive
- Management and configuration support via UEFI and OS-based tools
- Supports monitoring and reporting of events and temperature
- Firmware update via Lenovo firmware update tools
- Supports SED drive encryption

The ThinkSystem SR650/a V4 M.2 RAID B540d-2HS SATA/NVMe Enablement Kit (4XH7B03857) has the following features:

- Supports one or two M.2 drives, either SATA or NVMe
- Each drive is installed in a hot-swap carrier
- Support M.2 2280 (80mm) drive form factor only
- RAID support via an onboard Broadcom SAS3808N RAID Controller
- With 1 drive, supports JBOD
- With 2 drives, supports 2-drive RAID-0, 2-drive RAID-1, or JBOD (default is RAID-1)
- PCIe 4.0 x2 host interface; PCIe 4.0 x1 connection to each drive
- Management and configuration support via UEFI and OS-based tools
- Supports monitoring and reporting of events and temperature
- Firmware update via Lenovo firmware update tools
- Supports SED drive encryption

### **M.2 field upgrades for internal M.2**

For field upgrades to add one of the supported internal M.2 adapters, the SR650 V4 also requires an additional M.2 cable kit. Ordering information is listed in the following table.

Table 49. M.2 Cable Kits for field upgrades to add an internal M.2 adapter

| Part number | Description                                              | Qty |
|-------------|----------------------------------------------------------|-----|
| 4XH7B03859  | ThinkSystem SR650/a V4 Internal M.2 Enablement Cable kit | 1   |

### **M.2 field upgrades for hot-swap M.2**

This section applies to both the front hot-swap M.2 and rear hot-swap M.2.

For field upgrades to add one of the supported hot-swap M.2 modules, order one of the hot-swap part numbers as listed in the [M.2 ordering information](#) section. Unlike internal M.2, an additional cable kit is not required for hot-swap M.2.

Hot-swap M.2 drive bays require hot-swap M.2 drives which are regular M.2 drives with a hot-swap assembly attached. For configure-to-order builds, the hot-swap drive assembly is automatically included in the order for all M.2 drives.

For the hot-swap drives, ordering is as follows:

- For B550 hot-swap M.2, only specific hot-swap M.2 drives are supported as listed in the [Internal drive options](#) section. These drives include the hot-swap assembly; no other components are needed.
- For B540 hot-swap M.2, order non-hot-swap drives as listed in the [Internal drive options](#) section, plus a separate kit for each drive, ThinkSystem V4 Hot Swap M.2 SATA/NVMe Drive Assembly Kit (4XH7A96837). Ordering information is listed in the following table

Table 50. M.2 kit for field upgrades to add hot-swap M.2 drives (for B540 only; not needed for B550)

| Part number | Description                                                                                                                                                                              | Qty         |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 4XH7A96837  | ThinkSystem V4 Hot Swap M.2 SATA/NVMe Drive Assembly Kit (see below) <ul style="list-style-type: none"><li>• M.2 adapter</li><li>• M.2 drive tray</li><li>• M.2 drive heatsink</li></ul> | 1 per drive |

The following figure shows the components of the ThinkSystem V4 Hot Swap M.2 SATA/NVMe Drive Assembly Kit (4XH7A96837), used for B540 hot-swap M.2 (for both front and rear drive bays; not for the B550 adapter and not for internal M.2). The M.2 drive needs to be ordered separately.

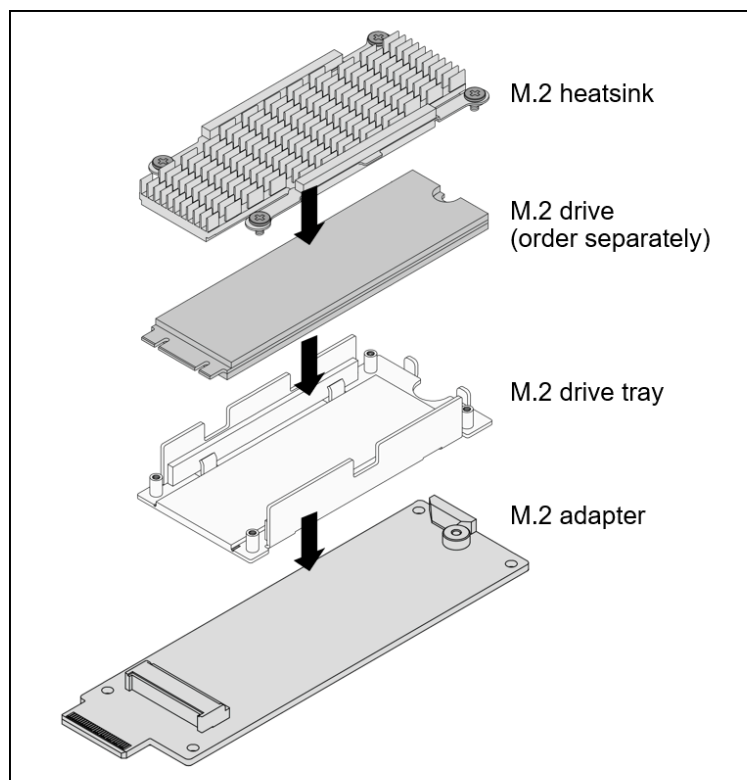


Figure 23. Components of the ThinkSystem V4 Hot Swap M.2 SATA/NVMe Drive Assembly Kit

### SED encryption key management with SKLM

The server supports self-encrypting drives (SEDs) as listed in the [Internal drive options](#) section. To effectively manage a large deployment of these drives in Lenovo servers, IBM Security Key Lifecycle Manager (SKLM) offers a centralized key management solution.

The IBM Security Key Lifecycle Manager software is available from Lenovo using the ordering information listed in the following table.

Table 51. IBM Security Key Lifecycle Manager licenses

| Part number                                  | Feature | Description                                                                                                                            |
|----------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>SKLM Basic Edition</b>                    |         |                                                                                                                                        |
| 7S0A007FWW                                   | S874    | IBM Security Key Lifecycle Manager Basic Edition Install License + SW Subscription & Support 12 Months                                 |
| 7S0A008VWW                                   | SDJR    | IBM Security Key Lifecycle Manager Basic Edition Install License + SW Subscription & 3 Years Of Support                                |
| 7S0A008WWW                                   | SDJS    | IBM Security Key Lifecycle Manager Basic Edition Install License + SW Subscription & 4 Years Of Support                                |
| 7S0A008XWW                                   | SDJT    | IBM Security Key Lifecycle Manager Basic Edition Install License + SW Subscription & 5 Years Of Support                                |
| <b>SKLM For Raw Decimal Terabyte Storage</b> |         |                                                                                                                                        |
| 7S0A007HWW                                   | S876    | IBM Security Key Lifecycle Manager For Raw Decimal Terabyte Storage Resource Value Unit License + SW Subscription & Support 12 Months  |
| 7S0A008YWW                                   | SDJU    | IBM Security Key Lifecycle Manager For Raw Decimal Terabyte Storage Resource Value Unit License + SW Subscription & 3 Years Of Support |

| Part number                              | Feature | Description                                                                                                                               |
|------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 7S0A008ZWW                               | SDJV    | IBM Security Key Lifecycle Manager For Raw Decimal Terabyte Storage Resource Value Unit License + SW Subscription & 4 Years Of Support    |
| 7S0A0090WW                               | SDJW    | IBM Security Key Lifecycle Manager For Raw Decimal Terabyte Storage Resource Value Unit License + SW Subscription & 5 Years Of Support    |
| SKLM For Raw Decimal Petabyte Storage    |         |                                                                                                                                           |
| 7S0A007KWW                               | S878    | IBM Security Key Lifecycle Manager For Raw Decimal Petabyte Storage Resource Value Unit License + SW Subscription & Support 12 Months     |
| 7S0A0091WW                               | SDJX    | IBM Security Key Lifecycle Manager For Raw Decimal Petabyte Storage Resource Value Unit License + SW Subscription & 3 Years Of Support    |
| 7S0A0092WW                               | SDJY    | IBM Security Key Lifecycle Manager For Raw Decimal Petabyte Storage Resource Value Unit License + SW Subscription & 4 Years Of Support    |
| 7S0A0093WW                               | SDJZ    | IBM Security Key Lifecycle Manager For Raw Decimal Petabyte Storage Resource Value Unit License + SW Subscription & 5 Years Of Support    |
| SKLM For Usable Decimal Terabyte Storage |         |                                                                                                                                           |
| 7S0A007MWW                               | S87A    | IBM Security Key Lifecycle Manager For Usable Decimal Terabyte Storage Resource Value Unit License + SW Subscription & Support 12 Months  |
| 7S0A0094WW                               | SDK0    | IBM Security Key Lifecycle Manager For Usable Decimal Terabyte Storage Resource Value Unit License + SW Subscription & 3 Years In Support |
| 7S0A0095WW                               | SDK1    | IBM Security Key Lifecycle Manager For Usable Decimal Terabyte Storage Resource Value Unit License + SW Subscription & 4 Years In Support |
| 7S0A0096WW                               | SDK2    | IBM Security Key Lifecycle Manager For Usable Decimal Terabyte Storage Resource Value Unit License + SW Subscription & 5 Years In Support |
| SKLM For Usable Decimal Petabyte Storage |         |                                                                                                                                           |
| 7S0A007PWW                               | S87C    | IBM Security Key Lifecycle Manager For Usable Decimal Petabyte Storage Resource Value Unit License + SW Subscription & Support 12 Months  |
| 7S0A0097WW                               | SDK3    | IBM Security Key Lifecycle Manager For Usable Decimal Petabyte Storage Resource Value Unit License + SW Subscription & 3 Years Of Support |
| 7S0A0098WW                               | SDK4    | IBM Security Key Lifecycle Manager For Usable Decimal Petabyte Storage Resource Value Unit License + SW Subscription & 4 Years Of Support |
| 7S0A0099WW                               | SDK5    | IBM Security Key Lifecycle Manager For Usable Decimal Petabyte Storage Resource Value Unit License + SW Subscription & 5 Years Of Support |

## Encryption Enablement

For CTO orders, the configurator allows you to specify that you want to create an "encryption-ready" server, and to ensure that only components that support encryption are selectable in the configurator. This is done in DCSC and x-config configurator tools by selecting the feature code listed in the following table.

Table 52. Encryption Enablement

| Part number | Feature code | Description                                           | Top Choice Express |
|-------------|--------------|-------------------------------------------------------|--------------------|
| CTO only    | CES1*        | Drive Encryption Enablement (not supported with vSAN) | TCE                |

\* In DCSC, this feature is in **Storage > Internal Storage** under the heading "Encryption Enablement"

**Tip:** This feature will not be selectable until an SED-capable drive is selected.

By selecting this feature code, the following requirements will be enabled in the configurator:

- SED drives will be required
- XCC3 Premier will be required

- VROC NVMe RAID cannot be selected
- 440-8i and 440-16i HBAs cannot be selected

Configuration notes:

- Do not use this feature code for VMware vSAN or Nutanix servers, since these virtualization offerings do not support HW RAID adapter-based encryption
- Selecting this feature code does not enable encryption at the factory
- If you don't select this feature code, you will still be able to enable encryption in your server, however you will need to ensure that you have suitable storage controllers and drives installed in the server

## Controllers for internal storage

The SR650 V4 offers a variety of controller options for internal drives:

- For 2.5-inch and 3.5-inch drives:
  - RAID adapters and HBAs for SAS/SATA drives (PCIe slot-based)
  - RAID adapters and HBAs for SAS/SATA drives (cabled in a dedicated space)
  - Onboard NVMe ports with RAID support using Intel VROC NVMe RAID
  - Tri-Mode support using RAID 940 adapters for NVMe drives, with RAID provided by the RAID adapter
  - RAID 960W-32i adapter for NVMe x2 support (max 16 drives per adapter)
- For E3.S EDSFF drives:
  - Onboard NVMe ports with RAID support using Intel VROC NVMe RAID
- For M.2 drives (see [M.2 drives](#) section)
  - SATA controller integrated on the M.2 adapters
  - NVMe controller integrated on the M.2 adapters (Intel VROC for RAID)

As well as supporting RAID adapters and HBAs that install in a PCIe slot, the SR650 V4 with 2.5-inch front drive bays supports a custom form factor (CFF) adapter that is mounted in the server and cabled to one of the onboard NVMe ports.

The following table lists the adapters used for the internal storage of the server. For VROC ordering information, see the [Intel VROC](#) section.

The table includes a Top Choice Express column; for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

**No Onboard SATA support:** The processors in the SR650 V4 do not offer onboard SATA support. For SATA drives, you will need either a RAID adapter or an HBA.

Table 53. Storage controller support for internal drives

| Part number                                                                             | Feature code | Description                                               | Top Choice Express | RAID level                         | Max Qty | Slots | Supercap |
|-----------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------|--------------------|------------------------------------|---------|-------|----------|
| Onboard NVMe - PCIe 5.0 - RAID using <a href="#">Intel VROC</a> .                       |              |                                                           |                    |                                    |         |       |          |
| CTO only                                                                                | BC4V         | Non RAID NVMe                                             | TCE                | Optional; See <a href="#">VROC</a> | 1       | None  | No       |
| SAS HBA - PCIe 4.0                                                                      |              |                                                           |                    |                                    |         |       |          |
| 4Y37A78602                                                                              | BM50         | ThinkSystem 440-16i SAS/SATA PCIe Gen4 12Gb HBA           | Not TCE            | None                               | 2       | 1-10  | No       |
| 4Y37A09725                                                                              | B8P1         | ThinkSystem 440-16i SAS/SATA PCIe Gen4 12Gb Internal HBA  | Not TCE            | None                               | 1       | CFF*  | No       |
| SAS Expander                                                                            |              |                                                           |                    |                                    |         |       |          |
| 4Y37A09736                                                                              | B8P6         | ThinkSystem 48 port 12Gb Internal Expander                | Not TCE            | None                               | 1       | CFF*  | No       |
| RAID Adapter - PCIe 4.0 (SAS/SATA; RAID 940 adapters support <a href="#">Tri-Mode</a> ) |              |                                                           |                    |                                    |         |       |          |
| 4Y37A93012                                                                              | C0TU         | ThinkSystem RAID 545-8i PCIe Gen4 12Gb Adapter            | TCE                | 0, 1, 10                           | 3       | 1-10  | No       |
| 4Y37A09728                                                                              | B8NY         | ThinkSystem RAID 940-8i 4GB Flash PCIe Gen4 12Gb Adapter  | TCE                | 0, 1, 10, 5, 50, 6, 60             | 3       | 1-10  | Included |
| 4Y37A78600                                                                              | BM35         | ThinkSystem RAID 940-16i 4GB Flash PCIe Gen4 12Gb Adapter | TCE                | 0, 1, 10, 5, 50, 6, 60             | 2       | 1-10  | Included |

| Part number                                                                                                                                             | Feature code | Description                                                                | Top Choice Express | RAID level             | Max Qty | Slots | Supercap |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------|--------------------|------------------------|---------|-------|----------|
| 4Y37A09730                                                                                                                                              | B8NZ         | ThinkSystem RAID 940-16i 8GB Flash PCIe Gen4 12Gb Adapter                  | TCE                | 0, 1, 10, 5, 50, 6, 60 | 2       | 1-10  | Included |
| 4Y37A09735                                                                                                                                              | B8P0         | ThinkSystem RAID 940-16i 8GB Flash PCIe Gen4 12Gb Internal Adapter         | Not TCE            | 0, 1, 10, 5, 50, 6, 60 | 1       | CFF*  | Included |
| RAID Adapter - PCIe 5.0 (NVMe U.2 only; no SAS/SATA)                                                                                                    |              |                                                                            |                    |                        |         |       |          |
| 4Y37B11710                                                                                                                                              | CECC         | ThinkSystem RAID 960W-32i NVMe PCIe Gen5 Adapter (16 drives max)           | Not TCE            | 0, 1, 10, 5, 50, 6, 60 | 1       | 3-8   | Included |
| NVMe (Tri-Mode support) ( <b>deprecated</b> - use SAS/SATA adapters B8NY, BM35, B8NZ, or B8P0 combined with CH5Z; see <a href="#">Tri-Mode</a> section) |              |                                                                            |                    |                        |         |       |          |
| 4Y37A09728†                                                                                                                                             | BGM1         | ThinkSystem RAID 940-8i 4GB Flash PCIe Gen4 12Gb Adapter for U.3           | Not TCE            | 0, 1, 10, 5, 50, 6, 60 | 3       | 1-10  | Included |
| 4Y37A78600†                                                                                                                                             | BM36         | ThinkSystem RAID 940-16i 4GB Flash PCIe Gen4 12Gb Adapter for U.3          | Not TCE            | 0, 1, 10, 5, 50, 6, 60 | 1       | 1-10  | Included |
| 4Y37A09730†                                                                                                                                             | BDY4         | ThinkSystem RAID 940-16i 8GB Flash PCIe Gen4 12Gb Adapter for U.3          | Not TCE            | 0, 1, 10, 5, 50, 6, 60 | 1       | 1-10  | Included |
| 4Y37A09735†                                                                                                                                             | BGM2         | ThinkSystem RAID 940-16i 8GB Flash PCIe Gen4 12Gb Internal Adapter for U.3 | Not TCE            | 0, 1, 10, 5, 50, 6, 60 | 1       | CFF*  | Included |

\* CFF (custom form factor) is the name for the cabled internal adapter form factor that doesn't occupy a standard PCIe slot

† with Tri-Mode enabled, the adapter supports PCIe 4.0 x1 connectivity to NVMe drives with U.3 interface  
Configuration notes:

- **Supercap support limits the number of RAID adapters installable:** The table lists whether the adapter includes a power module (supercap) to power the flash memory. The server supports between 1 and 3 supercaps depending on the server configuration as described in the [RAID flash power module \(supercap\) support](#) section. The number of supercaps supported also determines the maximum number of RAID adapters with flash that can be installed in the server.
- **Field upgrades:** If you are adding a RAID adapter with supercap to the server as a field upgrade, you may need a supercap holder as described in the [RAID flash power module \(supercap\) support](#) section.
- **Tri-Mode support and U.3 drives:** The use of a RAID adapter with Tri-Mode enabled requires U.3 drives (U.2 drives are not supported). Also, in the DCSC configurator, 7500 PRO and 7500 MAX U.3 drives can only be selected when a Tri-Mode RAID adapter is also selected.

The onboard NVMe support has the following features:

- Support integrated into the Intel processor
- Each drive has PCIe 5.0 x4 host interface
- Supports JBOD
- Supports RAID using Intel VROC

For specifications about the RAID adapters and HBAs supported by the SR650 V4, see the ThinkSystem RAID Adapter and HBA Comparison, available from:

<https://lenovopress.com/lp1288-lenovo-thinksystem-raid-adapter-and-hba-reference#sr650-v4-support=SR650%2520V4>

For details about these adapters, see the relevant product guide:

- SAS HBAs: <https://lenovopress.com/servers/options/hba>
- RAID adapters: <https://lenovopress.com/servers/options/raid>

## Tri-Mode support - RAID 940 adapters

The RAID 940 adapters support NVMe through a feature named Tri-Mode support (or Trimode support). This feature enables the use of NVMe U.3 drives at the same time as SAS and SATA drives. Tri-Mode requires an AnyBay backplane. Cabling of the controller to the backplanes is the same as with SAS/SATA drives, and the NVMe drives are connected via a PCIe x1 link to the controller.

NVMe drives connected using Tri-Mode support provide better performance than SAS or SATA drives: A SATA SSD has a data rate of 6Gbps, a SAS SSD has a data rate of 12Gbps, whereas an NVMe U.3 Gen 4 SSD with a PCIe x1 link will have a data rate of 16Gbps. NVMe drives typically also have lower latency and higher IOPS compared to SAS and SATA drives. Tri-Mode is supported with U.3 NVMe drives and requires an AnyBay backplane.

**Tri-Mode requires U.3 drives:** Only NVMe drives with a U.3 interface are supported. U.2 drives are not supported. See the [Internal drive options](#) section for the U.3 drives supported by the server.

**New as of July 2026:** To enable Tri-Mode in CTO orders of the SR650 V4, use the following feature code:

Table 54. Tri-Mode Enablement

| Feature code | Description         |
|--------------|---------------------|
| CH5Z         | Tri-Mode Enablement |

The process to configure Tri-Mode for a CTO order in DCSC is as follows:

1. In the **Storage** tab, select one of the 940-8i or 940-16i controllers from the **SAS/SATA RAID Controller** group
2. In the **Internal Storage** tab, select select **Tri-Mode Enablement** (feature CH5Z)
3. You can then select either NVMe or SATA or SATA drives or any combination of these drives, up to the number supported by the server and the backplanes selected.

### Tips:

- Feature CH5Z is not selectable until one of the 940-8i or 940-16i adapters in **SAS/SATA RAID Controller** group is selected.
- Feature CH5Z is not selectable if **Non RAID NVMe** (BC4V) is selected.
- The existing Tri-Mode "for U.3" adapter feature codes (features BGM1, BGM0, BM36, BDY4) in the **Tri-Mode Controller** group are for existing orders only and are being phased out (deprecated)

## Intel VROC onboard RAID

Intel VROC (Virtual RAID on CPU) is a feature of the Intel processor that enables Integrated RAID support.

On the SR650 V4, Intel VROC provides RAID functions for the onboard NVMe controller (Intel VROC NVMe RAID).

VROC NVMe RAID offers RAID support for any NVMe drives directly connected to the ports on the server's system board or via adapters such as NVMe retimers or NVMe switch adapters. On the SR650 V4, RAID levels implemented are based on the VROC feature selected as indicated in the following table. RAID 1 is limited to 2 drives per array, and RAID 10 is limited to 4 drives per array. Hot-spare functionality is also supported.

**Performance tip:** For best performance with VROC NVMe RAID, the drives in an array should all be connected to the same processor. Spanning processors is possible however performance will be unpredictable and should be evaluated based on your workload.

The SR650 V4 supports the VROC NVMe RAID offerings listed in the following table.

Table 55. Intel VROC NVMe RAID ordering information and feature support

| Part number | Feature code | Description                         | Top Choice Express | Intel NVMe SSDs | Non-Intel NVMe SSDs | RAID 0 | RAID 1 | RAID 10 | RAID 5 |
|-------------|--------------|-------------------------------------|--------------------|-----------------|---------------------|--------|--------|---------|--------|
| 4L47A92670  | BZ4W         | Intel VROC RAID1 Only               | TCE                | Yes             | Yes                 | No     | Yes    | No      | No     |
| 4L47A83669  | BR9B         | Intel VROC (VMD NVMe RAID) Standard | TCE                | Yes             | Yes                 | Yes    | Yes    | Yes     | No     |
| 4L47A39164  | B96G         | Intel VROC (VMD NVMe RAID) Premium  | TCE                | Yes             | Yes                 | Yes    | Yes    | Yes     | Yes    |

Configuration notes:

- If a feature code is ordered in a CTO build, the VROC functionality is enabled in the factory. For field upgrades, order a part number and it will be fulfilled as a Feature on Demand (FoD) license which can then be activated via the XCC management processor user interface.
- SED functionality is not supported with VROC NVMe RAID

**Virtualization support:** Virtualization support for Intel VROC is as follows:

- **VROC (VMD) NVMe RAID:** VROC (VMD) NVMe RAID is supported by ESXi, KVM, Xen, and Hyper-V. ESXi support is limited to RAID 1 only; other RAID levels are not supported. Windows and Linux OSes support VROC RAID NVMe, both for host boot functions and for guest OS function, and RAID-0, 1, 5, and 10 are supported. On ESXi, VROC is supported with both boot and data drives.

## Internal drive options

**VMware vSAN and Microsoft S2D configurations:** Vendor Agnostic (VA) drives are based on drives from multiple vendors, and not all vendor variants are VMware vSAN or Microsoft S2D certified. Additionally, ISVs require a single drive vendor per host and cluster to ensure optimal performance; a requirement that VA drives may not consistently meet. To ensure certification compliance, hardware uniformity, and alignment with VMware and Microsoft best practices, use ThinkAgile VX or ThinkAgile FX for all vSAN configurations and ThinkAgile MX for all S2D configurations, instead of general purpose ThinkSystem servers.

The following tables list the drive options for internal storage of the server.

2.5-inch hot-swap drives:

- [2.5-inch hot-swap 12 Gb SAS HDDs](#)
- [2.5-inch hot-swap 24 Gb SAS SSDs](#)
- [2.5-inch hot-swap 6 Gb SATA SSDs](#)
- [2.5-inch hot-swap PCIe 5.0 NVMe SSDs](#)
- [2.5-inch hot-swap PCIe 4.0 NVMe SSDs](#)

EDSFF hot-swap drives:

- [E3.S 1T EDSFF hot-swap PCIe 5.0 NVMe SSDs](#)

3.5-inch hot-swap drives:

- [3.5-inch hot-swap 12 Gb SAS HDDs](#)
- [3.5-inch hot-swap 6 Gb SATA HDDs](#)
- [3.5-inch hot-swap 24 Gb SAS SSDs](#)
- [3.5-inch hot-swap 6 Gb SATA SSDs](#)
- [3.5-inch hot-swap PCIe 5.0 NVMe SSDs](#)
- [3.5-inch hot-swap PCIe 4.0 NVMe SSDs](#)

M.2 drives for B550 and B350 adapters:

- [M.2 PCIe 4.0 NVMe hot-swap drives \(hot-swap M.2 adapters\)](#)
- [M.2 PCIe 4.0 NVMe non-hot-swap drives \(internal M.2 adapters\)](#)

M.2 drives (for M.2 adapters except B350/B550):

- [M.2 SATA non-hot-swap drives](#)

**M.2 drive support:** The use of M.2 drives requires an additional adapter as described in the [M.2 drives](#) subsection.

**SED support:** The tables include a column to indicate which drives support SED encryption. The encryption functionality can be disabled if needed. Note: Not all SED-enabled drives have "SED" in the description.

Table 56. 2.5-inch hot-swap 12 Gb SAS HDDs

| Part number                                   | Feature code | Description                                              | Top Choice Express | SED support | Max Qty |
|-----------------------------------------------|--------------|----------------------------------------------------------|--------------------|-------------|---------|
| <b>2.5-inch hot-swap HDDs - 12 Gb SAS 10K</b> |              |                                                          |                    |             |         |
| 7XB7A00025                                    | AULZ         | ThinkSystem 2.5" 600GB 10K SAS 12Gb Hot Swap 512n HDD    | Not TCE            | No          | 40      |
| 7XB7A00027                                    | AUM1         | ThinkSystem 2.5" 1.2TB 10K SAS 12Gb Hot Swap 512n HDD    | TCE                | No          | 40      |
| 4XB7A83970                                    | BRG7         | ThinkSystem 2.5" 2.4TB 10K SAS 12Gb Hot Swap 512e HDD v2 | TCE                | No          | 40      |

Table 57. 2.5-inch hot-swap 24 Gb SAS SSDs

| Part number                                                      | Feature code | Description | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------|--------------|-------------|--------------------|-------------|---------|
| <b>2.5-inch hot-swap SSDs - 24 Gb SAS - Mixed Use (3-5 DWPD)</b> |              |             |                    |             |         |

| Part number                                                             | Feature code | Description                                                | Top Choice Express | SED support | Max Qty |
|-------------------------------------------------------------------------|--------------|------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7B07612                                                              | CABL         | ThinkSystem 2.5" VA 800GB Mixed Use SAS 24Gb HS SSD        | TCE                | Support     | 40      |
| 4XB7B07613                                                              | CABR         | ThinkSystem 2.5" VA 1.6TB Mixed Use SAS 24Gb HS SSD        | TCE                | Support     | 40      |
| 4XB7B07614                                                              | CABQ         | ThinkSystem 2.5" VA 3.2TB Mixed Use SAS 24Gb HS SSD        | TCE                | Support     | 40      |
| 4XB7B07615                                                              | CABK         | ThinkSystem 2.5" VA 6.4TB Mixed Use SAS 24Gb HS SSD        | TCE ⓘ              | Support     | 40      |
| <b>2.5-inch hot-swap SSDs - 24 Gb SAS - Read Intensive (&lt;3 DWPD)</b> |              |                                                            |                    |             |         |
| 4XB7B07600                                                              | CABS         | ThinkSystem 2.5" VA 960GB Read Intensive SAS 24Gb HS SSD   | TCE                | Support     | 40      |
| 4XB7B07601                                                              | CABV         | ThinkSystem 2.5" VA 1.92TB Read Intensive SAS 24Gb HS SSD  | TCE                | Support     | 40      |
| 4XB7B07602                                                              | CABT         | ThinkSystem 2.5" VA 3.84TB Read Intensive SAS 24Gb HS SSD  | TCE                | Support     | 40      |
| 4XB7B07603                                                              | CABY         | ThinkSystem 2.5" VA 7.68TB Read Intensive SAS 24Gb HS SSD  | TCE                | Support     | 40      |
| 4XB7B07604                                                              | CABX         | ThinkSystem 2.5" VA 15.36TB Read Intensive SAS 24Gb HS SSD | TCE                | Support     | 40      |
| 4XB7B07605                                                              | CABW         | ThinkSystem 2.5" VA 30.72TB Read Intensive SAS 24Gb HS SSD | TCE                | Support     | 40      |

Table 58. 2.5-inch hot-swap 6 Gb SATA SSDs

| Part number                                                             | Feature code | Description                                                  | Top Choice Express | SED support | Max Qty |
|-------------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------------------|-------------|---------|
| <b>2.5-inch hot-swap SSDs - 6 Gb SATA - Mixed Use (3-5 DWPD)</b>        |              |                                                              |                    |             |         |
| 4XB7B09972                                                              | CBV9         | ThinkSystem 2.5" VA 480GB Mixed Use SATA 6Gb HS SSD SED      | TCE ⓘ              | Support     | 40      |
| 4XB7B09973                                                              | CBV8         | ThinkSystem 2.5" VA 960GB Mixed Use SATA 6Gb HS SSD SED      | TCE ⓘ              | Support     | 40      |
| 4XB7B09974                                                              | CBV7         | ThinkSystem 2.5" VA 1.92TB Mixed Use SATA 6Gb HS SSD SED     | Not TCE            | Support     | 40      |
| 4XB7B09975                                                              | CBV6         | ThinkSystem 2.5" VA 3.84TB Mixed Use SATA 6Gb HS SSD SED     | TCE                | Support     | 40      |
| 4XB7A93091                                                              | C1X3         | ThinkSystem 2.5" PM897a 480GB Mixed Use SATA 6Gb HS SSD      | Not TCE            | Support     | 40      |
| 4XB7A93092                                                              | C1X4         | ThinkSystem 2.5" PM897a 960GB Mixed Use SATA 6Gb HS SSD      | Not TCE            | Support     | 40      |
| 4XB7A90884                                                              | BYM2         | ThinkSystem 2.5" VA 480GB Mixed Use SATA 6Gb HS SSD v2       | TCE                | No          | 40      |
| 4XB7A90885                                                              | BYM4         | ThinkSystem 2.5" VA 960GB Mixed Use SATA 6Gb HS SSD v2       | TCE                | No          | 40      |
| 4XB7A90886                                                              | BYM5         | ThinkSystem 2.5" VA 1.92TB Mixed Use SATA 6Gb HS SSD v2      | Not TCE            | No          | 40      |
| 4XB7A90887                                                              | BYM6         | ThinkSystem 2.5" VA 3.84TB Mixed Use SATA 6Gb HS SSD v2      | TCE                | No          | 40      |
| <b>2.5-inch hot-swap SSDs - 6 Gb SATA - Read Intensive (&lt;3 DWPD)</b> |              |                                                              |                    |             |         |
| 4XB7B09967                                                              | CBVE         | ThinkSystem 2.5" VA 480GB Read Intensive SATA 6Gb HS SSD SED | TCE ⓘ              | Support     | 40      |

| Part number | Feature code | Description                                                   | Top Choice Express | SED support | Max Qty |
|-------------|--------------|---------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7B09968  | CBVD         | ThinkSystem 2.5" VA 960GB Read Intensive SATA 6Gb HS SSD SED  | TCE                | Support     | 40      |
| 4XB7B09969  | CBVC         | ThinkSystem 2.5" VA 1.92TB Read Intensive SATA 6Gb HS SSD SED | TCE                | Support     | 40      |
| 4XB7B09970  | CBVB         | ThinkSystem 2.5" VA 3.84TB Read Intensive SATA 6Gb HS SSD SED | TCE                | Support     | 40      |
| 4XB7B09971  | CBVA         | ThinkSystem 2.5" VA 7.68TB Read Intensive SATA 6Gb HS SSD SED | TCE ⓘ              | Support     | 40      |
| 4XB7A90873  | BYLR         | ThinkSystem 2.5" VA 480GB Read Intensive SATA 6Gb HS SSD v2   | TCE                | No          | 40      |
| 4XB7A90874  | BYLS         | ThinkSystem 2.5" VA 960GB Read Intensive SATA 6Gb HS SSD v2   | TCE                | No          | 40      |
| 4XB7A90875  | BYLT         | ThinkSystem 2.5" VA 1.92TB Read Intensive SATA 6Gb HS SSD v2  | TCE                | No          | 40      |
| 4XB7A90876  | BYLU         | ThinkSystem 2.5" VA 3.84TB Read Intensive SATA 6Gb HS SSD v2  | TCE                | No          | 40      |
| 4XB7A90877  | BYLV         | ThinkSystem 2.5" VA 7.68TB Read Intensive SATA 6Gb HS SSD v2  | TCE                | No          | 40      |

Table 59. 2.5-inch hot-swap PCIe 5.0 NVMe SSDs

| Part number                                                            | Feature code | Description                                                             | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------|--------------------|-------------|---------|
| <b>2.5-inch SSDs - U.2 PCIe 5.0 NVMe - Mixed Use (3-5 DWPD)</b>        |              |                                                                         |                    |             |         |
| 4XB7B14615                                                             | CGFZ         | ThinkSystem 2.5" U.2 CD9P 1.6TB Mixed Use NVMe PCIe 5.0 x4 HS SSD       | Not TCE            | Support     | 36      |
| 4XB7B14616                                                             | CGFY         | ThinkSystem 2.5" U.2 CD9P 3.2TB Mixed Use NVMe PCIe 5.0 x4 HS SSD       | Not TCE            | Support     | 36      |
| 4XB7B14617                                                             | CGFX         | ThinkSystem 2.5" U.2 CD9P 6.4TB Mixed Use NVMe PCIe 5.0 x4 HS SSD       | Not TCE            | Support     | 36      |
| 4XB7B14618                                                             | CGFW         | ThinkSystem 2.5" U.2 CD9P 12.8TB Mixed Use NVMe PCIe 5.0 x4 HS SSD      | Not TCE            | Support     | 36      |
| 4XB7A93097                                                             | C1WM         | ThinkSystem 2.5" U.2 PM9D5a 800GB Mixed Use NVMe PCIe 5.0 x4 HS SSD     | Not TCE            | Support     | 36      |
| 4XB7A93127                                                             | C0ZR         | ThinkSystem 2.5" U.2 VA 1.6TB Mixed Use NVMe PCIe 5.0 x4 HS SSD         | Not TCE            | Support     | 36      |
| 4XB7A93128                                                             | C0ZQ         | ThinkSystem 2.5" U.2 VA 3.2TB Mixed Use NVMe PCIe 5.0 x4 HS SSD         | TCE                | Support     | 36      |
| 4XB7A93129                                                             | C0ZP         | ThinkSystem 2.5" U.2 VA 6.4TB Mixed Use NVMe PCIe 5.0 x4 HS SSD         | TCE                | Support     | 36      |
| 4XB7A93130                                                             | C0ZN         | ThinkSystem 2.5" U.2 VA 12.8TB Mixed Use NVMe PCIe 5.0 x4 HS SSD        | Not TCE            | Support     | 36      |
| <b>2.5-inch SSDs - U.3 PCIe 5.0 NVMe - Mixed Use (3-5 DWPD)</b>        |              |                                                                         |                    |             |         |
| 4XB7A94637                                                             | C4D4         | ThinkSystem 2.5" U.3 PM1745 1.6TB Mixed Use NVMe PCIe 5.0 x4 HS SSD     | Not TCE            | Support     | 36      |
| <b>2.5-inch SSDs - U.2 PCIe 5.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                         |                    |             |         |
| 4XB7B14455                                                             | CGP9         | ThinkSystem 2.5" U.2 CD9P 1.92TB Read Intensive NVMe PCIe 5.0 x4 HS SSD | Not TCE            | Support     | 36      |

| Part number                                                            | Feature code | Description                                                               | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7B14456                                                             | CGP8         | ThinkSystem 2.5" U.2 CD9P 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD   | Not TCE            | Support     | 36      |
| 4XB7B14457                                                             | CGP7         | ThinkSystem 2.5" U.2 CD9P 7.68TB Read Intensive NVMe PCIe 5.0 x4 HS SSD   | Not TCE            | Support     | 36      |
| 4XB7B14458                                                             | CGP6         | ThinkSystem 2.5" U.2 CD9P 15.36TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B14459                                                             | CGPA         | ThinkSystem 2.5" U.2 CD9P 30.72TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B14460                                                             | CGP5         | ThinkSystem 2.5" U.2 CD9P 61.44TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B10396                                                             | CFD6         | ThinkSystem 2.5" U.2 SN861 1.92TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B10397                                                             | CFD5         | ThinkSystem 2.5" U.2 SN861 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B10398                                                             | CFD4         | ThinkSystem 2.5" U.2 SN861 7.68TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B10399                                                             | CFD3         | ThinkSystem 2.5" U.2 SN861 15.36TB Read Intensive NVMe PCIe 5.0 x4 HS SSD | Not TCE            | Support     | 36      |
| 4XB7B08330                                                             | CBSW         | ThinkSystem 2.5" U.2 6550 30.72TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7B08331                                                             | CBSX         | ThinkSystem 2.5" U.2 6550 61.44TB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7A93066                                                             | C0GK         | ThinkSystem 2.5" U.2 PM9D3a 960GB Read Intensive NVMe PCIe 5.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7A93122                                                             | C0ZV         | ThinkSystem 2.5" U.2 VA 1.92TB Read Intensive NVMe PCIe 5.0 x4 HS SSD     | Not TCE            | Support     | 36      |
| 4XB7A93123                                                             | C0ZU         | ThinkSystem 2.5" U.2 VA 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD     | TCE                | Support     | 36      |
| 4XB7A93124                                                             | C0ZT         | ThinkSystem 2.5" U.2 VA 7.68TB Read Intensive NVMe PCIe 5.0 x4 HS SSD     | TCE                | Support     | 36      |
| 4XB7A93125                                                             | C0ZS         | ThinkSystem 2.5" U.2 VA 15.36TB Read Intensive NVMe PCIe 5.0 x4 HS SSD    | Not TCE            | Support     | 36      |
| 4XB7A93126                                                             | C1W7         | ThinkSystem 2.5" U.2 VA 30.72TB Read Intensive NVMe PCIe 5.0 x4 HS SSD    | Not TCE            | Support     | 36      |
| <b>2.5-inch SSDs - U.3 PCIe 5.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                           |                    |             |         |
| 4XB7A82367                                                             | BTQ0         | ThinkSystem 2.5" U.3 PM1743 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD | Not TCE            | Support     | 36      |

Table 60. 2.5-inch hot-swap PCIe 4.0 NVMe SSDs

| Part number                                                     | Feature code | Description                                                                 | Top Choice Express | SED support | Max Qty |
|-----------------------------------------------------------------|--------------|-----------------------------------------------------------------------------|--------------------|-------------|---------|
| <b>2.5-inch SSDs - U.2 PCIe 4.0 NVMe - Mixed Use (3-5 DWPD)</b> |              |                                                                             |                    |             |         |
| 4XB7B01880                                                      | C6M3         | ThinkSystem 2.5" U.2 Solidigm P5620 3.2TB Mixed Use NVMe PCIe 4.0 x4 HS SSD | Not TCE            | Support     | 36      |
| <b>2.5-inch SSDs - U.3 PCIe 4.0 NVMe - Mixed Use (3-5 DWPD)</b> |              |                                                                             |                    |             |         |
| 4XB7A95054                                                      | C2BG         | ThinkSystem 2.5" U.3 7500 MAX 800GB Mixed Use NVMe PCIe 4.0 x4 HS SSD       | Not TCE            | Support     | 36      |

| Part number                                                            | Feature code | Description                                                                  | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7A95055                                                             | C2BV         | ThinkSystem 2.5" U.3 7500 MAX 1.6TB Mixed Use NVMe PCIe 4.0 x4 HS SSD        | Not TCE            | Support     | 36      |
| 4XB7A95056                                                             | C2BW         | ThinkSystem 2.5" U.3 7500 MAX 3.2TB Mixed Use NVMe PCIe 4.0 x4 HS SSD        | Not TCE            | Support     | 36      |
| 4XB7A95057                                                             | C2BF         | ThinkSystem 2.5" U.3 7500 MAX 6.4TB Mixed Use NVMe PCIe 4.0 x4 HS SSD        | Not TCE            | Support     | 36      |
| 4XB7A95058                                                             | C2BX         | ThinkSystem 2.5" U.3 7500 MAX 12.8TB Mixed Use NVMe PCIe 4.0 x4 HS SSD       | Not TCE            | Support     | 36      |
| <b>2.5-inch SSDs - U.3 PCIe 4.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                              |                    |             |         |
| 4XB7A95049                                                             | C2BY         | ThinkSystem 2.5" U.3 7500 PRO 960GB Read Intensive NVMe PCIe 4.0 x4 HS SSD   | Not TCE            | Support     | 36      |
| 4XB7A95050                                                             | C2BR         | ThinkSystem 2.5" U.3 7500 PRO 1.92TB Read Intensive NVMe PCIe 4.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7A95051                                                             | C2BS         | ThinkSystem 2.5" U.3 7500 PRO 3.84TB Read Intensive NVMe PCIe 4.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7A95052                                                             | C2BT         | ThinkSystem 2.5" U.3 7500 PRO 7.68TB Read Intensive NVMe PCIe 4.0 x4 HS SSD  | Not TCE            | Support     | 36      |
| 4XB7A95053                                                             | C2BU         | ThinkSystem 2.5" U.3 7500 PRO 15.36TB Read Intensive NVMe PCIe 4.0 x4 HS SSD | Not TCE            | Support     | 36      |

**U.3 drives:** In the DCSC configurator, 7500 PRO and 7500 MAX U.3 drives can only be selected when a Tri-Mode controller is also selected. See the [Controllers for internal storage](#) section.

Table 61. E3.S 1T EDSFF hot-swap PCIe 5.0 NVMe SSDs

| Part number                                                                | Feature code | Description                                                             | Top Choice Express | SED support | Max Qty |
|----------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------|--------------------|-------------|---------|
| <b>E3.S 1T hot-swap SSDs - PCIe 5.0 NVMe - Mixed Use (3-5 DWPD)</b>        |              |                                                                         |                    |             |         |
| 4XB7A93105                                                                 | C1WW         | ThinkSystem E3.S PM9D5a 1.6TB Mixed Use NVMe PCIe 5.0 x4 HS SSD         | Not TCE            | Support     | 32      |
| 4XB7A93107                                                                 | C1WY         | ThinkSystem E3.S PM9D5a 6.4TB Mixed Use NVMe PCIe 5.0 x4 HS SSD         | Not TCE            | Support     | 32      |
| 4XB7A93108                                                                 | C1WZ         | ThinkSystem E3.S PM9D5a 12.8TB Mixed Use NVMe PCIe 5.0 x4 HS SSD        | Not TCE            | Support     | 32      |
| 4XB7A93136                                                                 | C1WD         | ThinkSystem E3.S 1T VA 1.6TB Mixed Use NVMe PCIe 5.0 x4 HS SSD v2       | Not TCE            | Support     | 32      |
| 4XB7A93137                                                                 | C1WE         | ThinkSystem E3.S 1T VA 3.2TB Mixed Use NVMe PCIe 5.0 x4 HS SSD v2       | Not TCE            | Support     | 32      |
| 4XB7A93138                                                                 | C1WF         | ThinkSystem E3.S 1T VA 6.4TB Mixed Use NVMe PCIe 5.0 x4 HS SSD v2       | Not TCE            | Support     | 32      |
| 4XB7A93139                                                                 | C1WG         | ThinkSystem E3.S 1T VA 12.8TB Mixed Use NVMe PCIe 5.0 x4 HS SSD v2      | Not TCE            | Support     | 32      |
| <b>E3.S 1T hot-swap SSDs - PCIe 5.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                         |                    |             |         |
| 4XB7A93131                                                                 | C1W8         | ThinkSystem E3.S 1T VA 1.92TB Read Intensive NVMe PCIe 5.0 x4 HS SSD v2 | Not TCE            | Support     | 32      |
| 4XB7A93132                                                                 | C1W9         | ThinkSystem E3.S 1T VA 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD v2 | Not TCE            | Support     | 32      |

| Part number | Feature code | Description                                                                | Top Choice Express | SED support | Max Qty |
|-------------|--------------|----------------------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7A93133  | C1WA         | ThinkSystem E3.S 1T VA 7.68TB Read Intensive NVMe PCIe 5.0 x4 HS SSD v2    | Not TCE            | Support     | 32      |
| 4XB7A93134  | C1WB         | ThinkSystem E3.S 1T VA 15.36TB Read Intensive NVMe PCIe 5.0 x4 HS SSD v2   | Not TCE            | Support     | 32      |
| 4XB7B11623  | CCZR         | ThinkSystem E3.S 1T 6550 30.72TB Read Intensive NVMe PCIe 5.0 x4 HS SSD v2 | Not TCE            | Support     | 32      |
| 4XB7B11624  | CCZQ         | ThinkSystem E3.S 1T 6550 61.44TB Read Intensive NVMe PCIe 5.0 x4 HS SSD v2 | Not TCE            | Support     | 32      |
| 4XB7A93078  | C1AF         | ThinkSystem E3.S PM9D3a 1.92TB Read Intensive NVMe PCIe 5.0 x4 HS SSD      | Not TCE            | Support     | 32      |
| 4XB7A93079  | C1AE         | ThinkSystem E3.S PM9D3a 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD      | Not TCE            | Support     | 32      |

Table 62. 3.5-inch hot-swap 12 Gb SAS HDDs

| Part number                                      | Feature code | Description                                               | Top Choice Express | SED support | Max Qty |
|--------------------------------------------------|--------------|-----------------------------------------------------------|--------------------|-------------|---------|
| <b>3.5-inch hot-swap HDDs - 12 Gb NL SAS</b>     |              |                                                           |                    |             |         |
| 4XB7B01233                                       | C5WY         | ThinkSystem 3.5" 2TB 7.2K SAS 12Gb Hot Swap 512e HDD v2   | Not TCE            | Support     | 16      |
| 4XB7B01235                                       | C5X9         | ThinkSystem 3.5" 4TB 7.2K SAS 12Gb Hot Swap 512e HDD v2   | TCE                | Support     | 16      |
| 4XB7B01239                                       | C5XD         | ThinkSystem 3.5" 8TB 7.2K SAS 12Gb Hot Swap 512e HDD v2   | Not TCE            | Support     | 16      |
| 4XB7A93788                                       | C4DA         | ThinkSystem 3.5" 12TB 7.2K SAS 12Gb Hot Swap 512e HDD v2  | TCE                | Support     | 16      |
| 4XB7A93786                                       | C4D8         | ThinkSystem 3.5" 16TB 7.2K SAS 12Gb Hot Swap 512e HDD v2  | Not TCE            | Support     | 16      |
| 4XB7A93784                                       | C4D6         | ThinkSystem 3.5" 20TB 7.2K SAS 12Gb Hot Swap 512e HDD v2  | Not TCE            | Support     | 16      |
| 4XB7A93007                                       | C2BH         | ThinkSystem 3.5" 24TB 7.2K SAS 12Gb Hot Swap 512e HDD     | TCE                | Support     | 16      |
| <b>3.5-inch hot-swap SED HDDs - 12 Gb NL SAS</b> |              |                                                           |                    |             |         |
| 7XB7A00066                                       | B0YQ         | ThinkSystem 3.5" 8TB 7.2K SAS 12Gb Hot Swap 512e HDD FIPS | Not TCE            | Support     | 16      |

Table 63. 3.5-inch hot-swap 6 Gb SATA HDDs

| Part number                                  | Feature code | Description                                             | Top Choice Express | SED support | Max Qty |
|----------------------------------------------|--------------|---------------------------------------------------------|--------------------|-------------|---------|
| <b>3.5-inch hot-swap HDDs - 6 Gb NL SATA</b> |              |                                                         |                    |             |         |
| 4XB7B01234                                   | C5X8         | ThinkSystem 3.5" 2TB 7.2K SATA 6Gb Hot Swap 512e HDD v2 | TCE                | Support     | 16      |
| 4XB7B01236                                   | C5XA         | ThinkSystem 3.5" 4TB 7.2K SATA 6Gb Hot Swap 512e HDD v2 | TCE                | Support     | 16      |
| 4XB7B01240                                   | C5XE         | ThinkSystem 3.5" 8TB 7.2K SATA 6Gb Hot Swap 512e HDD v2 | Not TCE            | Support     | 16      |

| Part number | Feature code | Description                                              | Top Choice Express | SED support | Max Qty |
|-------------|--------------|----------------------------------------------------------|--------------------|-------------|---------|
| 4XB7A93787  | C4D9         | ThinkSystem 3.5" 12TB 7.2K SATA 6Gb Hot Swap 512e HDD v2 | Not TCE            | Support     | 16      |
| 4XB7A93785  | C4D7         | ThinkSystem 3.5" 16TB 7.2K SATA 6Gb Hot Swap 512e HDD v2 | TCE                | Support     | 16      |
| 4XB7A93783  | C4D5         | ThinkSystem 3.5" 20TB 7.2K SATA 6Gb Hot Swap 512e HDD v2 | TCE                | Support     | 16      |
| 4XB7A93008  | C2BJ         | ThinkSystem 3.5" 24TB 7.2K SATA 6Gb Hot Swap 512e HDD    | Not TCE            | Support     | 16      |

Table 64. 3.5-inch hot-swap 24 Gb SAS SSDs

| Part number                                                                      | Feature code | Description                                                  | Top Choice Express | SED support | Max Qty |
|----------------------------------------------------------------------------------|--------------|--------------------------------------------------------------|--------------------|-------------|---------|
| <b>3.5-inch hot-swap SSDs - 24 Gb SAS - Mixed Use (3-5 DWPD)</b>                 |              |                                                              |                    |             |         |
| 4XB7B07617                                                                       | CABP         | ThinkSystem 3.5" VA 1.6TB Mixed Use SAS 24Gb HS SSD          | TCE                | Support     | 16      |
| 4XB7B07618                                                                       | CABN         | ThinkSystem 3.5" VA 3.2TB Mixed Use SAS 24Gb HS SSD          | TCE ⓘ              | Support     | 16      |
| 4XB7B07619                                                                       | CAC8         | ThinkSystem 3.5" VA 6.4TB Mixed Use SAS 24Gb HS SSD          | TCE                | Support     | 16      |
| 4XB7A80344                                                                       | BNW7         | ThinkSystem 3.5" PM1655 800GB Mixed Use SAS 24Gb HS SSD      | TCE                | Support     | 16      |
| <b>3.5-inch hot-swap SSDs - 24 Gb SAS - Read Intensive/Capacity (&lt;3 DWPD)</b> |              |                                                              |                    |             |         |
| 4XB7B07607                                                                       | CAFN         | ThinkSystem 3.5" VA 1.92TB Read Intensive SAS 24Gb HS SSD    | TCE                | Support     | 16      |
| 4XB7B07608                                                                       | CAFP         | ThinkSystem 3.5" VA 3.84TB Read Intensive SAS 24Gb HS SSD    | TCE                | Support     | 16      |
| 4XB7B07609                                                                       | CAFM         | ThinkSystem 3.5" VA 7.68TB Read Intensive SAS 24Gb HS SSD    | TCE                | Support     | 16      |
| 4XB7B07610                                                                       | CAFR         | ThinkSystem 3.5" VA 15.36TB Read Intensive SAS 24Gb HS SSD   | TCE                | Support     | 16      |
| 4XB7A80324                                                                       | BNWD         | ThinkSystem 3.5" PM1653 960GB Read Intensive SAS 24Gb HS SSD | Not TCE            | Support     | 16      |

Table 65. 3.5-inch hot-swap 6 Gb SATA SSDs

| Part number                                                      | Feature code | Description                                              | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------|--------------|----------------------------------------------------------|--------------------|-------------|---------|
| <b>3.5-inch hot-swap SSDs - 6 Gb SATA - Mixed Use (3-5 DWPD)</b> |              |                                                          |                    |             |         |
| 4XB7B11652                                                       | CCZK         | ThinkSystem 3.5" VA 480GB Mixed Use SATA 6Gb HS SSD SED  | Not TCE            | Support     | 16      |
| 4XB7B11653                                                       | CCZJ         | ThinkSystem 3.5" VA 960GB Mixed Use SATA 6Gb HS SSD SED  | Not TCE            | Support     | 16      |
| 4XB7B11655                                                       | CCZG         | ThinkSystem 3.5" VA 3.84TB Mixed Use SATA 6Gb HS SSD SED | Not TCE            | Support     | 16      |
| 4XB7A90888                                                       | BYM3         | ThinkSystem 3.5" VA 480GB Mixed Use SATA 6Gb HS SSD v2   | TCE                | No          | 16      |
| 4XB7A90889                                                       | BYM7         | ThinkSystem 3.5" VA 960GB Mixed Use SATA 6Gb HS SSD v2   | Not TCE            | No          | 16      |

| Part number                                                             | Feature code | Description                                                   | Top Choice Express | SED support | Max Qty |
|-------------------------------------------------------------------------|--------------|---------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7A90890                                                              | BYM8         | ThinkSystem 3.5" VA 1.92TB Mixed Use SATA 6Gb HS SSD v2       | Not TCE            | No          | 16      |
| 4XB7A90891                                                              | BYLX         | ThinkSystem 3.5" VA 3.84TB Mixed Use SATA 6Gb HS SSD v2       | TCE                | No          | 16      |
| <b>3.5-inch hot-swap SSDs - 6 Gb SATA - Read Intensive (&lt;3 DWPD)</b> |              |                                                               |                    |             |         |
| 4XB7B11648                                                              | CCZP         | ThinkSystem 3.5" VA 480GB Read Intensive SATA 6Gb HS SSD SED  | Not TCE            | Support     | 16      |
| 4XB7B11649                                                              | CCZN         | ThinkSystem 3.5" VA 960GB Read Intensive SATA 6Gb HS SSD SED  | Not TCE            | Support     | 16      |
| 4XB7B11651                                                              | CCZL         | ThinkSystem 3.5" VA 3.84TB Read Intensive SATA 6Gb HS SSD SED | Not TCE            | Support     | 16      |
| 4XB7A90879                                                              | BYLJ         | ThinkSystem 3.5" VA 480GB Read Intensive SATA 6Gb HS SSD v2   | TCE                | No          | 16      |
| 4XB7A90880                                                              | BYLY         | ThinkSystem 3.5" VA 960GB Read Intensive SATA 6Gb HS SSD v2   | TCE                | No          | 16      |
| 4XB7A90881                                                              | BYLZ         | ThinkSystem 3.5" VA 1.92TB Read Intensive SATA 6Gb HS SSD v2  | TCE                | No          | 16      |
| 4XB7A90882                                                              | BYM0         | ThinkSystem 3.5" VA 3.84TB Read Intensive SATA 6Gb HS SSD v2  | TCE                | No          | 16      |
| 4XB7A90883                                                              | BYM1         | ThinkSystem 3.5" VA 7.68TB Read Intensive SATA 6Gb HS SSD v2  | Not TCE            | No          | 16      |

Table 66. 3.5-inch hot-swap PCIe 5.0 NVMe SSDs

| Part number                                                            | Feature code | Description                                                               | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------|--------------------|-------------|---------|
| <b>3.5-inch SSDs - U.2 PCIe 5.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                           |                    |             |         |
| 4XB7B13984                                                             | CFA6         | ThinkSystem 3.5" U.2 VA 1.92TB Read Intensive NVMe PCIe 5.0 x4 HS SSD     | Not TCE            | Support     | 4       |
| 4XB7B13985                                                             | CFA5         | ThinkSystem 3.5" U.2 VA 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD     | Not TCE            | Support     | 4       |
| 4XB7B13986                                                             | CFA4         | ThinkSystem 3.5" U.2 VA 7.68TB Read Intensive NVMe PCIe 5.0 x4 HS SSD     | Not TCE            | Support     | 4       |
| 4XB7B13987                                                             | CFA3         | ThinkSystem 3.5" U.2 VA 15.36TB Read Intensive NVMe PCIe 5.0 x4 HS SSD    | Not TCE            | Support     | 4       |
| 4XB7A93070                                                             | C3Q6         | ThinkSystem 3.5" U.2 PM9D3a 3.84TB Read Intensive NVMe PCIe 5.0 x4 HS SSD | Not TCE            | Support     | 4       |

Table 67. 3.5-inch hot-swap PCIe 4.0 NVMe SSDs

| Part number                                                            | Feature code | Description                                                                       | Top Choice Express | SED support | Max Qty |
|------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------|--------------------|-------------|---------|
| <b>3.5-inch SSDs - U.2 PCIe 4.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                                   |                    |             |         |
| 4XB7B01873                                                             | C6MF         | ThinkSystem 3.5" U.2 Solidigm P5520 7.68TB Read Intensive NVMe PCIe 4.0 x4 HS SSD | Not TCE            | Support     | 4       |
| <b>3.5-inch SSDs - U.3 PCIe 4.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                                                   |                    |             |         |

| Part number | Feature code | Description                                                                  | Top Choice Express | SED support | Max Qty |
|-------------|--------------|------------------------------------------------------------------------------|--------------------|-------------|---------|
| 4XB7B05733  | C9ZW         | ThinkSystem 3.5" U.3 7500 PRO 1.92TB Read Intensive NVMe PCIe 4.0 x4 HS SSD  | Not TCE            | Support     | 4       |
| 4XB7B05734  | C9ZX         | ThinkSystem 3.5" U.3 7500 PRO 3.84TB Read Intensive NVMe PCIe 4.0 x4 HS SSD  | Not TCE            | Support     | 4       |
| 4XB7B05735  | C9ZY         | ThinkSystem 3.5" U.3 7500 PRO 7.68TB Read Intensive NVMe PCIe 4.0 x4 HS SSD  | Not TCE            | Support     | 4       |
| 4XB7B05736  | C9ZZ         | ThinkSystem 3.5" U.3 7500 PRO 15.36TB Read Intensive NVMe PCIe 4.0 x4 HS SSD | Not TCE            | Support     | 4       |

Table 68. M.2 PCIe 4.0 NVMe hot-swap drives (hot-swap M.2 adapters)

| Part number                                                   | Feature code | Description                                          | Top Choice Express | SED support | Max Qty |
|---------------------------------------------------------------|--------------|------------------------------------------------------|--------------------|-------------|---------|
| <b>M.2 SSDs - PCIe 4.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                      |                    |             |         |
| 4XB7B14665                                                    | C286         | ThinkSystem M.2 VA 480GB Read Intensive NVMe HS SSD  | TCE                | Support     | 2       |
| 4XB7B14666                                                    | C287         | ThinkSystem M.2 VA 960GB Read Intensive NVMe HS SSD  | TCE                | Support     | 2       |
| 4XB7B14667                                                    | C288         | ThinkSystem M.2 VA 1.92TB Read Intensive NVMe HS SSD | Not TCE            | Support     | 2       |

Table 69. M.2 PCIe 4.0 NVMe non-hot-swap drives (internal M.2 adapters)

| Part number                                                   | Feature code | Description                                           | Top Choice Express | SED support | Max Qty |
|---------------------------------------------------------------|--------------|-------------------------------------------------------|--------------------|-------------|---------|
| <b>M.2 SSDs - PCIe 4.0 NVMe - Read Intensive (&lt;3 DWPD)</b> |              |                                                       |                    |             |         |
| 4XB7A93140                                                    | C286         | ThinkSystem M.2 VA 480GB Read Intensive NVMe NHS SSD  | TCE                | Support     | 2       |
| 4XB7A93141                                                    | C287         | ThinkSystem M.2 VA 960GB Read Intensive NVMe NHS SSD  | TCE                | Support     | 2       |
| 4XB7A93142                                                    | C288         | ThinkSystem M.2 VA 1.92TB Read Intensive NVMe NHS SSD | Not TCE            | Support     | 2       |

Table 70. M.2 SATA non-hot-swap drives

| Part number                                               | Feature code | Description                                              | Top Choice Express | SED support | Max Qty |
|-----------------------------------------------------------|--------------|----------------------------------------------------------|--------------------|-------------|---------|
| <b>M.2 SSDs - 6 Gb SATA - Read Intensive (&lt;3 DWPD)</b> |              |                                                          |                    |             |         |
| 4XB7B07588                                                | CABU         | ThinkSystem M.2 VA 480GB Read Intensive SATA 6Gb NHS SSD | TCE                | Support     | 2       |
| 4XB7B07589                                                | CACA         | ThinkSystem M.2 VA 960GB Read Intensive SATA 6Gb NHS SSD | Not TCE            | Support     | 2       |

## Internal backup units

The server does not supports any internal backup units, such as tape drives or RDX drives. External backup units are available as described in the [External backup units](#) section.

## Optical drives

The server supports the external USB optical drive listed in the following table.

Table 71. External optical drive

| Part number | Feature code | Description                                        | Top Choice Express |
|-------------|--------------|----------------------------------------------------|--------------------|
| 7XA7A05926  | AVV8         | ThinkSystem External USB DVD RW Optical Disk Drive | TCE                |

The drive is based on the Lenovo Slim DVD Burner DB65 drive and supports the following formats: DVD-RAM, DVD-RW, DVD+RW, DVD+R, DVD-R, DVD-ROM, DVD-R DL, CD-RW, CD-R, CD-ROM.

## I/O expansion

The SR650 V4 supports a total of up to 10x PCIe slots plus 2x OCP slots, all at the rear of the server. Slot availability is based on riser selection and drive bays configured.

**Internal (CFF) RAID adapter/HBA:** For configurations with 2.5-inch front drive bays, an internal RAID adapter or HBA (also known as CFF or custom form factor) can be installed in a dedicated space and cabled to a PCIe 4.0 x8 connector, thereby freeing up a slot for other purposes. E3.S configurations do not support a RAID adapter.

Topics in this section:

- [Slot ordering information](#)
- [Slot configurations](#)
- [Slot field upgrades](#)
- [OCP slot filler](#)
- [Serial port](#)

The following figure shows the locations of the rear-accessible slots. The rear OCP slots are located below the PCIe slots. All slots are PCIe Gen 5.

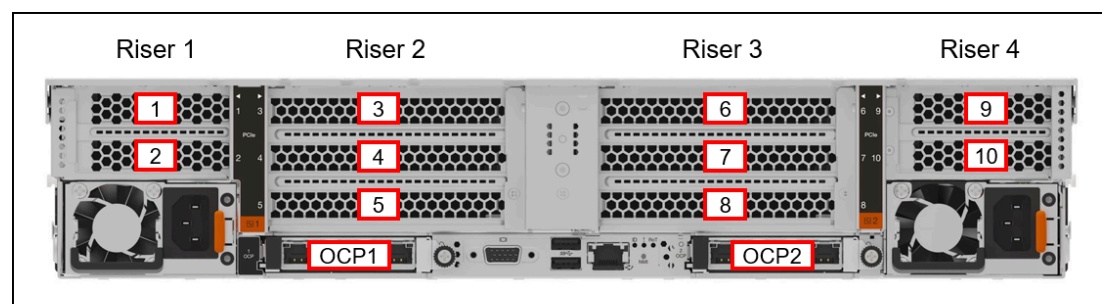


Figure 24. SR650 V4 rear slots

The supported slot and riser card combinations are as follows:

- Riser 1: Slots 1 & 2 (connects to CPU 1), low-profile slots (full height when configured with 8x 2.5-inch rear drive bays)
  - x8, x8
  - x8, x16 (only supported when configured with 8x 2.5-inch rear drive bays)
- Riser 2: Slots 3, 4 & 5 (connects to CPU 1), full-height slots
  - x8, x16, x16 (most configurations)
  - Empty, x16, x16 (use with double-wide GPU in slot 4)
  - x16, x16, x16 (for configurations with 4x 2.5-inch rear drive bays)

- Empty, Empty, x16 (use with 4x3.5-inch or 8x2.5-inch rear drive bays)
  - Other configurations available
- Riser 3: Slots 6, 7 & 8 (connects to CPU 2), full-height slots
  - x8, x16, x16 (most configurations)
  - Empty, x16, x16 (use with double-wide GPU in slot 7)
  - Empty, Empty, x16 (use with any 2.5-inch or 3.5-inch rear drive bays)
  - Other configurations available
- Riser 4: Slots 9 & 10 (connects to CPU 2), low-profile slots
  - x8, x8
  - x8, Empty

For details, see the [Slot configurations](#) section.

All x8 slots are open-ended slots, which means they can physically support x16 adapters even though only 8 lanes (x8) will be connected.

In addition, the server has two OCP slots:

- OCP1 (connects to CPU 1)
  - Choice 1: x8
  - Choice 2: x16 (requires ThinkSystem SR650 V4/SR630 V4 x16 OCP Cable Kit, feature C1YK)
- OCP2 (connects to CPU 2)
  - Choice 1: x8
  - Choice 2: x16 (requires ThinkSystem SR650 V4/SR630 V4 x16 OCP Cable Kit, feature C1YK)

Slots can be used in conjunction with rear 2.5-inch or 3.5-inch drive bays. Slot 8 can instead be configured as a pair of hot-swap M.2 drives. Combinations are shown in the following figure.

**Riser 1 slots:** With most configurations, slots 1 and 2 (Riser 1) are Low Profile slots, however with the 8x 2.5-inch drive configuration, the slots in Riser 1 are Full Height slots.

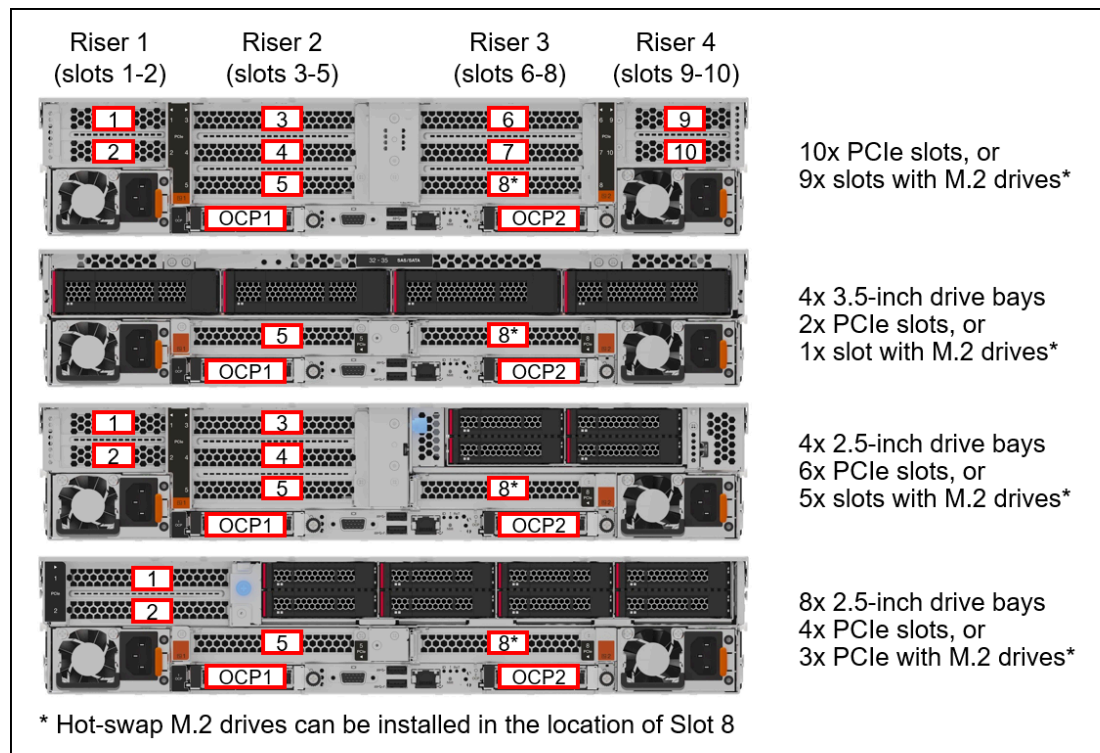


Figure 25. PCIe slots with rear drive bays

## Slot ordering information

In the SR650 V4, slots are configured one slot at a time, based on customer need for PCIe slots. This is different to prior generations, where slots were configured at the complete riser level, not the slot level.

The [DCSC configurator](#) includes a new slot configuration wizard which will automate the selection of slots and riser cages, based on the adapters and rear drive bays selected. The wizard includes a visual representation of where each adapter is to be placed.

The SR650 V4 uses a modular riser card design for Riser 2 and Riser 3, in that risers start with only 1 slot (slot 5 or slot 8) and can be configured with additional slots as needed. Additional slots can be configured CTO and built in the factory, or can be added as field upgrades after delivery. Slots 5 and 8 are connected to riser slots on the system board. The other slots (slots 4 & 3 for Riser 2 and slots 7 & 6 for Riser 3) are connected via cables to additional PCIe connectors on the system board. This "pay-as-you-go" design ensures that your server configuration only includes the adapter slots you need, thereby minimizing the cost of the overall server solution.

**Tip:** For Riser 2 and Riser 3, you do not need to select the lower slots before selecting upper slots. For example, you can select slots for Slot 3 and Slot 4, but leave Slot 5 empty.

The two slots in Riser 1 and Riser 4 are installed as a pair. Both must be selected and installed at the same time.

The following table lists the ordering information for the rear slots. The table includes a Top Choice Express column; for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

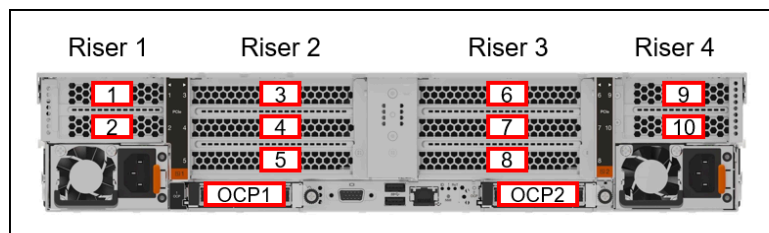


Figure 26. SR650 V4 rear slots

Table 72. Ordering information for rear slots

| Part number                                              | Feature | Description of feature code                                                | Top Choice Express | Purpose                                             |
|----------------------------------------------------------|---------|----------------------------------------------------------------------------|--------------------|-----------------------------------------------------|
| Riser 1 (slots 1 & 2)                                    |         |                                                                            |                    |                                                     |
| 4XH7B03717                                               | C3S7    | ThinkSystem SR650 V4 Riser Cage for Riser 1 or 4 Option Kit                | Not TCE            | Cage for Riser 1 or 4                               |
| 4XH7B03718                                               | C3R9    | ThinkSystem SR650 V4 x8 PCIe Gen5 Cable Riser for Slot 1/2/9/10 Option Kit | TCE                | Riser for Slot 1, x8                                |
| 4XH7B03718                                               | C4U8    | ThinkSystem SR650 V4 x8 PCIe Gen5 Cable Riser for Slot 1/2/9/10 Option Kit | TCE                | Riser for Slot 2, x8                                |
| Riser 1 (slots 1 & 2) for use with 8x2.5-inch drive bays |         |                                                                            |                    |                                                     |
| 4XH7B03848*                                              | C3RY    | ThinkSystem SR650 V4 8x2.5" SAS/SATA Rear Backplane option Kit             | Not TCE            | Cage for Riser 1 for use with 8x2.5-inch drive bays |
| 4X97B02316                                               | C4TW    | ThinkSystem SR650/a V4 x8 Gen5 Cable Riser Slot 3 & 6 Option Kit           | TCE                | Riser for Slot 1, x8                                |
| 4X97B02318                                               | C4TZ    | ThinkSystem SR650/a V4 x16 Gen5 Cable Riser Slot 4 & 7 Option Kit          | TCE                | Riser for Slot 2, x16                               |

| Part number             | Feature | Description of feature code                                                      | Top Choice Express | Purpose                                                                                                |
|-------------------------|---------|----------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------|
| Riser 2 (slots 3, 4, 5) |         |                                                                                  |                    |                                                                                                        |
| 4XH7B03839              | C3S5    | ThinkSystem SR650/a V4 Riser Cage for Riser 2 or 3 w/ 3x PCIe Slots installation | Not TCE            | Cage for Riser 2 or 3 without drive bays                                                               |
| 4XH7B03719              | C3S2    | ThinkSystem SR650 V4 Riser Cage for Riser 2 or 3 w/ Rear Backplane               | Not TCE            | Cage for Riser 2 or 3 for use with all rear drive bays                                                 |
| 4X97B02316              | C3R4    | ThinkSystem SR650/a V4 x8 Gen5 Cable Riser Slot 3 & 6 Option Kit                 | TCE                | Riser for Slot 3, x8                                                                                   |
| 4X97B02319              | C3R8    | ThinkSystem SR650/a V4 x16 Gen5 High Power Cable Riser Slot 3 & 6 Option Kit     | TCE                | Riser for Slot 3, x16                                                                                  |
| 4X97B02318              | C3R3    | ThinkSystem SR650/a V4 x16 Gen5 Cable Riser Slot 4 & 7 Option Kit                | TCE                | Riser for Slot 4, x16, 75W                                                                             |
| 4X97B02317              | C3R7    | ThinkSystem SR650/a V4 x16 Gen5 High Power Cable Riser Slot 4 & 7 Option Kit     | Not TCE            | Riser for Slot 4, x16, >75W                                                                            |
| 4XH7B02315              | C62D    | ThinkSystem SR650/a V4 x16 Gen5 Rigid Riser Slot 5 & 8 Option Kit                | TCE                | Riser for Slot 5, x16, 75W                                                                             |
| 4XH7B04304              | C62E    | ThinkSystem SR650 V4 x16 Rear Direct Riser Slot 5 for CPU+MEM Liquid Cooling     | Not TCE            | Riser for Slot 5, x16, 75W for Compute Complex Neptune water cooling (machine type 7DK2)               |
| Riser 3 (slots 6, 7, 8) |         |                                                                                  |                    |                                                                                                        |
| 4XH7B03839              | C3S5    | ThinkSystem SR650/a V4 Riser Cage for Riser 2 or 3 w/ 3x PCIe Slots installation | Not TCE            | Cage for Riser 2 or 3 without drive bays                                                               |
| 4XH7B03719              | C3S2    | ThinkSystem SR650 V4 Riser Cage for Riser 2 or 3 w/ Rear Backplane               | Not TCE            | Cage for Riser 2 or 3 for use with all rear drive bays                                                 |
| 4X97B02316              | C4TX    | ThinkSystem SR650/a V4 x8 Gen5 Cable Riser Slot 3 & 6 Option Kit                 | TCE                | Riser for Slot 6, x8                                                                                   |
| 4X97B02319              | C4U9    | ThinkSystem SR650/a V4 x16 Gen5 High Power Cable Riser Slot 3 & 6 Option Kit     | TCE                | Riser for Slot 6, x16                                                                                  |
| 4X97B02318              | C4TV    | ThinkSystem SR650/a V4 x16 Gen5 Cable Riser Slot 4 & 7 Option Kit                | TCE                | Riser for Slot 7, x16, 75W                                                                             |
| 4X97B02317              | C4UA    | ThinkSystem SR650/a V4 x16 Gen5 High Power Cable Riser Slot 4 & 7 Option Kit     | Not TCE            | Riser for Slot 7, x16, >75W                                                                            |
| 4XH7B02315              | C4U0    | ThinkSystem SR650/a V4 x16 Gen5 Rigid Riser Slot 5 & 8 Option Kit                | TCE                | Riser for Slot 8, 16                                                                                   |
| Riser 4 (slots 9, 10)   |         |                                                                                  |                    |                                                                                                        |
| 4XH7B03717              | C3S7    | ThinkSystem SR650 V4 Riser Cage for Riser 1 or 4 Option Kit                      | Not TCE            | Cage for Riser 1 or 4                                                                                  |
| 4XH7B03718              | C4U7    | ThinkSystem SR650 V4 x8 PCIe Gen5 Cable Riser for Slot 1/2/9/10 Option Kit       | TCE                | Riser for Slot 9, x8                                                                                   |
| 4XH7B03718              | C4U6    | ThinkSystem SR650 V4 x8 PCIe Gen5 Cable Riser for Slot 1/2/9/10 Option Kit       | TCE                | Riser for Slot 10, x8                                                                                  |
| OCP cable               |         |                                                                                  |                    |                                                                                                        |
| 4X97A97300              | C1YK    | ThinkSystem SR650 V4/SR630 V4 x16 OCP Cable Kit                                  | TCE                | With this cable, an OCP slot is PCIe x16. Without this cable, the OCP slot is PCIe x8. 1 per OCP slot. |

\* 4XH7B03848 also includes backplane and cage for 8x2.5-inch SAS/SATA drive bays

## Slot configurations

The following table lists the slot combinations that are supported for each drive configuration. The locations of the risers and slots is shown in the following figure.

**Tip:** The configuration numbers match the storage configurations listed in the [Storage configurations](#) section.

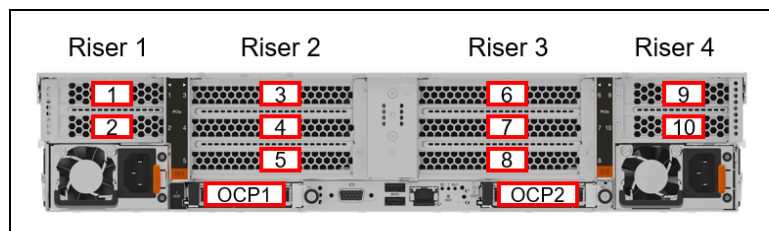


Figure 27. SR650 V4 rear slots

Table 73. Slot configurations ("-" means the slot is not available)

| Cfg |      | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           |             |                 | Riser 1 slots |    | Riser 2 slots |     |     | Riser 3 slots |     |     | Riser 4 slots |    | OCP slots |     |
|-----|------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-------------|-----------------|---------------|----|---------------|-----|-----|---------------|-----|-----|---------------|----|-----------|-----|
|     | CPUs | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | PCIe slot count | 1             | 2  | 3             | 4   | 5   | 6             | 7   | 8   | 9             | 10 | 1         | 2   |
|     |      |             |                |                    |                  |        |      |         |         |               |           |                 |               |           |             |                 |               |    |               |     |     |               |     |     |               |    |           |     |
| 1-1 | 1-2  | Y           | Y              | N                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 1-2 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 1-3 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 1-4 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 2-1 | 2    | Y           | Y              | N                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 2-2 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 3-1 | 2    | Y           | Y              | N                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 3-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 3-3 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 9               | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | -  | x8*       | x16 |
| 3-4 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 3-5 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 3-6 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 2               | -             | -  | -             | x16 | x16 | -             | -   | -   | -             | -  | x8        | -   |
| 5-1 | 1-2  | Y           | Y              | N                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 5-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 5-3 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 6-1 | 1-2  | Y           | Y              | N                  | 16               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 6-2 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 6-3 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 6-4 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 6-5 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 6-6 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |

| Cfg  | CPUs | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           |             | PCIe slot count | Riser 1 slots |    | Riser 2 slots |     |     | Riser 3 slots |     |     | Riser 4 slots |    | OCP slots |     |
|------|------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-------------|-----------------|---------------|----|---------------|-----|-----|---------------|-----|-----|---------------|----|-----------|-----|
|      |      | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                 | 1             | 2  | 3             | 4   | 5   | 6             | 7   | 8   | 9             | 10 | 1         | 2   |
|      |      |             |                |                    |                  |        |      |         |         |               |           |                 |               |           |             |                 |               |    |               |     |     |               |     |     |               |    |           |     |
| 7-1  | 2    | Y           | Y              | N                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 6               | -             | -  | x8            | x16 | x16 | x8            | x16 | x16 | -             | -  | x8*       | x16 |
| 7-2  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 1               | -             | -  | x8            | -   | -   | -             | -   | -   | -             | -  | x8*       | -   |
| 8-1  | 2    | Y           | Y              | N                  | 0                | 8      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 6               | -             | -  | x8            | x16 | x16 | x8            | x16 | x16 | -             | -  | x8*       | x16 |
| 8-2  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | -             | -  | x8            | x16 | x16 | x8            | x16 | x16 | -             | -  | x8*       | x16 |
| 8-3  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | -             | -  | x8            | x16 | x16 | -             | x16 | x16 | -             | -  | x8*       | x16 |
| 8-4  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 1               | -             | -  | x8            | -   | -   | -             | -   | -   | -             | -  | x8*       | -   |
| 8-5  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 1               | -             | -  | x8            | -   | -   | -             | -   | -   | -             | -  | x8*       | -   |
| 8-6  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 0               | -             | -  | -             | -   | -   | -             | -   | -   | -             | -  | x8        | -   |
| 9-1  | 2    | Y           | Y              | N                  | 8                | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 9-2  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 9-3  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 9-4  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 9-5  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 9               | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | -  | x8*       | x16 |
| 9-6  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 9-7  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 9-8  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 9-9  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 9-10 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 2               | -             | -  | -             | x16 | x16 | -             | -   | -   | -             | -  | x8        | -   |
| 10-1 | 2    | Y           | Y              | N                  | 8                | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 10-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 10-3 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 9               | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | -  | x8*       | x16 |
| 10-4 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 10-5 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 10-6 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 2               | -             | -  | -             | x16 | x16 | -             | -   | -   | -             | -  | x8        | -   |
| 11-1 | 1-2  | Y           | Y              | N                  | 0                | 16     | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 11-2 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 11-3 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 11-4 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 12-1 | 1-2  | Y           | Y              | N                  | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 12-2 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 12-3 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 12-4 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 12-5 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 12-6 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 13-1 | 2    | Y           | Y              | N                  | 0                | 0      | 24   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 4               | -             | -  | x8            | -   | x16 | x8            | -   | x16 | -             | -  | x8*       | x16 |
| 13-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 4               | -             | -  | x8            | x16 | -   | x8            | x16 | -   | -             | -  | x8*       | x16 |

| Cfg   | CPUs | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           |             | PCIe slot count | Riser 1 slots |    | Riser 2 slots |     |     | Riser 3 slots |     |     | Riser 4 slots |    | OCP slots |     |
|-------|------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-------------|-----------------|---------------|----|---------------|-----|-----|---------------|-----|-----|---------------|----|-----------|-----|
|       |      | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                 | 1             | 2  | 3             | 4   | 5   | 6             | 7   | 8   | 9             | 10 | 1         | 2   |
| 14-1  | 2    | Y           | Y              | N                  | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 14-2  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 14-3  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 14-4  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 14-5  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 9               | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | -  | x8*       | x16 |
| 14-6  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 14-7  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 14-8  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 14-9  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 14-10 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 2               | -             | -  | -             | x16 | x16 | -             | -   | -   | -             | -  | x8        | -   |
| 15-1  | 2    | Y           | Y              | N                  | 16               | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 15-2  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 15-3  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 15-4  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 15-5  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 9               | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | -  | x8*       | x16 |
| 15-6  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 15-7  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 15-8  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 15-9  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -  | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 15-10 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 2               | -             | -  | -             | x16 | x16 | -             | -   | -   | -             | -  | x8        | -   |
| 16-1  | 1-2  | Y           | Y              | N                  | 0                | 24     | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 16-2  | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 16-3  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8 | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 16-4  | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 17-3  | 1-2  | Y           | Y              | N                  | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 4               | 0             | 0         | 0           | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-4  | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-5  | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-6  | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-7  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-8  | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-9  | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-10 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-11 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8 | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 17-12 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 17-13 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 17-14 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | x8 | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |

| Cfg   |      | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           |             | PCIe slot count | Riser 1 slots |    | Riser 2 slots |    |     | Riser 3 slots |    |     | Riser 4 slots |    | OCP slots |     |     |
|-------|------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-------------|-----------------|---------------|----|---------------|----|-----|---------------|----|-----|---------------|----|-----------|-----|-----|
|       | CPUs | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                 | 1             | 2  | 3             | 4  | 5   | 6             | 7  | 8   | 9             | 10 | 1         | 2   |     |
|       |      |             |                |                     |                  |        |      |         |         |               |           |                 |               |           |             |                 |               |    |               |    |     |               |    |     |               |    |           |     |     |
| 18-3  | 2    | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 4               | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 18-4  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 18-5  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 18-6  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 18-7  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 18-8  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-3  | 2    | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 4             | 0         | 0           |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-4  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-5  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-6  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-7  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-8  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 19-9  | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 19-10 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 19-11 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 19-12 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 19-13 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 2             | -  | -             | -  | x16 | x16           | -  | -   | -             | -  | -         | x8  | -   |
| 19-14 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 2             | -  | -             | -  | x16 | x16           | -  | -   | -             | -  | -         | x8  | -   |
| 20-5  | 1-2  | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0               | 0             | 0         | 0           |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 20-6  | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 20-9  | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 20-10 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 20-11 | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 20-12 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | x8 | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 21-5  | 2    | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 8             | 0         | 0               | 0             | 0         | 0           |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 21-6  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 8             | x8 | -             | x8 | x16 | x16           | x8 | x16 | x16           | x8 | -         | x8* | x16 |
| 21-9  | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 21-10 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8 | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 22-5  | 2    | Y           | Y              | N                   | 16               | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 8             | 0         | 0           |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 22-6  | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 22-11 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | x8 | x8            | -  | -   | x16           | -  | -   | -             | -  | -         | x8† | -   |
| 22-12 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | x8 | x8            | -  | -   | x16           | -  | -   | -             | -  | -         | x8† | -   |
| 23-5  | 1-2  | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 8             | 0         | 0           |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 23-6  | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 23-15 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 23-16 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 23-17 | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | x8 | x8            | -  | -   | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 23-18 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | x8 | x8            | -  | -   | x16           | -  | -   | -             | -  | -         | x8† | -   |
| 24-1  | 2    | Y           | N              | N                   | 0                | 0      | 24   | 0       | 0       | 0             | 8         | 0               | 0             | 0         | 0           |                 | 2             | -  | -             | x8 | -   | -             | x8 | -   | -             | -  | -         | x8* | x8  |
| 25-1  | 2    | Y           | N              | N                   | 0                | 0      | 24   | 0       | 0       | 0             | 8         | 0               | 0             | 4         | 0           |                 | 1             | -  | -             | x8 | -   | -             | -  | -   | -             | -  | -         | x8* | x8  |

| Cfg  |      | CPU cooling |                |                     | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           |             |                 | Riser 1 slots |     | Riser 2 slots |     |     | Riser 3 slots |     |     | Riser 4 slots |    | OCP slots |     |
|------|------|-------------|----------------|---------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-------------|-----------------|---------------|-----|---------------|-----|-----|---------------|-----|-----|---------------|----|-----------|-----|
|      | CPUs | Air cooled  | Open loop CPUs | Open loop CPUs+ Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay | PCIe slot count | 1             | 2   | 3             | 4   | 5   | 6             | 7   | 8   | 9             | 10 | 1         | 2   |
|      |      |             |                |                     |                  |        |      |         |         |               |           |                 |               |           |             |                 |               |     |               |     |     |               |     |     |               |    |           |     |
| 26-1 | 1-2  | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0               | 4             | 0         | 0           | 5               | x8            | -   | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8†       | x16 |
| 26-2 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | -   | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8†       | x16 |
| 26-3 | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 5               | x8            | -   | x8            | x16 | x16 | -             | -   | x16 | -             | -  | x8†       | x16 |
| 26-4 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 4               | x8            | -   | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8†       | -   |
| 27-1 | 1-2  | Y           | Y              | N                   | 24               | 0      | 0    | 0       | 0       | 8             | 0         | 0               | 8             | 0         | 0           | 4               | x8            | x16 | -             | -   | x16 | -             | -   | x16 | -             | -  | x8†       | x16 |
| 27-2 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 4               | x8            | x16 | -             | -   | x16 | -             | -   | x16 | -             | -  | x8†       | x16 |
| 27-3 | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 4               | x8            | x16 | -             | -   | x16 | -             | -   | x16 | -             | -  | x8†       | x16 |
| 27-4 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 4               | x8            | x16 | -             | -   | x16 | -             | -   | x16 | -             | -  | x8†       | -   |
| 35-1 | 1-2  | Y           | Y              | N                   | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 35-2 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 36-1 | 1-2  | Y           | Y              | N                   | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 36-2 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 37-1 | 1-2  | Y           | Y              | N                   | 8                | 0      | 4    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 37-2 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 38-1 | 1-2  | Y           | Y              | N                   | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 4               | 0             | 0         | 0           | 2               | -             | -   | -             | -   | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 38-2 | 1-2  | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 2               | -             | -   | -             | -   | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 41-1 | 2    | Y           | Y              | N                   | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 4           | 6               | x8            | x8  | x16           | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 41-2 | 2    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 6               | x8            | x8  | x16           | x16 | x16 | -             | -   | x16 | -             | -  | x8*       | x16 |
| 42-1 | 2    | Y           | N              | N                   | 12               | 0      | 0    | 0       | 0       | 0             | 8         | 0               | 0             | 0         | 0           | 7               | x8            | -   | x8            | x16 | -   | x16           | x16 | x16 | x8            | -  | x8*       | x8  |
| 42-2 | 2    | Y           | N              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 7               | x8            | -   | x8            | x16 | -   | x16           | x16 | x16 | x8            | -  | x8*       | x8  |
| 59-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 8       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 59-2 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -   | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 60-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 16      | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 6               | -             | -   | x8            | x16 | x16 | x8            | x16 | x16 | -             | -  | x8*       | x16 |
| 60-2 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 1               | -             | -   | x8            | -   | -   | -             | -   | -   | -             | -  | x8*       | -   |
| 61-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 24      | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 4               | -             | -   | x8            | -   | x16 | x8            | -   | x16 | -             | -  | x8*       | x16 |
| 62-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 32      | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 2               | -             | -   | x8            | -   | -   | x8            | -   | -   | -             | -  | x8*       | x16 |
| 63-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 0       | 4       | 0             | 0         | 0               | 0             | 0         | 0           | 10              | x8            | x8  | x8            | x16 | x16 | x8            | x16 | x16 | x8            | x8 | x8*       | x16 |
| 63-2 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 3               | -             | -   | x8            | x16 | x16 | -             | -   | -   | -             | -  | x8*       | -   |
| 64-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 0       | 8       | 0             | 0         | 0               | 0             | 0         | 0           | 6               | -             | -   | x8            | x16 | x16 | x8            | x16 | x16 | -             | -  | x8*       | x16 |
| 65-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 0       | 12      | 0             | 0         | 0               | 0             | 0         | 0           | 4               | -             | -   | x8            | -   | x16 | x8            | -   | x16 | -             | -  | x8*       | x16 |
| 66-1 | 1    | Y           | Y              | N                   | 0                | 0      | 0    | 4       | 2       | 0             | 0         | 0               | 0             | 0         | 0           | 3               | -             | -   | x8            | x16 | x16 | -             | -   | -   | -             | -  | -         | -   |
| 67-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 8       | 4       | 0             | 0         | 0               | 0             | 0         | 0           | 6               | -             | -   | x8            | x16 | x16 | x8            | x16 | x16 | -             | -  | x8*       | x16 |
| 67-2 | 1    | Y           | Y              | N                   |                  |        |      |         |         |               |           |                 |               |           |             | 1               | -             | -   | x8            | -   | -   | -             | -   | -   | -             | -  | -         | -   |
| 68-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 8       | 8       | 0             | 0         | 0               | 0             | 0         | 0           | 4               | -             | -   | x8            | -   | x16 | x8            | -   | x16 | -             | -  | x8*       | x16 |
| 69-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 8       | 12      | 0             | 0         | 0               | 0             | 0         | 0           | 2               | -             | -   | x8            | -   | -   | x8            | -   | -   | -             | -  | x8*       | x16 |
| 70-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 16      | 4       | 0             | 0         | 0               | 0             | 0         | 0           | 4               | -             | -   | x8            | -   | x16 | x8            | -   | x16 | -             | -  | x8*       | x16 |
| 71-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 16      | 8       | 0             | 0         | 0               | 0             | 0         | 0           | 2               | -             | -   | x8            | -   | -   | x8            | -   | -   | -             | -  | x8*       | x16 |
| 72-1 | 2    | Y           | Y              | N                   | 0                | 0      | 0    | 24      | 4       | 0             | 0         | 0               | 0             | 0         | 0           | 2               | -             | -   | x8            | -   | -   | x8            | -   | -   | -             | -  | x8*       | x16 |
| 87-1 | 2    | N           | N              | Y                   | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 4               | -             | -   | x8            | x16 | x16 | x8            | -   | -   | -             | -  | x8*       | x8  |

| Cfg   |      | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           |             | PCIe slot count | Riser 1 slots |    | Riser 2 slots |     |     | Riser 3 slots |    |     | Riser 4 slots |    | OCP slots |     |     |
|-------|------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-------------|-----------------|---------------|----|---------------|-----|-----|---------------|----|-----|---------------|----|-----------|-----|-----|
|       | CPUs | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe | 2.5" AnyBay |                 | 1             | 2  | 3             | 4   | 5   | 6             | 7  | 8   | 9             | 10 | 1         | 2   |     |
|       |      |             |                |                    |                  |        |      |         |         |               |           |                 |               |           |             |                 |               |    |               |     |     |               |    |     |               |    |           |     |     |
| 88-1  | 2    | N           | N              | Y                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 4             | -  | -             | x8  | x16 | x16           | x8 | -   | -             | -  | -         | x8* | x8  |
| 88-2  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | -  | -             | x8  | x16 | x16           | x8 | -   | -             | -  | -         | x8* | x8  |
| 88-3  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8  | x16 | x16           | -  | -   | -             | -  | -         | x8* | x8  |
| 89-1  | 2    | N           | N              | Y                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 3             | -  | -             | x8  | -   | x16           | x8 | -   | -             | -  | -         | x8* | x8  |
| 90-1  | 2    | N           | N              | Y                  | 0                | 8      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 3             | -  | -             | x8  | -   | x16           | x8 | -   | -             | -  | -         | x8* | x8  |
| 90-2  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8  | -   | x16           | x8 | -   | -             | -  | -         | x8* | x8  |
| 90-3  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 2             | -  | -             | x8  | -   | x16           | -  | -   | -             | -  | -         | x8* | x8  |
| 91-1  | 2    | N           | N              | Y                  | 8                | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 4             | -  | -             | x8  | -   | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 91-2  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | -  | -             | x8  | -   | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 91-3  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | -  | -             | x8  | -   | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 91-4  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | -  | -             | x8  | -   | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 91-5  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8  | -   | x16           | -  | x16 | -             | -  | -         | x8* | x8  |
| 92-1  | 2    | N           | N              | Y                  | 8                | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 4             | -  | -             | x8  | -   | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 92-2  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 4             | -  | -             | x8  | -   | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 92-3  | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 3             | -  | -             | x8  | -   | x16           | -  | x16 | -             | -  | -         | x8* | x8  |
| 97-1  | 2    | N           | N              | Y                  | 0                | 0      | 0    | 8       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 5             | x8 | -             | x8  | -   | x16           | x8 | -   | -             | x8 | -         | x8* | x8  |
| 98-1  | 2    | N           | N              | Y                  | 0                | 0      | 0    | 16      | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 4             | x8 | -             | x8  | -   | -             | x8 | -   | -             | x8 | -         | x8* | x8  |
| 101-1 | 2    | Y           | Y              | N                  | 0                | 0      | 24   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 4           | 0               | 2             | -  | -             | x8  | -   | x16           | -  | -   | -             | -  | -         | x8* | x16 |
| 102-1 | 1    | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 2       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 5             | x8 | x8            | x8  | x16 | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 103-1 | 2    | N           | N              | Y                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 103-2 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 103-3 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 104-1 | 2    | N           | N              | Y                  | 0                | 8      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 104-2 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 105-1 | 2    | N           | N              | Y                  | 16               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 105-2 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 105-3 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 105-4 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 105-5 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 106-1 | 2    | N           | N              | Y                  | 0                | 16     | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 106-2 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 7             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | x8 | -         | x8* | x8  |
| 106-3 | 2    | N           | N              | Y                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | -             | x8  | x16 | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 117-1 | 1-2  | Y           | Y              | N                  | 8                | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 10            | x8 | x8            | x8  | x16 | x16           | x8 | x16 | x16           | x8 | x8        | x8* | x16 |
| 117-2 | 1-2  | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 10            | x8 | x8            | x8  | x16 | x16           | x8 | x16 | x16           | x8 | x8        | x8* | x16 |
| 118-1 | 2    | Y           | Y              | N                  | 12               | 0      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 4         | 0           | 0               | 6             | x8 | x8            | x16 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 118-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |             |                 | 6             | x8 | x8            | x16 | x16 | x16           | -  | -   | x16           | -  | -         | x8* | x16 |
| 119-1 | 1    | Y           | Y              | N                  | 0                | 0      | 0    | 0       | 6       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 2             | -  | -             | x8  | -   | x16           | -  | -   | -             | -  | -         | x8* | -   |
| 120-1 | 1    | Y           | Y              | N                  | 0                | 0      | 0    | 4       | 6       | 0             | 0         | 0               | 0             | 0         | 0           | 0               | 1             | -  | -             | x8  | -   | -             | -  | -   | -             | -  | -         | x8* | -   |

| Cfg   | CPUs | CPU cooling |                |                    | Front drive bays |        |      |         |         | Mid bays      |           | Rear drive bays |               |           | PCIe slot count | Riser 1 slots |    | Riser 2 slots |    |     | Riser 3 slots |    |     | Riser 4 slots |    | OCP slots |     |     |
|-------|------|-------------|----------------|--------------------|------------------|--------|------|---------|---------|---------------|-----------|-----------------|---------------|-----------|-----------------|---------------|----|---------------|----|-----|---------------|----|-----|---------------|----|-----------|-----|-----|
|       |      | Air cooled  | Open loop CPUs | Open loop CPUs+Mem | SAS/SATA         | AnyBay | NVMe | E3.S 1T | E3.S 2T | 2.5" SAS/SATA | 2.5" NVMe | 3.5" SAS/SATA   | 2.5" SAS/SATA | 2.5" NVMe |                 | 2.5" AnyBay   | 1  | 2             | 3  | 4   | 5             | 6  | 7   | 8             | 9  | 10        | 1   | 2   |
| 121-1 | 2    | Y           | Y              | N                  | 8                | 0      | 16   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0               | 6             | -  | -             | x8 | x16 | x16           | x8 | x16 | x16           | -  | -         | x8* | x16 |
| 121-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |                 | 6             | -  | -             | x8 | x16 | x16           | x8 | x16 | x16           | -  | -         | x8* | x16 |
| 121-3 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |                 | 5             | -  | -             | x8 | x16 | x16           | -  | x16 | x16           | -  | -         | x8* | x16 |
| 121-4 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |                 | 1             | -  | -             | x8 | -   | -             | -  | -   | -             | -  | -         | x8* | -   |
| 121-5 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |                 | 1             | -  | -             | x8 | -   | -             | -  | -   | -             | -  | -         | x8* | -   |
| 121-6 | 1    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |                 | 0             | -  | -             | -  | -   | -             | -  | -   | -             | -  | -         | x8  | -   |
| 122-1 | 2    | Y           | Y              | N                  | 8                | 4      | 0    | 0       | 0       | 0             | 0         | 0               | 0             | 4         | 0               | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 122-2 | 2    | Y           | Y              | N                  |                  |        |      |         |         |               |           |                 |               |           |                 | 6             | x8 | x8            | x8 | x16 | x16           | -  | -   | x16           | -  | -         | x8† | x16 |
| 151-1 | 1-2  | Y           | Y              | N                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0               | 10            | x8 | x8            | x8 | x16 | x16           | x8 | x16 | x16           | x8 | x8        | x8* | x16 |
| 152-1 | 1-2  | Y           | Y              | N                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0               | 10            | x8 | x8            | x8 | x16 | x16           | x8 | x16 | x16           | x8 | x8        | x8* | x16 |
| 155-1 | 2    | N           | N              | Y                  | 0                | 0      | 8    | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0               | 5             | -  | -             | x8 | x16 | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |
| 156-1 | 2    | N           | N              | Y                  | 0                | 0      | 16   | 0       | 0       | 0             | 0         | 0               | 0             | 0         | 0               | 5             | -  | -             | x8 | x16 | x16           | x8 | x16 | -             | -  | -         | x8* | x8  |

\* OCP slot 1 can be configured as x16 if slot 3 is not configured

† OCP slot 1 can be configured as x16 if slot 1 is not configured

## Slot field upgrades

Slot configurations can also be ordered as field upgrades using option part numbers, as listed in the following table.

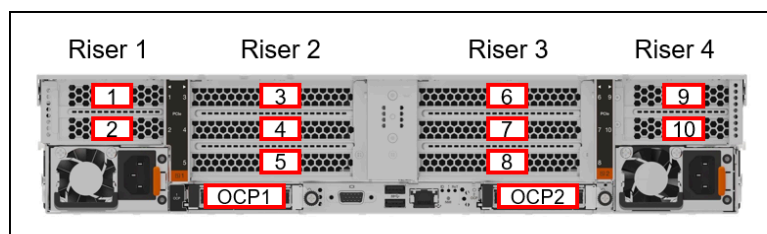


Figure 28. SR650 V4 rear slots

Table 74. Slot field upgrades

| Part number                                                              | Description                                                                           | Contents & purpose                                                                           | Max Qty |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------|
| <b>Riser 1 (slots 1 &amp; 2) or<br/>Riser 4 (slots 9, 10)</b>            |                                                                                       |                                                                                              |         |
| 4XH7B03718                                                               | ThinkSystem SR650 V4 x8 PCIe Gen5 Cable Riser for Slot 1/2/9/10 Option Kit            | Riser with x8 slot for use in slot 1, 2, 9, or 10                                            | 4       |
| 4XH7B03717                                                               | ThinkSystem SR650 V4 Riser Cage for Riser 1 or 4 Option Kit                           | Cage for Riser 1 or Riser 4 (2x LP slot cage)                                                | 2       |
| <b>Riser 1 (slots 1 &amp; 2) only for use with 8x2.5-inch drive bays</b> |                                                                                       |                                                                                              |         |
| 4X97B02316                                                               | ThinkSystem SR650/a V4 x8 Gen5 Cable Riser Slot 3 & 6 Option Kit                      | Riser with x8 slot for use in slot 1                                                         | 1       |
| 4X97B02318                                                               | ThinkSystem SR650/a V4 x16 Gen5 Cable Riser Slot 4 & 7 Option Kit                     | Riser with x16 slot for use in slot 2                                                        | 1       |
| 4XH7B03848                                                               | ThinkSystem SR650 V4 8x2.5" SAS/SATA Rear Backplane option Kit                        | Cage for Riser 1 with 2 full-height slots<br>Backplane & cage for 8x2.5-inch SAS/SATA drives | 1       |
| <b>Riser 2 (slots 3, 4, 5) or<br/>Riser 3 (slots 6, 7, 8)</b>            |                                                                                       |                                                                                              |         |
| 4XH7B03719                                                               | ThinkSystem SR650 V4 Riser Cage for Riser 2 or 3 w/ Rear Backplane                    | Cage for Riser 2 or 3 for use with any rear drive bays                                       | 2       |
| 4XH7B03839                                                               | ThinkSystem SR650/a V4 Riser Cage for Riser 2 or 3 w/ 3x PCIe Slots installation      | Cage for Riser 2 or 3 without any rear drive bays                                            | 2       |
| 4X97B02316                                                               | ThinkSystem SR650/a V4 x8 Gen5 Cable Riser Slot 3 & 6 Option Kit                      | Riser with x8 slot for Slot 3 or 6, 75W max                                                  | 2       |
| 4X97B02319                                                               | ThinkSystem SR650/a V4 x16 Gen5 High Power Cable Riser Slot 3 & 6 Option Kit          | Riser with x16 slot for Slot 3 or 6, >75W                                                    | 2       |
| 4X97B02318                                                               | ThinkSystem SR650/a V4 x16 Gen5 Cable Riser Slot 4 & 7 Option Kit                     | Riser with x16 slot for Slot 4 or 7, 75W max                                                 | 2       |
| 4X97B02317                                                               | ThinkSystem SR650/a V4 x16 Gen5 High Power Cable Riser Slot 4 & 7 Option Kit          | Riser with x16 slot for Slot 4 or 7, >75W                                                    | 2       |
| 4XH7B02315                                                               | ThinkSystem SR650/a V4 x16 Gen5 Rigid Riser Slot 5 & 8 Option Kit                     | Riser with x16 slot for Slot 5 or 8, 75W max                                                 | 2       |
| 4XH7B04304                                                               | ThinkSystem SR650 V4 x16 Gen5 Rigid Riser Option Kit for CPU+MEM Direct Water Cooling | Riser with x16 slot for Slot 8, 75W for use with Compute Complex Neptune Core liquid cooling | 1       |
| <b>OCP slots</b>                                                         |                                                                                       |                                                                                              |         |

| Part number | Description                                       | Contents & purpose                                                                                        | Max Qty |
|-------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------|
| 4X97A97300  | ThinkSystem V4 1U/2U OCP x16 Enablement Cable Kit | With this cable, an OCP slot is PCIe x16.<br>Without this cable, the OCP slot is PCIe x8. 1 per OCP slot. | 2       |

### OCP slot filler

If customers or partners remove an OCP adapter from the server, we recommend that a slot cover (slot filler) be installed in its place to ensure proper airflow in the server. Ordering information is listed in the following table.

**Tip:** For CTO orders and preconfigured models, slot fillers are automatically installed in slots where an OCP adapter is not installed.

Table 75. OCP slot filler

| Part number | Description             |
|-------------|-------------------------|
| 4XF7B06188  | ThinkSystem OCP3 FILLER |

## Serial port

The SR650 V4 optionally supports a RS-232 serial port by adding a COM port bracket to either slot 5 or slot 8.

Ordering information is shown in the following table.

Table 76. Serial port

| Part number | Feature code | Description                               | Top Choice Express |
|-------------|--------------|-------------------------------------------|--------------------|
| 4X97A97253  | C3FB         | ThinkSystem V4 1U/2U COM Port Upgrade Kit | Not TCE            |

The bracket is shown in the following figure. The option part number includes both Low Profile and Full Height brackets.

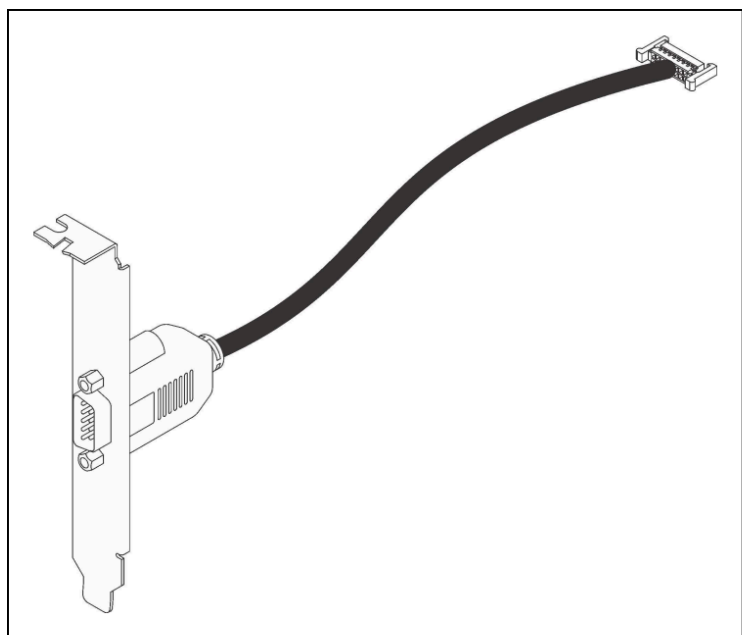


Figure 29. ThinkSystem V4 1U/2U COM Port Upgrade Kit

## Network adapters

This section lists the supported network adapters:

- [OCP network adapters](#)
- [PCIe network adapters](#)
- [BlueField-3 B3220 DPU cable kit](#)
- [ConnectX-8 auxiliary cable](#)
- [Adapters with Generic firmware](#)

### OCP network adapters

The server has two dedicated OCP 3.0 SFF slots each with either a PCIe x8 or x16 host interface. The OCP slots are both located at the rear of the server. See [Figure 3](#) for the location of the OCP slots.

The following table lists the supported OCP adapters. One port can optionally be shared with the XCC management processor for Wake-on-LAN and NC-SI support. The table includes a Top Choice Express column; for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

Table 77. Supported OCP adapters

| Part number                | Feature code | Description                                                                          | Top Choice Express | Maximum supported | PCIe width |
|----------------------------|--------------|--------------------------------------------------------------------------------------|--------------------|-------------------|------------|
| Gigabit Ethernet           |              |                                                                                      |                    |                   |            |
| 4XC7A08235                 | B5T1         | ThinkSystem Broadcom 5719 1GbE RJ45 4-port OCP Ethernet Adapter                      | TCE                | 2                 | PCIe x4    |
| 4XC7A96731                 | C4HR         | ThinkSystem Intel E610-T4 1GBase-T 4-Port OCP Ethernet Adapter(Generic FW)*          | TCE                | 2                 | PCIe x8    |
| 10 Gb Ethernet - 10GBASE-T |              |                                                                                      |                    |                   |            |
| 4XC7A95696                 | C4GB         | ThinkSystem Broadcom 57412 10GBase-T 4-Port OCP Ethernet Adapter                     | TCE                | 2                 | PCIe x8    |
| 4XC7A08236                 | B5ST         | ThinkSystem Broadcom 57416 10GBASE-T 2-port OCP Ethernet Adapter                     | TCE                | 2                 | PCIe x8    |
| 4XC7A96732                 | C4HS         | ThinkSystem Intel E610-T2 10GBase-T 2-Port OCP Ethernet Adapter(Generic FW)*         | TCE                | 2                 | PCIe x4    |
| 4XC7A96734                 | C4HU         | ThinkSystem Intel E610-T4 10GBase-T 4-Port OCP Ethernet Adapter(Generic FW)*         | TCE                | 2                 | PCIe x4    |
| 25 Gb Ethernet             |              |                                                                                      |                    |                   |            |
| 4XC7A08237                 | BN2T         | ThinkSystem Broadcom 57414 10/25GbE SFP28 2-Port OCP Ethernet Adapter                | TCE                | 2                 | PCIe x8    |
| 4XC7A80567                 | BPPW         | ThinkSystem Broadcom 57504 10/25GbE SFP28 4-Port OCP Ethernet Adapter                | TCE                | 2                 | PCIe x16†  |
| 4XC7A08294                 | BCD4         | ThinkSystem Intel E810-DA2 10/25GbE SFP28 2-Port OCP Ethernet Adapter                | TCE                | 2                 | PCIe x8    |
| 4XC7A62582                 | BE4T         | ThinkSystem Mellanox ConnectX-6 Lx 10/25GbE SFP28 2-port OCP Ethernet Adapter        | Not TCE            | 2                 | PCIe x8    |
| 100 Gb Ethernet            |              |                                                                                      |                    |                   |            |
| 4XC7A08243                 | BPPX         | ThinkSystem Broadcom 57508 100GbE QSFP56 2-Port OCP Ethernet Adapter                 | Not TCE            | 2                 | PCIe x16†  |
| 4XC7A99190                 | C62H         | ThinkSystem Nvidia ConnectX-6 Dx 100GbE QSFP56 2-port OCP Ethernet Adapter(Generic)* | Not TCE            | 2                 | PCIe x16†  |

| Part number     | Feature code | Description                                                                         | Top Choice Express | Maximum supported | PCIe width |
|-----------------|--------------|-------------------------------------------------------------------------------------|--------------------|-------------------|------------|
| 400 Gb Ethernet |              |                                                                                     |                    |                   |            |
| 4XC7A95695      | C4CQ         | ThinkSystem Broadcom 57608 2x200/1x400GbE QSFP112 OCP Ethernet Adapter(Generic FW)* | Not TCE            | 2                 | PCIe x16†  |

\* See the [Adapters with Generic firmware](#) section

† OCP x16 enablement cable required; see below

The table above indicates the PCIe width of host interface for each adapter. All adapters with a PCIe x16 interface will require that the OCP slots have a x16 connection, using the cable kit listed in the following table.

Table 78. ThinkSystem V4 1U/2U OCP x16 Enablement Cable Kit

| Part number | Feature code | Description                                       | Top Choice Express |
|-------------|--------------|---------------------------------------------------|--------------------|
| 4X97A97300  | C1YK         | ThinkSystem V4 1U/2U OCP x16 Enablement Cable Kit | TCE                |

**Note:** NC-SI support (also known as “Shared NIC” functionality) on ConnectX-6 OCP adapters is supported with XCC firmware IHX424P or later.

## PCIe network adapters

The following table lists additional supported network adapters that can be installed in the regular PCIe slots. The table includes a Top Choice Express column; for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

Table 79. Supported PCIe Network Adapters

| Part number                | Feature code | Description                                                                         | Top Choice Express | Maximum supported | Slots | PCIe width |
|----------------------------|--------------|-------------------------------------------------------------------------------------|--------------------|-------------------|-------|------------|
| Gigabit Ethernet           |              |                                                                                     |                    |                   |       |            |
| 7ZT7A00484                 | AUZV         | ThinkSystem Broadcom 5719 1GbE RJ45 4-Port PCIe Ethernet Adapter                    | TCE                | 10                | 1-10  | PCIe x8    |
| 10 Gb Ethernet - 10GBASE-T |              |                                                                                     |                    |                   |       |            |
| 4XC7A95697                 | C4GC         | ThinkSystem Broadcom 57412 10GBase-T 4-Port PCIe Ethernet Adapter                   | TCE                | 10                | 1-10  | PCIe x8    |
| 7ZT7A00496                 | AUKP         | ThinkSystem Broadcom 57416 10GBASE-T 2-Port PCIe Ethernet Adapter                   | TCE                | 10                | 1-10  | PCIe x8    |
| 4XC7A96733                 | C4HT         | ThinkSystem Intel E610-T4 10GBase-T 4-Port PCIe Ethernet Adapter(Generic FW)*       | TCE                | 6                 | 2-8   | PCIe x8    |
| 25 Gb Ethernet             |              |                                                                                     |                    |                   |       |            |
| 4XC7A08238                 | BK1H         | ThinkSystem Broadcom 57414 10/25GbE SFP28 2-port PCIe Ethernet Adapter              | TCE                | 10                | 1-10  | PCIe x8    |
| 4XC7A80566                 | BNWM         | ThinkSystem Broadcom 57504 10/25GbE SFP28 4-port PCIe Ethernet Adapter              | TCE                | 4                 | 3-8   | PCIe x16   |
| 4XC7A08295                 | BCD6         | ThinkSystem Intel E810-DA2 10/25GbE SFP28 2-Port PCIe Ethernet Adapter              | TCE                | 10                | 1-10  | PCIe x8    |
| 4XC7A62580                 | BE4U         | ThinkSystem Mellanox ConnectX-6 Lx 10/25GbE SFP28 2-port PCIe Ethernet Adapter      | Not TCE            | 10                | 1-10  | PCIe x8    |
| 4XC7A99191                 | C62J         | ThinkSystem Nvidia ConnectX-7 10/25GbE SFP28 4-Port PCIe Ethernet Adapter(Generic)* | Not TCE            | 4                 | 3-8   | PCIe x16   |

| Part number                               | Feature code | Description                                                                                       | Top Choice Express | Maximum supported | Slots | PCIe width |
|-------------------------------------------|--------------|---------------------------------------------------------------------------------------------------|--------------------|-------------------|-------|------------|
| 100 Gb Ethernet and HDR100 InfiniBand     |              |                                                                                                   |                    |                   |       |            |
| 4XC7A08297                                | BK1J         | ThinkSystem Broadcom 57508 100GbE QSFP56 2-Port PCIe 4 Ethernet Adapter                           | TCE                | 4                 | 3-8   | PCIe x16   |
| 4XC7A08248                                | B8PP         | ThinkSystem Mellanox ConnectX-6 Dx 100GbE QSFP56 2-port PCIe Ethernet Adapter                     | TCE                | 4                 | 3-8   | PCIe x16   |
| 200 Gb Ethernet and HDR/NDR200 InfiniBand |              |                                                                                                   |                    |                   |       |            |
| 4XC7A81883                                | BQBN         | ThinkSystem NVIDIA ConnectX-7 NDR200/200GbE QSFP112 2-port PCIe Gen5 x16 InfiniBand Adapter       | TCE                | 4                 | 3-8   | PCIe x16   |
| 4XC7B08942                                | CCVX         | ThinkSystem NVIDIA BlueField-3 B3220 VPI QSFP112 2P 200G PCIe Gen5 x16 with Tin Plating Connector | Not TCE            | 2                 | 3-8   | PCIe x16   |
| 400 Gb Ethernet and NDR InfiniBand        |              |                                                                                                   |                    |                   |       |            |
| 4XC7A95572                                | C4GA         | ThinkSystem Broadcom 57608 2x200/1x400GbE QSFP112 PCIe Ethernet Adapter                           | Not TCE            | 4                 | 3-8   | PCIe x16   |
| 4XC7A95508                                | C51C         | ThinkSystem NVIDIA ConnectX-7 NDR400 OSFP 1-port PCIe Gen5 VPI Adapter                            | Not TCE            | 4                 | 3-8   | PCIe x16   |
| 4XC7B03668                                | C9AQ         | ThinkSystem NVIDIA ConnectX-8 8240 400GbE / 400Gb/s IB QSFP112 2-port PCIe Gen6 x16 (Generic FW)* | Not TCE            | 2                 | 5,7   | PCIe x16   |
| 400Gb Omni-Path                           |              |                                                                                                   |                    |                   |       |            |
| 4XC7B00020                                | C5MY         | ThinkSystem Cornelis CN5000 Omni-Path 1-Port QSFP112 HFI SuperNIC (Generic FW)                    | Not TCE            | 4                 | 3-8   | PCIe x16   |
| 800 Gb XDR InfiniBand / 400 Gb Ethernet   |              |                                                                                                   |                    |                   |       |            |
| 4XC7B03667                                | C9AP         | ThinkSystem NVIDIA ConnectX-8 C8180 XDR / 2x400GbE OSFP 1-Port PCIe Gen6 x16 (Generic FW)*        | Not TCE            | 2                 | 5,7   | PCIe x16   |

\* See the [Adapters with Generic firmware](#) section

Configuration rules:

- The Cornelis CN5000 Omni-Path adapter cannot be used in conjunction with an NVIDIA or Mellanox network adapter due to a driver incompatibility
- NVIDIA ConnectX-8 and BlueField B3220 adapters require additional cables; see below

For more information, including the transceivers and cables that each adapter supports, see the list of Lenovo Press Product Guides in the Networking adapters category:

<https://lenovopress.com/servers/options/ethernet>

### BlueField-3 B3220 DPU cable kit

When adding a BlueField-3 B3220 adapter as a field upgrade, the power cable kit listed in the following table will also need to be ordered. For CTO orders, the appropriate cable is automatically selected when the adapter is selected, based on the slot the adapter is installed in.

Table 80. Power cable for BlueField-3 B3220 DPU

| Part number | Feature code | Description of part number                                                                                                                                           | Top Choice Express |
|-------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 4X97B04861  | C6R3         | ThinkSystem SR650/a V4 DPU Adapter Power Cable Kit <ul style="list-style-type: none"><li>1x 220mm power cable for adapters in slot 5, 8 (SBB7A95591, C6R3)</li></ul> | Not TCE            |
|             | C6R1         | <ul style="list-style-type: none"><li>1x 400mm power cable for adapters in slots 4, 6, 7 (SBB7A95592, C6R1)</li></ul>                                                | Not TCE            |

### ConnectX-8 auxiliary cable

The NVIDIA ConnectX-8 8240 adapter (2x 400Gb) and ConnectX-8 8180 adapter (1x 800Gb) both require the use of an Auxiliary cable which plugs into a second PCIe x16 connection. The combination of the x16 host interface of the adapter plus the x16 connection of the Auxiliary cable results in a PCIe 5.0 x32 connection, needed for 800 Gb networking connectivity. Ordering information for the Auxiliary cable is listed in the following table. For CTO orders, the cable is automatically selected when one of the ConnectX-8 is selected.

Table 81. Auxiliary cable needed for ConnectX-8 adapters

| Part number | Feature code | Description                                          | Top Choice Express |
|-------------|--------------|------------------------------------------------------|--------------------|
| 4X97B05994  | C8WC         | ThinkSystem V4 1U/2U NVIDIA ConnectX-8 Aux Cable Kit | Not TCE            |

**Field upgrades:** When adding the ConnectX-8 adapter and Aux cable as a field upgrade, you will also need to change the settings of the adapter to enable the x32 host connection. See the following Lenovo Press paper for instructions: [Configuring the NVIDIA ConnectX-8 Adapter and Auxiliary Cable on ThinkSystem Servers](#).

### Adapters with Generic firmware

As indicated in the tables of supported adapters, some adapters are now offered by Lenovo with standard vendor firmware (look for "Generic FW" or "Generic" in the adapter names). These adapters are supported in Lenovo servers and with Lenovo management tools.

Support in Lenovo XClarity management tools for adapters with generic firmware is per the following table.

**Tip:** Always use firmware that is obtained from Lenovo sources to ensure the firmware is fully tested by Lenovo and is supported. You should not use firmware that is obtained from the vendor web site, unless directed to do so by Lenovo support.

Table 82. Lenovo XClarity management tools support for adapters with generic firmware

| Function              | Lenovo XClarity Provisioning Manager | Lenovo XClarity OneCLI (out-of-band) | Lenovo XClarity OneCLI (in-band)     | Lenovo XClarity Administrator        |
|-----------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| Adapter configuration | Supported (in-band via UEFI)         | Supported with latest UEFI firmware* | Supported with latest UEFI firmware* | Supported with latest UEFI firmware* |

\* Minimum supported UEFI firmware is as follows: SR630 V4, SR650 V4, SR650a V4: [Release IHE116I](#); SR850 V4, SR860 V4: [Release RVE106I](#).

## GPU adapters

This section describes the supported GPUs.

- [GPU part numbers](#)
- [GPU Ready configurations](#)
- [GPU configuration rules](#)
- [GPU Full Length Thermal Option Kit](#)
- [GPU cable kits](#)

**4x DW GPU support?** If you are looking for support for 4 double-wide GPUs with NVLink in a 2U rack form factor, consider the [ThinkSystem SR650a V4 server](#).

## GPU part numbers

The SR650 V4 supports the following graphics processing units (GPUs).

**Additional GPU support:** Support for additional GPUs such as the NVIDIA RTX PRO 4000 Blackwell SFF Edition and NVIDIA RTX PRO 4000 Blackwell 24GB PCIe Gen5 Active GPU may be offered via Special Bid. Please contact your sales representative for more information.

Table 83. Supported GPUs

| Part number      | Feature code | Description                                          | Top Choice Express | Controlled GPU | Form factor | Aux power | Maximum supported | Slots supported |
|------------------|--------------|------------------------------------------------------|--------------------|----------------|-------------|-----------|-------------------|-----------------|
| Single-wide GPUs |              |                                                      |                    |                |             |           |                   |                 |
| 4X67A84824       | BS2C         | ThinkSystem NVIDIA L4 24GB PCIe Gen4 Passive GPU     | Not TCE            | Controlled     | LP          | No        | 10                | 1-10            |
| Double-wide GPUs |              |                                                      |                    |                |             |           |                   |                 |
| 4X67A90669       | BYFH         | ThinkSystem NVIDIA L40S 48GB PCIe Gen4 Passive GPU   | Not TCE            | Controlled     | FHFL        | Yes       | 2                 | 4,7*            |
| 4X67A96491       | C4RX         | ThinkSystem NVIDIA RTX 4500 Ada 24GB PCIe Active GPU | Not TCE            | No             | FHFL        | Yes       | 2                 | 4,7*            |

\* When a double-wide GPU is installed in slot 4, slot 3 is unavailable; similarly when a double-wide GPU is installed in slot 7, slot 6 is unavailable

For information about these GPUs, see the ThinkSystem GPU Summary, available at: <https://lenovopress.com/lp0768-thinksystem-thinkagile-gpu-summary>

## GPU Ready configurations

For CTO orders, the SR650 V4 also supports the selection of a placeholder for a GPU. This selection results in a "GPU ready" configuration which ensures that the server ships with the components needed for GPU installation (GPU power cables, air ducts, power supplies, fans, etc) without actually including the GPUs themselves. The following table lists the ordering information for CTO orders.

Table 84. GPU-Ready configurations - ordering information

| Part number | Feature code | Description                                                                       | Top Choice Express |
|-------------|--------------|-----------------------------------------------------------------------------------|--------------------|
| CTO only    | CBFQ         | ThinkSystem NVIDIA 4000 Ada GPU-Ready Installation                                | Not TCE            |
| CTO only    | CBFP         | ThinkSystem NVIDIA 4500 Ada GPU-Ready Installation                                | Not TCE            |
| CTO only    | BP4X         | ThinkSystem Double Width GPU-Ready Installation (for all other supported DW GPUs) | TCE                |

## GPU configuration rules

The following configuration requirements must be met when installing GPUs:

- The table includes a Controlled GPU column. If a GPU is listed as Controlled, that means the GPU is not offered in certain markets, as determined by the US Government. If a GPU is listed as No, that means the GPU is not controlled and is available in all markets.
- All GPUs installed must be identical
- When a double-wide GPU is installed in slot 4 or 7, the adjacent slot 3 and 6 respectively is not available
- Middle drive bays and Rear drive bays are not supported
- For configurations with 3.5-inch front drive bays, the 8x 3.5-inch backplane (C4DB) is the only backplane supported (configurations 117-1 and 117-2 in the [Storage configurations](#) section); Backplanes supporting 12x 3.5-inch drive bays are not supported. In DCSC, the selection of this backplane requires a GPU.
- GPUs are not supported with the ThinkSystem V4 1U/2U Compute Complex Neptune Core Module (machine type 7DK2)

For additional GPU requirements, see the Thermal rules page in the User Guide:

[https://pubs.lenovo.com/sr650-v4/thermal\\_rules](https://pubs.lenovo.com/sr650-v4/thermal_rules)

## GPU Full Length Thermal Option Kit

When installing any full-length GPU as a field upgrade, you will also need to order the GPU Enablement Kit as listed in the following table. This kit is only required for full-length GPUs and is not required for low profile GPUs such as the NVIDIA L4. See the GPU part number table above for GPU form factors.

Table 85. ThinkSystem SR650 V4 Full Length GPU Thermal Option Kit

| Part number | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Maximum supported |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 4X67B04303  | ThinkSystem SR650 V4 Full Length GPU Thermal Option Kit <ul style="list-style-type: none"> <li>• 2x performance heatsinks - replace existing 2U heatsinks (SBB7A85750, C3QR)</li> <li>• 1x ThinkSystem 2U GPU air duct - replaces main air baffle (SBB7A86004, C3RP)</li> <li>• 2x Air duct fillers - needed in each riser zone if no GPU is installed in that zone (SBB7A17338, B8MB)</li> <li>• 2x Air duct lower filler (SBB7A92685, C4S3)</li> <li>• 2x Air duct upper filler (SBB7A86338, C3SK)</li> <li>• 4x 16-pin GPU power cables for double-wide GPUs (SBB7A87740, C3QU)</li> </ul> | 1                 |

## GPU cable kits

The following cable kits are offered to provide auxiliary power cables for GPUs that require one. See the [GPU part numbers](#) section to see which GPUs require an auxiliary power cable.

Configuration notes:

- This cable kit is only required for field upgrades; factory (CTO) orders will automatically include any required cables.
- This cable kit is only required for GPUs that require an auxiliary power cable. See the GPU part number table.
- The cable kit is only required if you are adding *additional* GPUs to a server that already has at least one GPU installed from a CTO order. If you are doing a field upgrade to install the *first* DW GPU to a server, order the ThinkSystem SR650 V4 Full Length GPU Thermal Option Kit, 4X67B04303 instead, since 4X67B04303 includes the necessary power cables.

Table 86. GPU cable kits

| Part number | Description                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 4X97B04860  | ThinkSystem SR650/a V4 GPU Power Cable Kit <ul style="list-style-type: none"><li>• 1x 16-pin GPU power cable (SBB7A87740, C3QU)</li></ul> |

## Fibre Channel host bus adapters

The following table lists the Fibre Channel HBAs supported by the SR650 V4.

The table includes a Top Choice Express column; for CTO orders, select a TCE component for faster delivery. See the [Top Choice](#) section for more information.

Table 87. Fibre Channel HBAs

| Part number        | Feature code | Description                                                                              | Top Choice Express | Maximum supported | Slots supported |
|--------------------|--------------|------------------------------------------------------------------------------------------|--------------------|-------------------|-----------------|
| 32Gb Fibre Channel |              |                                                                                          |                    |                   |                 |
| 4XC7A96457         | C5FC         | ThinkSystem Emulex LPe37102 32Gb 2-port SecureHBA PCIe Fibre Channel Adapter(Generic FW) | TCE                | 10                | 1-10            |
| 4XC7A08279         | BA1G         | ThinkSystem QLogic QLE2770 32Gb 1-Port PCIe Fibre Channel Adapter                        | TCE                | 10                | 1-10            |
| 4XC7A08276         | BA1F         | ThinkSystem QLogic QLE2772 32Gb 2-Port PCIe Fibre Channel Adapter                        | TCE                | 10                | 1-10            |
| 64Gb Fibre Channel |              |                                                                                          |                    |                   |                 |
| 4XC7A96458         | C5FD         | ThinkSystem Emulex LPe38102 64Gb 2-port SecureHBA PCIe Fibre Channel Adapter(Generic FW) | TCE                | 10                | 1-10            |
| 4XC7A96744         | C4L3         | ThinkSystem QLogic QLE2872 64Gb 2-Port PCIe Fibre Channel Adapter(Generic FW)            | Not TCE            | 10                | 1-10            |

For more information, see the list of Lenovo Press Product Guides in the Host bus adapters category:  
<https://lenovopress.com/servers/options/hba>

## SAS adapters for external storage

The following table lists SAS HBAs and RAID adapters supported by SR650 V4 server for use with external storage.

Table 88. Adapters for external storage

| Part number             | Feature code | Description                                              | Top Choice Express | Maximum supported | Slots supported |
|-------------------------|--------------|----------------------------------------------------------|--------------------|-------------------|-----------------|
| SAS HBA - PCIe 4.0      |              |                                                          |                    |                   |                 |
| 4Y37A09724              | B8P7         | ThinkSystem 440-16e SAS/SATA PCIe Gen4 12Gb HBA          | TCE                | 10                | 1-10            |
| RAID Adapter - PCIe 4.0 |              |                                                          |                    |                   |                 |
| 4Y37A78836              | BNWJ         | ThinkSystem RAID 940-8e 4GB Flash PCIe Gen4 12Gb Adapter | TCE                | 4                 | 1-10            |

\* The RAID adapter use a flash power module (supercap), which needs to be installed in one of the available locations in the server. For field upgrades, ensure the server configuration supports the required number of supercaps. See the [RAID flash power module \(supercap\) support](#) section for details.

For a comparison of the functions of the supported storage adapters, see the ThinkSystem RAID Adapter and HBA Reference:

<https://lenovopress.lenovo.com/lp1288#sr650-v4-support=SR650%2520V4&internal-or-external-ports=External>

For more information, see the list of Lenovo Press Product Guides in the Host bus adapters and RAID adapters categories:

<https://lenovopress.com/servers/options/hba>

<https://lenovopress.com/servers/options/raid>

## Cooling

The SR650 V4 optionally supports open-loop water cooling to remove heat from the processors and memory. For details of available cooling methods, see the [Processor cooling](#) section.

The SR650 V4 server has up to six 60 mm hot-swap variable-speed fans. Five fans are needed when one processor is installed and six fans are required when two processors are installed. The server offers N+1 rotor redundancy. The server also has one or two additional fans integrated in each of the two power supplies.

System fan choices are as follows:

- Standard fans, single-rotor 24K RPM
- High performance fans, dual-rotor, 20K RPM
- Ultra performance fans, dual-rotor, 21K RPM

Fan selection is based on the configuration of the server, as follows:

**Standard fans:** Under the following conditions, Standard fans can be used:

- Processor with any TDP with Lenovo Neptune Core liquid cooling (Processor or Compute Complex cooling)
  - 2.5-inch drives: 16xSAS/SATA+8xAnyBay, 8xSAS/SATA+16xAnyBay, 24xSAS/SATA, 24xAnyBay
  - 3.5-inch drives: 12xSAS/SATA, 12xAnyBay
- Processor TDP ≤ 205W with air-cooled heatsink with any of these front drive bays
  - 2.5-inch drives: 16xSAS/SATA+8xAnyBay, 8xSAS/SATA+16xAnyBay, 24xSAS/SATA, 24xAnyBay

- Processor TDP  $\leq$  185W with air-cooled heatsink with any of these front drive bays
  - 3.5-inch drives: 12xSAS/SATA, 12xAnyBay
- Memory  $\leq$  32GB RDIMMs (any quantity)
- No mid-chassis or rear drive backplanes
- No GPUs
- Ambient temperature 35°C or lower

**Performance fans:** If any of the above conditions are not met, Performance fans are required.

**Ultra fans:** Under the following conditions, Ultra fans are required:

- E3.S front drives
- 36x 2.5-inch NVMe drive bays (24 front, 8 mid, 4 rear)

Ordering information for the fans is listed in the following table.

Table 89. Fan ordering information

| Part number | Feature code | Description                                              | Top Choice Express | Qty 1 CPU | Qty 2 CPUs |
|-------------|--------------|----------------------------------------------------------|--------------------|-----------|------------|
| 4H47B03708  | C3RQ         | ThinkSystem SR650/a V4 Standard Fan Module Option Kit    | TCE                | 5         | 6          |
| 4H47B03709  | C3RD         | ThinkSystem SR650/a V4 Performance Fan Module Option Kit | TCE                | 5         | 6          |
| 4H47B03710  | C3RF         | ThinkSystem SR650/a V4 Ultra Fan Module Option Kit       | TCE                | 5         | 6          |

## Power supplies

The SR650 V4 supports up to two redundant hot-swap power supplies.

The power supply choices are listed in the following table. If two power supplies are installed, both power supplies used in server must be identical.

Topics in this section:

- [Power supply LEDs](#)
- [Zero-output mode](#)
- [Power cords \(C13 connectors\)](#)
- [Power cords \(C19 connectors\)](#)
- [-48V DC power cord](#)
- [HVAC/HVDC power cord](#)

**Tip:** When configuring a server in the DCSC configurator, power consumption is calculated precisely by interfacing with Lenovo Capacity Planner. You can therefore select the appropriate power supply for your configuration. However, do consider future upgrades that may require additional power needs.

Table 90. Power supply options

| Part number                               | Feature code | Description                                                             | Top Choice Express | Min Qty | Max Qty | Capacity (230V) | Capacity (115V) | Voltage   | Conn. |
|-------------------------------------------|--------------|-------------------------------------------------------------------------|--------------------|---------|---------|-----------------|-----------------|-----------|-------|
| Titanium AC power supplies - CRPS Premium |              |                                                                         |                    |         |         |                 |                 |           |       |
| 4P57A88687                                | C0U7         | ThinkSystem 800W 230V/115V Titanium CRPS Premium Hot-Swap Power Supply  | TCE                | 1       | 2       | 800W            | 800W            | 230V/115V | C14   |
| 4P57A88621                                | C0U4         | ThinkSystem 1300W 230V/115V Titanium CRPS Premium Hot-Swap Power Supply | TCE                | 1       | 2       | 1300W           | 1000W           | 230V/115V | C14   |
| 4P57A88689                                | C0U3         | ThinkSystem 2000W 230V Titanium CRPS Premium Hot-Swap Power Supply      | TCE                | 2       | 2       | 2000W           | -               | 230V      | C14   |
| 4P57A88622                                | C0UC         | ThinkSystem 2700W 230V Titanium CRPS Premium Hot-Swap Power Supply      | Not TCE            | 2       | 2       | 2700W           | -               | 230V      | C20   |
| 4P57A88623                                | C0UD         | ThinkSystem 3200W 230V Titanium CRPS Premium Hot-Swap Power Supply      | TCE                | 2       | 2       | 3200W           | -               | 230V      | C20   |
| Titanium AC power supplies - CRPS         |              |                                                                         |                    |         |         |                 |                 |           |       |
| 4P57A87056                                | BWM3         | ThinkSystem 800W 230V/115V Titanium CRPS Hot-Swap Power Supply v1.4     | TCE                | 2       | 2       | 800W            | 800W            | 230V/115V | C14   |
| 4P57A87628                                | C2Y9         | ThinkSystem 1300W 230V/115V Titanium CRPS Hot-Swap Power Supply v2.4    | TCE                | 2       | 2       | 1300W           | 1000W           | 230V/115V | C14   |
| Platinum AC power supplies - CRPS         |              |                                                                         |                    |         |         |                 |                 |           |       |

| Part number                             | Feature code | Description                                                             | Top Choice Express | Min Qty | Max Qty | Capacity (230V) | Capacity (115V) | Voltage                    | Conn.   |
|-----------------------------------------|--------------|-------------------------------------------------------------------------|--------------------|---------|---------|-----------------|-----------------|----------------------------|---------|
| 4P57A89306                              | C0U8         | ThinkSystem 800W 230V/115V Platinum CRPS Hot-Swap Power Supply v1.5     | TCE                | 2       | 2       | 800W            | 800W            | 230V/115V                  | C14     |
| 4P57A89307                              | C0U6         | ThinkSystem 1300W 230V/115V Platinum CRPS Hot-Swap Power Supply v1.5    | Not TCE            | 2       | 2       | 1300W           | 1000W           | 230V/115V                  | C14     |
| 4P57A88636                              | C0U5         | ThinkSystem 1300W 230V/115V Platinum CRPS Hot-Swap Power Supply v2.4    | TCE                | 2       | 2       | 1300W           | 1000W           | 230V/115V                  | C14     |
| 4P57A89308                              | C0UA         | ThinkSystem 2700W 230V Platinum CRPS Hot-Swap Power Supply v2.3         | Not TCE            | 2       | 2       | 2700W           | -               | 230V                       | C20     |
| 4P57A88628                              | C0UB         | ThinkSystem 2700W 230V Platinum CRPS Hot-Swap Power Supply v2.4         | Not TCE            | 2       | 2       | 2700W           | -               | 230V                       | C20     |
| HVAC/HVDC power supplies - CRPS Premium |              |                                                                         |                    |         |         |                 |                 |                            |         |
| 4P57A88627                              | C0TR         | ThinkSystem 1300W HVAC/HVDC Platinum CRPS Premium Hot-Swap Power Supply | Not TCE            | 2       | 2       | 1300W           | -               | 200-277V AC<br>240-380V DC | HVAC    |
| -48V DC power supplies - CRPS Premium   |              |                                                                         |                    |         |         |                 |                 |                            |         |
| 4P57A88625                              | C0TS         | ThinkSystem 1300W -48V DC CRPS Premium Hot-Swap Power Supply            | Not TCE            | 2       | 2       | 1300W           | -               | -48V                       | -48V DC |

Configuration rules:

- When 2x power supplies are installed, they form a redundant pair; power supplies must be identical part numbers
- Most power supplies require that 2x PSUs be installed - See the Minimum Quantity column for specifics.

Supported voltage ranges are as follows:

- The 230V/115V AC power supplies support both low-range (100-127V 50/60 Hz) and high-range (200-240V 50/60 Hz) power, except where noted. For China customers, all power supplies support 240V DC.
- The -48V DC power supply supports voltage range -44V to -54V DC.
- The HVAC/HVDC power supply supports voltage ranges 200-277V AC single phase, and 240-380V DC

For inlet current requirements, see the [Physical and electrical specifications](#) section.

Power supply options do not include a line cord. See the tables below for details about supported line cords, including the power cords for the DC power supplies. For server configurations, the inclusion of a power cord is model dependent. Configure-to-order models can be configured without power cords if desired.

The SR650 V4 supports both CRPS and CRPS Premium power supplies. CRPS Premium power supplies offer the following additional features:

- Over-subscription
- More accurate power metering
- Virtual reseal
- Enhanced fault detection
- System cooling assist (fan override)
- Fault LEDs
- VPD support
- Zero-output mode support (cold redundancy mode) (see the [Zero-output mode](#) section)

## Power supply LEDs

CRPS Premium power supplies have two LEDs:

- Power output/fault LED:
  - Green: The server is on and the power supply is working normally
  - Green, slow blinking (1 flash/sec): The power supply is in Zero-output/Standby mode (see below)
  - Green, fast blinking (5 flashes/sec): The power supply unit is in firmware update mode
  - Yellow: The power supply unit may have failed.
  - Off: The server is powered off, or the power supply is not working properly
- Power input LED:
  - Green: The power supply is connected to the input power source
  - Off: The power supply is disconnected from the AC power source or a power problem has occurred

CRPS power supplies have one LED:

- Green: The server is on and the power supply is working normally
- Green, blinking: The power supply unit is in firmware update mode
- Yellow: Either the power supply has failed, or two power supplies are installed but one is not connected to the input power source
- Yellow, blinking: The power supply is indicating a warning such as over-temperature warning, over-current warning, or a slow fan speed
- Off: The server is powered off, or the power supply is not working properly

## Zero-output mode

Zero-output mode (also known as Standby mode or Cold Redundancy mode) allows a power supply to enter a low-power mode when it is not needed. Zero-output mode is only supported on CRPS Premium power supplies.

When Zero-output mode is configured in XCC and the server power load is sufficiently low, one of the installed power supplies enters into the Standby state while the other one delivers entire load. When the power load increases, the standby power supply will switch to Active state to provide sufficient power to the server.

Zero-output mode can be enabled or disabled in the XClarity Controller web interface, **Server Configuration > Power Policy**. If you select Disable, then both power supplies will be in the Active state.

**Supported with CRPS Premium only:** Zero-output mode is supported with CRPS Premium power supplies, but not with CRPS non-Premium power supplies

## Power cords (C13 connectors)

Line cords and rack power cables with C13 connectors can be ordered as listed in the following table.

**115V customers:** If you plan to use the 1300W power supply with a low-range (100-127V) power source, select a power cable that is rated above 10A. Power cables that are rated at 10A or below are not supported with low-range power.

Table 91. Power cords

| Market/region                      | Part number | Feature code | Description                                                    | Top Choice Express |
|------------------------------------|-------------|--------------|----------------------------------------------------------------|--------------------|
| Rack cables - C13 to C14           |             |              |                                                                |                    |
| Worldwide                          | 00Y3043     | A4VP         | 1.0m, 10A/100-250V, C13 to C14 Jumper Cord                     | TCE ⓘ              |
| Worldwide                          | 4L67A08367  | B0N5         | 1.0m, 13A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 39Y7937     | 6201         | 1.5m, 10A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 4L67A08368  | B0N6         | 1.5m, 13A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 4L67A08365  | B0N4         | 2.0m, 10A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 4L67A08369  | 6570         | 2.0m, 13A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 4L67A08366  | 6311         | 2.8m, 10A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 4L67A08370  | 6400         | 2.8m, 13A/100-250V, C13 to C14 Jumper Cord                     | TCE                |
| Worldwide                          | 39Y7932     | 6263         | 4.3m, 10A/100-250V, C13 to C14 Jumper Cord                     | TCE ⓘ              |
| Worldwide                          | 4L67A08371  | 6583         | 4.3m, 13A/100-250V, C13 to C14 Rack Power Cable                | TCE ⓘ              |
| Rack cables for India - C13 to C14 |             |              |                                                                |                    |
| India                              | 4L67B10326  | CC6R         | 2.0m, 10A/100-250V, C13 to C14 Jumper Cord (India)             | Not TCE            |
| India                              | 4L67B10327  | CC6S         | 2.8m, 10A/100-250V, C13 to C14 Jumper Cord (India)             | TCE ⓘ              |
| India                              | 4L67B10328  | CC6T         | 4.3m, 10A/100-250V, C13 to C14 Jumper Cord (India)             | Not TCE            |
| Rack cables - C13 to C20           |             |              |                                                                |                    |
| Worldwide                          | 39Y7938     | 6204         | 2.8m, 10A/100-250V, C13 to IEC 320-C20 Rack Power Cable        | TCE                |
| Rack cables - C13 to C20 (Y-cable) |             |              |                                                                |                    |
| Worldwide                          | 47C2491     | A3SW         | 1.2m, 16A/100-250V, 2 Short C13s to Short C20 Rack Power Cable | Not TCE            |
| Worldwide                          | 47C2492     | A3SX         | 2.5m, 16A/100-250V, 2 Long C13s to Short C20 Rack Power Cable  | Not TCE            |
| Worldwide                          | 47C2493     | A3SY         | 2.8m, 16A/100-250V, 2 Short C13s to Long C20 Rack Power Cable  | Not TCE            |
| Worldwide                          | 47C2494     | A3SZ         | 4.1m, 16A/100-250V, 2 Long C13s to Long C20 Rack Power Cable   | Not TCE            |
| Line cords                         |             |              |                                                                |                    |
| Argentina                          | 39Y7930     | 6222         | 2.8m, 10A/250V, C13 to IEC 2073 (Argentina) Line Cord          | TCE ⓘ              |
| Argentina                          | 81Y2384     | 6492         | 4.3m, 10A/250V, C13/IEC 2073 (Argentina) Line Cord             | TCE ⓘ              |
| Australia/NZ                       | 39Y7924     | 6211         | Australia/NZ 10A line C13 to SAA-AS C112 (2.8M)                | TCE ⓘ              |
| Australia/NZ                       | 81Y2383     | 6574         | 4.3m, 10A/250V, C13/AS/NZS 3112/2000 (Australia) Line Cord     | TCE ⓘ              |
| Brazil                             | 69Y1988     | 6532         | 2.8m, 10A/250V, C13 to NBR 14136 (Brazil) Line Cord            | TCE ⓘ              |
| Brazil                             | 81Y2387     | 6404         | 4.3m, 10A/250V, C13/NBR 14136 (Brazil) Line Cord               | TCE ⓘ              |
| China                              | 39Y7928     | 6210         | 2.8m, 10A/220V, C13 to GB 2099.1 (China) Line Cord             | Not TCE            |
| China                              | 81Y2378     | 6580         | 4.3m, 10A/250V, C13/GB2099.1 (China) Line Cord                 | Not TCE            |

| Market/region | Part number | Feature code | Description                                             | Top Choice Express |
|---------------|-------------|--------------|---------------------------------------------------------|--------------------|
| Europe        | 39Y7917     | 6212         | European 10A line C13 to CEE 7/7 (2.8M)                 | TCE                |
| Europe        | 81Y2376     | 6572         | 4.3m,10A/230V, C13 to CEE7-VII (Europe) Line Cord       | Not TCE            |
| India         | 39Y7927     | 6269         | 2.8m, 6A Line C13 to Fig 68 (India) Line Cord           | TCE ⓘ              |
| India         | 81Y2386     | 6567         | 4.3m, 10A/250V, C13 to IS 6538 (India) Line Cord        | TCE ⓘ              |
| Israel        | 39Y7920     | 6218         | Israel 10A line C13 to SI 32 (2.8M)                     | TCE                |
| Italy/Chile   | 39Y7921     | 6217         | Italy 10A line C13 to CEE 7/7 (2.8M)                    | TCE ⓘ              |
| Italy/Chile   | 81Y2380     | 6493         | 4.3m,10A/250V, C13/CEI 23-16 (IT) Line cord             | TCE ⓘ              |
| Japan         | 46M2593     | A1RE         | Japan 12A/125V C13 to JIS C-8303 2.8m line cord         | TCE ⓘ              |
| Japan         | 39Y7926     | 6335         | 4.3m, 12A/100V, C13 to JIS C-8303 (Japan) Line Cord     | TCE                |
| South Africa  | 39Y7922     | 6214         | South Africa 10A line C13 to SABS 164/1 (2.8M)          | TCE ⓘ              |
| South Africa  | 81Y2379     | 6576         | 4.3m,10A/250V, C13/SANS 164-1 (South Africa) Line Cord  | TCE ⓘ              |
| South Korea   | 39Y7925     | 6219         | Korea 7A line C13 to KETI 15A/250V (2.8M)               | TCE ⓘ              |
| South Korea   | 81Y2385     | 6494         | 4.3m, 12A/250V C13/KSC 8305 (Korea) Line Cord           | TCE ⓘ              |
| Switzerland   | 39Y7919     | 6216         | Switzerland 10A line C13 to SEV 1011 (2.8M)             | TCE ⓘ              |
| Switzerland   | 81Y2390     | 6578         | 4.3m,10A/250V, C13/SEV 1011 (Switzerland)Line Cord      | Not TCE            |
| Taiwan        | 23R7158     | 6386         | Line cord - 2.8m, 10A/125V, C13 to CNS 10917-3 (Taiwan) | TCE ⓘ              |
| Taiwan        | 81Y2375     | 6317         | Taiwan 10A/250V C13/CNS 10917 2.8m line cord            | TCE ⓘ              |
| Taiwan        | 81Y2374     | 6402         | 2.8m, 13A/125V, C13 to CNS 60799 (Taiwan) Line Cord     | TCE ⓘ              |
| Taiwan        | 4L67A08363  | AX8B         | 4.3m, 10A/125V, C13 to CNS 10917 (Taiwan) Line Cord     | TCE ⓘ              |
| Taiwan        | 81Y2389     | 6531         | 4.3m, 10A/250V, C13/CNS 10917 (Taiwan) Line Cord        | TCE ⓘ              |
| Taiwan        | 81Y2388     | 6530         | Taiwan 13A/125V C13/CNS 10917 4.3m line cord            | TCE ⓘ              |
| UK            | 39Y7923     | 6215         | United Kingdom 10A line C13 to BS 1363 (2.8M)           | TCE                |
| UK            | 81Y2377     | 6577         | 4.3m,10A/230V, C13 to BS 1363/A (UK) Line Cord          | TCE                |
| USA/Canada    | 90Y3016     | 6313         | 2.8m, 10A/120V, C13 to NEMA 5-15P (US) Line Cord        | TCE                |
| USA/Canada    | 46M2592     | A1RF         | 10A/250V C13 to NEMA 6-15P 2.8m line cord               | TCE                |
| USA/Canada    | 00WH545     | 6401         | 2.8m, 13A/120V, C13 to NEMA 5-15P (US) Line Cord        | TCE                |
| USA/Canada    | 4L67A08359  | 6370         | 4.3M, 10A/125V, C13 to NEMA 5-15P (US) Line Cord        | TCE                |
| USA/Canada    | 4L67A08361  | 6373         | 4.3m, 10A/250V, C13 to NEMA 6-15P (US) Line Cord        | TCE                |
| USA/Canada    | 4L67A08360  | AX8A         | 4.3m, 13A/125V, C13 to NEMA 5-15P (US) Line Cord        | TCE                |

### Power cords (C19 connectors)

Line cords and rack power cables with C19 connectors can be ordered as listed in the following table.

Table 92. Power cords (C19 connectors)

| Part number            | Feature code | Description                                | Top Choice Express |
|------------------------|--------------|--------------------------------------------|--------------------|
| C19 to C20 Rack cables |              |                                            |                    |
| 4L67A86677             | BV8K         | 0.5m, 16A/100-250V, C19 to C20 Jumper Cord | Not TCE            |
| 4L67A86678             | B4L0         | 1.0m, 16A/100-250V, C19 to C20 Jumper Cord | Not TCE            |

| Part number                      | Feature code | Description                                             | Top Choice Express |
|----------------------------------|--------------|---------------------------------------------------------|--------------------|
| 4L67A86679                       | B4L1         | 1.5m, 16A/100-250V, C19 to C20 Jumper Cord              | TCE                |
| 4L67A86680                       | B4L2         | 2.0m, 16A/100-250V, C19 to C20 Jumper Cord              | Not TCE            |
| 39Y7916                          | 6252         | 2.5m, 16A/100-240V, C19 to IEC 320-C20 Rack Power Cable | TCE                |
| 4L67A86681                       | B4L3         | 4.3m, 16A/100-250V, C19 to C20 Jumper Cord              | Not TCE            |
| C19 to C20 Rack cables for India |              |                                                         |                    |
| 4L67B10323                       | CC6N         | 2.0m, 16A/100-250V, C19 to C20 Jumper Cord (India)      | Not TCE            |
| 4L67B10324                       | CC6P         | 2.8m, 16A/100-250V, C19 to C20 Jumper Cord (India)      | Not TCE            |
| 4L67B10325                       | CC6Q         | 4.3m, 16A/100-250V, C19 to C20 Jumper Cord (India)      | Not TCE            |
| C19 Line cords                   |              |                                                         |                    |
| 40K9777                          | 6525         | C19 4.3 meter Line Cord - Argentina                     | Not TCE            |
| 40K9773                          | 6521         | C19 4.3 meter Line Cord - Australia/NZ                  | Not TCE            |
| 40K9775                          | 6523         | C19 4.3 meter Line Cord - Brazil                        | Not TCE            |
| 40K9774                          | 6522         | C19 4.3 meter Line Cord - China                         | Not TCE            |
| 40K9766                          | 6514         | C19 4.3 meter Line Cord - Europe                        | Not TCE            |
| 40K9776                          | 6524         | C19 4.3 meter Line Cord - India                         | Not TCE            |
| 40K9768                          | 6516         | C19 4.3 meter Line Cord - Italy                         | Not TCE            |
| 41Y9231                          | 6289         | 4.3m, 15A/250V, C19 to KSC 8305 (S. Korea) Line Cord    | TCE ⓘ              |
| 40K9767                          | 6515         | C19 4.3 meter Line Cord - UK                            | Not TCE            |
| 40K9772                          | 6275         | C19 4.3m cord - NEMA L6-20P                             | Not TCE            |
| 00D7197                          | A1NV         | 4.3m, 15A/250V, C19 to NEMA 6-15P (US) Line Cord        | Not TCE            |

### -48V DC power cord

For the -48V DC Power Supply, the following power cable is supported.

Table 93. -48V DC power cable

| Part number | Feature code | Description                           |
|-------------|--------------|---------------------------------------|
| 4L67A97438  | C682         | 2.8m, 38A /-48V, -48V (3V3) Line Cord |

The following figure shows the power cable.

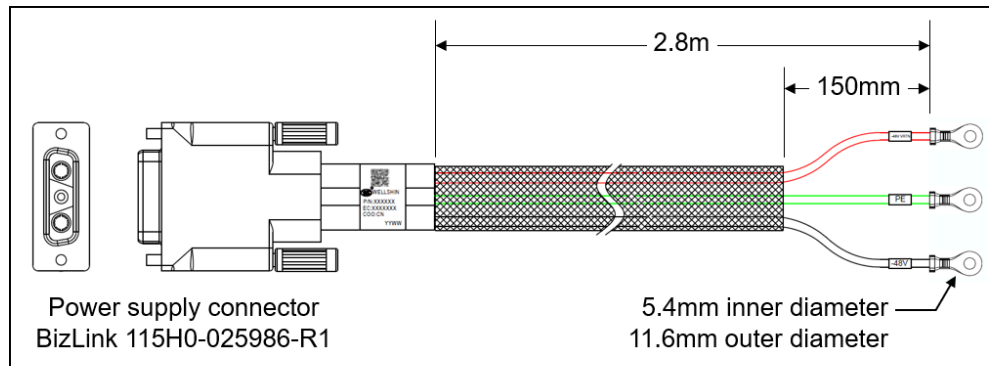


Figure 30. -48V DC power cord

## HVAC/HVDC power cord

For the HVDC Power Supply, the following power cable is supported.

Table 94. -48V DC power cable

| Part number | Feature code | Description                  |
|-------------|--------------|------------------------------|
| 4L67A97238  | C683         | 2.8M,10A/400V,HVDC Line Cord |

The following figure shows the power cable.

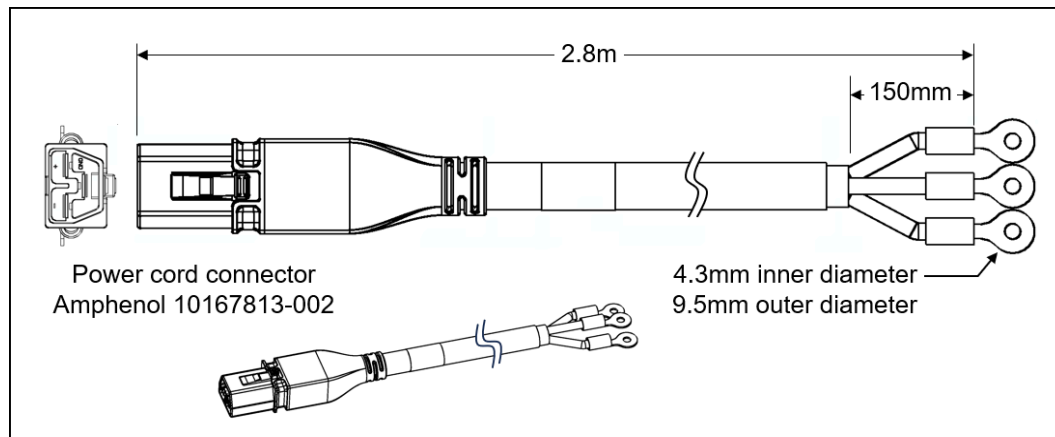


Figure 31. HVDC power cord

## Systems management

The SR650 V4 contains an integrated service processor, XClarity Controller 3 (XCC3), which provides advanced control, monitoring, and alerting functions. The XCC3 is based on an OpenBMC design, using the AST2600 baseboard management controller (BMC) with a dual-core ARM Cortex A7 32-bit RISC service processor running at 1.2 GHz.

Topics in this section:

- [System I/O Board \(DC-SCM\)](#)
- [Local management](#)
- [System status with XClarity One Mobile](#)
- [Remote management](#)
- [Shared connectivity for remote management](#)
- [MicroSD for XCC local storage](#)
- [USB flash drive](#)
- [XCC3 Premier](#)
- [Lenovo XClarity Provisioning Manager](#)
- [Lenovo XClarity One](#)
- [Lenovo XClarity Essentials](#)
- [Lenovo XClarity Energy Manager](#)
- [Lenovo Capacity Planner](#)

### System I/O Board (DC-SCM)

The SR650 V4 implements a separate System I/O Board, also known as the DC-SCM (Data Center Secure Control Module, DCSCM), that connects to the system board as shown in the Internal view in the [Components and connectors](#) section. The System I/O Board contains connectors that are accessible from the exterior of the server as shown in the following figure.

**Note:** The NMI (non-maskable interrupt) button is no longer present on the board. Lenovo recommends using the NMI function that is part of the XCC user interfaces instead.

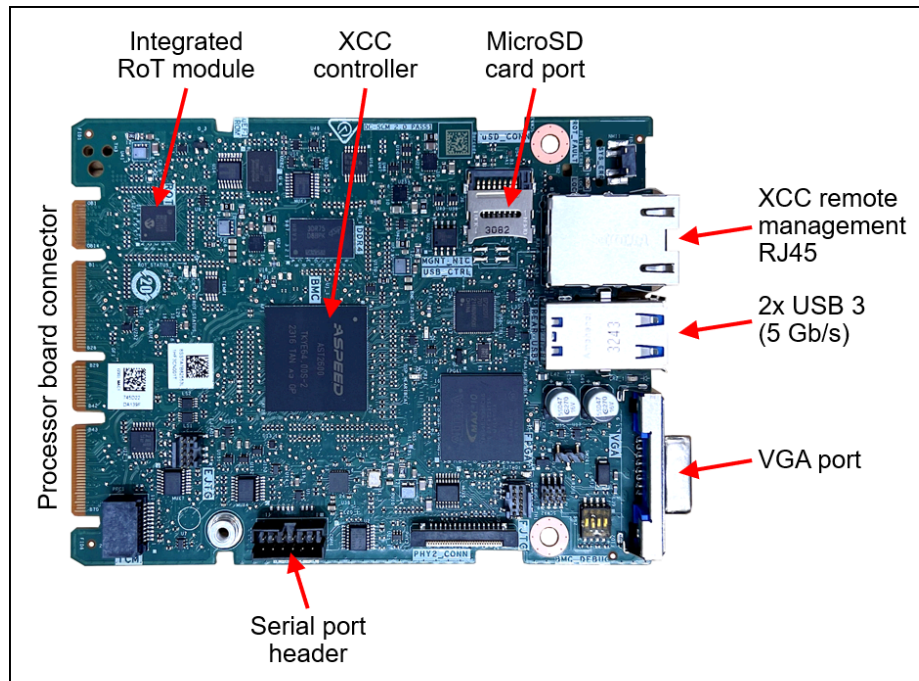


Figure 32. System I/O Board

The board also has the following components:

- XClarity Controller 3, implemented using the ASPEED AST2600 baseboard management controller (BMC).
- Root of Trust (RoT) module - implements Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT) which enables the server to be NIST SP800-193 compliant. For more details about PFR, see the [Security](#) section.
- MicroSD card port to enable the use of a MicroSD card for additional storage for use with the XCC3 controller. XCC3 can use the storage as a Remote Disc on Card (RDOC) device (up to 4GB of storage). It can also be used to store firmware updates (including N-1 firmware history) for ease of deployment.

**Tip:** Without a MicroSD card installed, the XCC controller will have 100MB of available RDOC storage.

Ordering information for the supported Micro SD cards are listed in the [MicroSD for XCC local storage](#) section.

## Local management

The SR650 V4 offers a front operator panel with key LED status indicators, as shown in the following figure.

**Tip:** The Network LED only shows network activity of an installed OCP network adapter. The LED shows activity from both OCP adapters if two are installed.

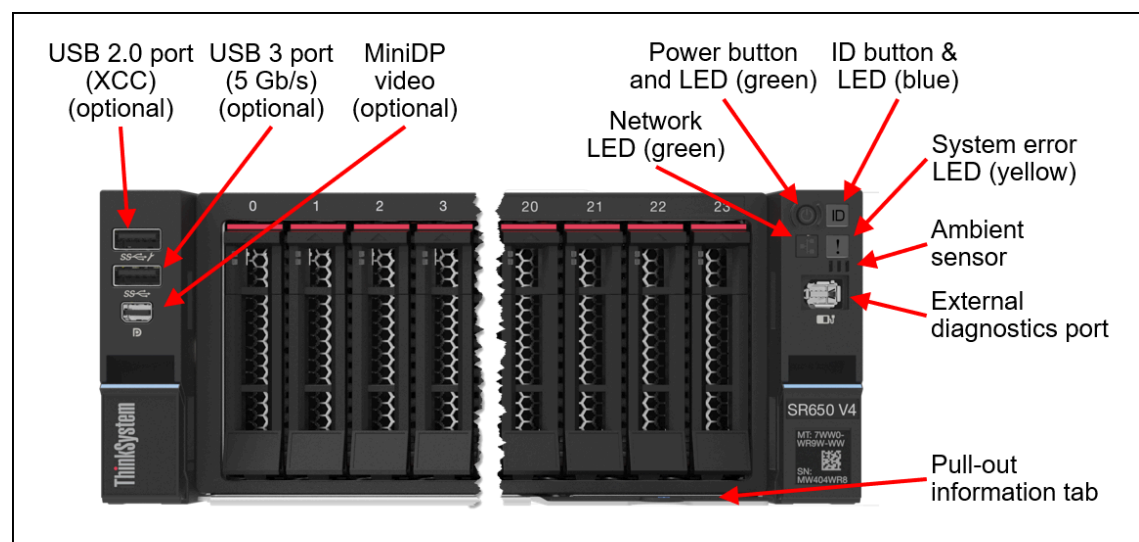


Figure 33. Front operator controls are on the left and right side of the server

### Front DisplayPort video port and Front/Internal USB ports

The rear USB ports and VGA video port are standard on all models of the SR650 V4, however the front and internal USB ports, and front MiniDP video port are optional, and can be configured in the factory in CTO orders, or as field upgrades using option part numbers.

**Internal USB port:** The internal USB port supports USB drives that have an overall length of less than 30mm. See the [USB flash drive](#) section for the supported drive.

For CTO orders, the feature codes are listed in the following table.

Table 95. CTO orders - Front & internal ports

| Feature code | Description                                          | Top Choice Express | Purpose                                                                                            |
|--------------|------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------|
| BQQ2         | ThinkSystem 2U V3 EIA Latch Standard                 | TCE                | No front USB ports, no MiniDP video port                                                           |
| C1YR         | ThinkSystem 1U/2U V4 USB Ports Extension Board       | TCE                | Provides the Internal USB 3 port (5 Gb/s)                                                          |
| C3RG*        | ThinkSystem SR650 V4 Left Rack Latch with USB/MiniDP | TCE                | Provides the 2x Front USB 3 ports (5 Gb/s) and MiniDP 1.1a port (requires Internal USB port, C1YR) |

\* Feature C1YR must also be selected

Configuration rules:

- The Internal USB port (C1YR) can be selected without also selecting the front USB/MiniDP ports
- The Front USB ports and MiniDP requires that the Internal USB port (C1YR) also be selected

- The front USB ports are required to use XClarity Mobile, as described in the [System status with XClarity Mobile](#) section.

For field upgrades, the part numbers listed in the following table are available.

Table 96. Local management

| Part number | Description                                                                                                                                                                                                                                                                                                   | Purpose                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4XF7B03892  | ThinkSystem SR650/a V4 Left Rack Latch with USB/MiniDP Option kit <ul style="list-style-type: none"> <li>• USB I/O board with Internal USB port</li> <li>• Front media bezel with USB ports and MiniDP port for SR650 V4</li> <li>• Front media bezel with USB ports and MiniDP port for SR650a V4</li> </ul> | Adds Internal USB 3 port (5 Gb/s), 2x Front USB 3 ports (5 Gb/s), MiniDP 1.1a video port<br><br>See the <a href="#">USB flash drive</a> section for supported USB drives for internal USB port. |
| 4XF7A99087  | ThinkSystem V4 Internal USB I/O Board Option kit <ul style="list-style-type: none"> <li>• USB I/O board with Internal USB port</li> </ul>                                                                                                                                                                     | Adds Internal USB 3 port (5 Gb/s) only.                                                                                                                                                         |

Configuration notes for field upgrades:

- If you order ThinkSystem 1U/2U V4 Front Media Bay Option kit (4X97A96850) for use in a server that already has the internal USB port installed (feature C1YR), the USB I/O board from the kit will not be needed as it is a duplicate of what is already installed.
- ThinkSystem 1U/2U V4 Front Media Bay Option kit (4X97A96850) is required to use XClarity Mobile, as described in the [System status with XClarity Mobile](#) section.

### External Diagnostics Handset

The SR650 V4 optionally has a port to connect an External Diagnostics Handset as described in the preceding section. The External Diagnostics Handset has the same functions as the Integrated Diagnostics Panel but has the advantages of not consuming space on the front of the server plus it can be shared among many servers in your data center. The handset has a magnet on the back of it to allow you to easily mount it on a convenient place on any rack cabinet.

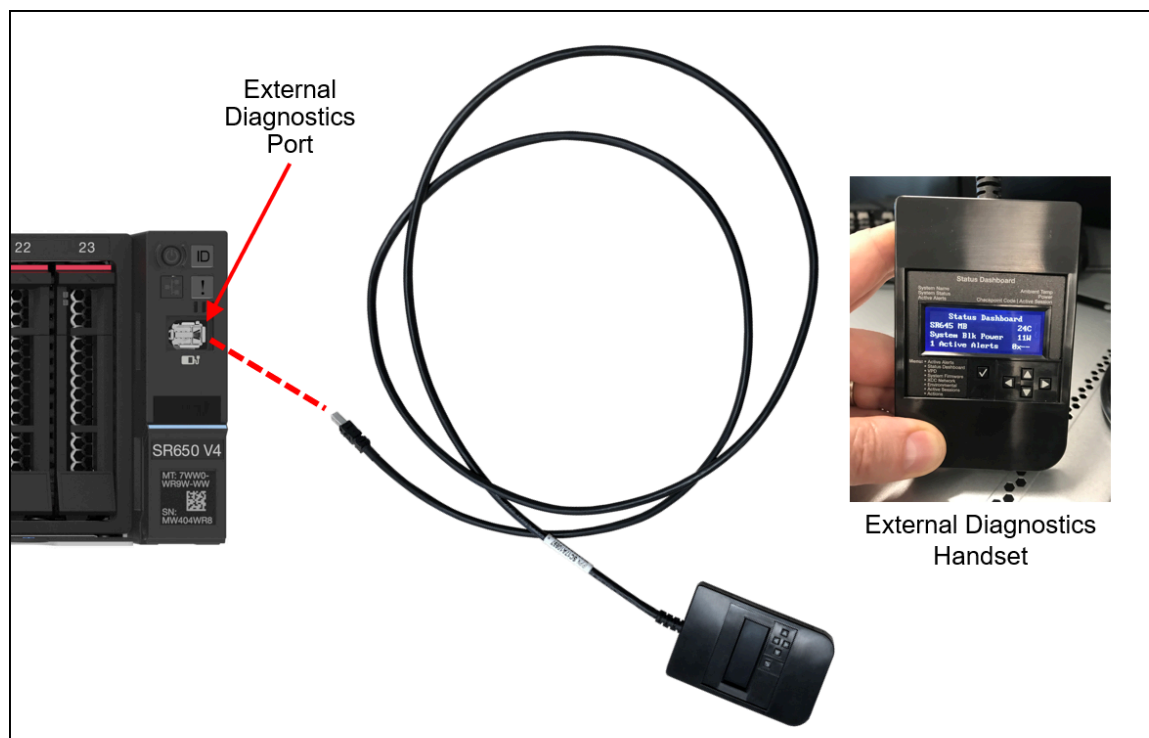


Figure 34. External Diagnostics Handset

The External Diagnostics Handset allows quick access to system status, firmware, network, and health information. The LCD display and the function buttons give you access to the following information:

- Active alerts
- Status Dashboard
- System VPD: machine type & mode, serial number, UUID string
- System firmware levels: UEFI and XCC firmware
- XCC network information: hostname, MAC address, IP address, DNS addresses
- Environmental data: Ambient temperature, CPU temperature, AC input voltage, estimated power consumption
- Active XCC sessions
- System reset action

Ordering information for the External Diagnostics Handset with is listed in the following table.

Table 97. External Diagnostics Handset ordering information

| Part number | Feature code | Description                              |
|-------------|--------------|------------------------------------------|
| 4TA7A64874  | BEUX         | ThinkSystem External Diagnostics Handset |

### Information pull-out tab

The front of the server also houses an information pull-out tab (also known as the network access tag). See [Figure 2](#) for the location. A label on the tab shows the network information (MAC address and other data) to remotely access the service processor.

### Light path diagnostics

The server offers light path diagnostics. If an environmental condition exceeds a threshold or if a system component fails, XCC lights LEDs inside the server to help you diagnose the problem and find the failing

part.

The server has fault LEDs next to the following components:

- Each memory DIMM
- Each drive bay
- Each power supply

### System status with XClarity One Mobile

The XClarity One Mobile app includes a tethering function where you can connect your Android or iOS device to the server via USB to see the status of the server.

Download the XClarity One Mobile app from these sites:

- [Google Play Store](#)
- [Apple App Store](#)

The steps to connect the mobile device to the server are as follows:

1. In the XCC web interface under **BMC Configuration > Network**, set the USB mode to **Shared mode: Owned by BMC**
2. Connect the mobile device via a USB cable to the server's USB port with the management symbol 
3. In iOS or Android settings, enable Personal Hotspot or USB Tethering
4. Launch the Lenovo XClarity Mobile app

Once connected you can see the following information:

- Server status including error logs (read only, no login required)
- Server management functions (XClarity login credentials required)

Configuration notes:

- The use of XClarity Mobile requires front USB ports. If your server doesn't already include front USB ports, order the field upgrade ThinkSystem SR650/a V4 Left Rack Latch with USB/MiniDP Option kit (4XF7B03892) as described in the [Local management](#) section

### Remote management

The server offers a dedicated RJ45 Ethernet port at the rear of the server for remote management via the XClarity Controller management processor. The port supports 10/100/1000 Mbps speeds.

Remote server management is provided through industry-standard interfaces:

- Intelligent Platform Management Interface (IPMI) Version 2.0
- Simple Network Management Protocol (SNMP) Version 3 (no SET commands; no SNMP v1)
- Common Information Model (CIM-XML)
- Representational State Transfer (REST) support
- Redfish support (DMTF compliant)
- Web browser - HTML 5-based browser interface (Java and ActiveX not required) using a responsive design (content optimized for device being used - laptop, tablet, phone) with NLS support

IPMI via the Ethernet port (IPMI over LAN) is supported, however it is disabled by default. For CTO orders you can specify whether you want the feature enabled or disabled in the factory, using the feature codes listed in the following table.

Table 98. IPMI-over-LAN settings

| Feature code | Description                     | Top Choice Express |
|--------------|---------------------------------|--------------------|
| B7XZ         | Disable IPMI-over-LAN (default) | TCE                |
| B7Y0         | Enable IPMI-over-LAN            | TCE                |

### Shared connectivity for remote management

To reduce the number of Ethernet connections needed for remote management, the SR650 V4 supports an adapter that installs in the OCP slot that allows four servers to share the one Ethernet connection. The adapter implements a 5-port Gigabit switch based on the Microchip KSZ9896 switch chip. Ordering information is listed in the following table.

Table 99. 4-to-1 Management Port Consolidation Adapter

| Part number | Feature code | Description                                                                                                                                                                                                                       |
|-------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4XC7A90299  | BZGE         | <p>ThinkSystem OCP 4 to 1 Management Port Consolidation Adapter</p> <p>Contains:</p> <ul style="list-style-type: none"> <li>• 1x OCP adapter</li> <li>• 2x 0.45m blue Cat5e cable</li> <li>• 1x 0.25m blue Cat5e cable</li> </ul> |

The adapter is installed in the OCP slot of one server, as shown in the figure below. The adapter connects to the remote management port of up to three other servers via the included (or other) Ethernet cables.

The OCP slot in the other three servers can be used for network connectivity, if desired.

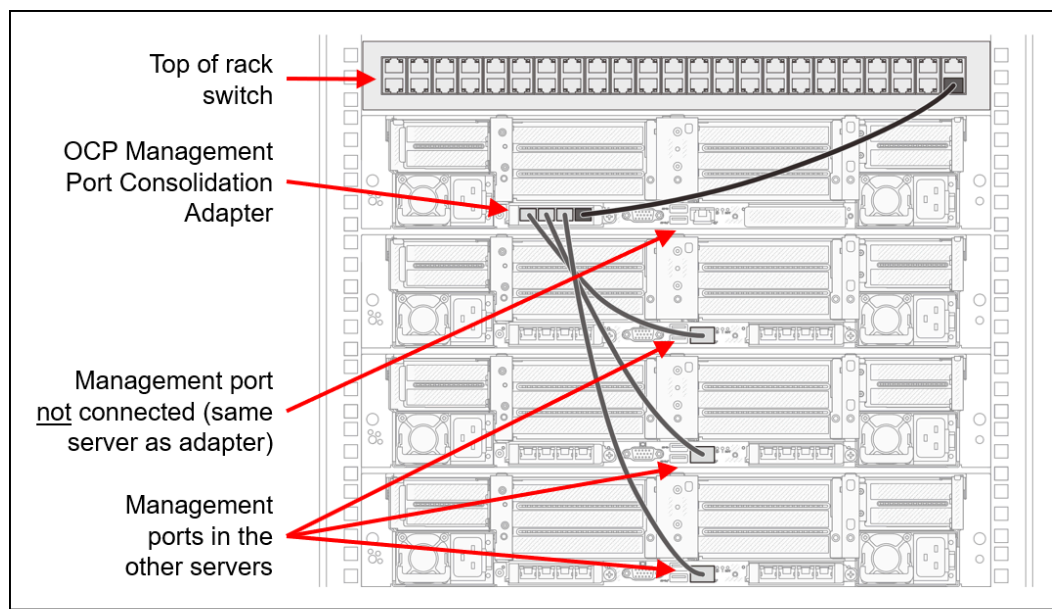


Figure 35. Connections to the ThinkSystem OCP 4 to 1 Management Port Consolidation Adapter

The adapter has four RJ45 ports. One port of the adapter connects to the local remote management port and the other three adapter ports connect to the remote management ports of three nearby servers. Either the included short Cat5e cables can be used or customer-supplied Cat5e Ethernet cables can be used.

Configuration notes:

- The adapter is only supported in OCP slot 1
- When the adapter is installed in slot 1, OCP slot 2 is disabled and cannot be used
- The OCP slot in the other three servers can be used for network connectivity, if desired.
- On the server with the adapter installed, in System Setup (F1 at boot), change **BMC settings > Network Settings > Network Interface port** to **Uplink MAC**.
- The connected servers can be any mix of servers that support the adapter. They do not all have to be SR650 V4 servers, for example.

For more information, see [https://pubs.lenovo.com/sr650-v4/multi-server\\_configuration](https://pubs.lenovo.com/sr650-v4/multi-server_configuration)

### MicroSD for XCC local storage

The server includes a MicroSD card port to enable the use of a MicroSD card for additional storage for use with the XCC controller. XCC can use the storage as a Remote Disc on Card (RDOC) device (up to 4GB of storage). It can also be used to store firmware updates (including N-1 firmware history) for ease of deployment.

**Tip:** Without a MicroSD card installed, the XCC controller will have 100MB of available RDOC storage.

Ordering information for the supported Micro SD cards is listed in the following table.

Table 100. Media for use with the MicroSD card port

| Part number | Feature code | Description                                         |
|-------------|--------------|-----------------------------------------------------|
| 4X77A92672  | C0BC         | ThinkSystem MicroSD 64GB Class 10 Flash Memory Card |

## USB flash drive

For general portable storage needs, the server also supports the USB flash drive option that is listed in the following table.

Table 101. USB memory key

| Part number | Feature | Description                              | Top Choice Express |
|-------------|---------|------------------------------------------|--------------------|
| 4X77A95465  | C44Q    | ThinkSystem USB 64GB USB 3.0 Flash Drive | Not TCE            |

## XCC3 Premier

The XCC3 service processor in the SR650 V4 supports an upgrade to the Premier level of features. XCC3 Premier in ThinkSystem V4 servers is equivalent to the XCC2 Premium offering in ThinkSystem V3 servers.

XCC3 Premier adds the following functions:

- System Guard - Monitor hardware inventory for unexpected component changes, and simply log the event or prevent booting
- Neighbor Group - Enables administrators to manage and synchronize configurations and firmware level across multiple servers
- Syslog alerting
- Lenovo SED security key management
- Boot video capture and crash video capture
- Virtual console collaboration - Ability for up to 6 remote users to be log into the remote session simultaneously
- Remote console
- System utilization data and graphic view
- Single sign on with Lenovo XClarity Administrator
- Update firmware from a repository
- Enterprise Strict Security mode - Enforces CNSA 1.0 level security
- Remotely viewing video with graphics resolutions up to 1600x1200 at 75 Hz with up to 23 bits per pixel, regardless of the system state
- Remotely accessing the server using the keyboard and mouse from a remote client
- International keyboard mapping support
- Redirecting serial console via SSH
- Component replacement log (Maintenance History log)
- Access restriction (IP address blocking)
- Displaying graphics for real-time and historical power usage data and temperature
- Mapping the ISO and image files located on the local client as virtual drives for use by the server
- Mounting the remote ISO and image files via HTTPS, SFTP, CIFS, and NFS
- Power capping
- License for XClarity Energy Manager

Ordering information is listed in the following table. XCC3 Premier is a software license upgrade - no additional hardware is required.

Table 102. XCC3 Premier license upgrade

| Part number | Feature code | Description                                 |
|-------------|--------------|---------------------------------------------|
| 7S0X000XWW  | SCY0         | Lenovo XClarity Controller 3 (XCC3) Premier |

With XCC3 Premier, for CTO orders, you can request that System Guard be enabled in the factory and the first configuration snapshot be recorded. To add this to an order, select feature code listed in the following table. The selection is made in the Security tab of the configurator.

Table 103. Enable System Guard in the factory (CTO orders)

| Feature code | Description          | Top Choice Express |
|--------------|----------------------|--------------------|
| BUT2         | Install System Guard | TCE                |

For more information about System Guard, see [https://pubs.lenovo.com/xcc3/nn1ia\\_c\\_systemguard](https://pubs.lenovo.com/xcc3/nn1ia_c_systemguard)

### Lenovo XClarity Provisioning Manager

Lenovo XClarity Provisioning Manager (LXPM) is a UEFI-based application embedded in ThinkSystem servers and accessible via the F1 key during system boot.

LXPM provides the following functions:

- Graphical UEFI Setup
- System inventory information and VPD update
- System firmware updates (UEFI and XCC)
- RAID setup wizard
- OS installation wizard (including unattended OS installation)
- Diagnostics functions

### Lenovo XClarity One

Lenovo XClarity One is a cloud-native unified Management-as-a-Service (MaaS) platform, engineered to modernize IT infrastructure. .

XClarity One is the next milestone in Lenovo's portfolio of systems management products. Now you can leverage the benefits of a true next-generation platform for the deployment, management, and maintenance of your infrastructure through a single, centralized platform that delivers a consistent user experience across all Lenovo products.

XClarity One's primary architectural value is divided into three key areas:

- How it predicts issues (AI/Analytics)
- How it securely connects (Secure Management Hubs),
- How it actually configures the hardware at scale (Template-Driven Lifecycle Control)

### AI-powered Automation

XClarity One harnesses the power of AI and predictive analytics to enhance the performance and reliability of your infrastructure with proactive protection.

- **AI-Powered Predictive Failure Analytics** - XC1 utilizes machine learning infrastructure to look at live device telemetries. The platform securely collects memory-diagnostic logs and pushes them into an active MPFA AI model infrastructure to analyze health patterns and predict probable hardware degradation before it manifests as a hard failure.
- **AI-Powered Call-Home** - A Call-Home serviceable event opens a support ticket automatically, leveraging AI technology (in the case of MPFA) for problem determination and fast resolution automatically dispatching parts and services, reducing service costs and minimizing downtime.

### Secure Management Hub

Lenovo's proprietary Management Hub is an on-premises virtual appliance that acts as the bridge between your infrastructure and the cloud.

- **On-Premises Security with Cloud Flexibility** - your infrastructure has no direct connection to the cloud, greatly reducing your attack surface from external threats while still having the deployment benefits, flexibility, and scalability of a cloud solution.

- **Authentication and Authorization** - built on a Zero Trust Architecture and requiring OTP Application authentication for all users to handle the support of all customers' servers and client devices. Role-based access controls help define and restrict permissions based on user roles.

### Template-Driven Lifecycle Control

Lenovo's unified management model eliminates fragmented server administration, allowing you to deploy, track, and maintain your entire infrastructure fleet from a single cloud-native dashboard.

- **Automated Configuration and Compliance:** Leverage scalable, repeatable device templates to rapidly deploy firmware updates, operating systems, and core baseboard management controller (BMC/UEFI) simultaneously across multiple sites. XClarity One continuously analyzes hardware compliance, shifting out-of-compliance systems to a critical status until a template re-deployment aligns them back to corporate standards.
- **Global Asset Visibility and Fleet Telemetry:** Centralize real-time tracking of component health, warranty status, and comprehensive hardware inventories. The platform aggregates live usage data, including memory allocation, power consumption, and processor thermal trends, giving you total operational awareness whether a server resides in a centralized data center or a remote edge environment.

Centralized management via Lenovo XClarity One can be integrated into your environment as an optional component. License information for XClarity One is listed in the following table.

Table 104. XClarity One license information

| Part number                                               | Feature code | Description                                         |
|-----------------------------------------------------------|--------------|-----------------------------------------------------|
| Starter licenses (limited to 10 devices per organization) |              |                                                     |
| 7S0X001EWW                                                | SF8Y         | XClarity One - Starter, Per Endpoint w/1 Yr SW S&S  |
| 7S0X001FWW                                                | SF8Z         | XClarity One - Starter, Per Endpoint w/2 Yr SW S&S  |
| 7S0X001GWW                                                | SF90         | XClarity One - Starter, Per Endpoint w/3 Yr SW S&S  |
| 7S0X001HWW                                                | SF91         | XClarity One - Starter, Per Endpoint w/4 Yr SW S&S  |
| 7S0X001JWW                                                | SF92         | XClarity One - Starter, Per Endpoint w/5 Yr SW S&S  |
| Standard licenses                                         |              |                                                     |
| 7S0X000LWW                                                | SCJC         | XClarity One - Standard, Per Endpoint w/1 Yr SW S&S |
| 7S0X001CWW                                                | SENW         | XClarity One - Standard, Per Endpoint w/2Yr SW S&S  |
| 7S0X000MWW                                                | SCJD         | XClarity One - Standard, Per Endpoint w/3 Yr SW S&S |
| 7S0X001DWW                                                | SENX         | XClarity One - Standard, Per Endpoint w/4Yr SW S&S  |
| 7S0X000NWW                                                | SCJE         | XClarity One - Standard, Per Endpoint w/5 Yr SW S&S |
| 7S0X000VWW                                                | SFQ1         | XClarity One - Standard, Per Endpoint w/6Yr SW S&S  |
| 7S0X000WWW                                                | SFQ2         | XClarity One - Standard, Per Endpoint w/7Yr SW S&S  |

Licensing notes:

- One license is needed per physical managed device.
- Starter license is functionally identical to Standard license but is limited to 10 devices per organization maximum.

For more information, see these resources:

- Lenovo XClarity One datasheet:  
<https://lenovopress.lenovo.com/ds0188-lenovo-xclarity-one>
- Lenovo XClarity One product guide:  
<https://lenovopress.lenovo.com/lp1992-lenovo-xclarity-one>
- Migrating to Lenovo XClarity One: An Overview:  
<https://lenovopress.lenovo.com/lp2429-migrating-to-lenovo-xclarity-one-an-overview>

## Lenovo XClarity Essentials

Lenovo offers the following XClarity Essentials software tools that can help you set up, use, and maintain the server at no additional cost:

- **Lenovo Essentials OneCLI**  
OneCLI is a collection of server management tools that uses a command line interface program to manage firmware, hardware, and operating systems. It provides functions to collect full system health information (including health status), configure system settings, and update system firmware and drivers.
- **Lenovo Essentials UpdateXpress**  
The UpdateXpress tool is a standalone GUI application for firmware and device driver updates that enables you to maintain your server firmware and device drivers up-to-date and help you avoid unnecessary server outages. The tool acquires and deploys individual updates and UpdateXpress System Packs (UXSPs) which are integration-tested bundles.
- **Lenovo Essentials Bootable Media Creator**  
The Bootable Media Creator (BOMC) tool is used to create bootable media for offline firmware update.

**ThinkSystem V4 servers:** The format of UEFI and BMC settings has changed for ThinkSystem V4 servers, to align with OpenBMC and Redfish requirements. See the documentation of these tools for details. As a result, the following versions are required for these servers:

- OneCLI 5.x or later
- UpdateXpress 5.x or later
- BOMC 14.x or later

For more information and downloads, visit the Lenovo XClarity Essentials web page:

<http://support.lenovo.com/us/en/documents/LNVO-center>

## Lenovo XClarity Energy Manager

Lenovo XClarity Energy Manager (LXEM) is a power and temperature management solution for data centers. It is an agent-free, web-based console that enables you to monitor and manage power consumption and temperature in your data center through the management console. It enables server density and data center capacity to be increased through the use of power capping.

LXEM is a licensed product. A single-node LXEM license is included with the XClarity Controller Premier upgrade as described in the [XCC3 Premier](#) section. If your server does not have the XCC Premier upgrade, Energy Manager licenses can be ordered as shown in the following table.

Table 105. Lenovo XClarity Energy Manager

| Part number | Description                                                               |
|-------------|---------------------------------------------------------------------------|
| 4L40E51621  | Lenovo XClarity Energy Manager Node License (1 license needed per server) |

For more information about XClarity Energy Manager, see the following resources:

- Lenovo Support page:  
<https://datacentersupport.lenovo.com/us/en/solutions/Invo-lxem>
- User Guide for XClarity Energy Manager:  
<https://pubs.lenovo.com/lxem/>

## Lenovo Capacity Planner

Lenovo Capacity Planner is a power consumption evaluation tool that enhances data center planning by enabling IT administrators and pre-sales professionals to understand various power characteristics of racks, servers, and other devices. Capacity Planner can dynamically calculate the power consumption, current, British Thermal Unit (BTU), and volt-ampere (VA) rating at the rack level, improving the planning efficiency for large scale deployments.

For more information, refer to the Capacity Planner web page:  
<http://datacentersupport.lenovo.com/us/en/solutions/Invo-lcp>

## Security

Topics in this section:

- [Security features](#)
- [Platform Firmware Resiliency - Lenovo ThinkShield](#)
- [Security standards](#)

### Security features

The SR650 V4 server offers the following electronic security features:

- Secure Boot function of the Intel Xeon processor
- Support for Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT) - see the [Platform Firmware Resiliency](#) section
- Firmware signature processes compliant with FIPS and NIST requirements
- System Guard (part of [XCC3 Premier](#)) - Proactive monitoring of hardware inventory for unexpected component changes
- Administrator and power-on password
- Integrated Trusted Platform Module (TPM) supporting TPM 2.0
- For China users, optional Nationz TPM 2.0 module
- Self-encrypting drives (SEDs) with support for enterprise key managers - see the [SED encryption key management](#) section

The server is NIST SP 800-147B compliant.

The SR650 V4 server also offers the following optional physical security features:

- Optional chassis intrusion switch
- Optional lockable front security bezel

The optional lockable front security bezel is shown in the following figure and includes a key that enables you to secure the bezel over the drives and system controls thereby reducing the chance of unauthorized or accidental access to the server.



Figure 36. Lockable front security bezel

The dimensions of the security bezel are:

- Width: 435 mm (17.1 in.)
- Height: 87 mm (3.4 in.)
- Width: 30 mm (1.2 in.)

The following table lists the security options for the SR650 V4.

Table 106. Security features

| Part number | Feature code | Description                                | Top Choice Express |
|-------------|--------------|--------------------------------------------|--------------------|
| 4X97B03890  | C3QT         | ThinkSystem SR650/a V4 Intrusion Cable kit | TCE                |
| 4XF7B03891  | C1S5         | 2U Bezel Option Kit                        | TCE                |
| CTO only    | C8UF         | ThinkSystem V4 PRC NationZ TPM 2.0 Module  | Not TCE            |

## Platform Firmware Resiliency - Lenovo ThinkShield

Lenovo's ThinkShield Security is a transparent and comprehensive approach to security that extends to all dimensions of our data center products: from development, to supply chain, and through the entire product lifecycle.

The ThinkSystem SR650 V4 includes Platform Firmware Resiliency (PFR) hardware Root of Trust (RoT) which enables the system to be NIST SP800-193 compliant. This offering further enhances key platform subsystem protections against unauthorized firmware updates and corruption, to restore firmware to an integral state, and to closely monitor firmware for possible compromise from cyber-attacks.

PFR operates upon the following server components:

- UEFI image – the low-level server firmware that connects the operating system to the server hardware
- XCC image – the management “engine” software that controls and reports on the server status separate from the server operating system
- FPGA image – the code that runs the server’s lowest level hardware controller on the motherboard

The Lenovo Platform Root of Trust Hardware performs the following three main functions:

- Detection – Measures the firmware and updates for authenticity
- Recovery – Recovers a corrupted image to a known-safe image
- Protection – Monitors the system to ensure the known-good firmware is not maliciously written

These enhanced protection capabilities are implemented using a dedicated, discrete security processor whose implementation has been rigorously validated by leading third-party security firms. Security evaluation results and design details are available for customer review – providing unprecedented transparency and assurance.

The SR650 V4 includes support for Secure Boot, a UEFI firmware security feature developed by the UEFI Consortium that ensures only immutable and signed software are loaded during the boot time. The use of Secure Boot helps prevent malicious code from being loaded and helps prevent attacks, such as the installation of rootkits. Lenovo offers the capability to enable secure boot in the factory, to ensure end-to-end

protection. Alternatively, Secure Boot can be left disabled in the factory, allowing the customer to enable it themselves at a later point, if desired.

The following table lists the relevant feature code(s).

Table 107. Secure Boot options

| Part number | Feature code | Description              | Top Choice Express | Purpose                                                                                              |
|-------------|--------------|--------------------------|--------------------|------------------------------------------------------------------------------------------------------|
| CTO only    | BPKQ         | TPM 2.0 with Secure Boot | TCE                | Configure the system in the factory with Secure Boot enabled.                                        |
| CTO only    | BPKR         | TPM 2.0                  | TCE                | Configure the system without Secure Boot enabled. Customers can enable Secure Boot later if desired. |

**Tip:** If Secure Boot is not enabled in the factory, it can be enabled later by the customer. However once Secure Boot is enabled, it cannot be disabled.

## Security standards

The SR650 V4 supports the following security standards and capabilities:

- **Industry Standard Security Capabilities**
  - Intel CPU Enablement
    - Intel Trust Domain Extensions (Intel TDX)
    - Intel Crypto Acceleration
    - Intel QuickAssist Software Acceleration
    - Intel Platform Firmware Resilience Support
    - Intel Control-Flow Enforcement Technology
    - Intel Total Memory Encryption - Multi Key
    - Intel Total Memory Encryption
    - Intel AES New Instructions (AES-NI)
    - Intel OS Guard
    - Execute Disable Bit (XD)
    - Intel Boot Guard
    - Mode-based Execute Control (MBEC)
    - Intel Virtualization Technology (VT-x)
    - Intel Virtualization Technology for Directed I/O (VT-d)
  - Microsoft Windows Security Enablement
    - Credential Guard
    - Device Guard
    - Host Guardian Service
  - TPM 2.0 (Trusted Platform Module 2.0)
  - UEFI (Unified Extensible Firmware Interface) Forum Secure Boot
- **Hardware Root of Trust and Security**
  - Independent security subsystem providing platform-wide NIST SP800-193 compliant Platform Firmware Resilience (PFR)
  - Management domain RoT provided by the Secure Boot feature of the AST2600 Baseboard Management Controller (BMC)
- **Platform Security**
  - Boot and run-time firmware integrity monitoring with rollback to known-good firmware (e.g., “self-healing”)

- Non-volatile storage bus security monitoring and filtering
- Resilient firmware implementation, such as to detect and defeat unauthorized flash writes or SMM (System Management Mode) memory incursions
- Patented IPMI KCS channel privileged access authorization (USPTO Patent# 11,256,810)
- Host and management domain authorization, including integration with CyberArk for enterprise password management
- KMIP (Key Management Interoperability Protocol) compliant, including support for IBM SKLM and Thales KeySecure
- Reduced “out of box” attack surface
- Configurable network services
- FIPS 140-3 (in progress) validated cryptography for XCC
- CNSA Suite 1.0 Quantum-resistant cryptography for XCC

For more information on platform security, see the paper “How to Harden the Security of your ThinkSystem Server and Management Applications” available from <https://lenovopress.com/lp1260-how-to-harden-the-security-of-your-thinksystem-server>.

- **Standards Compliance and/or Support**

- NIST SP800-131A rev 2 “Transitioning the Use of Cryptographic Algorithms and Key Lengths”
- NIST SP800-147B “BIOS Protection Guidelines for Servers”
- NIST SP800-193 “Platform Firmware Resiliency Guidelines”
- ISO/IEC 11889 “Trusted Platform Module Library”
- Common Criteria TCG Protection Profile for “PC Client Specific TPM 2.0”
- European Union Commission Regulation 2019/424 (“ErP Lot 9”) “Ecodesign Requirements for Servers and Data Storage Products” Secure Data Deletion
- Optional FIPS 140-2 validated Self-Encrypting Disks (SEDs) with external KMIP-based key management

- **Product and Supply Chain Security**

- Suppliers validated through Lenovo’s Trusted Supplier Program
- Developed in accordance with Lenovo’s Secure Development Lifecycle (LSDL)
- Continuous firmware security validation through automated testing, including static code analysis, dynamic network and web vulnerability testing, software composition analysis, and subsystem-specific testing, such as UEFI security configuration validation
- Ongoing security reviews by US-based security experts, with attestation letters available from our third-party security partners
- Digitally signed firmware, stored and built on US-based infrastructure and signed on US-based Hardware Security Modules (HSMs)
- TAA (Trade Agreements Act) compliant manufacturing, by default in Mexico for North American markets with additional US and EU manufacturing options
- US 2019 NDAA (National Defense Authorization Act) Section 889 compliant

## Rack installation

The following table lists the rack installation options that are available for the SR650 V4.

**Tip:** Long Travel rail kits are not supported with the SR650 V4.

For supported racks, see the [Rack cabinets](#) section.

Table 108. Rack installation options

| Part number                                                                                                        | Feature Code | Description                                                 | Top Choice Express | CMA support           |
|--------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------|--------------------|-----------------------|
| Rail kits                                                                                                          |              |                                                             |                    |                       |
| 4XF7A97379                                                                                                         | C2DG         | ThinkSystem Toolless Friction Rail V4                       | Not TCE            | No support            |
| 4XF7A97370                                                                                                         | C2DH         | ThinkSystem Toolless Slide Rail Kit V4                      | TCE                | Optional - 7M27A05698 |
| 4XF7A97374                                                                                                         | C3QS         | ThinkSystem Toolless Slide Rail Kit V4 with 2U CMA          | TCE                | Included              |
| 4XF7A97371                                                                                                         | C2DJ         | ThinkSystem Advanced Toolless Slide Rail Kit V4             | TCE                | Optional - 7M27A05698 |
| 4XF7A97376                                                                                                         | C3UF         | ThinkSystem Advanced Toolless Slide Rail Kit V4 with 2U CMA | Not TCE            | Included              |
| Cable management arm for field upgrades only (for CTO orders, use one of the above kits where the CMA is included) |              |                                                             |                    |                       |
| 7M27A05698                                                                                                         | -            | ThinkSystem 2U CMA Upgrade Kit for Toolless Slide Rail      | -                  | Yes                   |

See the Rail Kit comparison for the specifications of each rail kit:

<https://lenovopress.lenovo.com/lp1838-thinksystem-and-thinkedge-rail-kit-reference#availability=Available&sr650-v4-support=SR650%2520V4>

**Tip:** The Advanced rail kits add support for threaded mounting holes.

## Operating system support

The SR650 V4 supports the following operating systems:

- Microsoft Windows Server 2022
- Microsoft Windows Server 2025
- Red Hat Enterprise Linux 9.4
- Red Hat Enterprise Linux 9.5
- Red Hat Enterprise Linux 9.6
- Red Hat Enterprise Linux 9.7
- Red Hat Enterprise Linux 9.8
- Red Hat Enterprise Linux 10.0
- Red Hat Enterprise Linux 10.1
- Red Hat Enterprise Linux 10.2
- SUSE Linux Enterprise Server 15 SP6
- SUSE Linux Enterprise Server 15 SP7
- SUSE Linux Enterprise Server 16
- Ubuntu 22.04 LTS 64-bit
- Ubuntu 24.04 LTS 64-bit
- Ubuntu 26.04 LTS 64-bit
- VMware ESXi 8.0 U3
- VMware ESXi 9.0
- VMware ESXi 9.1

For a complete list of supported, certified and tested operating systems, plus additional details and links to relevant web sites, see the Operating System Interoperability Guide: <https://lenovopress.lenovo.com/osig>

For configure-to-order configurations, the SR650 V4 can be preloaded with VMware ESXi. Ordering information is listed in the following table.

Table 109. VMware ESXi preload

| Part number | Feature code | Description                            | Top Choice Express |
|-------------|--------------|----------------------------------------|--------------------|
| CTO only    | BZ97         | VMware ESXi 8.0 U3 (Factory Installed) | TCE                |
| CTO only    | C91V         | VMware ESXi 9.0 (Factory Installed)    | TCE                |
| CTO only    | CHGN         | VMware ESXi 9.1 (Factory Installed)    | Not TCE            |

Configuration rule:

- An ESXi preload cannot be selected if the configuration includes an NVIDIA GPU (ESXi preload cannot include the NVIDIA driver)

You can download supported VMware vSphere hypervisor images from the following web page and install it using the instructions provided:

[https://vmware.lenovo.com/content/custom\\_iso/](https://vmware.lenovo.com/content/custom_iso/)

## Configuring the server for Nutanix Compute Cluster support

The SR650 V4 supports Nutanix AHV to be installed by the customer. To help ensure that the server is a supported configuration, feature code CK17 can be selected in the DCSC configurator.

Table 110. Nutanix Compute-Only node

| Part number | Feature code | Description               |
|-------------|--------------|---------------------------|
| CTO only    | CK17*        | Nutanix Compute-Only node |

\* In DCSC, this feature code is listed in the Base tab.

Selecting CK17 in DCSC ensures that only components qualified by Nutanix can be selected for the server:

- Storage configuration - Only M.2 adapters and drives that are HX-validated can be selected. See the equivalent ThinkAgile HX product guide for specific M.2 adapter and drive support. No other storage can be selected.
- Memory configuration - A minimum of 64GB of memory per processor
- Networking - A minimum of one Ethernet adapter, 10GbE or higher, is required. The adapter can be OCP or PCIe form factor. Only adapters that are HX-validated (certain Broadcom or NVIDIA adapters) are supported. See the equivalent ThinkAgile HX product guide for specific adapter support.
- Lenovo XCC3 Premier (V4 servers) or XCC2 Platinum (V3 servers) upgrade is required
- IPMI-over-LAN is required to be enabled

Notes:

- Only Ethernet adapters can be selected as described above. GPUs, storage controllers (other than M.2 adapter), Fibre Channel adapters, and other non-Ethernet networking adapters cannot be selected.
- Only M.2 drives can be selected. Other internal drives (2.5-inch SSDs etc) cannot be selected
- Nutanix Acropolis and AHV preload is not offered on these servers. Use ThinkAgile HX for Nutanix-preload support.
- Acropolis OS install, upgrades, patching, and cluster lifecycle are managed by Nutanix tools, not Lenovo, and are the responsibility of the customer
- Nutanix certification information is available at this page:  
<https://portal.nutanix.com/page/documents/details?targetId=Lenovo-Compute-Server-HW-FW-Compatibility:len-lenovo-co-chassis-r.html>

## Physical and electrical specifications

The SR650 V4 has the following overall physical dimensions, excluding components that extend outside the standard chassis, such as EIA flanges, front security bezel (if any), and power supply handles:

- Width: 445 mm (17.5 inches)
- Height: 87 mm (3.4 inches)
- Depth: 796 mm (31.3 inches)

The following table lists the detailed dimensions. See the figure below for the definition of each dimension.

Table 111. Detailed dimensions

| Dimension | Description                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| 482 mm    | $X_a$ = Width, to the outsides of the front EIA flanges                                                                  |
| 435 mm    | $X_b$ = Width, to the rack rail mating surfaces                                                                          |
| 445 mm    | $X_c$ = Width, to the outer most chassis body feature                                                                    |
| 87 mm     | $Y_a$ = Height, from the bottom of chassis to the top of the chassis                                                     |
| 723 mm    | $Z_a$ = Depth, from the rack flange mating surface to the rearmost I/O port surface                                      |
| 761 mm    | $Z_b$ = Depth, from the rack flange mating surface to the rearmost feature of the chassis body                           |
| 761 mm    | $Z_c$ = Depth, from the rack flange mating surface to the rearmost feature such as power supply handle                   |
| 35 mm     | $Z_d$ = Depth, from the forwardmost feature on front of EIA flange to the rack flange mating surface                     |
| 51 mm     | $Z_e$ = Depth, from the front of security bezel (if applicable) or forwardmost feature to the rack flange mating surface |

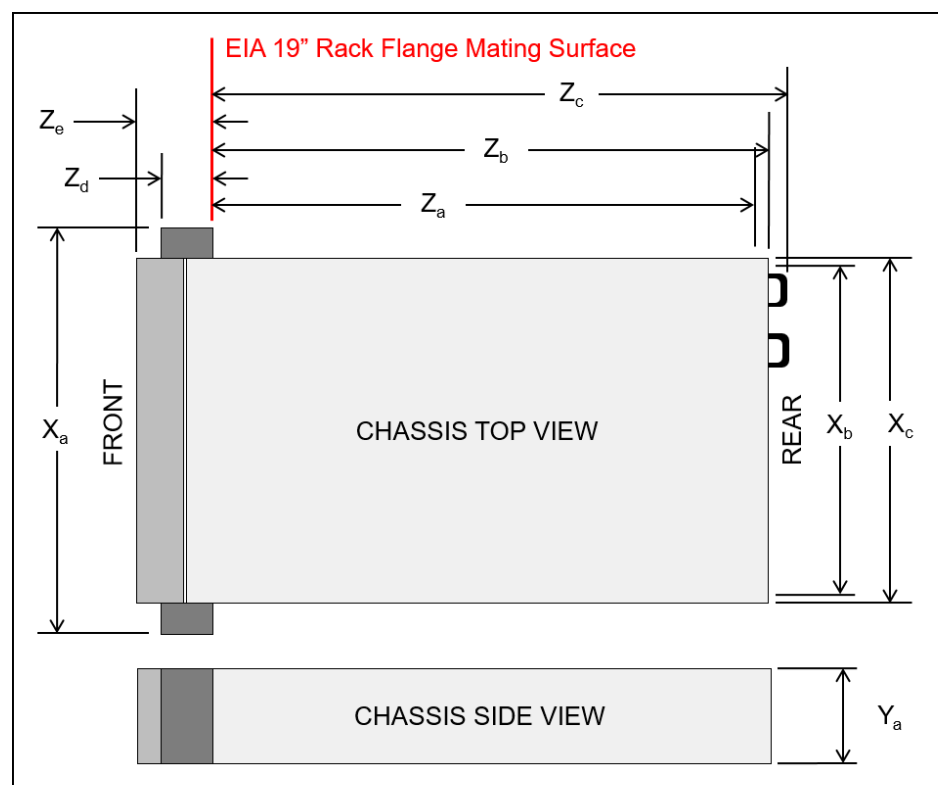


Figure 37. Server dimensions

The shipping (cardboard packaging) dimensions of the SR650 V4 are as follows:

- Width: 592 mm (23.3 inches)
- Height: 282 mm (11.1 inches)
- Depth: 992 mm (39.1 inches)

The server has the following weight:

- Maximum weight: 38.8 kg (85.5 lb)

The server has the following electrical specifications for AC input power supplies:

- Input voltage:
  - 100 to 127 (nominal) Vac, 50 Hz or 60 Hz
  - 200 to 240 (nominal) Vac, 50 Hz or 60 Hz
  - 180 to 300 Vdc (China only)
- Inlet current: see the following table.

Table 112. Maximum inlet current

| Part number                | Description                                                                 | 100V AC    | 200V AC | 240V DC |
|----------------------------|-----------------------------------------------------------------------------|------------|---------|---------|
| Titanium AC power supplies |                                                                             |            |         |         |
| 4P57A88687                 | ThinkSystem 800W 230V/115V Titanium CRPS Premium Hot-Swap Power Supply      | 9.3A       | 4.5A    | 4A      |
| 4P57A87056                 | ThinkSystem 800W 230V/115V Titanium CRPS Hot-Swap Power Supply v1.4         | 10A        | 5A      | 5A      |
| 4P57A88621                 | ThinkSystem 1300W 230V/115V Titanium CRPS Premium Hot-Swap Power Supply     | 11.4A      | 7.2A    | 6.2A    |
| 4P57A89421                 | ThinkSystem 1300W 230V/115V Titanium CRPS Premium Hot-Swap Power Supply DSC | 11.4A      | 7.2A    | 6.2A    |
| 4P57A87628                 | ThinkSystem 1300W 230V/115V Titanium CRPS Hot-Swap Power Supply v2.4        | 12A        | 8A      | 8A      |
| 4P57A88689                 | ThinkSystem 2000W 230V Titanium CRPS Premium Hot-Swap Power Supply          | No support | 11A     | 9.1A    |
| 4P57A92648                 | ThinkSystem 2000W 230V Titanium CRPS Premium Hot-Swap Power Supply v2       | No support | 11A     | 9.1A    |
| Platinum AC power supplies |                                                                             |            |         |         |
| 4P57A89306                 | ThinkSystem 800W 230V/115V Platinum CRPS Hot-Swap Power Supply v1.5         | 10A        | 5A      | 4.5A    |
| 4P57A89307                 | ThinkSystem 1300W 230V/115V Platinum CRPS Hot-Swap Power Supply v1.5        | 12A        | 8A      | 6.5A    |
| 4P57A88636                 | ThinkSystem 1300W 230V/115V Platinum CRPS Hot-Swap Power Supply v2.4        | 12A        | 8A      | 8A      |
| HVAC/HVDC power supplies   |                                                                             |            |         |         |
| 4P57A88627                 | ThinkSystem 1300W HVAC/HVDC Platinum CRPS Premium Hot-Swap Power Supply     | No support | 7.2A    | 6.2A    |

Electrical specifications for -48V DC input power supply, 4P57A88625:

- Input voltage: -48 to -60 Vdc
- Inlet current: 29.8 A

Electrical specifications for HVAC/HVDC power supply, 4P57A88627:

- Input voltage ranges:

- 200-277V AC single phase
  - 240-380V DC
- Inlet current:
  - AC: 7.2 A
  - DC: 6.2 A

## Operating environment

The SR650 V4 server complies with ASHRAE Class A2 specifications with most configurations, and depending on the hardware configuration, also complies with ASHRAE Class A3 and Class A4 specifications. System performance may be impacted when operating temperature is outside ASHRAE A2 specification.

Depending on the hardware configuration, the SR650 V4 server also complies with ASHRAE Class H1 specification. System performance may be impacted when operating temperature is outside ASHRAE H1 specification.

Topics in this section:

- [Ambient temperature requirements](#)
- [Temperature and humidity](#)
- [Acoustical noise emissions](#)
- [Shock and vibration](#)
- [Particulate contamination](#)

### Ambient temperature requirements

The restrictions to ASHRAE support are as follows.

#### Air-cooled configurations - ambient requirements

The restrictions to ASHRAE support are as follows (cooling by air):

- The ambient temperature must be no more than 35°C in any of the following configurations:
  - standard configurations with  $\geq 64$  GB DIMMs or  $> 205$  W processors
  - storage configurations without middle or rear drive bays
  - GPU configurations
  - configurations with special parts listed in [Thermal rules](#)
  - configurations with Intel Xeon 6725P processor
- The ambient temperature must be no more than 30°C in any of the following configurations:
  - configurations with processor 6732P
  - storage configurations with middle or rear drive bays
  - storage or GPU configurations with MRDIMMs or 256 GB 3DS RDIMMs
  - configurations with special parts listed in [Thermal rules](#)
  - some standard or GPU configurations with DPU adapter, which are specified in [Thermal rules](#)
- The ambient temperature must be no more than 25°C in any of the following configurations:
  - storage configurations with DPU adapter
  - 16 x 2.5" standard configurations with DPU adapter
  - 16 x 2.5" AnyBay GPU configurations with DPU adapter
  - storage configurations with a ConnectX-8 adapter used with ThinkSystem NDR/NDR200 QSFP112 IB Multi Mode Solo-Transceiver
  - 24 x 2.5" AnyBay GPU configurations with a ConnectX-8 adapter used with ThinkSystem NDR/NDR200 QSFP112 IB Multi Mode Solo-Transceiver
  - configurations with special parts listed in [Thermal rules](#)

#### Processor Neptune Core open-loop water configurations - ambient requirements

The restrictions to ASHRAE support are as follows (cooling by ThinkSystem V4 1U/2U Processor Neptune Core Module, feature C1XH):

- The ambient temperature must be no more than 30°C in any of the following configurations:
  - 36 NVMe configurations with standard fans
  - standard configurations with standard fans and MRDIMMs or 256 GB 3DS RDIMMs
  - GPU configurations with MRDIMMs or 256 GB 3DS RDIMMs
  - standard configurations with DPU adapter

- 8 x 2.5" AnyBay GPU configurations with DPU adapter
- GPU configurations with DPU adapter and 0–4 E3.S backplanes
- storage configurations with ConnectX-8 adapter
- The ambient temperature must be no more than 25°C in any of the following configurations:
  - storage configurations with DPU adapter
  - 16 x 2.5" AnyBay GPU configurations with DPU adapter
  - storage configurations with 12 x 3.5" front drive bays, standard fans, and MRDIMMs or 256 GB 3DS RDIMMs

### **Compute Complex Neptune Core Module open-loop water configurations - ambient requirements**

The restrictions to ASHRAE support are as follows (cooling by Compute Complex Neptune Core Module):

- The ambient temperature must be no more than 30°C in configurations with DPU adapter.

For additional information, see the Environmental specifications and Thermal rules sections in the product documentation:

[https://pubs.lenovo.com/sr650-v4/server\\_specifications\\_environmental](https://pubs.lenovo.com/sr650-v4/server_specifications_environmental)

[https://pubs.lenovo.com/sr650-v4/thermal\\_rules](https://pubs.lenovo.com/sr650-v4/thermal_rules)

### **Water requirements**

The SR650 V4 has the following requirements for open-loop liquid cooling:

- Maximum pressure: 3 bars
- Water inlet temperature and flow rates for Compute Complex Neptune Core Module:
  - 45°C (113°F) inlet temperature: 1 liter per minute (lpm) per server
- Water inlet temperature and flow rates for Processor Neptune Core Module:
  - 50°C (122°F) inlet temperature: 1.5 liters per minute (lpm) per server
  - 45°C (113°F) inlet temperature: 1 liter per minute (lpm) per server
  - 40°C (104°F) or lower inlet temperature: 0.5 liters per minute (lpm) per server

The water required to initially fill the system side cooling loop must be reasonably clean, bacteria-free water (<100 CFU/ml) such as de-mineralized water, reverse osmosis water, de-ionized water, or distilled water. The water must be filtered with an in-line 50 micron filter (approximately 288 mesh). The water must be treated with anti-biological and anti-corrosion measures. Environment quality must be maintain over the lifetime of the system to receive warranty and support on affecting components. For more information, see [Lenovo Neptune Direct Water-Cooling Standards](#).

### **Temperature and humidity**

The server is supported in the following environment:

- Air temperature:
  - Operating:
    - ASHRAE Class A2: 10°C to 35°C (50°F to 95°F); the maximum ambient temperature decreases by 1°C for every 300 m (984 ft) increase in altitude above 900 m (2,953 ft).
    - ASHRAE Class A3: 5°C to 40°C (41°F to 104°F); the maximum ambient temperature decreases by 1°C for every 175 m (574 ft) increase in altitude above 900 m (2,953 ft).
    - ASHRAE Class A4: 5°C to 45°C (41°F to 113°F); the maximum ambient temperature decreases by 1°C for every 125 m (410 ft) increase in altitude above 900 m (2,953 ft).
    - ASHRAE Class H1: 5 °C to 25 °C (41 °F to 77 °F); Decrease the maximum ambient temperature by 1°C for every 500 m (1640 ft) increase in altitude above 900 m (2,953 ft).
  - Server off: -10°C to 60°C (14°F to 140°F)
  - Shipment/storage: -40°C to 70°C (-40°F to 158°F)
- Maximum altitude: 3,050 m (10,000 ft)
- Relative Humidity (non-condensing):
  - Operating

- ASHRAE Class A2: 8% to 80%; maximum dew point: 21°C (70°F)
- ASHRAE Class A3: 8% to 85%; maximum dew point: 24°C (75°F)
- ASHRAE Class A4: 8% to 90%; maximum dew point: 24°C (75°F)
- ASHRAE Class H1: 8% to 80%; Maximum dew point: 17°C (63°F)
- Shipment/storage: 8% to 90%

## Acoustical noise emissions

The server has the following acoustic noise emissions declaration.

Table 113. Acoustic noise emissions declaration

| Acoustic performance @ 25°C ambient                                                                       | Configuration    | Min | Typical | Storage | GPU rich |
|-----------------------------------------------------------------------------------------------------------|------------------|-----|---------|---------|----------|
| Declared mean A-weighted sound power level, LWA,m (B)<br>Statistical adder for verification, Kv (B) = 0.4 | Idle mode        | 5.6 | 5.9     | 6.8     | 6.8      |
|                                                                                                           | Operating mode 1 | 5.9 | 6.2     | 6.8     | 7.3      |
|                                                                                                           | Operating mode 2 | 6.4 | 6.7     | 8.4     | 8.7      |
| Declared mean A-weighted emission sound pressure level, LpA,m (dB)<br>Bystander position                  | Idle mode        | 44  | 47      | 56      | 56       |
|                                                                                                           | Operating mode 1 | 47  | 50      | 56      | 62       |
|                                                                                                           | Operating mode 2 | 52  | 54      | 71      | 75       |

Modes:

- **Idle mode:** The steady-state condition in which the server is powered-on but not operating any intended function.
- **Operating mode 1:** The maximum acoustic output of 50% CPU TDP or active storage drives
- **Operating mode 2:** The maximum acoustical output of 100% CPU TDP or GPU TDP

The declared acoustic sound levels are based on the following configurations, which may change depending on configuration or conditions.

- **Min:** 2 x 250 W CPUs, 8 x 64 GB RDIMMs, 8 x 2.5" SAS HDDs, 1 x RAID 940-8i, 1 x ThinkSystem Broadcom 5719 1GbE RJ45 4-port OCP Ethernet Adapter, 2 x 1300 W PSUs
- **Typical:** 2 x 270 W CPUs, 16 x 64 GB RDIMMs, 16 x 2.5" SAS HDDs, 1 x RAID 940-16i, 2 x ThinkSystem Broadcom 57414 10/25GbE SFP28 2-port OCP Ethernet Adapter, 2 x 1300W PSUs
- **GPU rich:** 2 x 350 W CPUs, 2 x H100 NVL GPUs, 16 x 64 GB RDIMMs, 8 x 2.5" NVMe drives, 2 x ThinkSystem Broadcom 57508 100GbE QSFP56 2-Port OCP Ethernet Adapter, 2 x 2700W PSUs
- **Storage rich:** 2 x 270 W CPUs, 16 x 64 GB RDIMMs, 16 x 3.5" SAS HDDs, 1 x RAID 940-16i, 2 x ThinkSystem Broadcom 57414 10/25GbE SFP28 2-port OCP Ethernet Adapter, 2 x 1300 W PSUs

Notes:

- These sound levels were measured in controlled acoustical environments according to procedures specified by ISO7779 and are reported in accordance with ISO 9296.
- The declared sound levels may change depending on configuration/conditions.
- Government regulations (such as those prescribed by OSHA or European Community Directives) may govern noise level exposure in the workplace and may apply to you and your server installation. The actual sound pressure levels in your installation depend upon a variety of factors, including the number of racks in the installation; the size, materials, and configuration of the room; the noise levels from other equipment; the room ambient temperature, and employee's location in relation to the equipment. Further, compliance with such government regulations depends on a variety of additional factors, including the duration of employees' exposure and whether employees wear hearing protection. Lenovo recommends that you consult with qualified experts in this field to determine whether you are in compliance with the applicable regulations.

## Shock and vibration

The server has the following vibration and shock limits:

- Vibration:
  - Operating: 0.21 G rms at 5 Hz to 500 Hz for 15 minutes across 3 axes
  - Non-operating: 1.04 G rms at 2 Hz to 200 Hz for 15 minutes across 6 surfaces
- Shock:
  - Operating: 15 G for 3 milliseconds in each direction (positive and negative X, Y, and Z axes)
  - Non-operating:
    - 12 kg - 22 kg: 50 G for 152 in./sec velocity change across 6 surfaces
    - 23 kg - 31 kg: 35 G for 152 in./sec velocity change across 6 surfaces (3x GPU config, 2.5" config)
    - 32 kg - 68 kg: 35 G for 136 in./sec velocity change across 6 surfaces (16x 3.5" HDD config)

## Particulate contamination

Airborne particulates (including metal flakes or particles) and reactive gases acting alone or in combination with other environmental factors such as humidity or temperature might damage the system that might cause the system to malfunction or stop working altogether.

The following specifications indicate the limits of particulates that the system can tolerate:

- Reactive gases:
  - The copper reactivity level shall be less than 200 Angstroms per month ( $\text{\AA}/\text{month}$ )
  - The silver reactivity level shall be less than 200  $\text{\AA}/\text{month}$
- Airborne particulates:
  - The room air should be continuously filtered with MERV 8 filters.
  - Air entering a data center should be filtered with MERV 11 or preferably MERV 13 filters.
  - The deliquescent relative humidity of the particulate contamination should be more than 60% RH
  - Environment must be free of zinc whiskers

For additional information, see the Specifications section of the documentation for the server, available from the Lenovo Documents site, <https://pubs.lenovo.com/>

## Water infrastructure for the Lenovo Processor Neptune Core Module

The Lenovo Processor Neptune Core Module is the liquid-based processor cooling offering for the SR650 V4, as described in the [Lenovo Processor Neptune Core Module](#) section.

The open-loop cooling module requires the following water infrastructure components in the rack cabinet and data center:

- Supported 42U or 48U rack cabinet

The 42U or 48U Heavy Duty Rack Cabinet (machine types 7D6D or 7D6E) are supported. Two 0U mounting points are required for the water manifolds, at the rear of the rack cabinet, one either side.

For information about the 42U and 48U Heavy Duty Rack Cabinets, see the product guide: <https://lenovopress.lenovo.com/lp1498-lenovo-heavy-duty-rack-cabinets>

- 38-port water manifold (machine type 7DE6), installed in the rear of the rack cabinet

The manifold provides quick-disconnect couplings that each server in the rack are connected to. Ordering information is in the table below.

- Coolant distribution unit (CDU), either in-rack or in-row

In-rack CDUs are installed at the bottom of the rack cabinet.

In-row CDUs are separate cabinets that are typically installed at the end of a row of rack cabinets. Examples of suitable in-row CDUs include (but not limited to):

- Vertiv XDU450 CDU
- Vertiv XDU600 CDU
- Vertiv XDU100 CDU

- Hose kit to connect to the CDU to the manifold

Ordering information is in the table below.

The following figure shows the major components of the solution.

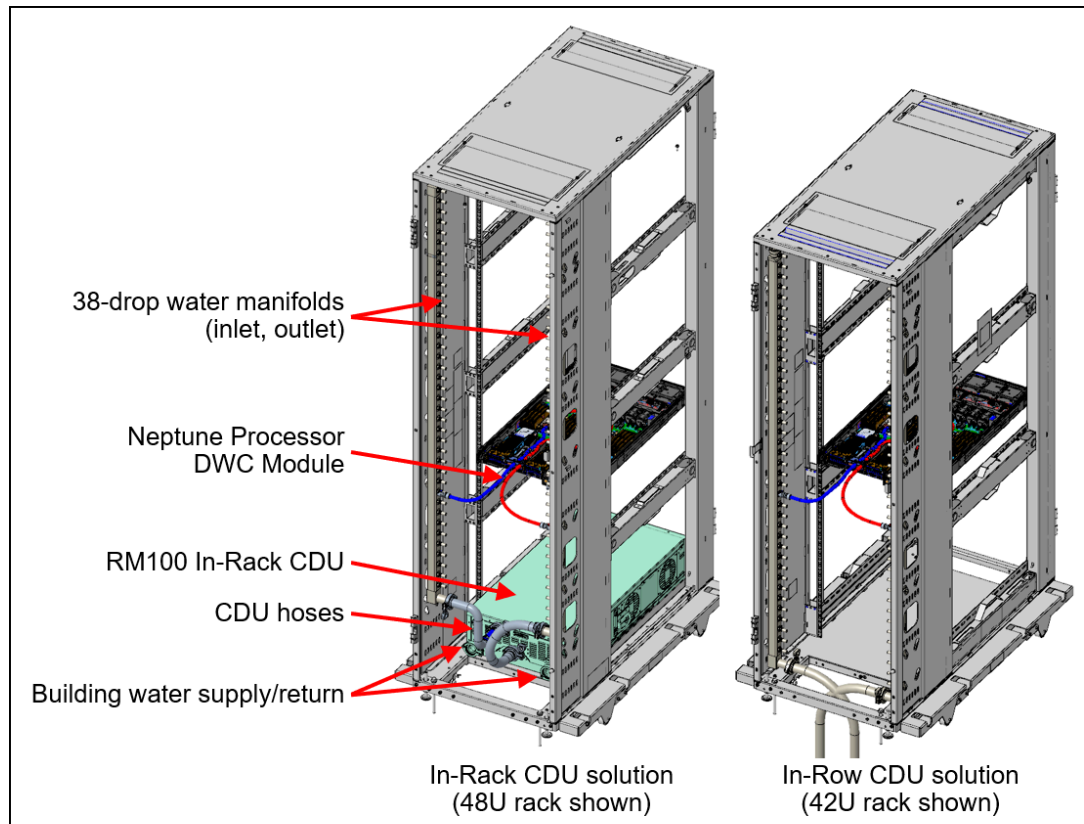


Figure 38. Water manifold connections

Configuration requirements:

- Maximum number of SR650 V4 servers supported in a rack:
  - 48U rack: 19 servers
  - 42U rack with in-rack CDU: 18 servers
  - 42U rack without in-rack CDU: 19 servers
- Inlet water flow rate:
  - 0.5 LPM: Maximum 40°C inlet water temperature
  - 1.0 LPM: Maximum 45°C inlet water temperature
  - 1.5 LPM: Maximum 50°C inlet water temperature
- Water pressure requirement:
  - Maximum operating node inlet pressure = 43.5 psi (3 bars)

**Note:** Water quality must be maintained over the lifetime of the system to receive warranty and support on affecting components. For water quality requirement, see [Lenovo Neptune Direct Water-Cooling Standards](#)

The 38-drop water manifold and hoses can be ordered as listed in the following table.

Table 114. Water infrastructure ordering information

| Part number                                     | Feature code | Description                               |
|-------------------------------------------------|--------------|-------------------------------------------|
| Manifold for 42U and 48U rack cabinet           |              |                                           |
| 4XF7A90061                                      | C5YW         | ThinkSystem Neptune 38-Port Rack Manifold |
| Hoses to connect the manifold to an in-rack CDU |              |                                           |

| Part number                                    | Feature code | Description                                                    |
|------------------------------------------------|--------------|----------------------------------------------------------------|
| 4XF7A90232                                     | C5YX         | Connection Set, 38/45 Ports Manifold with in-rack CDU          |
| 4XF7A90233                                     | C5YY         | Connection Set, 38-Port Manifold with in-rack CDU for 48U Rack |
| Hoses to connect the manifold to an in-row CDU |              |                                                                |
| 4XF7A90234                                     | C5YZ         | Hose Set, 1 inch EPDM, 1.3m, for Rack Manifold with in-row CDU |
| 4XF7A90235                                     | C5Z0         | Hose Set, 1 inch EPDM, 2.3m, for Rack Manifold with in-row CDU |

Configuration notes:

- This water connection solution described here cannot be used with the DW612S and N1380 enclosures as the water requirements are different.
- The hoses for in-row CDUs that are listed in the table above have Eaton FD83 quick-disconnect couplings

## Warranty upgrades and post-warranty support

The SR650 V4 has a 1-year or 3-year warranty based on the machine type of the system:

- 7DGC, - 1 year warranty
- 7DGD, 7DK2, 7DLN, - 3 year warranty

Our global network of regional support centers offers consistent, local-language support enabling you to vary response times and level of service\* to match the criticality of your support needs:

- **Standard Next Business Day** – Best choice for non-essential systems requiring simple maintenance.
- **Premier Next Business Day** – Best choice for essential systems requiring technical expertise from senior-level Lenovo engineers.
- **Premier 24x7 4-Hour Response** – Best choice for systems where maximum uptime is critical.

For more information, consult the brochure [Lenovo Operational Support Services for Data Centers Services](#).

\* Some service levels may not be available in all markets. Contact your sales representative for more information.

## Services

Lenovo Data Center Services empower you at every stage of your IT lifecycle. From expert advisory and strategic planning to seamless deployment and ongoing support, we ensure your infrastructure is built for success. Our comprehensive services accelerate time to value, minimize downtime, and free your IT staff to focus on driving innovation and business growth.

**Note:** Some service options may not be available in all markets or regions. For more information, go to <https://lenovocator.com/>. For information about Lenovo service upgrade offerings that are available in your region, contact your local Lenovo sales representative or business partner.

In this section:

- [Lenovo Advisory Services](#)
- [Lenovo Plan & Design Services](#)
- [Lenovo Deployment, Migration, and Configuration Services](#)
- [Lenovo Support Services](#)
- [Lenovo Managed Services](#)
- [Lenovo Sustainability Services](#)

### Lenovo Advisory Services

Lenovo Advisory Services simplify the planning process, enabling customers to build future-proofed strategies in as little as six weeks. Consultants provide guidance on projects including VM migration, storage, backup and recovery, and cost management to accelerate time to value, improve cost efficiency, and build a flexibly scalable foundation.

- **Assessment Services**

An Assessment helps solve your IT challenges through an onsite, multi-day session with a Lenovo technology expert. We perform a tools-based assessment which provides a comprehensive and thorough review of a company's environment and technology systems. In addition to the technology based functional requirements, the consultant also discusses and records the non-functional business requirements, challenges, and constraints. Assessments help organizations like yours, no matter how large or small, get a better return on your IT investment and overcome challenges in the ever-changing technology landscape.

- **Design Services**

Professional Services consultants perform infrastructure design and implementation planning to support your strategy. The high-level architectures provided by the assessment service are turned into low level designs and wiring diagrams, which are reviewed and approved prior to implementation. The implementation plan will demonstrate an outcome-based proposal to provide business capabilities through infrastructure with a risk-mitigated project plan.

### Lenovo Plan & Design Services

Unlock faster time to market with our tailored, strategic design workshops to align solution approaches with your business goals and technical requirements. Leverage our deep solution expertise and end-to-end delivery partnership to meet your goals efficiently and effectively.

### Lenovo Deployment, Migration, and Configuration Services

Optimize your IT operations by shifting labor-intensive functions to Lenovo's skilled technicians for seamless on-site or remote deployment, configuration, and migration. Enjoy peace of mind, faster time to value, and

comprehensive knowledge sharing with your IT staff, backed by our best-practice methodology.

- **Deployment Services for Storage and ThinkAgile**

A comprehensive range of remote and onsite options tailored specifically for your business needs to ensure your storage and ThinkAgile hardware are fully operational from the start.

- **Hardware Installation Services**

A full-range, comprehensive setup for your hardware, including unpacking, inspecting, and positioning components to ensure your equipment is operational and error-free for the most seamless and efficient installation experience, so you can quickly benefit from your investments.

- **DM/DG File Migration Services**

Take the burden of file migration from your IT's shoulders. Our experts will align your requirements and business objectives to the migration plans while coordinating with your team to plan and safely execute the data migration to your storage platforms.

- **DM/DG/DE Health Check Services**

Our experts perform proactive checks of your Firmware and system health to ensure your machines are operating at peak and optimal efficiency to maximize up-time, avoid system failures, ensure the security of IT solutions and simplify maintenance.

- **Factory Integrated Services**

A suite of value-added offerings provided during the manufacturing phase of a server or storage system that reduces time to value. These services aim at improving your hardware deployment experience and enhance the quality of a standard configuration before it arrives at your facility.

## **Lenovo Support Services**

In addition to response time options for hardware parts, repairs, and labor, Lenovo offers a wide array of additional support services to ensure your business is positioned for success and longevity. Our goal is to reduce your capital outlays, mitigate your IT risks, and accelerate your time to productivity.

Our commitment to ensuring the fastest, most seamless resolution times for mission-critical systems that require immediate attention to ensure minimal downtime and risk for your business. This service is only available for machines under the Premier 4-Hour Response SLA.

- **Premier Support for Data Centers**

Your direct line to the solution that promises the best, most comprehensive level of support to help you fully unlock the potential of your data center.

- **Premier Support Plus (PSP) for Servers 24x7 4-Hour Response**

Helping customers reduce unplanned downtime by moving from reactive support to proactive and preventative care

- **Premier Support Plus (PSP) for Storage 24x7 4-Hour Response**

Helping storage customers reduce unplanned downtime by moving from reactive support to proactive and preventative care

- **Committed Service Repair (CSR)**

Our commitment to ensuring the fastest, most seamless resolution times for mission-critical systems that require immediate attention to ensure minimal downtime and risk for your business. This service is only available for machines under the Premier 4-Hour Response SLA.

- **Multivendor Support Services (MVS)**

Your single point of accountability for resolution support across vast range of leading Server, Storage, and Networking OEMs, allowing you to manage all your supported infrastructure devices seamlessly from a single source.

- **Keep Your Drive (KYD)**

Protect sensitive data and maintain compliance with corporate retention and disposal policies to ensure your data is always under your control, regardless of the number of drives that are installed in your Lenovo server.

- **Technical Account Manager (TAM)**

Your single point of contact to expedite service requests, provide status updates, and furnish reports to track incidents over time, ensuring smooth operations and optimized performance as your business grows.

- **Enterprise Software Support (ESS)**

Gain comprehensive, single-source, and global support for a wide range of server operating systems and Microsoft server applications.

For more information, consult the brochure [Lenovo Operational Support Services for Data Centers](#).

## **Lenovo Managed Services**

Achieve peak efficiency, high security, and minimal disruption with Lenovo's always-on Managed Services. Our real-time monitoring, 24x7 incident response, and problem resolution ensure your infrastructure operates seamlessly. With quarterly health checks for ongoing optimization and innovation, Lenovo's remote active monitoring boosts end-user experience and productivity by keeping your data center's hardware performing at its best.

Lenovo Managed Services provides continuous 24x7 remote monitoring (plus 24x7 call center availability) and proactive management of your data center using state-of-the-art tools, systems, and practices by a team of highly skilled and experienced Lenovo services professionals.

Quarterly reviews check error logs, verify firmware & OS device driver levels, and software as needed. We'll also maintain records of latest patches, critical updates, and firmware levels, to ensure you systems are providing business value through optimized performance.

## **Lenovo Sustainability Services**

- **Asset Recovery Services**

Lenovo Asset Recovery Services (ARS) provides a secure, seamless solution for managing end-of-life IT assets, ensuring data is safely sanitized while contributing to a more circular IT lifecycle. By maximizing the reuse or responsible recycling of devices, ARS helps businesses meet sustainability goals while recovering potential value from their retired equipment. For more information, see the [Asset Recovery Services offering page](#).

- **CO2 Offset Services**

Lenovo's CO2 Offset Services offer a simple and transparent way for businesses to take tangible action on their IT footprint. By integrating CO2 offsets directly into device purchases, customers can easily support verified climate projects and track their contributions, making meaningful progress toward their sustainability goals without added complexity.

- **Lenovo Certified Refurbished**

Lenovo Certified Refurbished offers a cost-effective way to support IT circularity without compromising on quality and performance. Each device undergoes rigorous testing and certification, ensuring reliable performance and extending its lifecycle. With Lenovo's trusted certification, you gain peace of mind while making a more sustainable IT choice.

## **Lenovo TruScale**

Lenovo TruScale XaaS is your set of flexible IT services that makes everything easier. Streamline IT procurement, simplify infrastructure and device management, and pay only for what you use – so your business is free to grow and go anywhere.

Lenovo TruScale is the unified solution that gives you simplified access to:

- The industry's broadest portfolio – from pocket to cloud – all delivered as a service
- A single-contract framework for full visibility and accountability
- The global scale to rapidly and securely build teams from anywhere
- Flexible fixed and metered pay-as-you-go models with minimal upfront cost
- The growth-driving combination of hardware, software, infrastructure, and solutions – all from one single provider with one point of accountability.

For information about Lenovo TruScale offerings that are available in your region, contact your local Lenovo sales representative or business partner.

## Regulatory compliance

The SR650 V4 conforms to the following standards:

- ANSI/UL 62368-1
- IEC 62368-1 (CB Certificate and CB Test Report)
- CSA C22.2 No. 62368-1
- Argentina IEC 62368-1
- Mexico NOM-019
- India BIS 13252 (Part 1)
- Germany GS
- TUV-GS (EN62368-1, and EK1-ITB2000)
- Brazil INMETRO
- South Africa NRCS LOA
- Ukraine UkrCEPRO
- Morocco CMIM Certification (CM)
- Russia, Belorussia and Kazakhstan, TP EAC 037/2016 (for RoHS)
- Russia, Belorussia and Kazakhstan, EAC: TP TC 004/2011 (for Safety); TP TC 020/2011 (for EMC)
- CE, UKCA Mark (EN55032 Class A, EN62368-1, EN55035, EN61000-3-2, EN61000-3-3 or EN61000-3-11, EN61000-3-12) , (EU) 2019/424, and EN IEC 63000 (RoHS))
- FCC - Verified to comply with Part 15 of the FCC Rules, Class A
- Canada ICES-003, issue 7, Class A
- CISPR 32, Class A, CISPR 35
- Korea KS C 9832 Class A, KS C 9835
- Japan VCCI, Class A
- Taiwan BSMI CNS15936, Class A; CNS15598-1; Section 5 of CNS15663
- Australia/New Zealand AS/NZS CISPR 32, Class A; AS/NZS 62368.1
- UL Green Guard, UL2819
- [Energy Star 4.0](#)
- EPEAT (NSF/ ANSI 426) Bronze
- EPEAT 2.0
- Japanese Energy-Saving Act
- EU2019/424 Energy Related Product (ErP Lot9)
- TCO Certified
- China CCC certificate, GB17625.1; GB4943.1; GB/T9254
- China CECP certificate, CQC3135
- China CELP certificate, HJ 2507-2011

## External drive enclosures

The server supports attachment to external drive enclosures using a RAID controller with external ports or a SAS host bus adapter. Adapters supported by the server are listed in the [SAS adapters for external storage](#) section.

**Note:** Information provided in this section is for ordering reference purposes only. For the operating system and adapter support details, refer to the interoperability matrix for a particular storage enclosure that can be found on the Lenovo Data Center Support web site:

<http://datacentersupport.lenovo.com>

Table 115. External drive enclosures

| Model      | Description                                                                             |
|------------|-----------------------------------------------------------------------------------------|
| 7DAHCTO1WW | Lenovo ThinkSystem D4390 Direct Attached Storage (4U enclosure with 90x LFF drive bays) |

For details about supported drives, adapters, and cables, see the following Lenovo Press Product Guides:

- Lenovo ThinkSystem D4390  
<https://lenovopress.lenovo.com/lp1681>

## External storage systems

Lenovo offers the ThinkSystem DE Series, ThinkSystem DG Series, ThinkSystem DM Series and ThinkSystem DS Series external storage systems for high-performance storage. See the DE Series, DG Series and DM Series product guides for specific controller models, expansion enclosures and configuration options:

- ThinkSystem DE Series Storage  
<https://lenovopress.com/storage/thinksystem/de-series#rt=product-guide>
- ThinkSystem DS Series Storage  
<https://lenovopress.lenovo.com/storage/thinksystem/ds-series>
- ThinkSystem DM Series Storage  
<https://lenovopress.com/storage/thinksystem/dm-series#rt=product-guide>
- ThinkSystem DG Series Storage  
<https://lenovopress.com/storage/thinksystem/dg-series#rt=product-guide>

## External backup units

The following table lists the external backup options that are offered by Lenovo.

Table 116. External backup options

| Part number                          | Description                              |
|--------------------------------------|------------------------------------------|
| External SAS tape backup drives      |                                          |
| 6160S8E                              | IBM TS2280 Tape Drive Model H8S          |
| 6160S9E                              | IBM TS2290 Tape Drive Model H9S          |
| External SAS tape backup autoloaders |                                          |
| 6171S8R                              | IBM TS2900 Tape Autoloader w/LTO8 HH SAS |
| 6171S9R                              | IBM TS2900 Tape Autoloader w/LTO9 HH SAS |
| External tape backup libraries       |                                          |

| Part number                               | Description                                         |
|-------------------------------------------|-----------------------------------------------------|
| 6741B1F                                   | IBM TS4300 3U Tape Library Base Unit - Max 48U      |
| 6741B3F                                   | IBM TS4300 3U Tape Library Expansion Unit - Max 48U |
| SAS backup drives for TS4300 Tape Library |                                                     |
| 01KP937                                   | LTO 7 HH SAS Drive                                  |
| 01KP953                                   | LTO 8 HH SAS Drive                                  |
| 02JH836                                   | LTO 9 HH SAS Drive                                  |
| Full High 8 Gb Fibre Channel for TS4300   |                                                     |
| 01KP938                                   | LTO 7 FH Fibre Channel Drive                        |
| 01KP954                                   | LTO 8 FH Fibre Channel Drive                        |
| 02JH837                                   | LTO 9 FH Fibre Channel Drive                        |
| Half High 8 Gb Fibre Channel for TS4300   |                                                     |
| 01KP936                                   | LTO 7 HH Fibre Channel Drive                        |
| 01KP952                                   | LTO 8 HH Fibre Channel Drive                        |
| 02JH835                                   | LTO 9 HH Fibre Channel Drive                        |
| Half High 6 Gb SAS for TS4300             |                                                     |
| 01KP937                                   | LTO 7 HH SAS Drive                                  |
| 01KP953                                   | LTO 8 HH SAS Drive                                  |
| 02JH836                                   | LTO 9 HH SAS Drive                                  |

For more information, see the list of Product Guides in the Backup units category:

<https://lenovopress.com/servers/options/backup>

## Fibre Channel SAN switches

Lenovo offers the ThinkSystem DB Series of Fibre Channel SAN switches for high-performance storage expansion. See the DB Series product guides for models and configuration options:

- ThinkSystem DB Series SAN Switches:  
<https://lenovopress.com/storage/switches#rt=product-guide>

## Uninterruptible power supply units

The following table lists the uninterruptible power supply (UPS) units that are offered by Lenovo.

Table 117. Uninterruptible power supply units

| Part number                                  | Description                                   |
|----------------------------------------------|-----------------------------------------------|
| Rack-mounted or tower UPS units - 200-240VAC |                                               |
| 7DD5A002WW                                   | RT1.5kVA 2U Rack or Tower UPS-G2 (200-240VAC) |
| 7DD5A005WW                                   | RT3kVA 2U Rack or Tower UPS-G2 (200-240VAC)   |

For more information, see the list of Product Guides in the UPS category:

<https://lenovopress.com/servers/options/ups>

## Power distribution units

The following table lists the power distribution units (PDUs) that are offered by Lenovo.

Table 118. Power distribution units

| Part number                                                 | Feature code | Description                                                                      | ANZ | ASEAN | Brazil | EET | MEA | RUCIS | WE | HTK | INDIA | JAPAN | LA | NA | PRC |
|-------------------------------------------------------------|--------------|----------------------------------------------------------------------------------|-----|-------|--------|-----|-----|-------|----|-----|-------|-------|----|----|-----|
| <b>0U Basic PDUs</b>                                        |              |                                                                                  |     |       |        |     |     |       |    |     |       |       |    |    |     |
| 4PU7A93176                                                  | C0DA         | 0U 36 C13 and 6 C19 Basic 32A 1 Phase PDU                                        | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | N     | Y  | Y  | Y   |
| 4PU7A93177                                                  | C0D9         | 0U 24 C13/C15 and 24 C13/C15/C19 Basic 32A 3 Phase WYE PDU v2                    | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| <b>0U Switched and Monitored PDUs</b>                       |              |                                                                                  |     |       |        |     |     |       |    |     |       |       |    |    |     |
| 4PU7B08148                                                  | CD15         | 0U 3 C13 and 18 C19 Monitored 48A 3 Phase WYE PDU                                | N   | N     | N      | N   | N   | N     | N  | N   | N     | N     | N  | Y  | N   |
| 4PU7B08149                                                  | CD16         | 0U 3 C13 and 18 C19 Monitored 63A 3 Phase WYE PDU                                | N   | N     | N      | Y   | N   | N     | Y  | Y   | N     | N     | N  | N  | N   |
| 4PU7A93181                                                  | C0D5         | 0U 21 C13/C15 and 21 C13/C15/C19 Switched and Monitored 60A 3 Phase Delta PDU v2 | N   | Y     | N      | N   | Y   | N     | N  | Y   | N     | Y     | N  | Y  | N   |
| 4PU7A93178                                                  | C0D8         | 0U 20 C13 and 4 C19 Switched and Monitored 32A 1 Phase PDU                       | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | N     | Y  | Y  | Y   |
| 4PU7A93182                                                  | C0CS         | 0U 18 C13/C15 and 18 C13/C15/C19 Switched and Monitored 63A 3 Phase WYE PDU      | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| 4PU7A93180                                                  | C0D6         | 0U 18 C13/C15 and 18 C13/C15/C19 Switched and Monitored 32A 3 Phase WYE PDU v2   | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| 4PU7A93179                                                  | C0D7         | 0U 16 C13/C15 and 16 C13/C15/C19 Switched and Monitored 30A 1 Phase PDU v2       | N   | Y     | N      | N   | N   | N     | N  | Y   | N     | Y     | Y  | Y  | N   |
| <b>1U Basic PDUs</b>                                        |              |                                                                                  |     |       |        |     |     |       |    |     |       |       |    |    |     |
| 4PU7B12339                                                  | CE6L         | 1U Basic 16A 1 Phase PDU                                                         | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | N     | Y  | Y  | Y   |
| <b>1U Switched and Monitored PDUs</b>                       |              |                                                                                  |     |       |        |     |     |       |    |     |       |       |    |    |     |
| 4PU7A90809                                                  | C0DE         | 1U 18 C19/C13 Switched and monitored 48A 3P WYE PDU V2 CE                        | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | N  | Y   |
| 4PU7A90810                                                  | C0DD         | 1U 18 C19/C13 Switched and monitored 80A 3P Delta PDU V2                         | N   | N     | N      | N   | N   | N     | N  | Y   | N     | Y     | Y  | Y  | N   |
| 4PU7A90811                                                  | C0DC         | 1U 12 C19/C13 Switched and monitored 32A 3P WYE PDU V2                           | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| 4PU7A90812                                                  | C0DB         | 1U 12 C19/C13 Switched and monitored 60A 3P Delta PDU V2                         | N   | N     | Y      | N   | N   | N     | N  | Y   | N     | Y     | Y  | Y  | N   |
| <b>Line cords for 1U PDUs that ship without a line cord</b> |              |                                                                                  |     |       |        |     |     |       |    |     |       |       |    |    |     |
| 40K9611                                                     | 6504         | 4.3m, 32A/380-415V, EPDU/IEC 309 3P+N+G 3ph wye (non-US) Line Cord               | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| 40K9612                                                     | 6502         | 4.3m, 32A/230V, EPDU to IEC 309 P+N+G (non-US) Line Cord                         | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| 40K9613                                                     | 6503         | 4.3m, 63A/230V, EPDU to IEC 309 P+N+G (non-US) Line Cord                         | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |
| 40K9614                                                     | 6500         | 4.3m, 30A/208V, EPDU to NEMA L6-30P (US) Line Cord                               | Y   | Y     | Y      | Y   | Y   | Y     | Y  | Y   | Y     | Y     | Y  | Y  | Y   |

| Part number | Feature code | Description                                         | ANZ | ASEAN | Brazil | EET | MEA | RUCIS | WE | HTK | INDIA | JAPAN | LA | NA | PRC |
|-------------|--------------|-----------------------------------------------------|-----|-------|--------|-----|-----|-------|----|-----|-------|-------|----|----|-----|
| 40K9615     | 6501         | 4.3m, 60A/208V, EPDU to IEC 309 2P+G (US) Line Cord | N   | N     | Y      | N   | N   | N     | Y  | N   | N     | Y     | Y  | Y  | N   |

For more information, see the Lenovo Press documents in the PDU category:

<https://lenovopress.com/servers/options/pdu>

## Rack cabinets

The following table lists the supported rack cabinets.

Table 119. Rack cabinets

| Model      | Description                                                    |
|------------|----------------------------------------------------------------|
| 7D6DA007WW | ThinkSystem 42U Onyx Primary Heavy Duty Rack Cabinet (1200mm)  |
| 7D6DA008WW | ThinkSystem 42U Pearl Primary Heavy Duty Rack Cabinet (1200mm) |
| 7D6EA009WW | ThinkSystem 48U Onyx Primary Heavy Duty Rack Cabinet (1200mm)  |
| 7D6EA00AWW | ThinkSystem 48U Pearl Primary Heavy Duty Rack Cabinet (1200mm) |
| 1410O42    | Lenovo EveryScale 42U Onyx Heavy Duty Rack Cabinet             |
| 1410P42    | Lenovo EveryScale 42U Pearl Heavy Duty Rack Cabinet            |
| 1410O48    | Lenovo EveryScale 48U Onyx Heavy Duty Rack Cabinet             |
| 1410P48    | Lenovo EveryScale 48U Pearl Heavy Duty Rack Cabinet            |
| 93072RX    | 25U Standard Rack (1000mm)                                     |
| 93072PX    | 25U Static S2 Standard Rack (1000mm)                           |
| 93074RX    | 42U Standard Rack (1000mm)                                     |
| 93604PX    | 42U 1200mm Deep Dynamic Rack                                   |
| 93614PX    | 42U 1200mm Deep Static Rack                                    |
| 93634PX    | 42U 1100mm Dynamic Rack                                        |
| 93634EX    | 42U 1100mm Dynamic Expansion Rack                              |

For specifications about these racks, see the Lenovo Rack Cabinet Reference, available from:

<https://lenovopress.com/lp1287-lenovo-rack-cabinet-reference>

For more information, see the list of Product Guides in the Rack cabinets category:

<https://lenovopress.com/servers/options/racks>

## KVM console options

The following table lists the supported KVM consoles.

Table 120. KVM console

| Part number | Description                                              |
|-------------|----------------------------------------------------------|
| 4XF7A84188  | ThinkSystem 18.5" LCD console (with US English keyboard) |

The following table lists the available KVM switches and the options that are supported with them.

Table 122. KVM switches and options

| Part number                             | Description                                 |
|-----------------------------------------|---------------------------------------------|
| KVM Console switches                    |                                             |
| 1754A1X                                 | Local 1x8 Console Manager (LCM8)            |
| Cables for GCM and LCM Console switches |                                             |
| 46M5383                                 | Virtual Media Conversion Option Gen2 (VCO2) |
| 46M5382                                 | Serial Conversion Option (SCO)              |

For more information, see the list of Product Guides in the KVM Switches and Consoles category:  
<http://lenovopress.com/servers/options/kvm>

## Lenovo Financial Services

Why wait to obtain the technology you need now? No payments for 90 days and predictable, low monthly payments make it easy to budget for your Lenovo solution.

- **Flexible**

Our in-depth knowledge of the products, services and various market segments allows us to offer greater flexibility in structures, documentation and end of lease options.

- **100% Solution Financing**

Financing your entire solution including hardware, software, and services, ensures more predictability in your project planning with fixed, manageable payments and low monthly payments.

- **Device as a Service (DaaS)**

Leverage latest technology to advance your business. Customized solutions aligned to your needs. Flexibility to add equipment to support growth. Protect your technology with Lenovo's Premier Support service.

- **24/7 Asset management**

Manage your financed solutions with electronic access to your lease documents, payment histories, invoices and asset information.

- **Fair Market Value (FMV) and \$1 Purchase Option Leases**

Maximize your purchasing power with our lowest cost option. An FMV lease offers lower monthly payments than loans or lease-to-own financing. Think of an FMV lease as a rental. You have the flexibility at the end of the lease term to return the equipment, continue leasing it, or purchase it for the fair market value. In a \$1 Out Purchase Option lease, you own the equipment. It is a good option when you are confident you will use the equipment for an extended period beyond the finance term. Both lease types have merits depending on your needs. We can help you determine which option will best meet your technological and budgetary goals.

Ask your Lenovo Financial Services representative about this promotion and how to submit a credit application. For the majority of credit applicants, we have enough information to deliver an instant decision and send a notification within minutes.

## Seller training courses

The following sales training courses are offered for employees and partners (login required). Courses are listed in date order.

### 1. **Think AI Weekly: Lenovo and Sickbay: Enabling AI in healthcare through physiological data at scale**

2026-09-17 | 22 minutes | Employees and Partners

Please watch this session to hear Paul Cefaratti sharing some fascinating insights about enabling AI in healthcare through physiological data at scale.

Topics include:

- Customer challenges addressed by Sickbay.
- High-level overview and key use cases of Sickbay.
- Customer personas and ideal customer profile.
- Engagement strategies for sellers and example customer success stories.

Published: 2026-09-17

Length: 22 minutes

#### **Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Partner link: [Lenovo 360 Learning Center](#)

Course code: DTAIW185

### 2. **Partner Technical Webinar: Intel Xeon 6 & Ethernet NIC update**

2026-07-14 | 60 minutes | Employees and Partners

In this 60 minute replay, Madhu Bramharouthu and Raj Pathak from Intel presented the latest information on Xeon 6 processors and Intel Ethernet adapters.

Published: 2026-07-14

Length: 60 minutes

#### **Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Partner link: [Lenovo 360 Learning Center](#)

Course code: JUL1026

### 3. **Lenovo Data Center Sales Certification Exam Study Guide**

2026-07-13 | 10 minutes | Employees and Partners

This guide includes information to help candidates prepare and register for the Data Center Sales practice and certification exams.

Published: 2026-07-13

Length: 10 minutes

#### **Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Partner link: [Lenovo 360 Learning Center](#)

Course code: LENU-122C-SG

#### 4. **Lenovo Data Center Technical Certification Exam Study Guide**

2026-07-13 | 10 minutes | Employees and Partners

This guide includes information to help candidates prepare and register for the Data Center Technical practice and certification exams.

Published: 2026-07-13

Length: 10 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: LENU-322C-SG

#### 5. **Think AI Weekly: AI Inference Servers**

2026-06-23 | 25 minutes | Employees Only

Please watch this session to hear Traci Parker, WW Senior Solutions Marketing, as she walks us through the exciting AI Inference Servers.

Topics include:

- Why AI Inference Servers?
- Target use cases and industries
- Resources for customer conversations

Published: 2026-06-23

Length: 25 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Course code: DTAIW177

#### 6. **FY27Q1 Solutions Launch AI Inferencing Quick Hit**

2026-06-23 | 12 minutes | Employees and Partners

This QH covers the new enterprise-grade agentic AI with autonomous AI agents built on NVIDIA NemoClaw using the Lenovo AI Library. By enabling one-click deployment and built-in governance, customers can accelerate AI innovation, business outcomes and time-to-value. In addition, a new CPU-only Hybrid AI platform, 201, combines efficient CPU-based AI inference high-performance outcomes for predictable workloads, enabling enterprise scalability and flexibility to grow as AI demands evolve. This platform is built on the SR650 V4 server, powered by Intel Xeon 6 CPUs with NPUs. This server is one of the compute foundations for the new ThinkAgile hyperconverged systems aimed at AI environments. New services are also available for ThinkSystem storage systems.

Published: 2026-06-23

Length: 12 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: DAIQ102

**7. Partner Technical Webinar - TCE, DCSC, Enterprise Solutions, & LBPe ISG Financing**

2026-06-09 | 60 minutes | Employees and Partners

In this 60-minute replay, we received updates from George Scarbrough on TCE, Amit Chandok on the DCSC, Tracie Heath Garcia on LBPe ISG Financing, and Alejandro Perez Quintero on Enterprise IT Solutions.

Published: 2026-06-09

Length: 60 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: JUN0526

**8. Partner Technical Webinar - mAlnstream Volume Enterprise AI**

2026-05-11 | 45 minutes | Employees and Partners

In this 45 minute replay, Alex “Doc” Docherty, Lenovo AI Technology for NA Channel, discussed Mainstream and volume Servers for enterprise AI solutions. He talked about the need, the supply and some new programs for the channel.

Published: 2026-05-11

Length: 45 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: MAY0826

**9. Partner Technical Webinar - AI Servers**

2026-04-20 | 30 minutes | Employees and Partners

In this 30 minute replay, Mark Bica, Lenovo NA ISG Server Product Manager, presented the Lenovo AI Servers. Mark also gave an update on M.2 changes the latest with TCE (Top Choice Express).

Published: 2026-04-20

Length: 30 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: APR1726

#### 10. **Partner Technical Webinar - Server Review Racks, Towers & Edge**

2026-03-16 | 60 minutes | Employees and Partners

In this 60-minute replay, the Channel Solution Architect team provides an overview of the Lenovo ISG server portfolio. Eugene Grandberry, Ian Sloan, and Chris Laffey cover Towers, Racks, Edge, and TCE.

Published: 2026-03-16

Length: 60 minutes

##### **Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: MAR1326

#### 11. **ISG Smarter Tools, Faster Deals and Better Outcomes**

2026-02-27 | 40 minutes | Employees and Partners

In this 40-minute video, we review several tools and programs from Lenovo ISG to help our Partners' Data Center selling efforts: Smarter Tools / Faster Deals / Better Outcomes. We cover Top Choice Express, our premier program for supply optimized solutions. We next demonstrated DCSC, the Data Center Solution Configurator, Compete Tool for comparisons with our competition, and XClarity One, the next evolution of our XClarity Systems Management tool.

Tags: DataCenter Products

Published: 2026-02-27

Length: 40 minutes

##### **Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: FEB2426

#### 12. **ThinkSystem Rackmount and Tower Servers powered by AMD**

2026-02-23 | 25 minutes | Employees and Partners

This course presents the key products and features of the ThinkSystem Rackmount and Tower server family powered by AMD processors. It describes customer benefits and will help you recognize when a specific product should be selected.

By the end of this course, you should be able to:

- Identify products and features within the family
- Describe customer benefits offered by this family
- Recognize when a specific product should be selected

Last updated: 2/24/2026

Tags: Server, ThinkSystem

Published: 2026-02-23

Length: 25 minutes

##### **Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW1216r11

### 13. **OneIQ Overview and Demo**

2026-02-10 | 35 minutes | Employees and Partners

In this 35-minute video, Peter Grant, Head of Customer Experience for OneIQ, presents an overview and demo of OneIQ for Lenovo Partners.

Tags: Sales Tools

Published: 2026-02-10

Length: 35 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: ONEIQ101

### 14. **Family Portfolio: ThinkSystem Rack and Tower Servers Powered by Intel**

2026-02-04 | 25 minutes | Employees and Partners

This course is designed to give Lenovo sales and partner representatives a foundation of the ThinkSystem Intel Rackmount and Tower server family.

After completing this course, you will be able to:

- Identify products within this ThinkSystem server family
- Describe features of this family
- Recognize when a specific product should be selected

Tags: Server, ThinkSystem

Published: 2026-02-04

Length: 25 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW1204r15

### 15. **ThinkSystem Rack & Tower Server Primer - Module 2**

2025-12-17 | 45 minutes | Employees Only

This course builds on the foundational concepts from the prework modules, helping Client Managers deepen their understanding of ThinkSystem Rack and Tower Servers and apply that knowledge to confidently initiate value-driven customer conversations. Please Note: The videos in this course are taken from the Module 2 live session led by Tikiri Wandregala.

By the end of this course, you will be able to:

- Demonstrate the ability to initiate a data center discussion with a customer.
- Analyze and evaluate potential data center opportunities.
- Apply criteria to effectively qualify an opportunity.
- Locate and differentiate key resources and contacts.

Tags: DataCenter Products, ThinkSystem

Published: 2025-12-17

Length: 45 minutes

#### **Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Course code: DSRTB101

### 16. **Partner Technical Webinar - Mainstream CPUs**

2025-09-29 | 40 minutes | Employees and Partners

In this 40-minute replay, Brad Violand, Lenovo ISG Category team, reviewed the strategy to optimize the CPU portfolio and ensuing changes to DCSC planned for October 1, 2025.

Tags: Server

Published: 2025-09-29

Length: 40 minutes

#### **Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SEP2625

### 17. **Partner Technical Webinar - ISG Portfolio - Wayne's World**

2025-09-25 | 60 minutes | Employees and Partners

In this 60-minute replay, Wayne Pecht, Lenovo Channel SA, shared his extensive experience and unique approach to the ISG Portfolio "Wayne's World for Lenovo ISG"!

Tags: Channel, Technical Sales

Published: 2025-09-25

Length: 60 minutes

#### **Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SEP1925

**18. Think AI Weekly: ISG & SSG Better Together: Uniting AI Solutions & Services for Smarter Outcomes**

2025-08-01 | 55 minutes | Employees Only

View this session to hear from our speakers Allen Holmes, AI Technologist, ISG and Balaji Subramaniam, AI Regional Leader-Americas, SSG.

Topics include:

- An overview of ISG & SSG AI CoE Offerings with Customer Case Studies
- The Enterprise AI Deal Engagement Flow with ISG and SSG
- How sellers can leverage this partnership to differentiate with Enterprise clients.
- NEW COURSE: From Inception to Execution: Evolution of an AI Deal

Tags: Artificial Intelligence (AI), Sales, Services, Technology Solutions, TruScale Infrastructure as a Service

Published: 2025-08-01

Length: 55 minutes

**Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Course code: DTAIW145

**19. VTT-HPC: Explaining HPC Benchmarking: Processes, Standards and Best Practices**

2025-07-30 | 87 minutes | Employees Only

Please join us as Eric Michel, HPC & AI Performance Team Coordinator and Kevin Dean, Sr Manager, HPC Performance, Projection & Operations speak with us about HPC Benchmarking.

Topics will include:

- Why sellers need to request benchmarks
- How to request benchmarks
- How the team performs a benchmark
- The performance database
- Demonstration of the tool

Tags: Advanced Data Center, High Performance Computing (HPC), Technical Sales

Published: 2025-07-30

Length: 87 minutes

**Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Course code: DVHPC224

## 20. **Family Portfolio: Server Component Options**

2025-07-21 | 20 minutes | Employees and Partners

This course covers opportunities that are available to upsell server components. By upselling the correct components, you can increase the size and revenue of the original sale.

After completing this course, you should be able to:

- Identify the opportunities for an upsell
- Recognize when specific products should be discussed
- Emphasize how a product benefits the customer
- Identify keywords that indicate upsell opportunities

Tags: Server, Storage

Published: 2025-07-21

Length: 20 minutes

### **Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW1109r2

## 21. **ThinkSystem Rack and Tower Introduction for ISO Client Managers**

2025-06-16 | 20 minutes | Employees Only

In this course, you will learn about Lenovo's Data Center Portfolio, its ThinkSystem Family and the key features of the Rack and Tower servers. It will equip you with foundational knowledge which you can then expand upon by participating in the facilitated session of the curriculum.

Tags: Server, ThinkSystem

Published: 2025-06-16

Length: 20 minutes

### **Start the training:**

Employee link: [Grow@Lenovo](#)

Course code: DSRT0101r2\_JP

## 22. **VTT HPC: AI and the Impact on the Environment**

2025-06-11 | 58 minutes | Employees Only

Please join us as Matthew Ziegler, Director of Lenovo Neptune and Sustainability speaks with us about AI and the Impact on the Environment.

Topics will include:

- Why is ESG essential for your customer?
- How to find and read an eco declaration
- What is a product carbon footprint?
- Demo of the Lenovo Capacity Planner

Tags: Advanced DataCenter, Artificial Intelligence (AI), Environmental Social Governance (ESG), High-Performance Computing (HPC), Server

Published: 2025-06-11

Length: 58 minutes

### **Start the training:**

Employee link: [Grow@Lenovo](#)

Course code: DVHPC223

## 23. **Lenovo Data Center Product Portfolio**

2025-06-11 | 20 minutes | Employees and Partners

This course introduces the Lenovo data center portfolio, and covers servers, storage, storage networking, and software-defined infrastructure products. After completing this course about Lenovo data center products, you will be able to identify product types within each data center family, describe Lenovo innovations that this product family or category uses, and recognize when a specific product should be selected.

Course objectives:

1. Identify product types within each data center family
2. Describe the features of the product family or category
3. Recognize when a specific product should be selected

Tags: Advanced DataCenter, DataCenter Products, Server, ThinkAgile, ThinkEdge, ThinkSystem

Published: 2025-06-11

Length: 20 minutes

### **Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW1110r8

**24. Lenovo Cloud Architecture VTT: Supercharge Your Enterprise AI with NVIDIA AI Enterprise on Lenovo Hybrid AI Platform**

2025-04-17 | 75 minutes | Employees and Partners

Join us for an in-depth webinar with Justin King, Principal Product Marketing Manager for Enterprise AI exploring the power of NVIDIA AI Enterprise, delivering Generative and Agentic AI outcomes deployed with Lenovo Hybrid AI platform environments.

In today's data-driven landscape, AI is evolving at high speed, with new techniques delivering more accurate responses. Enterprises are seeking not just an understanding but also how they can achieve AI-driven business outcomes.

With this, the demand for secure, scalable, and high-performing AI operations-and the skills to deliver them-is top of mind for many. Learn how NVIDIA AI Enterprise, a comprehensive software suite optimized for NVIDIA GPUs, provides the tools and frameworks, including NVIDIA NIM, NeMo, and Blueprints, to accelerate AI development and deployment while reducing risk-all within the control and security of your Lenovo customer's hybrid AI environment.

Tags: Artificial Intelligence (AI), Cloud, Data Management, Nvidia, Technical Sales

Published: 2025-04-17

Length: 75 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: DVCLD221

**25. Lenovo VTT Cloud Architecture: Intel Xeon 6: Performance and Efficiency**

2025-03-05 | 60 minutes | Employees and Partners

Intel Xeon 6 processors introduce a robust computing platform that excels at both performance and efficiency, crucial for meeting the evolving demands of modern data centers. Addressing the broadest array of uses and workloads, from compute-intensive HPC & AI to traditional enterprise apps to scalable cloud-native applications, Intel Xeon 6 provides versatility for diverse operational and workload requirements.

Please join our guest speakers, Bhanu Jaiswal, Intel Product Line Manager and Rakib Sarwar, Intel Technical Sales Manager as they discuss Intel's latest family of processors.

Tags: Artificial Intelligence (AI), Server, Technical Sales, ThinkAgile, ThinkSystem

Published: 2025-03-05

Length: 60 minutes

**Start the training:**

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: DVCLD220

## 26. Family Portfolio: Storage Controller Options

2025-03-03 | 25 minutes | Employees and Partners

This course covers the storage controller options available for use in Lenovo servers. The classes of storage controller are discussed, along with a discussion of where they are used, and which to choose.

After completing this course, you will be able to:

- Describe the classes of storage controllers
- Discuss where each controller class is used
- Describe the available options in each controller class

Tags: Sales, Storage

Published: 2025-03-03

Length: 25 minutes

### Start the training:

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW1111r2

## 27. Q4 Solutions Launch ThinkSystem SR630 V4, SR650 V4, SR650a V4 Update Quick Hit

2025-02-12 | 5 minutes | Employees and Partners

This Quick Hit covers 3 new Lenovo servers – the ThinkSystem SR630 V4, SR650 V4, and SR650a V4 - designed to drive today's workloads and elevate data centers of any size to AI-enabled powerhouses. With designed-in and proven reliability, ThinkSystem V4 servers with Intel® Xeon 6 processors deliver a secure, AI-ready, scalable data center building block. Your customers will enjoy efficient, high-performing AI and other compute-intensive app execution with any one of these servers.

Tags: Cloud, Server, Storage, ThinkSystem, XClarity

Published: 2025-02-12

Length: 5 minutes

### Start the training:

Employee link: [Grow@Lenovo](#)

Partner link: [Lenovo 360 Learning Center](#)

Course code: SXXW1204r14a

**28. ThinkSystem Rack and Tower Introduction for ISO Client Managers**

2024-12-10 | 20 minutes | Employees Only

In this course, you will learn about Lenovo's Data Center Portfolio, its ThinkSystem Family and the key features of the Rack and Tower servers. It will equip you with foundational knowledge which you can then expand upon by participating in the facilitated session of the curriculum.

Course Objectives:

- By the end of this course, you should be able to:
- Identify Lenovo's main data center brands.
- Describe the key components of the ThinkSystem Family servers.
- Differentiate between the Rack and Tower servers of the ThinkSystem Family.
- Understand the value Rack and Tower servers can provide to customers.

Tags: Server, ThinkSystem

Published: 2024-12-10

Length: 20 minutes

**Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Course code: DSRT0101r2

**29. SAP Webinar for Lenovo Sellers: Lenovo Portfolio Update for SAP Landscapes**

2024-06-04 | 60 minutes | Employees Only

Join Mark Kelly, Advisory IT Architect with the Lenovo Global SAP Center of Competence as he discusses:

- Challenges in the SAP environment
- Lenovo On-premise Solutions for SAP
- Lenovo support resources for SAP solutions

Tags: SAP, ThinkAgile, ThinkEdge, ThinkSystem

Published: 2024-06-04

Length: 60 minutes

**Start the training:**

Employee link: [Grow@Lenovo](mailto:Grow@Lenovo)

Course code: DSAPF101

## Related publications and links

For more information, see these resources:

- Lenovo SR650 V4 product page:  
<https://www.lenovo.com/us/en/p/servers-storage/servers/racks/thinksystem-sr650-v4/LEN21TS0042>
- ThinkSystem SR650 V4 datasheet  
<https://lenovopress.lenovo.com/datasheet/ds0194-lenovo-thinksystem-sr650-v4>
- ThinkSystem SR650 V4 drivers and support  
<http://datacentersupport.lenovo.com/products/servers/thinksystem/sr650v4/7dgd/downloads>
- Lenovo ThinkSystem SR650 V4 product publications:  
<https://pubs.lenovo.com/sr650-v4/>
  - User Guide, which includes:
    - System Configuration Guide
    - Hardware Maintenance Guide
  - Rack Installation Guides
  - Messages and Codes Reference
  - UEFI Manual for ThinkSystem Servers
- SR650 V4 hardware repair & replacement videos:  
<https://www.youtube.com/playlist?list=PLYV5R7hVcs-BjeTQPScjJcBe8degdSAXG>
- User Guides for options:  
<https://serveroption.lenovo.com>
- ServerProven hardware compatibility:  
<http://serverproven.lenovo.com>

## Related product families

Product families related to this document are the following:

- [2-Socket Rack Servers](#)
- [ThinkSystem SR650 V4 Server](#)

## Notices

Lenovo may not offer the products, services, or features discussed in this document in all countries. Consult your local Lenovo representative for information on the products and services currently available in your area. Any reference to a Lenovo product, program, or service is not intended to state or imply that only that Lenovo product, program, or service may be used. Any functionally equivalent product, program, or service that does not infringe any Lenovo intellectual property right may be used instead. However, it is the user's responsibility to evaluate and verify the operation of any other product, program, or service. Lenovo may have patents or pending patent applications covering subject matter described in this document. The furnishing of this document does not give you any license to these patents. You can send license inquiries, in writing, to:

Lenovo (United States), Inc.  
8001 Development Drive  
Morrisville, NC 27560  
U.S.A.  
Attention: Lenovo Director of Licensing

LENOVO PROVIDES THIS PUBLICATION "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Some jurisdictions do not allow disclaimer of express or implied warranties in certain transactions, therefore, this statement may not apply to you.

This information could include technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in new editions of the publication. Lenovo may make improvements and/or changes in the product(s) and/or the program(s) described in this publication at any time without notice.

The products described in this document are not intended for use in implantation or other life support applications where malfunction may result in injury or death to persons. The information contained in this document does not affect or change Lenovo product specifications or warranties. Nothing in this document shall operate as an express or implied license or indemnity under the intellectual property rights of Lenovo or third parties. All information contained in this document was obtained in specific environments and is presented as an illustration. The result obtained in other operating environments may vary. Lenovo may use or distribute any of the information you supply in any way it believes appropriate without incurring any obligation to you.

Any references in this publication to non-Lenovo Web sites are provided for convenience only and do not in any manner serve as an endorsement of those Web sites. The materials at those Web sites are not part of the materials for this Lenovo product, and use of those Web sites is at your own risk. Any performance data contained herein was determined in a controlled environment. Therefore, the result obtained in other operating environments may vary significantly. Some measurements may have been made on development-level systems and there is no guarantee that these measurements will be the same on generally available systems. Furthermore, some measurements may have been estimated through extrapolation. Actual results may vary. Users of this document should verify the applicable data for their specific environment.

**© Copyright Lenovo 2026. All rights reserved.**

This document, LP2127, was created or updated on September 12, 2026.

Send us your comments in one of the following ways:

- Use the online Contact us review form found at:  
<https://lenovopress.lenovo.com/LP2127>
- Send your comments in an e-mail to:  
[comments@lenovopress.com](mailto:comments@lenovopress.com)

This document is available online at <https://lenovopress.lenovo.com/LP2127>.

## Trademarks

Lenovo and the Lenovo logo are trademarks or registered trademarks of Lenovo in the United States, other countries, or both. A current list of Lenovo trademarks is available on the Web at <https://www.lenovo.com/us/en/legal/copytrade/>.

The following terms are trademarks of Lenovo in the United States, other countries, or both:

Lenovo®  
AnyBay®  
Lenovo TruScale®  
Neptune®  
ServerProven®  
System x®  
ThinkAgile®  
ThinkShield®  
ThinkSystem®  
TruScale®  
XClarity®

The following terms are trademarks of other companies:

AMD® is a trademark of Advanced Micro Devices, Inc.

Intel®, the Intel logo, Intel Core®, and Xeon® are trademarks of Intel Corporation or its subsidiaries.

Linux® is the trademark of Linus Torvalds in the U.S. and other countries.

Microsoft®, ActiveX®, Excel®, Hyper-V®, PowerPoint®, PowerShell®, SharePoint®, Windows 11™, Windows Server®, and Windows® are trademarks of Microsoft Corporation in the United States, other countries, or both.

TPC® is a trademark of Transaction Processing Performance Council.

SPEC®, SPEC ACCEL®, SPEC CPU®, SPEC MPI®, SPEC OMP®, SPECchpc™, SPECjbb®, SPECpower®, SPECrate®, and SPECspeed® are trademarks of the Standard Performance Evaluation Corporation (SPEC).

IBM® and IBM Security® are trademarks of IBM in the United States, other countries, or both.

NVIDIA® and NVIDIA RTX® are trademarks of NVIDIA Corporation.

Other company, product, or service names may be trademarks or service marks of others.